



2024 Sustainability Report

ENVIRONMENT, SOCIAL AND GOVERNANCE

Powerchip Semiconductor Manufacturing Corp.



CONTENTS

Message from the Chairman and Sustainable Development Committee Chairman 3

About Powerchip Semiconductor Manufacturing Corporation 4

Company Profile	4
Economic Performance	7
Sustainability Performance and Honors	8

Commitment to Sustainability Management 10

Organizational Structure for Sustainable Development	10
Sustainable Development Strategy	10
Materiality Analysis	13
Stakeholder Communication	21

Appendix

About this Report	139
Sustainability Information Disclosure Guidelines	141
Independent Third Party Verification Statement	154

1 Dedicated to Sustainability | Innovation, Quality and Customers |

1.1 Innovation and Intellectual Property Management	25
Column : Navigator for Intelligent and Innovative Manufacturing	30
1.2 Product Liability and Quality	33
Column : Continuous Improvement Process (CIP) Competition	38
1.3 Customer Relationship Management	40

4 Collaborative Sustainability | Talent |

4.1 Talent Attraction and Retention	85
Column - Pre-employment Internship Program	88
Column - Pre-employment Internship Program	89
4.2 Talent Nurturing and Development	94
4.3 Human Rights	99
4.4 Occupational Safety and Health	100

2 Synergized Sustainability | Supply Chain |

2.1 Supply Chain Overview	44
2.2 Sustainable Supply Chain Management	46
2.3 Conflict Mineral Management	49

5 Integrating Sustainability | Society |

5.1 Social Influence	110
5.2 Social Co-Prosperity Development	111

3 Commitment to Sustainability | Environment |

3.1 Climate Strategy	52
3.2 Energy Management	66
3.3 Water Resources Management	70
3.4 Waste Management	75
3.5 Air Pollution Emissions Management	79
3.6 Biodiversity Management	81

6 Efficient Sustainability | Corporate |

6.1 Corporate Governance	119
6.2 Integrity Management	123
Column - Implement RBA Core Spirits and Fulfill Ethical Management Culture	126
6.3 Regulatory Compliance	127
6.4 Responsible Taxation	128
6.5 Risk Management	129
6.6 Information Security	135



Message from the Chairman and Sustainable Development Committee Chairman

The year 2024 marks a turning point for global sustainable development. From the adoption and implementation of the IFRS Sustainability Disclosure Standards, to policy shifts driven by leadership changes in major global economies, they are all making it so that businesses must transition from past development models to a focused approach on sustainability issues closely tied to their operations. Only by concentrating resources, developing new markets, and creating more collaborative opportunities can companies maintain their competitive edge in an ever-evolving and challenging global landscape.

In its Emissions Gap Report 2024, the United Nations Environment Program (UNEP) emphasized that the world is significantly behind in achieving the Paris Agreement goal of limiting global warming to 1.5°C. In the face of both sluggish progress in global sustainability and the intensifying competition in semiconductor applications, PSMC recognizes that a sustainable semiconductor supply chain and product innovation are no longer merely corporate strategies. Instead, they represent opportunities and risks that impact social prosperity, industry development, and numerous emerging and not-yet-fully-defined issues. The fundamental essence of sustainability lies in navigating challenges by seizing opportunities and mitigating risks.

Under these circumstances, PSMC takes on the responsibility of driving sustainable development and is committed to collaborating with our partners across all ESG (environmental, social, and governance) domains to advance low-carbon transformation. Guided by our core values of integrity, service, quality, and innovation, we strive to contribute even greater value to society. Building on past management efforts and resources, PSMC delivered an outstanding sustainability performance in 2024. We not only achieved leadership-level recognition in the Carbon Disclosure Project (CDP) water security questionnaire, but also ranked 5th globally in the semiconductor industry in the Corporate Sustainability Assessment (CSA). We firmly believe that by continuing to implement our three major implementation strategies – "Low-Carbon Operations & Innovation Development", "Employee Participation & Community-Building", and "Diverse Values & A Friendly Workplace" – ESG principles will become even more deeply embedded in our corporate governance. Through these concrete actions, we demonstrate our unwavering commitment to sustainability. We are fully dedicated to realizing our sustainability vision of "creating a symbiotic, mutually beneficial relationships between business, the environment, and society".

Chairman

Frank Huang

Sustainable Development
Committee Chairman

Brian Shieh

Low-Carbon Operations ♦ Innovative Development

How do we adapt to structural changes in the industry, while simultaneously achieving low-carbon operations? Our solution: Transformation. PSMC has collaborated with the Industrial Technology Research Institute (ITRI) to develop 3D AI Foundry technology, producing 3D AI chips that have won the 2024 R&D 100 Awards. These chips, applied in artificial intelligence inference (AI Inference) systems, have demonstrated data transmission bandwidth 10 times that of traditional AI chips while consuming only about 10% of the power, for exceptionally efficient performance. This innovative technology has also attracted the attention of major international companies. Furthermore, leveraging Logic-DRAM multi-layer wafer stacking technology, we have developed high-bandwidth, high-capacity, low-power consumption 3D AI chips that provide cost-effective, high-performance solutions for large language model AI (LLM AI) applications and AI PCs. These innovations also offer a more energy-efficient choice for power-intensive AI applications. To meet the high-speed transmission demands of GPUs and HBM (high bandwidth memory), we have introduced a high-density capacitor IPD 2.5D Interposer, which has already passed certification by major international companies 2024 and is set to enter mass production at our new Tongluo fab.

Regarding operational innovation strategy, PSMC has officially launched our Fab IP program, leveraging technology transfer in wafer fab construction to assist countries such as India in establishing advanced semiconductor manufacturing capabilities. This initiative aims to drive local innovation, technological advancement, and job creation in developing countries, fostering more balanced development for the global market.

Employee Participation ♦ Community-Building

PSMC remains dedicated to fostering deep collaboration between business and local communities from a long-term perspective. Over the years, through our Public Welfare Committee and collaboration with the Powerchip Cultural Foundation, we have integrated internal/external resources and influence to actively contribute to environmental and ecological protection, community revitalization and engagement, as well as arts and industry promotion. We work hand in hand with employees, suppliers, contractors, and regulatory authorities to drive meaningful impact. In 2024, we launched a local ecological protection project near Tongluo fab, has completed the construction of tiger-friendly facilities,, aimed at minimizing disturbances to native species and their habitats. In addition, we collaborated with the Taiwan Hakka Museum, the Leopard Cat Association of Taiwan, and the Tongluo Farmers' Association. Through a variety of outreach activities, we raise awareness among all stakeholders and promote environmental protection and ecological sustainability.

Diverse Values ♦ A Friendly Workplace

Diversity, Equity, and Inclusion (DEI) are at the core of PSMC's labor and human rights policies. We are committed to fostering a zero-discrimination, diverse, and inclusive workplace free from human rights violations, while enhancing employee well-being through a talent-based approach to workforce development. Since the official launch of the RBA project in early 2021, PSMC has systematically reviewed risk assessments related to labor and human rights, health and safety, environment, business ethics, and management systems. We have steadily worked toward ensuring that all operations comply with established standards. As of 2024, Fabs P1/2, P3, 8A, and 8B have successfully passed the RBA Validated Audit Process (VAP), highlighting our commitment and achievements in these areas. Our goal is to complete re-certification of all fabs by the end of 2025, ensuring that our management systems meet or exceed international standards.

Amid the rapid growth of the global semiconductor market and the increasing emphasis on sustainability, PSMC fully recognizes our vital role in driving technological advancements while ensuring environmental sustainability. We will continue to deepen strategic collaborations and foster innovation to respond to market and societal demands. Upholding our commitment to environmental and social responsibility, we are dedicated to advancing sustainable development; working alongside our supply chain, society, and all stakeholders, we are creating a future of shared prosperity.

About Powerchip Semiconductor Manufacturing Corporation

Company Profile

With the three major foundry services of advanced memory, custom logic ICs, and discrete components as the main pillars, PSMC continues its Open Foundry business model, from wafer design and manufacturing services to equipment and capacity sharing, and works together with our clients to establish a close and flexible cooperation mechanism according to different customers' attributes and needs. As of the report issuance date, PSMC has two 8-inch wafer fab (8A; 8B) and three 12-inch wafer fab (P1; P2; P3; P5) in Taiwan. PSMC adheres to continuous improvement in technology, strict quality control, and efficient manufacturing. The Company continues to promote international cooperation strategies, introduce cutting-edge technology, develop independent technology, and expand into new markets. PSMC is committed to providing specialized wafer foundry services and creating win-wins with customers. The Company also follows the highest international environmental, social, and governance (ESG) standards, establishing itself as a responsible corporate citizen committed to sustainable and profitable operations.

Company Name	Powerchip Semiconductor Manufacturing Corporation
Year of Establishment	2008
Company Headquarters Address	No. 18, Li-Hsin 1st Road, Hsinchu Science Park, Hsinchu, Taiwan
Listing Date / Stock Code	December 6, 2021 / TSE:6770
Main Products	Wafer Fabrication
Chairman	Huang Chongren (Frank Huang)
Number of employees	A total of approximately 8,387 people worldwide
Capital	NT\$41.721 billion
Operating Income	About NT\$44.726 billion of 2024

Note: Fab 8AD ceased production in 2024 and obtained closure approval in January 2025.

Hsinchu Science Park



Zhunan Science Park



Tongluo Science Park



External Public Association Participation

In the rapidly changing semiconductor industry, PSMC continuously improves production technology, and also participates actively in a variety of industry guilds and associations, holding important positions such as chairman, director, and supervisor. This involvement allows PSMC to collaborate with peers in promoting industry environment and policy optimization, thereby precisely staying on top of industry dynamics. PSMC has also taken the initiative to call for the establishment of industry associations in various professional fields to promote cross-industry exchanges and development. The topics of concern that are of interest to the guilds and associations that PSMC participates in include corporate sustainability, technological innovation, environmental sustainability, and climate change. PSMC is also actively involved in proposing policies to the government that cover land, water, electricity, and human resources that are relevant to industrial competitiveness and support environmental sustainability.

Public Affairs Governance

Public Policy Engagement – Climate Policy Alignment

PSMC's climate policy is centered on achieving net-zero emissions by 2050, has established net-zero targets, commitments, and carbon reduction pathways to build a sustainable ecosystem. The Company actively supports national policies and participates in the global low-carbon transition, while ensuring alignment with the goals of the Paris Agreement. PSMC encourages the Company and among our supplier partners to jointly implement energy conservation and carbon reduction. In addition to implementing internal net-zero actions, PSMC has also developed a management system for participating in climate change-related public policy engagement and industry associations. Related departments regularly review and monitor whether the activities and industry associations PSMC participates in align with the Paris Agreement. If any misalignment is identified, PSMC will actively engage in discussions and negotiations to address the issue. If no improvements are made, PSMC will withdraw from the activity or industry association.

Industry Association Management Mechanism

PSMC adheres to the Sustainable Development Best Practice Principles as the highest guiding principle for corporate sustainability. With the vision of "Innovative Technology, Serving Society", the Company is committed to environmental sustainability, social prosperity, and long-term corporate success as part of its mission to contribute to society. We are dedicated to advancing corporate sustainability while ensuring business operations and protecting shareholder interests. PSMC actively participates in industry unions, associations, and related organizations, engaging in close exchanges with the industry, staying informed on public issues, and contributing to the formulation and promotion of industry-related policies.

- **Industry Associations:** PSMC's participation in public affairs and public policy through industry associations follows a structured review process. Related departments first evaluate the association's relevance to the Company, its scale and influence, and whether its advocacy aligns with PSMC's development goals. Upon approval by department heads, the final decision to participate is authorized by the President, who designates the appropriate internal unit to engage with the association and support the advancement of public policy initiatives. If, during participation, PSMC finds that the association or its activities deviate from the Company's original intent or stance, an internal meeting will be convened to discuss the issue, clarify concerns, and express the Company's position. If a misalignment with PSMC's principles is confirmed, the Company will withdraw from the relevant association or activity.
- **When engaging in advocacy activities and industry associations,** related departments are responsible for reviewing and managing different policy discussions to ensure compliance with the Paris Agreement. This is done through meetings or formal approvals, providing timely feedback based on practical considerations and reaching a consensus. The conclusions are then reported to senior management and governance bodies for review.

In 2024, PSMC participated in a total of 24 domestic and international industry associations related to industrial development, among PSMC's position in external associations related to climate change was fully aligned with the Paris Agreement goals, and the Company did not engage in any direct lobbying activities.



Domestic Associations - Industry Development and Technology Innovation

Organization Name	Sustainability Action	Role
Taiwan Semiconductor Industry Association	PSMC participates in the Taiwan Semiconductor Industry Association (TSIA) to show concern for industry development. Through the association's activities and consensus-building with peers, PSMC facilitates cooperative efforts amid competition. This collaboration aims to promote the sound development of the industry as a whole, and to strengthen the overall competitiveness of the industry, with the hope that the Company can maintain its advantage in the ever-changing international environment. In 2023, the chairman of TSIA led the members in publicly to declare the goal of net-zero emissions and disclose the net-zero target of carbon reduction plan to mitigate climate change. The TSIA directors and supervisors unanimously approved the TSIA Carbon Reduction Pathway and held a declaration ceremony. This initiative aims to encourage the industry and relevant units to cooperate with national policies in reducing carbon emissions by reducing greenhouse gas emissions, promoting green energy industries, and developing energy-saving and carbon-reducing technologies in order to achieve the goals of the Paris Agreement.	Executive Director
The Allied Association for Science Park Industries	PSMC participates in the Allied Association for Science Park Industries, to broaden the communication with those in the industry and manufacturers of other industries in the Science Park, to establish an effective communication mechanism with the government, and to protect the common rights and interests of the manufacturers through the organization of the association. PSMC participated in the White Paper on industrial development strategy proposed by the association for Taiwan's science park. In response to the risks posed by climate change leading to extreme weather events and the industry's challenges with energy and water resource shortages, the Hsinchu Science Park Bureau collaborated across government departments. They utilized green technology to drive a circular economy, encouraged companies to develop energy-saving and low-carbon measures, and collaborated publicly and privately to construct a smart green ecological science park. This effort aimed to optimize the entrepreneurial and sustainable environment within the park and lead Taiwan's industries in accelerating the achievement of net-zero carbon emissions goals.	Executive Director
Taiwan Internet of Things Technology and Industry Association	By joining the Taiwan IOT Technology and Industry Association (TwIoTA) and taking on management roles, PSMC gains decisive influence in setting standards and applications in the IoT industry. This ensures that the Company's existing technologies can be widely applied across a variety of sectors within the IoT industry, further enhancing PSMC's core competitiveness in the application of emerging industries. In response to the global trend toward achieving net-zero emissions by 2050, and to comply with ESG and green supply chain standards, industries are transitioning their energy sources. The development of new and renewable energy sources has become an international consensus, combining new energy technologies with industry applications and demands. This not only enhances the energy resilience of companies but also reduces carbon emissions through a variety of measures and technologies. Making new energy applications a new driving force for Taiwan's industrial development, therefore, in November 2023, representatives from five major associations including the TwIoTA, TADA, THFCP, TBA, and Taiwan Green Energy Association, jointly signed MOU. With the assistance of the Taipei Computer Business Association, we jointly organized the "Prospects for the Application and Development of New Energy in Taiwan" forum to discuss in depth the cross-domain application opportunities of new energy in Taiwan's industrial development, in preparation for the future trend of ESG and carbon reduction.	Executive Director Director Supervisor
Taiwan Advanced Automotive Technology Development Association	Through the participation in the Taiwan Advanced Automotive Technology Development Association (TAATDA), PSMC will be able to communicate with the automotive industry and integrate their knowledge areas, which will enable the company's technology development to connect to the considerable EV and smart car business opportunities in the future. In the face of the automotive industry's carbon reduction regulations, the expanded development of electric vehicles, including the need for a large number of semiconductors and automotive electronic modules, are increasing. PSMC's participation in the association will enable the electronics industry and automotive industry to move into the advanced automotive field together and transform the automotive industry into a smart one, deepening the cooperation between the two industries and establishing a complete supply chain for the electric vehicle industry in line with the country's future goal of net zero sustainability.	Chair Supervisor

Domestic Associations - ESG Interactive Exchange

Organization Name	Role	Organization Name	Role
Taipei Computer Association	Executive Director	Computer Audit Association	Member
Taiwan IC Industry & Academia Research Alliance	Executive Director	Taiwan Chemical Engineering Association	Member
Friends of the Police Association of R.O.C	Director	Taiwan Stock Affairs Association	Member
RISC-V Taiwan Alliance	Member	The General Assembly of Chinese Culture	Member
Cross-Strait CEO Summit	Member	Internal Audit Association of the Republic of China	Member
Taiwan Environmental Management Association	Member	Taiwan India Business Association	Member
PM2.5 Control Association	Member	Miaoli County Nurses Association	Member
Hsinchu City Human Resource Management Association (HCHRMA)	Member	Hsinchu City Nurses Association	Member

International Associations - Mastering International Trends

Organization Name	Role
Global Semiconductor Alliance	Member
JEDEC Solid State Technology Association (JC-42)	Committee Member
UCIe (Universal Chiplet Interconnect Express)	Adopter Member
ISIG (International Semiconductor Industry Group)	Associate Membership

Economic Performance

In 2024, Powerchip Semiconductor Manufacturing Corporation (below, "PSMC") achieved total revenue of NT\$44.7 billion, for 1.6% growth over the previous year. However, the Company faced challenges from aggressive capacity expansion and price competition by Chinese semiconductor manufacturers in the mature process foundry market, as well as depreciation expenses from the newly established Tongluo fab, which has not yet reached economies of scale. As a result, PSMC recorded a net loss of NT\$6.7 billion for the year.

To avoid engaging in low-price competition with Chinese competitors, PSMC has actively transitioned to AI-related product lines. Since 2019, PSMC has been developing Wafer on Wafer (WoW) 3D stacking technology, which integrates logic processes with memory to provide high-bandwidth and low-power edge AI solutions. The development of PSMC's WoW 3D stacking technology has been completed and has enjoyed adoption and trial production by multiple major international companies.

Looking ahead to 2025, as the global semiconductor supply chain restructuring progresses and 5G networks, the Internet of Things (IoT), artificial intelligence (AI), and autonomous driving technologies continue to advance rapidly, the global semiconductor market is expected to maintain steady growth in the coming years. The widespread adoption of these technologies will drive increasing demand

for semiconductor products, particularly in high-performance computing, data centers (cloud), and smart devices (edge). PSMC is committed to seizing these market opportunities by continuously advancing R&D innovation and strengthening collaboration with customers to deliver more efficient and high-quality semiconductor solutions.

Despite the financial performance in 2024 falling short of expectations due to market demand fluctuations, PSMC continues to adhere its belief in environmental friendliness and giving back to society. Through sound corporate governance, PSMC seeks to balance the interests of stakeholders, fulfill its commitment to sustainable development, and adhere to the values of integrity, service, quality, and innovation in its corporate culture. PSMC aims to achieve stable profitability and set a benchmark for ESG sustainability by fostering relationships of coexistence with customers, society, and the environment.

Please refer to the Company's financial report for detailed financial data:
https://www.powerchip.com/en-global/financials/quarterly?target=target_title



Unit: NT\$ thousands

Category	Item	2021	2022	2023	2024
Economic Value Generated	Operating income	65,622,945	76,086,619	44,022,552	44,725,710
Economic Value Distributed	Operating Costs	38,038,331	40,507,654	38,730,581	44,200,285
	Employee Salaries and Benefits (Operating Expenses)	5,253,129	6,058,260	4,202,513	4,340,611
	Distribution of dividends to shareholders	3,739,245	7,124,836	1,216,676	0
	Interest on Loans	454,720	415,151	386,024	1,333,147
	Payments to the government	3,122,441	5,538,058	0	0
	Total	50,607,866	59,443,959	44,535,794	49,874,043

Sustainability Performance and Honors

S&P Global 2024 Sustainability Yearbook

PSMC was ranked **5th** globally in the **semiconductor industry** and listed in the **S&P Global Sustainability Yearbook** for two consecutive years

S&P Global

Taiwan Corporate Sustainability Awards (TCSA)

Won the **Platinum Award** of the Sustainability Report Award.
Won the "Single Performance - Workplace **Welfare Leadership Award**: Manufacturing and Energy Industry"

Corporate Governance Evaluation

In the 11th Corporate Governance Evaluation (2024), PSMC ranks in the **top 5%** of listed companies

 臺灣證券交易所

 **PSMC**

 **TCSA**

CDP Evaluation

Participated in the Water Security and Climate Change questionnaire assessments. The water security questionnaire received an **A** leadership grade, and the climate change questionnaire received a **B** management grade

 **CDP**
DISCLOSURE INSIGHT ACTION

Ministry of Environment 6th National Enterprise Environmental Protection Awards Silver Award (Feb 8B)



Ministry of Environment Outstanding Award for Regional Joint Prevention Organization Operations (P3 Feb)



Ministry of Environment 2023 Green Procurement Outstanding Organization Award



Hsinchu Science Park Business Waste Reduction and Circular Economic Achievements Excellence Award (Feb 8B)



Hsinchu Science Park Outstanding Occupational Safety and Health Organization Excellence Award (Feb P3)



Hsinchu Science Park Outstanding Personnel Award, Occupational Safety and Health Management Category (Feb 8B)
Hsinchu Science Park Outstanding Personnel Award, Health Management Category (Feb P3)



Hsinchu Science Park Outstanding Environmental Protection Officer Award, Toxic Chemical Substances Management Category (Feb P1/2)
Hsinchu Science Park Outstanding Environmental Protection Officer Award, Waste Management Category (Feb 8A)



Supported the Emergency Response and Safety Protection Capability Enhancement Program for Enterprises (Feb 8A)



Ministry of Economic Affairs' Industrial Development Administration, Certified Clean Production Assessment System (Feb P3)



Ministry of Health and Welfare's Health Promotion Administration, Outstanding Healthy Workplace and Excellent Promoter, Health Management Award



Hsinchu City, 2023 Outstanding Green Procurement Organization Award



Hsinchu City Certificate of Appreciation for adopting Nanliao Ecological Park



Ministry of Labor Certificate of Appreciation for active participation in Taiwan Occupational Safety and Health Management System North District Promotion Association activities



Miaoli County Workplace Health Screening Outstanding Implementer Award



Miaoli County Workplace Health Screening Workplace Implementation Award



Miaoli County Workplace Employee Health Health Innovation Award



1111 Job Bank 2024 Happy Business Awards Silver Award



Certified by CommonWealth Magazine as meeting the Paris Agreement's 1.5°C temperature target through the Temperature Rising Index for Pathways (TRIPs) platform



Commitment to Sustainability Management

Organizational Structure for Sustainable Development

PSMC Sustainable Governance Structure

Highest Governance

Board of Directors

Supervisory Unit

Audit Committee

Meeting Frequency: Regularly held every year
Work Items:
• Assists Board of Directors in fulfilling risk management responsibilities

Assessment Unit

Sustainable Development Committee

Chairman: Vice Chairman
Vice Chairman: Vice President & Chief Financial Officer (CFO)

Meeting Frequency: Held annually
Work Items:
• Formulate countermeasures and review
• Assist/Guide all departments
• Regularly report to Audit Committee and Board of Directors

To practice corporate sustainable development and promote progress in the economy, environment, and society, beginning in 2022, PSMC established the Sustainable Development Committee as dedicated departments for driving sustainable development. In order to deepen supervision and review responsibilities and enhance the effectiveness of sustainable governance, the Audit Committee assist the Board of Directors in supervising the Company's sustainable development initiatives.

The Sustainable Development Committee oversees three major implementation teams in environmental, social, governance (ESG). The Chairperson is appointed by the Chairman of the Board. In addition to the corporate governance officer, who serves as an ex officio member, the Chairperson selects vice president-level or higher personnel as committee members based on ESG implementation needs and appoints related department heads to form the committee. These teams report annually to the Audit Committee and the Board of Directors on sustainable development goals, execution effectiveness, and plans for the next year. In 2024, the Sustainable Development Committee held four meetings; they also submitted reports to the Board of Directors on May 21, 2024, and December 17, 2024 respectively, detailing implementation status. There are five key and important proposals to be communicated with the highest governance unit, the Board of Directors, including stakeholder negotiation, corporate governance, sustainability report compilation, carbon reduction methods and response measures, and material topics identified through materiality assessments.

Sustainable Development Strategy

Sustainable Development Commitment

PSMC continued to implement three major implementation guidelines set by the Sustainable Development Committee, focusing on strengthening corporate governance, pursuing a balanced interest of stakeholders, implementing environmental protection and energy conservation efforts, and contributing to society.

Vision

- Innovation in Technology
- Service to Society

Mission

- Environmental Sustainability
- Social Co-Prosperity
- Corporate Perpetuity

Corporate Culture



Integrity



Service



Quality



Innovation

Implementation Guidelines

With the goal of low carbon, green energy and net zero emissions, we will continue to cooperate with industry, government and academia, explore new knowledge and new technologies, refine green operations, and share experiences and results with the supply chain through mutual encouragement

Encourage employees to participate and make good use of corporate resources to support education in remote areas, sponsor cultural activities, assist the weak and vulnerable, and participate in community building

Recognize diversified values, build a friendly workplace, and create a win-win situation with stakeholders by being fair, open, and law-abiding

Corresponding SDGs

SUSTAINABLE
DEVELOPMENT
GOALS



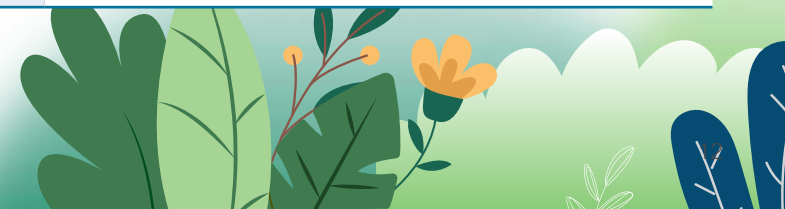
Sustainable Development Goals

Goals	Corresponding Chapters	Action and Effectiveness
Goal 3 	Collaborative Sustainability · Talent	Based on the annual health examination results of colleagues, we promote a series of health promotion activities including ultrasound examinations, vaccinations, low-calorie healthy meals and health professional consultations, and set up a Powerchip clinic in the factory to provide 24-hour services of professional health and safety care to all employees and non-employees. In addition, to promptly address employees' needs and guide them to appropriate resources, employees are entitled to five free psychological counseling sessions per year (SDG 3.4 achieved)
	Commitment to Sustainability · Environment	- In 2024, electronic information boards and PA systems were installed, and dedicated control software was developed to automatically play promotional messages and warning announcements at scheduled times. According to survey results, 95% of employees stated that the newly installed information boards and broadcast systems effectively provide real-time reminders and warnings while driving, helping to reduce accident rates and establish a culture of traffic safety (SDG 3.6 achieved) - Successfully passed ISO 14001 Environmental Management System certification and continued to employ the best feasible air pollution emission management technologies (SDG 3.9 achieved)
Goal 4 	Collaborative Sustainability · Talent	Collaborated with domestic universities to implement the Pre-Employment Internship Program for Current Students, providing semester and summer internship opportunities for graduating students interested in joining the Company. In 2024, a total of 29 interns participated, with 27 of them obtaining pre-employment qualifications. This program allows students to pre-familiarize themselves with PSMC's work environment and team culture, gain an understanding of wafer fab operations, and receive relevant practical training (SDG 4.4 achieved)
Goal 5 	Collaborative Sustainability · Talent	- Due to the characteristics of the semiconductor industry, management and professional engineering personnel are predominantly male, and technical and administrative personnel are mostly female. However, PSMC has set a goal for female supervisors (including junior, mid-level, and senior management positions) to reach a 30% ratio by 2025 (the target for junior management has already been achieved at 39.87% in 2024) (SDG 5.5 achieved) - The talent recruitment and compensation systems adhere to principles of fairness and reasonableness, ensuring equal treatment for all employees. In addition, 100% of employees who have served for more than three months during the 2024 performance evaluation period were required to undergo regular assessments (SDG 5.5 achieved)

Goals	Corresponding Chapters	Action and Effectiveness
Goal 6 	Commitment to Sustainability · Environment	- Continuing our commitment to technological improvements and equipment investments, PSMC has increased water recovery rates and process water efficiency. We became the first semiconductor company in the Hsinchu Science Park to commit to a process water recovery rate of 85% or better. In 2024, the water recovery rates across all fabs reached 88% (SDG 6.4 achieved) - Since the official production launch of Fab P5 in 2024, it has utilized 600 cubic meters of reclaimed water per day (SDG 6.b achieved) - PSMC participated in 40 water-saving guidance projects for suppliers, achieving a cumulative water-saving result of 1.3862 million metric tons. In 2024, the suppliers receiving guidance are expected to save 382,000 metric tons of water annually. Additionally, a follow-up on the 10 suppliers that participated in the 2023 water-saving guidance program showed a water-saving volume of c. 341,000 metric tons, for an achievement rate of 83.7% (SDG 6.5 achieved) - The manufacturing process wastewater is pre-treated in the wastewater treatment plant, then discharged to the wastewater treatment plant in the park for subsequent treatment and discharge, and divide the discharge from the manufacturing process into a total of 20 different drainage water systems, such as recyclable and non-recyclable, according to their different characteristics, such as type, concentration, and conductivity. In addition to increasing the recovery rate of water, some waste acids and organic waste liquids have economic value for recycling, so separate diversion will not only reduce the amount of treatment dosage in wastewater plants, but also reduce the difficulty and environmental burden of waste treatment at the back end (SDG 6.3, SDG 6.4 achieved)
Goal 7 	Dedicated to Sustainability · Innovation, Quality and Customers	- The development of 24nm NAND Flash and 1.2V NOR Flash technology platforms has significantly reduced the number of photomasks by 16% and the total number of processes by 9% while continuing to increase output per unit area by 14%, effectively reducing production energy consumption and meeting the future low-power product market needs (SDG 7.3 achieved) - Developed at both the 8-inch and 12-inch wafer fabs, the separated MOSFET, Bipolar-CMOS-DMOS (BCD), and Ultra-High Voltage (UHV) technologies have enabled the end product applications to stabilize power supply and reduce energy loss (SDG 7.3 achieved) - Launched IGBT products for applications in electric vehicles and variable-frequency home appliances, serving as an essential component in energy-saving products. In 2024, PSMC assisted customers in reducing approximately 31.22 million metric tons of CO₂e greenhouse gas emissions, accounting for 3.4% of the Company's total revenue (SDG 7.3 achieved)
	Commitment to Sustainability · Environment	The implement of the ISO 50001 energy management system to analyze the efficiency of energy consumption through energy management, seeks the effective use of energy (SDG 7.3 achieved)

Goals	Corresponding Chapters	Action and Effectiveness
Goal 8 	Dedicated to Sustainability · Innovation, Quality and Customers	Developed wafer stacking technology to create a 3D AI accelerator that integrates logic and memory, significantly enhancing computational efficiency while reducing power consumption. This provides the AI industry with a high-performance, flexible, and cost-effective alternative solution, earning the 2024 R&D 100 Award (SDG 8.2 achieved)
	Collaborative Sustainability · Talent	- As one of the few companies fully participating in and funding all four major semiconductor colleges in Taiwan, PSMC collaborated on 52 research projects with these colleges in 2024, investing nearly NT\$60 million. Total scholarships we awarded reached NT\$3.24 million (SDG 8.6 achieved) - PSMC has implemented ISO 45001 Occupational Health and Safety Management Systems, and formulated the Environmental Safety and Health Management Manual to effectively manage internal and external issues related to activities, products and services, as well as the operation of environmental safety and health at the plant site (SDG 8.8 achieved)
Goal 9 	Dedicated to Sustainability · Innovation, Quality and Customers	The IGBT products launched by PSMC are produced on 0.4μm 8-inch wafers, which are applied in electric vehicles and variable-frequency home appliances. They are essential components in energy-saving products. In 2024, they contributed to reducing greenhouse gas emissions by 31.22 billion kg CO₂e collectively for customers (SDG 9.4 achieved)
Goal 11 	Commitment to Sustainability · Environment	- Regarding the management philosophy of introducing waste management into the lifecycle, we will continue to promote the following objectives: (1) source reduction, (2) end point waste reduction, (3) enhancing the value of waste reuse, and (4) proper disposal and the tracking of waste flow (SDG 11.6 achieved) - To reduce the risk of traffic accidents during employee commutes, electronic information boards and PA systems were installed at fab area entrances and exits to enhance alertness during peak commuting hours (SDG 11.2 achieved)
Goal 12 	Commitment to Sustainability · Environment	The waste generated is classified by type and disposal method, and is cleaned up and reused by qualified waste vendors; waste cleaning is strictly required to comply with waste cleaning laws and international regulations, and waste generated on site is attended to at all times and evaluated for the appropriateness in the waste cleaning market (SDG 12.5 achieved)
	Dedicated to Sustainability · Innovation, Quality, and Customers	To understand the environmental impacts of semiconductor foundry products across raw material acquisition and manufacturing stages, identify environmental impact hotspots at each stage, facilitate performance improvement strategies, and communicate with customers, a comprehensive product lifecycle assessment mechanism was implemented in 2023 (SDG 12.2 achieving)

Goals	Corresponding Chapters	Action and Effectiveness
Goal 12 	Synergized Sustainability · Supply Chain	- The key suppliers have signed 100% of the Responsible Business Alliance's RBA pledge to not accept metals from conflict mined areas and to require raw material suppliers to fulfill their social and environmental responsibilities (SDG 12.7 achieved) - In 2024, the total amount of green procurement reached NT\$200 million, with priority given to evaluating and adopting products and services that have domestic and international environmental protection/energy-saving labels and certifications. The proportion of local suppliers reached approximately 97% in 2024, maintaining a steadily increasing trend year by year (SDG 12.b achieved) - A Supplier Code of Conduct statement has been formulated to require suppliers to truly abide by business conduct in various aspects such as labor, health and safety, environment, business ethics and management systems. By 2024, 100% of qualified and active suppliers, as well as new suppliers, had completed signing the Supplier Code of Conduct (SDG 12.4 achieved)
Goal 13 	Commitment to Sustainability · Environment	To ensure the ability to resume normal operations within the shortest possible time in the event of a major accident or crisis, PSMC has established business continuity plans for specific crisis events such as prolonged power supply interruptions or disruptions in raw material supply (SDG 13.1 achieved)
Goal 16 	Competent Sustainability · Corporate Governance	- The Company has established the "Code of Conduct for Credible Management" and the "Procedures and Conduct Guidelines for Credible Management", which explicitly provide for the recusal of conflicts of interest, and the relevant recusal is recorded in the minutes of the meetings (SDG 16.5 achieved) - The Board of Directors is the highest corporate governance unit of the Company, the all directors are elected by a vote of all shareholders (SDG 16.6 achieved) - PSMC has established the Whistleblower Protection and Retaliation Management Procedures to safeguard the rights of employees, suppliers, and other internal and external whistleblowers to exercise their whistleblowing rights in accordance with the law (SDG 16.6 achieved)
Goal 17 	Integrating Sustainability · Society	PSMC has established three major social co-prosperity strategies: Environmental Conservation, Public Welfare Investment, and Arts & Cultural Festivals, supporting social welfare activities through donations, sponsorships, and strategic partnerships. The Company contributed 452 volunteers, totaling 1,511 hours (SDG 17.17 achieved)

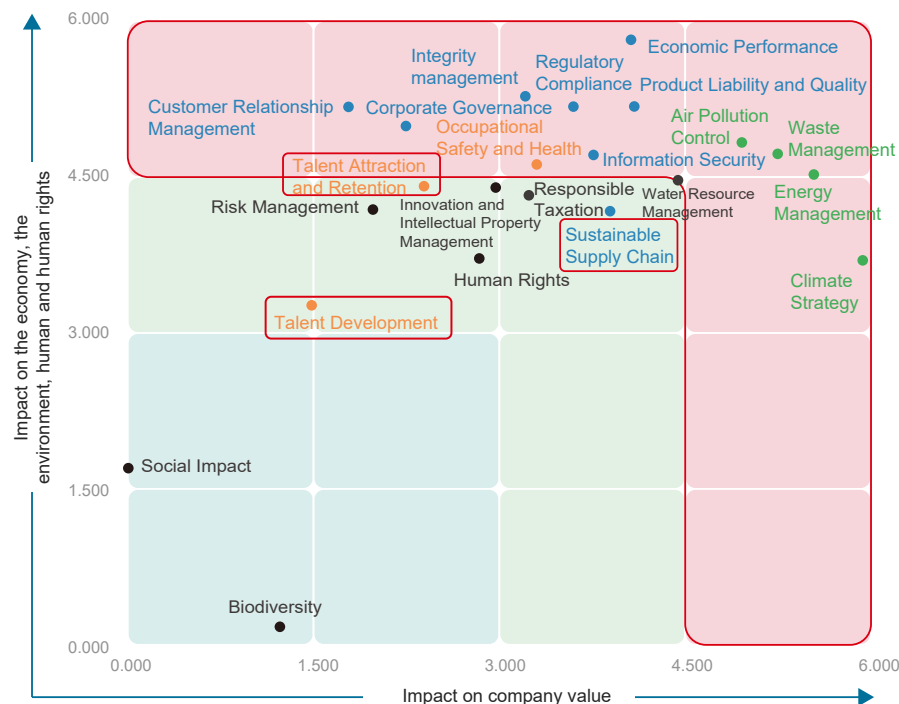


Materiality Analysis

Every year, PSMC regularly uses the "Double Materiality" principle proposed by the European Union to conduct materiality analysis of sustainability issues, and follows the four-stage materiality identification process of GRI Standards (2021), and invites internal senior management and external stakeholders to evaluate the actual and potential positive and negative impact of sustainability issues, using "impact level" as the evaluation principle. After comprehensively assessing the impact on "company value" and "on the economy, environment, people and human rights", a total of 15 major themes were identified in 2024. In addition, PSMC also incorporates the results of materiality assessments and sustainability issues that have a significant impact on the company's operations into the company's corporate risk management processes and systems, and reviews them regularly.



Matrix of Material Topics Analysis for 2024



Note: After a comprehensive assessment of sustainability issues and their relevance to PSMC's operations by the Sustainable Development Committee, the decision was to include "Sustainable Supply Chain", "Talent Attraction and Retention", and "Talent Development" as material topics for this year.

Material Topics for 2024

Environmental Sustainability

- Energy Management
- Waste Management
- Air Pollution Control
- Climate Strategy

Social Co-prosperity

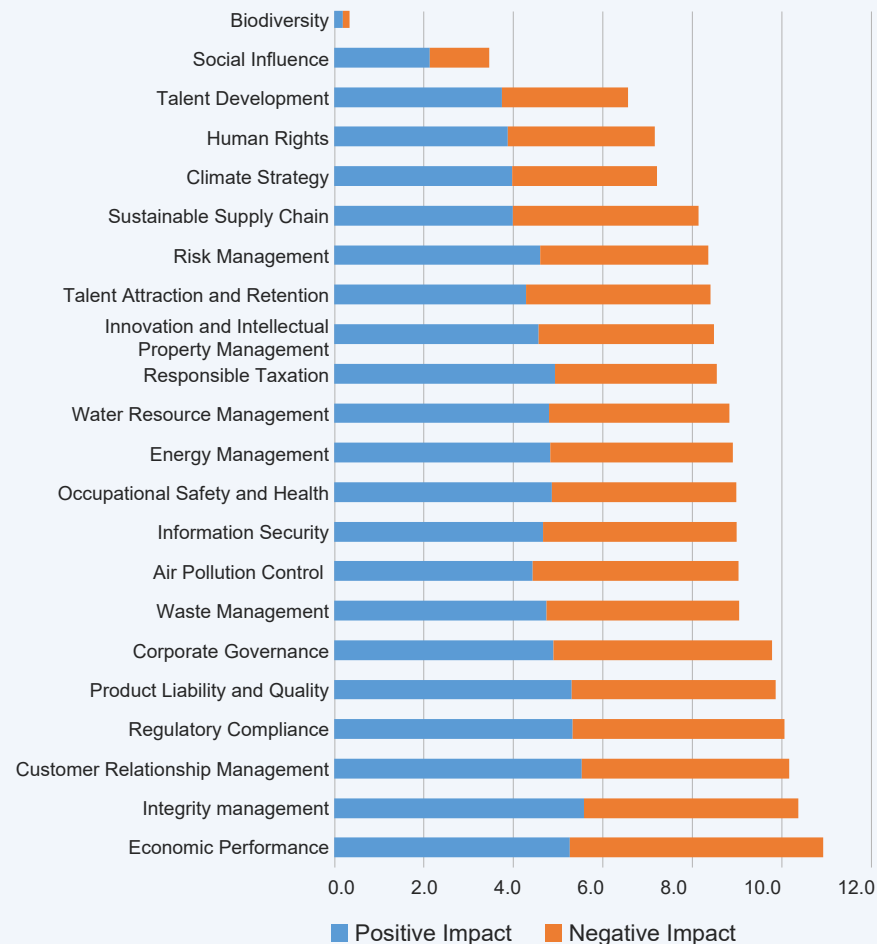
- Occupational Safety and Health
- Talent Attraction and Retention
- Talent Development

Corporate Prosperity

- Economic Performance
- Product Liability and Quality
- Regulatory Compliance
- Information Security
- Ethical management
- Corporate Governance
- Customer Relationship Management
- Sustainable Supply Chain

When assessing the positive and negative impact of sustainability issues, the impact indicators of sustainability issues on PSMC's operating income, operating costs, supply chain management, corporate image, goodwill, and legal liability will be taken into consideration. The comprehensive assessments on the management of sustainability issues will have positive and negative impacts on PSMC's operating value and ESG development, which serves as an important basis for PSMC's issue management.

Ranking of impact of sustainability issues



Management of Material Topic Impact

● : Direct impact ▲ : Indirect impact

Material Topics Categories	Material Topics	Positive, Negative Impacts Statement	Corresponding GRI theme standards, SASB Standards, and TCFD Framework	Value Chain Impact Hotspots			Management results in 2024 (please refer to the following section)
				Upstream	PSMC Operation	Downstream	
				Suppliers / Contractors	PSMC and consolidated financial reporting subsidiaries	Customers	
Environmental Sustainability	Energy Management	Positive impact: Examine corporate energy policies, management practices, goals and achievements. Effective energy management will help the company reduce carbon emission. Negative impact: The company may face high electricity and carbon emission costs, which will affect the amount of investment and the number of customers.	GRI 302: Energy TC-SC-130a.1: (1) Total energy consumed (2) Proportion of grid electricity (3) Proportion of renewable energy		●	▲	3.2 Energy Management
	Waste Management	Positive impact: Review the amount of waste generated from business operations, disposal methods, and recycling practices, and establish waste management goals and measures. Negative impact: The company may face regulatory violations and penalties, increasing external and environmental cost risks while also weakening product competitiveness in sustainability-related issues.	GRI 306: Waste Material TC-SC-150a.1: (1) Hazardous waste generated during the manufacturing process weight (2) Proportion recycled		●	▲	3.4 Waste Management
	Air Pollution Control	Positive impact: Assess potential regulatory violation risks faced by the company, proactively formulate prevention plans, and install or upgrade equipment accordingly. Negative impact: The company may exceed regulatory standards for pollutant emissions, leading to penalties and negatively impacting its reputation.	GRI 305: Emissions		●	▲	3.5 Air Pollution Emissions Management
	Climate Strategy	Positive impact: The company should establish a climate change governance structure and carbon reduction-related strategies, assess climate-related risks and opportunities, and formulate corresponding strategies, and reduce the impact of climate change on business while exploring new business opportunities to increase corporate revenue. Negative impact: Facing the risks and challenges brought about by climate change, if mitigation measures are not formulated, it will have a significant impact on the overall operation.	GRI 201: Economic Performance GRI 305: Emissions TC-SC-110a.1: (1) Gross Global Scope 1 emissions (2) Amount of total emissions from perfluorinated compounds TC-SC-110a.2: Discussion of long- and short-term strategy or plan to manage scope 1 emissions, emissions reduction targets and an analysis of performance against those targets TCFD Governance, Strategy, Risk Management, Metrics and Targets		●	●	3.1 Climate Strategy

● : Direct impact ▲ : Indirect impact

Material Topics Categories	Material Topics	Positive, Negative Impacts Statement	Corresponding GRI theme standards, SASB Standards, and TCFD Framework	Value Chain Impact Hotspots			Management results in 2024 (please refer to the following section)
				Upstream	PSMC Operation	Downstream	
				Suppliers / Contractors	PSMC and consolidated financial reporting subsidiaries	Customers	
Social Co-Prosperity	Occupational Safety and Health	Positive impact: Healthy and safe working conditions can prevent physical and mental injuries, promote the health of workers and maintain the health and safety of colleagues. Negative impact: The company lacks employee physical and mental health care and related services, resulting in employee turnover. Failure of the Company to safeguard the health and safety of its employees may also lead to occupational disasters.	GRI 403: Occupational Health and Safety TC-SC-320a.1: Description of efforts to assess, monitor, and reduce exposure of employees to human health hazards TC-SC-320a.2: Total amount of monetary losses as a result of legal proceedings associated with employee safety and health violations		●		4.4 Occupational Safety and Health
	Talent Attraction and Retention	Positive impact: In response to changes in the employment environment, continuous adjustments in talent recruitment and employee salaries, etc., it can attract more talent to join the company and increase retention rates. Negative impact: If the company does not continuously adjust salaries and benefits, it will affect talent recruitment decisions, and the company will be unable to recruit outstanding talents and reduce retention rates.	GRI 201: Economic Performance GRI 401: Employment GRI 405: Diversity and Equal Opportunity TC-SC-330a.1: Percentage of employees requiring a work visa		●		4.1 Talent Attraction and Retention
	Talent Development	Positive impact: Planning comprehensive foundational training for talent and developing their specialized skills, as well as providing assistance for employee career development. Negative impact: If relevant cultivation or training is not provided, the company will face the problem of declining retention rates, which will affect the operations.	GRI 402: Labor/Management Relations GRI 404: Training and Education		●		4.2 Talent Training and Education
Corporate Prosperity	Economic Performance	Positive impact: Good economic performance steadily can create values for the company through the growth of operating income and the continuous improvement of profitability. Negative impact: The financial difficulties in a company can lead to operational disruptions.	GRI 201: Economic Performance	●	●	▲	About PSMC
	Product Liability and Quality	Positive impact: The company should manage and supervise product quality to ensure that customers receive the best quality products. Negative impact: The quality of defective product will affect the company's reputation. Losing customer trust will impact the company's revenue and market share.	GRI 416: Customer Health and Safety TC-SC-410a.1: Proportion of products by revenue that contain IEC 62474-declarable substances TC-SC-410a.2: Processor energy efficiency at system level for: (1) servers, (2) desktops, and (3) laptops		●	●	1.2 Product Liability and Quality
	Regulatory Compliance	Positive impact: Compliance with government regulations, policies and procedures, and the requirement for all employees to comply with all business-related laws and regulations. Negative impact: Violation can lead to penalties and affect the reputation of the company.	GRI 205: Anti-corruption GRI 206: Anti-competitive behavior Topic TC-SC-520a.1: Total monetary losses as a result of legal proceedings associated with anticompetitive behavior regulations		●	▲	6.3 Regulatory Compliance

● : Direct impact ▲ : Indirect impact

Material Topics Categories	Material Topics	Positive, Negative Impacts Statement	Corresponding GRI theme standards, SASB Standards, and TCFD Framework	Value Chain Impact Hotspots			Management results in 2024 (please refer to the following section)
				Upstream	PSMC Operation	Downstream	
				Suppliers / Contractors	PSMC and consolidated financial reporting subsidiaries	Customers	
Corporate Posterity	Information Security	Positive impact: Establishing information security mechanisms, protective measures, and emergency response procedures ensures the security of operational information and protects assets. Negative impact: DThe personal information of enterprises, customers or suppliers will not be protected and the risk of information leakage will be increased, thus affecting the reputation of enterprises.	GRI 418: Customer Privacy	●	●	●	6.6 Information Security
	Ethical management	Positive impact: The company demonstrates the commitment to integrity and responsible governance management strategies. Negative impact: Dishonest and illegal behavior will face penalties and will also affect the company's reputation.	GRI 205: Anti-Corruption GRI 206: Anti-Competitive Behavior	●	●	▲	2.1 Supply Chain Overview 6.2 Ethical management 6.3 Regulatory Compliance
	Corporate Governance	Positive impact: Establishing an effective corporate governance framework will maintain the effective operation of the Board of Directors and all functional committees, strengthen corporate governance, and protect shareholders' rights. Negative impact: The rights and interests of shareholders, investors and stakeholders will not be protected and the competitiveness of the enterprise will be reduced.	GRI 405: Diversity and Equal Opportunity		●	▲	6.1 Corporate Governance
	Customer Relationship Management	Positive impact: Communicating services, product information and service markets with customers, understanding their needs, and establishing good relationships will bring positive economic performance to the company. Negative impact: Customer loyalty will be significantly reduced, which will damage the company's reputation and further affect the company's revenue.	Self-Declared Topic		●	●	1.3 Customer Relationship Management
	Sustainable Supply Chain	Positive impact: Establishing supply procurement policies and mechanisms with suppliers, the comprehensive supply chain policies and audits will effectively ensure product quality. Negative impact: If the company fails to establish procurement-related policies, it will not be able to guarantee the quality of purchased products, which will affect the company's reputation and cause the company to face financial losses.	GRI 204: Procurement Practices GRI 205: Anti-corruption GRI 308: Supplier Environmental Assessment GRI 408: Child Labor GRI 409: Forced or Compulsory Labor GRI 414: Supplier Social Assessment TC-SC-440a.1: Description of management for risks associated with the use of critical materials	●	●		2.1 Supply Chain Overview 2.2 Sustainable Supply Chain Management

Note: In response to the requirements of the GRI Standards (2021) for investigating and analyzing material topics, two new material topics, "Waste Management" and "Air Pollution Control", have been added as compared to the 2023 report.

Material Topics Development Objectives

✓ Indicates that the 2024 target has been achieved

✗ Indicates that the 2024 target was not achieved and formulated relevant improvement plans

Aspect	Material Topic	Policy and Management	Action Plan	2024 Target	2024 Implementation results	Execution Status	2025 Short-Term Objectives	2027 Mid-Term Objectives	2030 Long-Term Objectives
Environmental Sustainability	Energy Management	Introduce ISO management system, improve energy efficiency through PDCA model, and set up a green energy team to plan the introduction of renewable energy.	Reduction in electricity consumption per unit wafer area or improve the average annual electricity saving rate (%; kWh/m ² ;GWH) (base year 2015)	14% (8,864 kWh/m ²)	-22% (12,573 kWh/m ²)	✓	15%	17%	20%
				99 GWH	122 GWH		110 GWH	132 GWH	160 GWH
			Improve the average annual electricity saving rate (%)	≥ 1	1.4	✓	≥ 1.5	≥ 1.5	≥ 1.5
			Install renewable energy facilities/purchase green electricity ratio (%)	>8	>8	✓	>8	-	≥ 30
			Performance Review: The reason for not achieving the target of reduction in electricity consumption per unit wafer area is that production capacity has not yet reached full utilization, and the launch of mass production at Fab P5 has led to an increase in electricity consumption. Although energy-saving measures have been continuously promoted to reduce base load electricity consumption, adjustments for air pollution improvement and new processes have made it impossible to achieve the targeted reduction in electricity consumption per unit. Continuous monitoring and attention will be maintained moving forward.						
	Waste Policy and Management	With the goals of increasing waste recycling rates and reducing landfill rates, a multi-layered management strategy and vendor guidance program has been implemented, incorporating ISO evaluations to minimize environmental impact to the greatest extent possible.	Waste recycling rate (%)	This is a newly added material topic for 2024			≥92	≥93	≥94
			Waste landfill rate (%)				≤0.5	≤0.4	≤0.3
			Proportion of on-site audits and guidance for key waste disposal vendors (%)				100	100	100
			Guidance for waste disposal vendors to obtain ISO 14001 certification or complete compliance evaluations (%)				≥60	≥70	≥80
			Waste reduction projects (projects /year)				40	45	50
	Air Pollution Prevention	Establish target indices for air pollution, with VOCs (volatile organic compounds) as the primary observation category, ensuring effective control of air pollutant emissions during operations and minimizing potential environmental impact.	Abnormal emission incidents (incidents)	This is a newly added material topic for 2024			<1	<1	<1
			Efficiency of VOCs treatment facilities across the company (%)				>94	>95	>96
			VOCs emissions per unit product area (%) (base year: 2015)				reduction 28	reduction 29	reduction 30
			Cumulative reduction of total air pollutant emissions (tons ;%)				5.86 tons (15%)	6.65 tons (17%)	9.78 tons (25%)
	Climate Strategy	Establish a "Sustainable Development Committee" that is responsible for formulating climate change strategies and reviewing the compliance and appropriateness of the strategies and objectives. Establish a "Risk Management Committee" that is responsible for identifying and analyzing the organization's climate related risks, and effectively managing the potential risks faced by the organization in the face of climate change.	Average annual reduction rate of greenhouse gases (%)	1.0	1.4	✓	1.0	1.0	1.0
			Reduction of total perfluorinated compound emissions from the manufacturing process (%)	84.5	93	✓	85.0	-	90.0
			Greenhouse gas direct emissions – Absolute reduction (%;tons) (base year 2020)	-	-	-	10% (10,000 tons)	12% (12,000 tons)	14% (15,000 tons)
			Greenhouse gas scopes 1 and 2, Absolute reduction (%) (base year 2020)	-	-	-	-	-	10

✓ Indicates that the 2024 target has been achieved ✗ Indicates that the 2024 target was not achieved and formulated relevant improvement plans

Aspect	Material Topic	Policy and Management	Action Plan	2024 Target	2024 Implementation results	Execution Status	2025 Short-Term Objectives	2027 Mid-Term Objectives	2030 Long-Term Objectives
Social Co-Prosperity	Occupational Safety and Health	Carry out daily operation according to the occupational safety and health management system, and review and adjust the overall system operation through regular audits in order to achieve the purpose of continuous improvement.	Improved the effective number of health proposals (%/cases)	10% (baseline of 55)	10% (baseline of 55)	✓	20% (baseline of 60)	30% (baseline of 65)	50% (baseline of 75)
			Cases of occupational illnesses caused by chemical exposure (cases)	0	0	✓	0	0	0
			Total health improvement achievement rate for key groups ^{Note} (%)	≥ 65	≥ 70	✓	≥ 70	≥ 75	≥ 80
	Talent Attraction and Retention	Establish a perfect communication channel, formulate an education and training plan that meets the development needs of employees, listen to the employees' needs and suggestion as the basis of reference for system revisions to create a friendly workplace where employees can work safely.	Connection rate of compensation with employee performance (%)	≥ 90	≥ 90	✓	95	95	95
			Response rate on Partner's Words/ employees care network (%)	100	100	✓	100	100	100
			Employee work engagement survey (point)	≥ 4.0	3.9	✗	≥ 4.0	≥ 4.2	≥ 4.5
			Performance Review: Employee engagement survey scores were lower, primarily in the areas of work engagement and salary satisfaction. 1. Work Engagement: Supervisors will be encouraged to communicate and interact more with their team members, understand their work challenges, and provide timely support and training. This will help enhance employees' sense of achievement, thereby improving work engagement and satisfaction. 2. Salary Satisfaction: Due to the Company's less-than-ideal operational performance 2024 year, salary satisfaction was affected.						
	Talent Development	Establish a comprehensive and transparent promotion mechanism, and organize related courses and seminars to assist employees in their career development within the Company.	Conduct senior executive courses and workshops (sessions)	≥ 2	≥ 2	✓	≥ 2	≥ 2	≥ 2
			Mid-level management training program (sessions)	-	-	-	≥ 2	-	-
			Professional skill lectures and courses (for all employees) (sessions)	-	-	-	≥ 2	-	-
			Establish succession planning and implementation for managers and supervisors at all levels (including divisions, department and sections) to effectively cultivate and promote suitable candidates	Continuous processing	Continuous processing	✓	Continuous processing	Continuous processing	Continuous processing
			MTP training participation rate for succeeding managers at department level and higher (%)	-	-	-	50	80	90
Corporate Prosperity	Economic Performance	Continuously improve corporate governance evaluation scores and participate in international competitions to enhance corporate visibility and economic performance.	Maintain Corporate Governance Evaluation ranking listed companies.	Top 6%~20%	Top 5%	✓	Top 6%~20%	Top 5%	Top 5%
			Maintain for the TCSA sustainability report assessment ranking in the Category 1 Electronics Manufacturing industry	Gold Grade	Platinum Grade	✓	Platinum Grade	Platinum Grade	Platinum Grade
			Maintain DJSI assessment ranking in the Semiconductor industry	Top 10	Fifth	✓	Top 10	Top 10	Top 10
	Product Liability and Quality	Regularly manage and monitor product quality to ensure customers receive the best quality products.	Stop & Fix registration rate (%)	98	98	✓	-	-	-
			Number of Six Sigma certification holders (Number of people)	101	92	✗	-	-	-
			Total number of entries for the CIP competition (cases)	275	271	✗	-	-	-
			Implement comprehensive testing or declaration for raw material suppliers	-	-	-	Achieve 100% Hazardous Substance-Free (HSF) quality target	Achieve 100% Hazardous Substance-Free (HSF) quality target	Achieve 100% Hazardous Substance-Free (HSF) quality target
			Number of products recalled by customers due to safety concerns (cases)	-	-	-	0	0	0
			Performance Review: 1. The number of Six Sigma certified personnel did not meet the target due to the need for re-certification every three years, employee turnover (2 people), and job reassignments (7 people). It is planned that in 2025, (1) a new GB training program will be launched for 24 employees, and (2) the conversion of GB/BB certifications to permanent validity will be implemented. 2. 2024, there were 4 fewer CIP competition entries than previously, a slight decline, as participation has reached a saturation point. In 2025, efforts will be made to promote CIP activities through encouragement and requirements set by unit supervisors.						

✓ Indicates that the 2024 target has been achieved

✗ Indicates that the 2024 target was not achieved and formulated relevant improvement plans

Aspect	Material Topic	Policy and Management	Action Plan	2024 Target	2024 Implementation results	Execution Status	2025 Short-Term Objectives	2027 Mid-Term Objectives	2030 Long-Term Objectives
Corporate Posterity	Regulatory Compliance	Establish a clear regulatory compliance and internal audit system to strictly comply with legal requirements in order to avoid potential financial losses that could have an impact on the company's operations.	Achieved a completion rate for participation in courses related to regulatory risk (%)	100	100	✓	100	100	100
	Information Security	Regularly conduct information security campaigns, purchase additional information security analysis software, and develop communication checking systems to ensure information security is implemented.	Enhance employee education and training to improve information security awareness	Continuous reinforcement	Continuous reinforcement	✓	Continuous reinforcement	Continuous reinforcement	Continuous reinforcement
			Strengthen drills to shorten the time required for restoration	Continue to strengthen	Continue to strengthen	✓	Continue to strengthen	-	-
			Identify and audit trade secrets	Continuous execution	Continuous execution	✓	Continuous execution	Strengthen trade secret management	Develop a trade secret registration system
	Ethical Management	With the belief of "integrity, service, quality, and innovation", the company has established the "Ethical Management Code", "Ethical Management Procedures and Behavior Guidelines" and related regulations. We are committed to building a corporate culture of ethical management.	Ethical management risk assessment operations	Continuous execution	Continuous execution	✓	Continuous execution	Continue promoting ethical management, to enhance employees' awareness of anti-bribery measures and deepen their understanding of business ethics.	Adhere to legal and business ethical standards, to build the Company's reputation for integrity and fulfill our social responsibility.
			Enhance defense against hacker attacks	-	-	-	-	Introduce Team Drills	Cultivate a red team drill team
	Corporate Governance	Establish a sound corporate governance structure and culture in accordance with the Code of Corporate Governance Practices to strengthen the functions of the Board of Directors and enhance operational effectiveness.	Develop action plans to address ESG evaluation indicators and the latest international sustainability trends	Continuous tracking and planning	Continuous tracking and planning	✓	Continuous tracking and draw up further training programs for directors and governance staff to enhance their governance capabilities.	Rolling revision of corporate governance-related regulations and management systems in accordance with ESG evaluation indicators	Rolling revision of corporate governance-related regulations and management systems in accordance with ESG evaluation indicators
			Conduct annual Board of Directors meetings, director self-assessments, and functional committee performance evaluations to strengthen board functions	Continuous processing	Continuous processing	✓	Continuous processing	Continue to complete internal evaluations and engage external institutions for assessments every three years.	Continue to complete internal evaluations and engage external institutions for assessments every three years.
	Customer Relationship Management	Communicate with customers regarding services, product information, and the service market to understand their needs, thereby maintaining customer relationships and enhancing customer satisfaction.	Customer satisfaction score (score ;%)	Average score 88; % ≥ 90%	Average score 95;93%	✓	Average score ≥ 90; % ≥ 92%	Average score ≥ 90; % ≥ 92%	Average score ≥ 92; % ≥ 92%
	Sustainable Supply Chain	Comply with PSMC's procurement policy to ensure seamless supply from suppliers.	A completion rate for CSR questionnaire survey of the Tier 1 suppliers (%)	100	100	✓	-	-	-
			Select 50 suppliers annually for the Sustainability Management Self-Assessment Questionnaire, with a target completion rate (%)	100	100	✓	-	-	-
			Select 50 contractors annually for the Sustainability Management Field Audit, with a target completion rate (%)	100	100	✓	-	-	-
			Achieved a completion rate for regular semi-annual assessment of suppliers (%)	100	100	✓	-	-	-
			Achieve a completion rate for regular annual assessment of Contractors (%)	100	100	✓	-	-	-
			For the RBA project plan for Tier-1 critical suppliers and their critical suppliers (Tier-2 critical suppliers)	-	-	-	Conduct RBA awareness promotion, with 100% target completion rate.	Obtain signed RBA pledge based on the target number of suppliers, with 100% target completion rate.	Conduct CSR self-assessments based on the target number of suppliers, with 100% target completion rate.
			Completion rate for supplier qualification reassessments, based on major environmental, safety, and health violations (%)	-	-	-	100	100	100

Note: key groups refer to general employees with abnormally high health examinations and special operations level three or above reported by the hospital.

Stakeholder Communication

In accordance with the five principles of the AA1000SES Stakeholder Engagement Standard, and after internal discussions and reference to the industry's stakeholder identification and engagement methods, six internal and external stakeholder groups were identified. To further master each stakeholder and assess the impact of each sustainability issue, PSMC designed questionnaires to be completed by internal and external stakeholder groups, and the results of stakeholder negotiations and impact assessments of significant issues are reviewed by the Sustainable Development Committee chairman and presented to the Board of Directors.

Stakeholders	Importance of Stakeholder	Top five impact issues (Total positive and negative impacts)	Communication channel and frequency	Effectiveness of communication in 2024
Employees	Employees are the most valuable assets of PSMC, and they are also the most solid elite forces driving our production and operation. We will dedicate ourselves to improving the benefits system and promoting workplace protection, seeking and retaining talents, creating a friendly workplace and a win-win situation for both labor and management.	- Ethical management - Waste Management - Information Security - Energy Management - Regulatory Compliance	[Regularly] - Health Consultation with Health Service Physicians and Occupational Specialists (Weekly) - Occupational Safety, Health and Environment Committee Meetings (Quarterly) - Quarterly Meeting (Quarterly) - Labor-management meetings (Quarterly), welfare committee meetings (Quarterly): to make suggestions and negotiate with the company [Irregularly] - Announcements and inquiries on the PSMC official website (in real-time) - Partner's Words /employees care network (at any time) - Grievance hotline (at any time): Assist and handle issues raised by and received from employees - Employee feedback (at any time) and setting up suggestion boxes - Human Resources Recruiting Line on the PSMC official website (at any time) (https://www.powerchip.com/en-global/contact) - Human Resources Interviewer Phone/e-mail (at any time)	- Accepted total of 549 "Partner's Words" - Accepted 1 cases through "Grievance hotline" - Received 5 case of "Employee Feedback" (The above all response rate are 100%)
Customers	The satisfaction of our customers and their feedback on our services are the driving force for our continuous efforts and growth.	- Product Liability and Quality - Information Security - Regulatory Compliance - Customer Relationship Management - Risk Management	[Regularly] - Customer Satisfaction Survey (Annually) [Irregularly] - Customer service telephone/mailbox (at any time) - Participate in relevant product exhibitions to gain first-hand knowledge of our customers and the direction of market development (at any time) - Marketing and sales colleagues visit customers (at any time)	Customers satisfaction score of 95
Government/ Authorities	The competent authorities are the supervisors of company operations related to regulations, and compliance governance is the primary responsibility of corporate management.	- Responsible taxation - Air Pollution Control - Waste Management - Ethical management, Occupational Safety and Health, Water Resource Management, Corporate Governance and Customer Relationship Management (above 5 topics have the same score)	[Irregularly] - Maintain good interaction with competent authorities and actively participate in seminars and workshops organized by the competent authorities (at any time) - Management System Regulations Identification (at any time) - Correspondence, project meetings, public information (at any time) - Participate in the operation of the functional organizations of the Hsinchu Science Park and the Hsinchu Science Park Administration (at any time)	- Won many awards from the competent authorities for environmental safety and health - No major incidents of violations occurred - Participated in conferences organized by the competent authorities

Stakeholders	Importance of Stakeholder	Top five impact issues (Total positive and negative impacts)	Communication channel and frequency	Effectiveness of communication in 2024
Shareholders / Investors	All of the capital of PSMC is funded by shareholders and investors, and it is the basic responsibility of the company to protect the interests of the shareholders.	- Regulatory Compliance - Economic Performance - Ethical management - Information Security - Corporate Governance	[Regularly] - Regular publication of operational information (Monthly) and financial reports (Quarterly) - Shareholders' meetings (Annually): Annual financial report prepared in accordance with regulations - Financial information, stock information, and annual reports [Irregularly] - A special area for investors is set up on the official website (in real time) (https://www.powerchip.com/en-global/staticpage/ir_contact) - Set up dedicated contact window and mailbox for shareholders (at any time) - A spokesperson contact window is available (at any time)	- Held regular shareholders' meeting on May 21, 2024 - No incidents of corruption occurred
Suppliers / Contractors	As a global team of professionals, PSMC maintain the quality of our supply chain through the evaluation of our supplier management system. The contractors are co-operative partners of PSMC, and share the prosperity with PSMC in terms of business promotion and work quality.	- Product Liability and Quality - Ethical management - Regulatory Compliance - Information Security - Risk Management	[Regularly] - Toolbox Meeting (Daily) - Contractor Agreement Meeting (Quarterly) - Supplier Evaluations (Semiannually) - Contractor Appraisal (Annually) [Irregularly] - Supplier Conferences (irregularly) - E-Supplier System (at any time) - Sustainability Management Self-Assessment Questionnaire (Implemented for the first time in 2023)	100% completion of raw material supplier evaluations 100% completion of preliminary survey and audit of new suppliers 100% use of conflict-free minerals 100% of critical suppliers signed the Responsible Business Alliance (RBA) pledge
Social / Local Communities	The operational sites in the Science Park are located in close proximity to each other and maintain good communication and contact with the neighboring plants to collectively maintain the safety and environment of the Science Park.	- Climate Strategy - Air Pollution Control - Responsible Taxation - Occupational Safety and Health - Talent Attraction and Retention	[Irregularly] - Weekday Visits and Telephone Communication (at any time) - Official Company Website (at any time) - Participate in Association Group Activities or Symposiums (at any time) - Participate in External Guild /Association Functions (at any time)	- Participated in Taiwan Semiconductor Industry Association annual meeting - Attended a variety of professional committee meetings organized by the Allied Association for Science Park Industries - Participated in 23 investor events in 2024, with approximately 900 participants - Engaged in 10 media-related events in 2024, reaching approximately 200,000 people

* Although not classified as a material topic for this year, issues that stakeholders consider to have a high level of impact are also disclosed in the sustainability report.

1 Dedicated to Sustainability

| Innovation, Quality and Customers |

1.1 Innovation and Intellectual Property Management

1.2 Product Liability and Quality

1.3 Customer Relationship Management



Dedicated to Sustainability

Innovation, Quality and Customers

PSMC continues to deepen its Open Foundry model by leveraging an open wafer foundry platform that enables customers to participate in advanced equipment deployment and capacity planning, offering highly customized and high-performance manufacturing services. In response to the rapid development of generative AI and the growing demand for high-performance computing (HPC) chips, PSMC utilizes 3D stacking technology to develop the MOSAIC 3D AI chip, which integrates logic and memory to significantly enhance computational efficiency while reducing power consumption. While meeting customers' high-performance requirements, PSMC remains committed to minimizing environmental impact and delivering top-quality, green, innovative manufacturing services.



Corresponding United Nations Sustainable Development Goals (SDGs)



Material Topics GRI Standards

GRI 416 : Customer Health and Safety



2024 Sustainability Performance Highlights

145

Number of valid patent acquisitions

100%

Obtain ISO9001 Quality Management System Certification of the entire factory

97.75%

Potential implement Hazardous Substance Substitution Program to reduce the use of NMP substances

100%

Customer Satisfaction

Level-A Certificate

Introduce the Taiwan Intellectual Property Management System (TIPS)



Corresponding Material Topics



Product responsibility and quality



customer relationship management

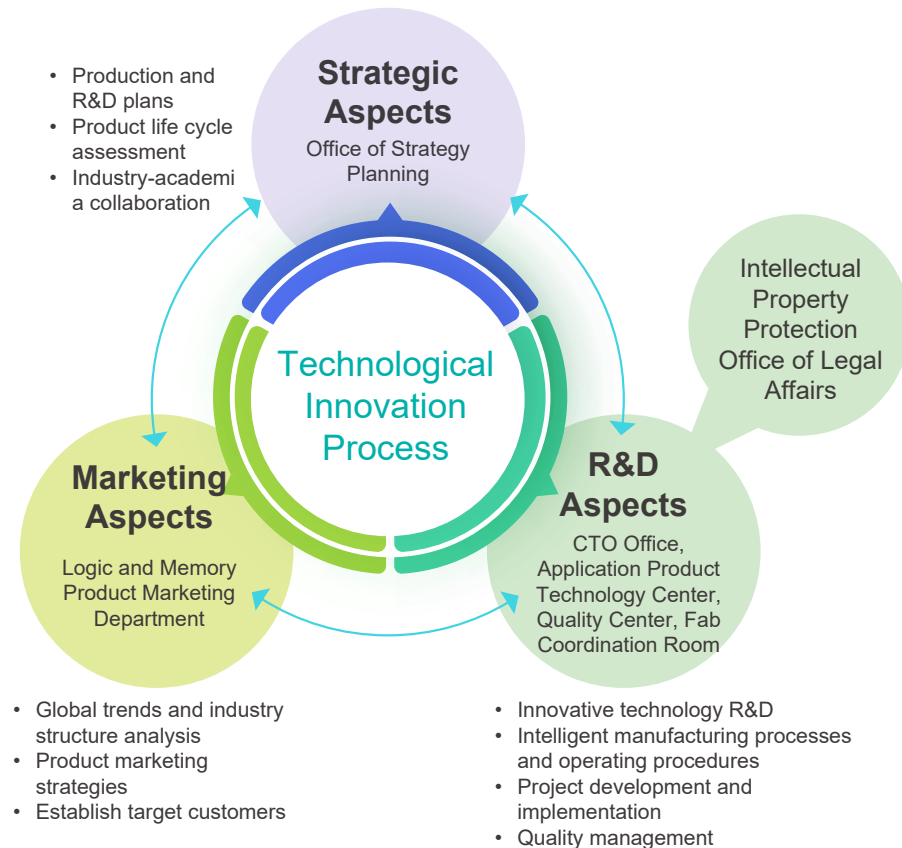


Stakeholders who have priority reading

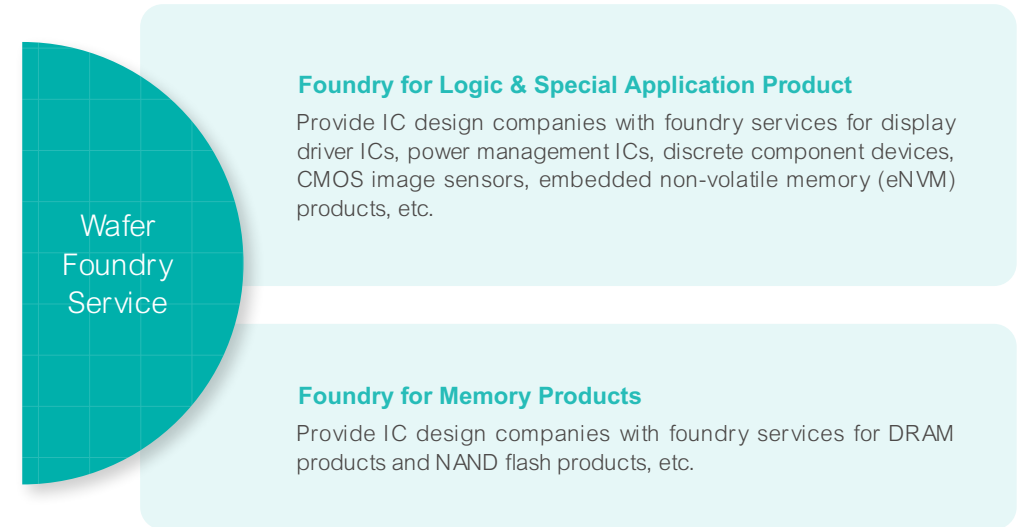
Customers, shareholders / investors, suppliers / contractors

1.1 Innovation and Intellectual Property Management

PSMC takes "Strategy", "R&D" and "Marketing" as the three pillars of the technological innovation process. It formulates production and R&D plans through the strategic planning office, and collects and analyzes product market intelligence and customer needs in conjunction with the marketing business department, formulate product layout strategies, and have the technical R&D team develop projects and implement them to carry out quality management and optimization of manufacturing processes and operating procedures, forming a smooth triangular interactive relationship.



PSMC provide 12" and 8" wafer foundry services, including the provision of logic and special application products and memory products foundry services:



(I) Foundry Services - Logic and Special Application Products Foundry Business

As Moore's Law stalls or slows down at the 28 nm node, PSMC has turned to the more profitable special application product market based on its excellent logic processes and techniques to provide a variety of customized special application processes (applications for the manufacturing of display driver ICs, power management ICs, discrete devices, CMOS image sensors, and embedded non-volatile memory products), and diverse foundry collaboration models to effectively shorten the production process.. Meanwhile, the market demand for mature processes at 28 nm or higher, including 5G, industrial, automotive electronics, and AI related applications. In contrast to the 12-inch standard logic wafer foundry services commonly available in the market which are carried out mainly based on copper processes, PSMC can provide lower-cost 12-inch aluminum process platforms, which, compared to 8-inch aluminum processes at the same technology node, can reduce costs by 30%.

Wafer Foundry Service- Main Product Lines of Logic and Specialty Application Products Foundry Business

Four main areas ; 160Foundry Platforms

Specialty Logic Foundry Field	2024 Main Product Lines of Specialty Logic Foundry	
MCU-based Logic IC	1. MCU(OTP, MTP, eFlash) 2. RFID/NFC	3. OIS 4. RF BlueTooth/Wifi 5. MRAM
Display Driver IC	1. STN 2. a-Si TFT	3. OLED 4. e-paper/e-tag
Image Sensors	1. CIS(FSI/BSI) 2. a-Si	3. FoD
Power Management IC/ Power Components	1. DC/DC controller, converter 2. AC/DC controller, converter 3. LED driver 4. Audio Amplifier	5. Battery Management/PD IC 6. Motor Gate Driver 7. MOSFET 8. IGBT 9. GaN 10. IPD

R&D Project Highlights in Logic and Specialized Application Wafer Foundry Business

Integrating storage and computing to improve display resolution: The Fourth Generation Oxide Semiconductor Material IGZO

With the bursts of AI applications, the integration of data storage and computing can effectively reduce data transfer and improve computing efficiency. In 2023, PSMC collaborated with Japan's Semiconductor Energy Laboratory (SEL) in developing storage-computing-integrated chips manufactured with the fourth-generation oxide semiconductor material IGZO (indium gallium zinc oxide). The content of the collaboration project has been cited and reported by IEEE Journal & Magazine. The fourth-generation oxide semiconductor materials can be used to produce metaverse display driver ICs with a resolution exceeding 3,200 ppi. They can also be applied in virtual reality(AR/VR) products to greatly increase the display resolution. The IGZO storage-computing-integrated chips can effectively meet the need for large-scale data processing in machine learning, artificial neural networks, biological systems, and more. The technological features of the IGZO storage-computing-integrated CPU developed by PSMC are listed as follows:

1. Metaverse Display Driver ICs with a resolution exceeding 5,000 ppi
2. It is a new-type storage-computing-integrated memory (Analog in Memory, AIM) with extremely low cut-off leakage current and ultra-low power consumption
3. The CPU performance can reach 143.95 tera-operations per second per watt (TOPS/W)
4. The Analog-in-Memory-Compute solutions can distinguish 64 conditions

Current Status

- Successfully developed 12-inch 40 nm LCD driver IC high-voltage process
- Successfully developed 12-inch 80 nm BCD G1 process
- Successfully developed 8-inch GaN power semiconductor 100V & 200V processes
- Initial mass production of power management IC (PMIC) products from two leading international companies (130 nm, 110 nm)

Short-term Goals

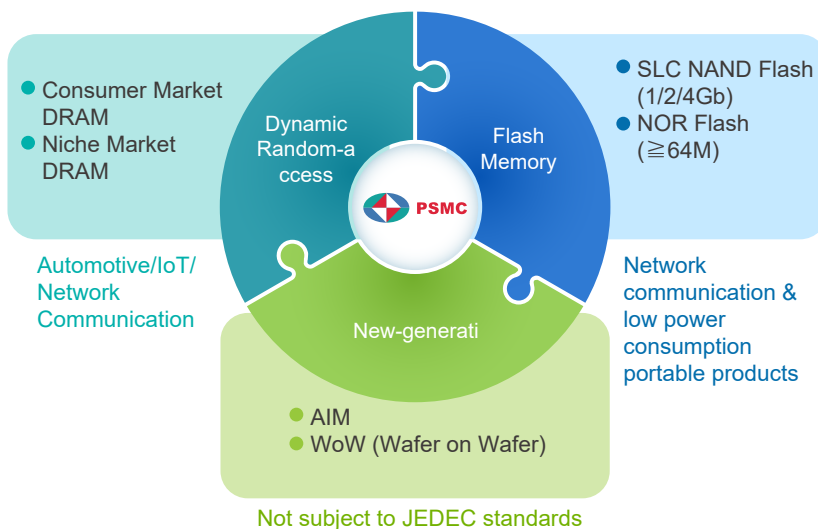
- Complete 8-inch GaN power semiconductor 650V foundry process
- Complete 12-inch PMIC 80 nm BCD G2 & 110 nm BCD foundry process
- Complete 12-inch CMOS image sensor product development with backside illumination (90 nm BSI) process
- Continue to integrate the 12-inch 55 nm RF (Wi-Fi, Bluetooth Low Energy) product line into P1/2 & P5 fabs, collaborating with global manufacturers to develop a broader power management IC platform spanning 150 nm, 130 nm, 110 nm, 90 nm, and 55 nm BCD processes

Long-term Goals

- Accelerate building a logic specialized foundry platform and develop silicon IP (FOT), including 28 nm and 22 nm HPC & HV process platforms, Interposer, and WoW-related processes
- Secure more world-class customers to adopt exclusive technologies in PSMC production (COT) and accelerate mass production of power management IC (PMIC) products from international companies across 150 nm, 130 nm, 110 nm, 90 nm, and 55 nm BCD processes
- Provide diversified, modular foundry models under the Open Foundry concept in response to 5G, EV, and AI-related application markets, expanding logic-related processes
- Accelerate building automotive-grade products and develop diverse embedded IP for power management IC BCD platforms
- Actively secure OOC (Out of China) customers and projects across multiple platforms
- Accelerate completing the Fab IP plan

(II) Foundry Services - Memory Wafer Foundry

Memory Foundry Major Product Lines



R&D Project Highlights in Memory Wafer Foundry Business

Continuous optimization of high efficiency and low power consumption process technologies in response to the needs for performance, thinness and energy efficiency

In consideration of the trend where endpoint electronic product specifications are evolving toward enhanced performance, thinner designs, and greater energy efficiency, chip design requires higher-end process memory products to meet demands for high performance and low power consumption. 2024, DRAM foundry services have adopted the 25 nm process and are accelerating the development of a new 19 nm process platform. Additionally, PSMC is developing In-Memory Computing (IMC), a memory-based computing technology for AI-driven neural network systems, as well as logic wafer stacking (Wafer on Wafer, WoW) memory products that can be stacked with logic chips. These innovations aim to meet customers' different customized memory requirements.

New-generation Flash processes: Capacity improvement, cost reduction, and quality guarantee

Our Flash products are targeted at mobile applications, consumer electronics and industrial applications, providing low-power, energy-efficient and highly reliable products. And Low-capacity NAND Flash with low-power DRAM is the main memory solution for entry-level wireless communication products. 2024, the 28 nm NAND Flash process has successfully entered the mass production stage. Being benefited from the rising demand for memory products such as AMOLED panels, True Wireless Stereo (TWS), 5G base station facilities, etc., the global NOR Flash market has been driven to expand. 2024, PSMC's new-generation 48 nm NOR Flash process has entered the mass production stage, through which Flash products with higher-level capacity, cost competitiveness, and upgraded quality reliability can be provided in the future.

After the success of 30/25 nm memory process mass production, PSMC continued to complete the development of 25E nm process, and 2024 has successfully assisted customers in introducing the 25E nm process. In terms of new memory, the Wafer-On-Wafer technology for logic wafers is continuously being advanced to further enhance the performance of products.

Current Status

- 30 nm and 25 nm DRAM processes matured for mass production
- Assisted customers in introducing the 25E nm process, which has entered mass production
- Developed new memory technology integrating logic and memory wafer stacking (Wafer on Wafer, WoW)
- Mass production of 28 nm NAND Flash and 48 nm NOR Flash

Short-term Goals

- Develop a technology platform for DRAM process 19 nm
- Develop a technology platform for NAND Flash process 24 nm and enhanced version of such

Long-term Goals

- Collaborate with customers in producing customized products to expand products' life cycle
- Closely collaborate with customers and other suppliers to develop products in the rapidly growing edge computing (Edge AI) market
- Continue the development of next-generation memory advanced process technology
- Actively explore the new market and new customers in out of China region

1.1.1 Innovative Research and Development Breakthroughs

PSMC's product applications span wireless communications, consumer electronics, automotive electronics, and 3C (computer, communication, and consumer electronics) applications. The Company not only provides advanced process services for niche memory, but is also the world's only 12-inch wafer foundry offering a comprehensive memory product line. In recent years, PSMC has collaborated with partners to develop 3D AI chips designed specifically for generative AI applications. By integrating 40 nm logic and 25 nm DRAM process technologies, the Company has leveraged Wafer-on-Wafer (WoW) stacking technology to create a 3D AI accelerator, which won a 2024 R&D 100 Award. This solution offers the AI industry a high-performance, flexible, and cost-effective alternative for high-bandwidth memory and can be applied to AI inference systems. Its data transmission bandwidth is 10 times that of traditional AI chips, while its power consumption is only one-seventh of conventional designs.

By extending the Open Foundry service model, PSMC has collaborated with strategic partners and industrial/academic organizations to establish an "Open Foundry Project" open-source platform, which helps engineers to efficiently develop prototypes through repeated attempts in the initial stage of technology development. This will not only save the resources invested in the process, but will also shorten a project's development time to three quarters (nine months). In addition, the Company also conducts regular technology transfer and related training programs with business partners to exchange technological knowledge and practical experiences with each other, thereby maximizing the progress of innovation and R&D process, and solving the technical problems encountered in heterogeneous 3D IC packaging to achieve the goal of low power consumption and high efficiency for products.

Significant Innovative Results

Category	Process Technology	Product Applications	Breakthroughs	Future Goals
Logic and Special Application Products Foundry	80 nm crystallite	Display driver ICs, Power management ICs, CMOS image sensor IC products	The back-end copper process has been successfully developed, which can reduce power consumption by 30% compared with the aluminum process	Continuous improvement of 55 nm, 40 nm, and 25 nm efficiencies
Memory Foundry	21 nm DRAM Yield Improvement	Network communication, Consumer electronic products	The yield rate has been successfully enhanced to 75%, the product also has passed the reliability verification, and the NTO mode collaboration with customers has been started	1 X nm DRAM development to increase the number of DDR4 (4th generation Double Data Rate Synchronous Dynamic Random Access Memory) chips by 30% per wafer
Memory Foundry	24 nm NAND Flash Technology Platform Development	Mobile application devices, Consumer electronic products	The per unit area output has been increased by 14%, and the reliability has been improved to the same level as that of 28 nm NAND Flash	Continuously improve 24 nm yield rate and capacity, and assist customers to expand into the 5G cellular base station and related applications
Memory Foundry + Logic and Special Application Products Foundry	3D AI chip Wafer on Wafer Technology	Artificial Intelligence(AI)	- Bandwidth: 10 times HBM3 (2.5D) - Power Consumption: One-seventh of HBM3 (2.5D)	- AI-related applications: Products with 4-layer stacked DRAM wafers are currently being validated by customers, and development of 8-layer stacked DRAM wafer technology is underway - Wafer stacking foundry services: PSMC provides foundry services for stacking customer-supplied DRAM and logic wafers
Memory Foundry + Logic and Special Application Products Foundry	Fourth Generation Oxide-Semiconductor Material IGZO (Indium Gallium Zinc Oxide)	AR/VR, 5G communication, Artificial Intelligence(AI)	300mm Wafer Output Demonstration: 1.03-inch IGZO-on-Si OLED Display - High Resolution: 3,500 ppi - High Brightness: 5,000 cd/m²	- Enhance display resolution and brightness - Develop ultra-low-power computing-in-memory (Analog In-Memory) next-generation memory
Logic and Special Application Products Foundry	0.18 μm Discrete Metal-Oxide-Semiconductor Field-Effect Transistor Technology	Consumer electronic products, Communication and Computer Products	A Double Gate MOSFET (discrete MOSFET) that can operate at 200V with a specific on-resistance (R _{sp}) of 280 mΩ·mm ² has been successfully developed	Continue development of components for higher operating voltages such as 150V, 200V, 250V
Memory Foundry + Logic and Special Application Products Foundry	2.5D Interposer (Interposer with IPD)	Artificial Intelligence (AI), High-Performance Computing (HPC), 5G communication, Internet of Things (IoT)	- Integrate high-density capacitors (~2,100 nF/mm²) into the interposer - Supports interposers with masks 3X+larger, 5-layer thick copper, and through-silicon via (TSV) technology	The 2.5D Interposer, combined with a high-capacitance IPD solution, supports GPU and high-bandwidth memory (HBM2E, HBM3) transmission

1.1.2 Green Product Development and Technology Breakthrough

Continuous Technology Breakthroughs : Reduce Process Consumption, Strengthen Performance of Product Application

Category	Process Technology	Process Power Consumption Reduction Achievements
Logic and Special Application Products Foundry	28 nm Process Technology	Targeting smartphones and consumer electronics applications, PSMC developed 28nm logic process technology in 2023, including 28 nm High Performance Compact (28HPC) process technology and HV28 high-voltage process. The 28 nm logic process adopts High-K Metal Gate (HKMG) technology, differing from the 40 nm Poly/SiON process. The core device voltage has been reduced from 1.1V (40 nm) to 0.9V (28 nm). Key advantages of the 28 nm process compared to the 40 nm process include reducing chip size by c. ~50%, reducing power consumption by ~50% across all operating speeds, while increasing speed by ~45% at the same power consumption level. The 28HPC and HV28 platform technology developments are planned for completion in the first half and second half of 2026, respectively.
Logic and Special Application Products Foundry	80 nm Technology Process	The number of electronic components that an 80 nm die can accommodate is twice that of a 110 nm die, and the power consumption of an 80 nm IC product in use or on standby is only about 70% of that of a 110 nm product, which is equivalent to a 2.8 times increase in performance per unit area. Our goal is to continue to improve the development of the 80 nm process and more advanced process technologies (e.g. 55 nm, 40 nm, 25 nm, etc.), and to increase the share of more energy-efficient processes below 80 nm in our wafer sales revenue.
Memory Foundry	24 nm NAND Flash Technology Platform	In 2020, PSMC started the development of the 24 nm NAND Flash technology platform with the new IP design that significantly reduced the mask count by 16% and the total process count by 9%, while continuing to increase the output per unit area by 14%, effectively reducing the production power consumption. In 2021, the development of the 1.2V NOR Flash technology platform was initiated to meet the future needs of the low power-consumption product market.
Logic and Special Application Products Foundry	Power Management IC(PMIC) Technology for Optimal Power Efficiency	Power management IC is the core component required for electronic products' power processing. PSMC has developed discrete MOSFET at 8-inch and 12-inch fab, Bipolar-CMOS-DMOS (BCD) and UHV (Ultra-High Voltage) technologies for endpoint product applications to stabilize power supply and reduce power consumption. PSMC will continue to increase PMIC products' shipment proportion, and develop or co-develop more PMIC-related BCD process technologies to improve BCD technical characteristics and Ron & Qgd efficiency.
Logic and Special Application Products Foundry	Discrete MOSFET and Discrete MOSFET Technology	Discrete MOSFETs are electronic components formed by many MOSFET elements of the same type, which act as switches for the circuit. The unit of Rsp is $[m\Omega \cdot mm^2]$, and it is an indicator of great concern for technological breakthroughs. The lower the Rsp value, the better the performance during voltage conversion, and PSMC continues to reduce the Rsp value every year to bring better product quality and performance to our customers. In addition, to meet application demand that continues moving toward higher current, there is an increasing emphasis not only on reducing resistance values but also on improving switching speed. Therefore, starting in 2024, PSMC began working on improving the platform's FoM (Figure of Merit, defined as $R_{dson} \times Q_{gd}$); we project completing development of low-FoM platforms for 60V, 80V, and 100V by the end of 2025. This aims to enhance conversion efficiency in high-current applications to meet energy-saving requirements.
Logic and Special Application Products Foundry	Insulated Gate Bipolar Transistor (IGBT)	The IGBT product launched by PSMC is produced using 8-inch wafers based on the 0.4 μm technology, which is applied in electric vehicles and variable frequency home appliances, and is an indispensable component in energy-efficient products with its advantages of high input impedance, high withstand voltage, and low turn-on resistance. In 2024, PSMC helped its customers reduce approximately 31.22 million metric tons of CO ₂ e greenhouse gas emissions. This accounted for 3.4% of the Company's total revenue. However, due to market demand fluctuations, this figure declined compared to the previous year, though the medium- and long-term demand trend remains unchanged. Additionally, PSMC completed the development of 12-inch wafer IGBT products in 2024, laying the foundation for future energy-saving product applications.

Column

Navigator for Intelligent and Innovative Manufacturing

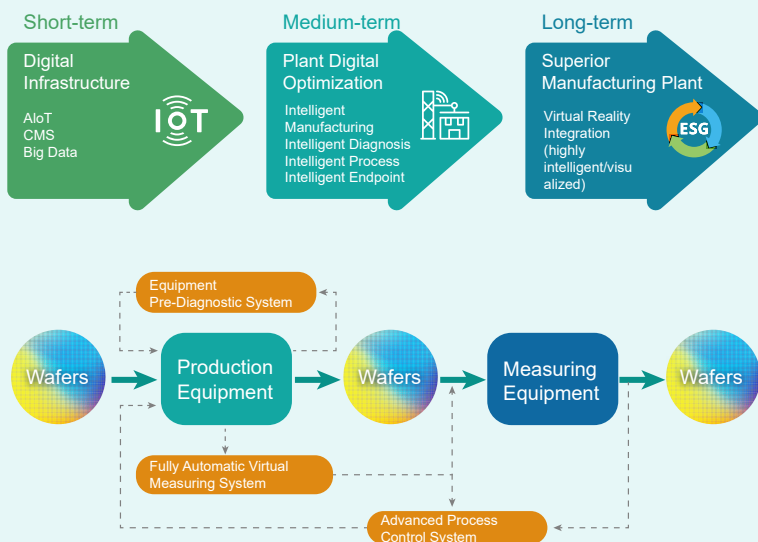


PSMC formally established the "Digital Transformation Committee" and is actively developing smart manufacturing technologies and applications, to focus on production intelligence, process intelligence, terminal intelligence and diagnostic intelligence, combining experts from various fields, paired with a deepened infrastructure. The AIoT project connects machines and sub-system parameters to execute digital transformation projects, mainly in optimizing production dispatching and scheduling, automating transmission paths, production quality prediction and monitoring, machine abnormalities detection and major parts residual life-span warning, personnel safety monitoring, energy-saving, improving personnel efficiency and machine productivity, and product yield quality.

In terms of quality control, the production quality system will use machine parameters (FDC/ED data) to conduct real-time anomaly monitoring and checkpoint control, then measuring machines will be used to confirm product quality. In addition, AI and big data technologies have been introduced into products' pre-production, in-production, and post-production stages to achieve machines' and products' 3P (Prediction, Prevention, and Protection). By flexibly utilizing AI technologies, the anomalous conditions of production machines and product quality can be predicted or controlled in real time for improvement measures to be taken in a timely manner.

PSMC started the implementation of smart-manufacturing-related industry-academia collaborative projects with the semiconductor academies at National Tsing Hua University and National Cheng Kung University in 2024. A total of 19 projects have been presented (including completed ones and those still in development), which will provide great help for the Company's product manufacturing process, quality, and energy efficiency. The concept of ESG green manufacturing is highly emphasized during the development of smart manufacturing technologies. Topics encompassed in the projects included using big data analysis and AI algorithms to construct air compressor prognosis-based maintenance and machine scheduling models, methods to optimize transmission order/allocation under limited transmission resources, and more. The Automatic Virtual Metrology (AVM) conduct production quality control and application Prognostics and Health Management (PHM) improve equipment maintenance efficiency, hoping that the academia's research and technology achievements can be introduced through the collaboration, and also apply big data analysis to different fields to cultivate more professional and research talents, and to help PSMC build a comprehensive smart factory environment.

PSMC Digital Transformation Roadmap



2024 Highlight Intelligent Projects

Category	Capacity Anomaly Management System (TC/EPT)	Photoresist Analysis System
Purpose	To solve the problem of overcomplicated query process.	To build a photoresist analysis system to provide machines with photoresist product WP (Wafer-In-Process) information so that users can determine the post-processing procedures for small amounts of WP photoresist.
Current status	Currently the TC query process needs to rely on different query systems, and tests need to be conducted at machine side's every position. The whole process lacks effective TC detection systems, and the average TC test time is as long as 2 hours.	There used to be numerous photoresists and photolithography production machines used in fab, and the numbers of photoresist thickness measurement categories and machine test items were as many as 215 and 1,137 respectively. Engineers needed to manually analyze and organize expired photoresist information and allocate production capacity based on related photoresist information, which had hampered effective operations.
Solution	To solve the problem of overcomplicated query process, PSMC established a TC/EPT Smart Report Statement System in 2023, through which the machine with anomalous production efficiency can be located, and the machine's anomalous position can be quickly identified for the station's engineers to take action to improve it. In 2024, the system was expanded to the P5 fab, with 84.1% of machines currently online.	In 2023, PSMC introduced a intelligent photoresist analysis system to efficiently screen out and analyze low-usage photoresist machines and invalid photoresist machine test items, which can not only reduce human calculation errors and engineers' work load, but can also help engineers quickly optimize photoresist pipeline layout and machine test items. In 2024, a photoresist usage analysis function was added to alert engineers about expired or excess photoresist, providing a reference for quick path adjustments. The entire system was expanded to the P5 fab in October 2024.
Improvement benefits	Through weekly outlier machine monitoring, and daily/weekly monitoring on the machines' variations, the time required for engineer testing has been shorten to 10 minutes. As of November 18, 2024, the accumulated production increase volume benefited from improved production efficiency in P1/2/3/5 Plants was 137,866 wafers.	1. PM test items reduced by 18% 2. Equipment engineer labor investment reduced by 0.3 people/day 3. Average machine uptime increased by 0.1% 4. Photoresist expiration analysis efficiency effectively improved by 1.2 hours/per analysis 5. Amount being saved: NT\$4.008 million in 2024

1.1.3 Intellectual Property Rights Management

By focusing on memory and logic circuit wafer foundry services and the Open Foundry operating model, PSMC has developed two major strategies, which are "Technological innovation" and "Patent Deployment". In accordance with the strategies, IP management plans are developed based on the Company's operating goals and technical goals. Additionally, resources are continuously invested for patent R&D, and a global patent deployment network is established to ensure that all PSMC R&D achievements are comprehensively protected by intellectual property rights. PSMC has formulated its "Patent Management Regulations" to systematically manage the invention proposals involving technologies related to the Company's R&D, production, and operations through the Company's internal "Patent Management System". All patented technologies have been strictly reviewed by internal technical reviewers to maintain PSMC's intellectual property quality.

Since 2023, PSMC has successfully introduced the Taiwan Intellectual Property Management System (TIPS), conducting a comprehensive review of the systematic management framework that links intellectual property management with operational goals, and has obtained TIPS Level-A verification.



2024 Goal Achievement Performance

- Successfully passed TIPS Level-A verification for second year in a row (valid for two years)
- 111 patents successfully passed internal review

Short-term Goals

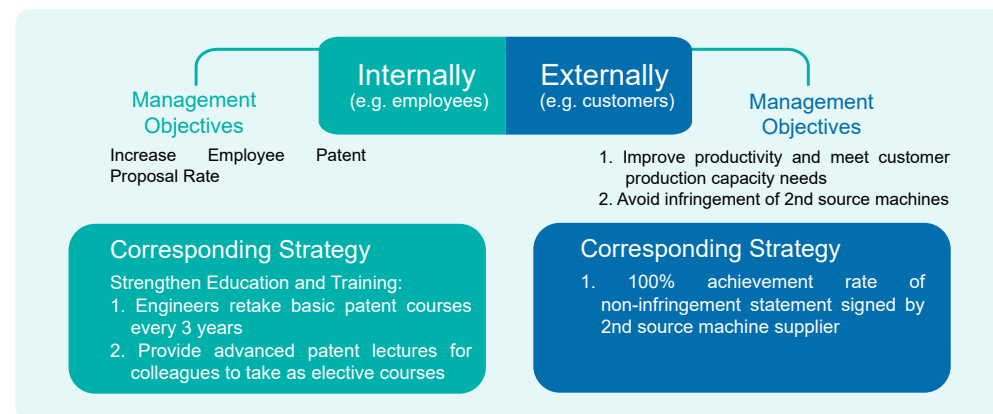
- Maintain TIPS Level-A verification and implement the intellectual property management system
- Number of patents passed internal patent approve review reaches 120

Medium-/Long-term Goals

- Diversify product technologies and high gross profit production-marketing portfolios, and upgrade process technologies to enhance profitability, including 5G, automotive products, big data, AIoT, and AI Chips. In addition, consult the Patent Map and use innovative analysis approaches to formulate patent deployment strategies, and follow R&D technical blueprint to explore innovative technologies for patent application to ensure that the critical technologies achieved in R&D results can be protected through patent acquisition

IP Management Measures

PSMC has established online intellectual property rights training sessions to strengthen employees' IP-related legal consciousness, and designed incentive mechanisms to encourage innovative technology proposals. In addition, the patent R&D team utilizes the Derwent Innovation patent database to analyze global technology trends and monitor competitors' patent intelligence, serving as a reference for setting internal proposal strategies and external patent application targets.

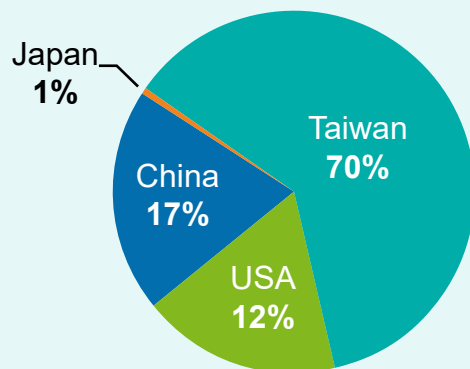


2024 Employee IP Management Educational Training

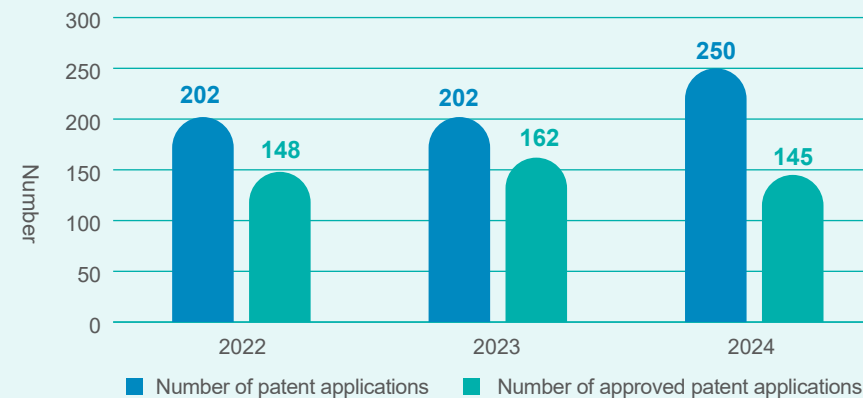
Course Title	Course format (online/in-person)	Participants	Number of participating employees	Training completion rate (%)	Participating hours (hours)
Innovative Proposal's Patentability Analysis(Advanced)	Online	All employees	209	98	1.5
Patent Infringement and Design around Existing Patents(Advanced)	Online	All employees	8	100	2.5
Fundamentals of Intellectual Property Management and IP Management System Awareness Course (TIPS Basic Training Course)	Online	TIPS-related departments	169	100	1.4
Practical Experience Sharing on Patent Application Defense (Advanced)	In-person	TIPS-related departments	41	100	2
Basic Patent-themed Lectures(Basic)	Online	Required for employees	135	98	2.1
Introduction to the New System for Intellectual Property Case Review (Advanced)	Online	TIPS-related departments	13	100	0.3

As of 2024, PSMC has obtained 1,355 valid patents, including 145 newly granted patents domestically and internationally in 2024.

Number of Patent Acquisitions in 2024



Patent Statistics in the past three years



Patent Incentive System and Effectiveness

PSMC has specified patent incentive measures in its "Patent Management Regulations". There were 300 patent applications being granted financial incentives in 2024, with the incentive amount totaling to NT\$4.42 million, representing a 5.74% increase over 2023. In particular, in line with carbon reduction goals, one of 2024 year's patents covers a method for controlling process parameters to reduce combustion waste in thermal processes. This demonstrates the Company's commitment to implementing carbon reduction strategies through R&D investments.

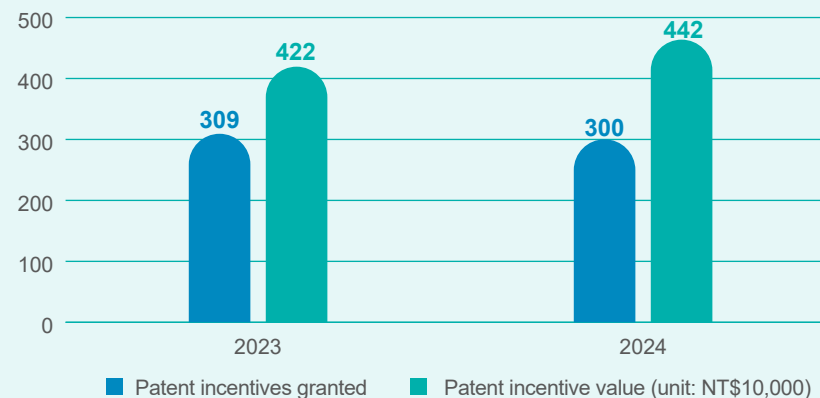
Obtain an official
application number
granted by an
external agency

- Awarded a bonus of NT\$10,000

Patent acquisition

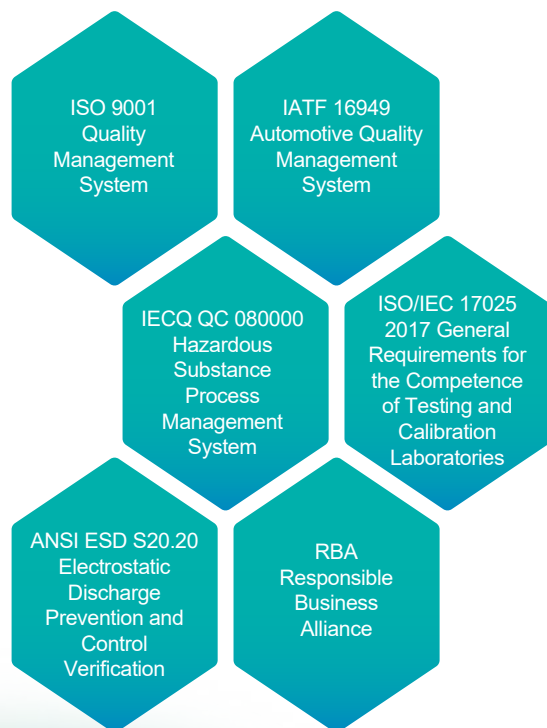
- US, Japan, EU: NT\$40,000
- Taiwan, China: NT\$20,000

Patent Incentive Distribution Over the Past Two Years



1.2 Product Liability and Quality

With professional and superb process technology and strict quality control process, PSMC provides wafer foundry services to our customers through the introduction of various management systems and clear processes to control all aspects of production, and adheres to the principle of strict quality control and insists on being customer-focused, continually pursuing excellence in quality and service.



1.2.1 Product Quality Management

PSMC has established the "Three Don'ts Policy" to provide customers with products and services of excellent quality, cost competitiveness and on-time delivery through the participation of all employees. We also have a professional laboratory quality team to provide accurate and reliable test results to ensure product quality no doubt.



PSMC is known for its diversified product technologies, and currently provides process technologies and products that include LCD Driver IC, Integrated Memory Chip, CMOS Image Sensor, Discrete Device and Power management. In order to effectively manage product quality, PSMC adopts an automated production management system and abides by the following three quality management principles to ensure that our production efficiency and quality meet the requirements and expectations of our stakeholders.

2024 Annual Certification Record

Management System	Plant passed Certification
ISO 9001 Quality Management Systems	P1/2、P3、P5、8A、8B
IATF 16949 Automotive Quality Management System	P1/2、P3、8A、8B
IECQ QC 080000 Hazardous Substance Process Management System	P1/2、P3、8A、8B
ISO/IEC 17025 General requirements for the competence of testing and calibration laboratories	P1/2
ANSI ESD S20.20 Electrostatic Discharge Prevention and Control Verification	P1/2、P3

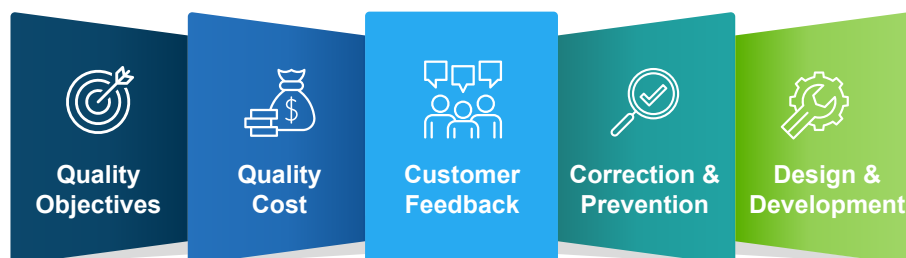
The Three Principles of Quality Management

Effective Yield Improvement	Optimal and Flexible Production Scheduling Management	On-time Delivery
Effectively shorten the learning curve, Improve product yield and shorten time-to-market for customers.	Continuously improve production management efficiency and optimize production process to meet customer needs in the shortest possible time, in the most efficient way, and reduce risks with flexible production scheduling	Fully automated production and strict online monitoring system, with full control of the product input and output schedule, to effectively complete the delivery.

Product Quality Management Objectives

In order to ensure that the products provided meet the expectations and needs of all stakeholders, PSMC has formulated five quality management review objectives to maintain a high level of stability and consistency in internal product quality. Through quarterly statistics on the performance of quality control projects, deficiencies can be discovered in a timely manner and corresponding improvement measures can be taken. Internal audit and quality management review meetings are held every six months to review various indicators and track improvement progress. Under the operation of PSMC's strict and high-standard quality control mechanism, no product recalls have occurred in the past four years as of the end of 2024.

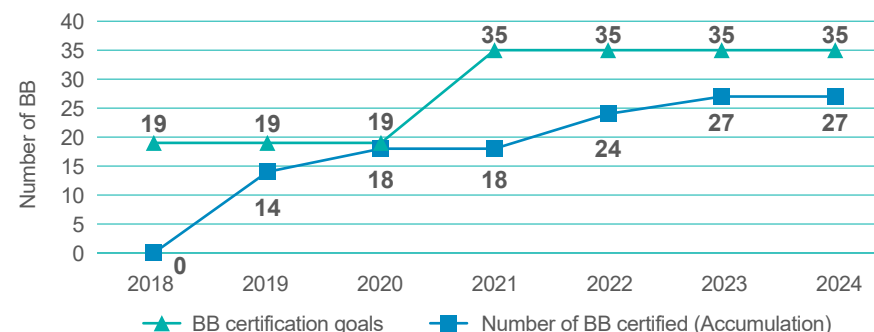
Quality Management Review Indicators



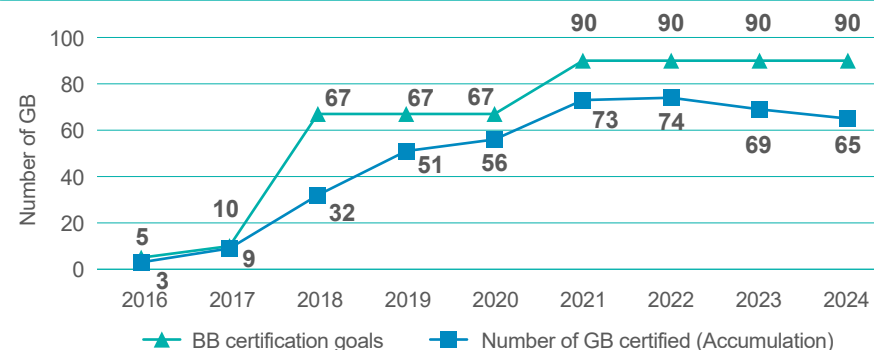
Six Sigma (6S) Training

Six Sigma is a set of systematic scientific management approaches aimed at reducing product defect rate and maintaining product quality. PSMC has required employees of engineering, R&D, and quality-related departments to participate in Six Sigma training courses to enhance team members' problem-solving skills and work performance. The training courses is classified into three levels, which are Yellow Belt (YB), Green Belt (GB), and the highest Black Belt (BB) level. As of the end of 2024, the cumulative number of obtained GB certificates holders at PSMC is 65, and the number of BB certificates holders is 27. The GB/BB permanent certificate conversion rates in 2023 are 18.8% and 48.1% respectively.

Accumulative number of BB (Black Belt) Certificate holders over the years



Accumulative number of GB (Green Belt) Certificate holders over the years



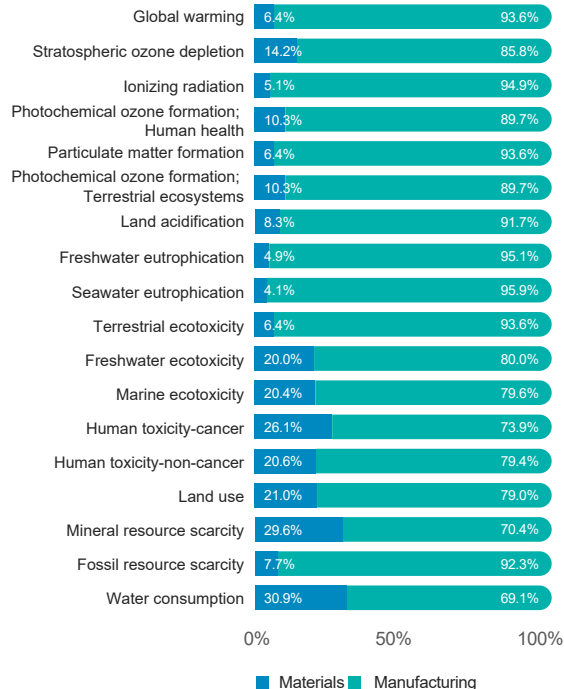
Note 1: In 2024, the number of GB certificate holders decreased by four compared to 2023, primarily due to work reassignment issues (GB/BB certification requires renewal every three years).

Note 2: In 2025, a new cohort of GB certificate holders will undergo training.

Life Cycle of Product

To assess the environmental impact of 12-inch wafer foundry products across the raw material acquisition and manufacturing stages, PSMC has identified key environmental impact hotspots at each stage. This helps in formulating strategies for performance improvement and enhancing communication with customers. In 2023, PSMC conducted a comprehensive product life cycle assessment (LCA) for its P1/2 fab, following the requirements of ISO 14040 and ISO 14044, using LCA SimaPro 9.5.0.0 software. The LCA assessment covers two stages: the raw material stage, and the manufacturing stage. The system boundary is defined as Cradle to Gate (from raw material to the customer's gate). The results of the product life cycle environmental impact assessment are illustrated in the figure below.

Environmental Impact Assessment Chart of Product Life Cycle



The assessment results provide R&D developers with an understanding of the sensitivity to the potential impacts of various design decisions, including material selection, packaging use, and the relevance of manufacturing to the use of renewable energy for that manufacturing. During the manufacturing stage, it probably should be made to reduce the use of materials, minimize the number of equipment used, and maximize the utilization of production equipment. In addition, potential environmental impacts during the manufacturing phase, PSMC will continue to through increase the use of renewable energy, reduce process waste, and improve production yield process improvement to reduce material use in the raw material stage, and also reduce the environmental impact of raw material transportation.

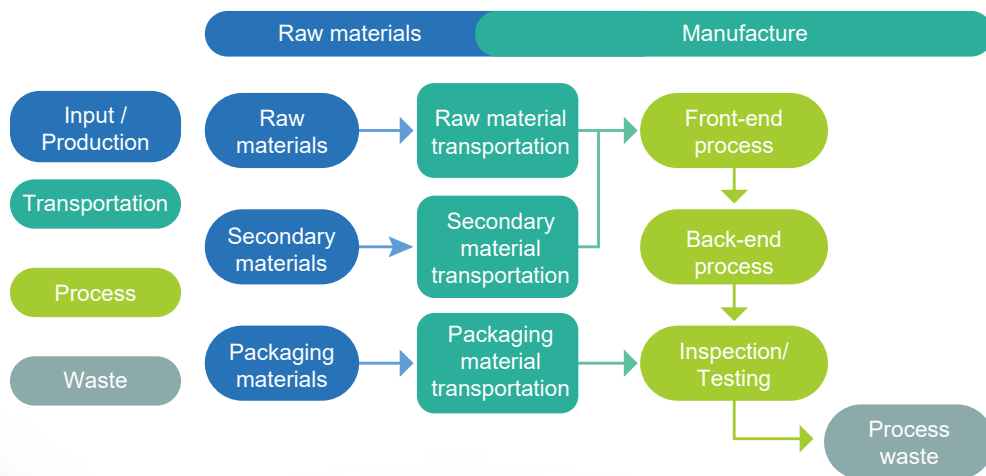
Impacts Encompassed in Life Cycle Assessment

Impact Category	Impacts
Resource consumption	Abiotic resource depletion (fossil fuels, minerals), land use, water resource depletion
Ecological consequences	Acidification, dust and particulate matter, ecotoxicity, eutrophication, global warming, ozone depletion, photochemical ozone formation, species richness
Human health	Human toxicity, Ionizing radiation

Life Cycle Stage	Friendly Environment Considerations	Improvement Program & Action at future
Product R&D Raw material procurement	Green Products Hazardous substances Green supply chain	- Material selection, use of packing materials, and production/manufacturing of renewable energy and the use of renewable energy during manufacturing - Collaborate with customers in developing low power-consumption products - Comply with environmental laws and regulations, and monitor & manage the use of hazardous substances to reduce the impact on the environment - Implement green procurement by choosing eco-friendly suppliers - Optimize delivery batch quantities and frequencies to reduce transportation energy consumption
Product manufacturing	Reduce carbon emissions Reduce energy and material consumption Reduce water consumption Increase recycling and reuse	- Increase the use of renewable energy - Promote equipment energy-saving projects - Implement manufacturing process water recovery, recycling, and reuse to reduce wastewater discharge - Save materials usage and reduce waste output - Process waste recycling and reuse - Improve production yield through process improvement to reduce raw material consumption
Product Packaging & Transportation	Reduce carbon emissions Increase recycling	- Use green packaging materials - Reuse the wafer cassettes originally used for wafer raw material packaging in outsourcing transportation or final product packaging
Waste Treatment & Disposal	Increase recycling Reduce all stages' packaging materials	- Improve the recycling and reuse rate of suppliers' packaging materials - Suppliers' packaging material reduction design to reduce waste

Product Carbon Footprint/Water Footprint

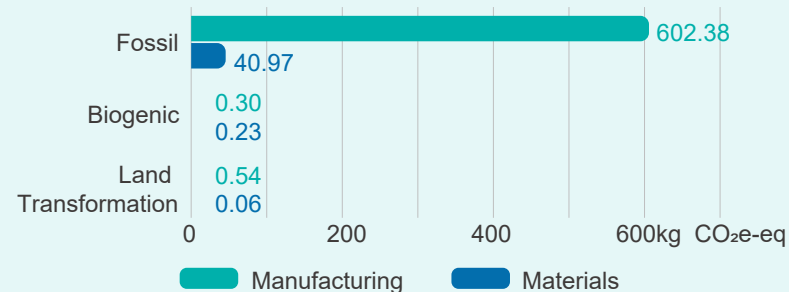
PSMC produces advanced, energy-saving and environmentally friendly products for customers and continues to promote new process technologies to reduce chip power consumption and save resources. In order to implement the "Net Zero by 2050" goal, PSMC has conducted a carbon footprint assessment for 12-inch products at the P1/2 fab in accordance with ISO 14067 standards, using SimaPro 9.5.0.0 analysis software. The assessment and calculations cover greenhouse gas emissions across different stages, including raw material manufacturing, raw material transportation, and energy/resource consumption during product manufacturing. Additionally, PSMC has conducted a water footprint assessment in compliance with ISO 14046 standards, measuring water consumption from raw material production to the manufacturing process. Both carbon footprint and water footprint assessments were conducted within a Cradle to Gate system boundary, covering raw material and manufacturing stages. The results have been verified by TÜV Rheinland, enhancing the quality and credibility of the assessment data, PSMC will continue to formulate improvement actions based on the carbon footprint and water footprint results in the future.



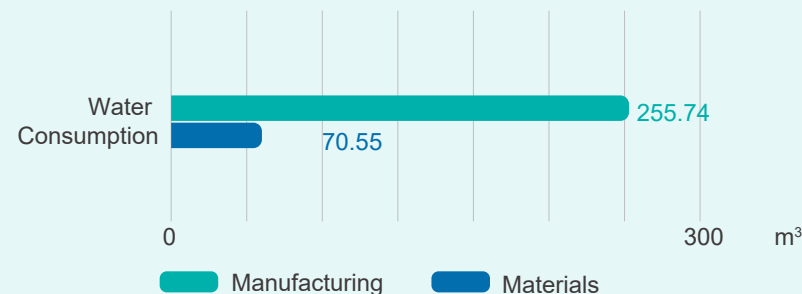
Product Carbon Footprint and Water Footprint Implementation

Category	Product range	Improvement Strategies and Actions at future
Foundry Product	Logic and Special Application Products Foundry	• Increase the use of renewable energy • Promote facilities energy-saving projects • Implement process water reclamation (recycling and reuse) to reduce wastewater discharge • Implement process waste recycling and reuse

Product Carbon Footprint



Product Water Footprint



1.2.2 Hazardous Substance Management

PSMC by complying with regulations, reducing risks, improving hazardous material process management, establish a comprehensive risk assessment process and promotes existing factories to pass IECQ QC 080000 certification. PSMC establishes a hazardous material management mechanism based on this standard, and subsequently promotes the P5 factory to complete the IECQ QC 080000 certification. In addition, in accordance with EU RoHS hazardous substances control regulations and other laws and regulations, PSMC automatically manages raw materials to comply with Hazardous Substance Free (HSF) regulations during the incoming quality control (IQC) stage, strictly controls test reports of the supply and the validity of declarations. The risk assessment process involves identifying potentially hazardous substances contained in raw materials and evaluating their potential impact on human health and the environment. The assessment results serve as a critical reference for decisions in procurement, R&D, and manufacturing, ensuring that products meet customer needs while complying with each region's regulatory restrictions.

When international or regional regulations are revised or new restricted substances are introduced, PSMC initiates a regulation tracking and impact assessment mechanism. Cross-functional meetings are held to develop response strategies and update the list of hazardous substances. Related departments are notified to identify alternative materials, in order to ensure that transitions are completed before the regulations take effect, thereby maintaining product compliance in the market. In 2024, PSMC did not use any banned or regulated hazardous substances, and the revenue generated from products containing regulated hazardous substances was 0%.

If substances controlled by regulations must be used due to process research and development needs, internal reviews will be implemented to ensure that environmental protection, safety and health and other related risks are effectively controlled, and the user unit will be required to evaluate alternative chemicals, and at the same time provide on-site operators with strong Protective equipment, and conduct regular operating environment risk analysis in accordance with the "Environmental Safety and Health Risk Assessment Operation Methods" to fully ensure the safety of colleagues at the work site. Under the operation of the hazardous substances management mechanism, no violations of relevant laws and regulations such as product quality or the use of hazardous substances occurred in 2024.

Responding to the international hazardous substances management situation

Legal/Regulatory	PSMC Compliance Status
Restriction of Hazardous Substances in Electrical and Electronic Equipment Directive 2011/65/EU, RoHS	Compliant
EU Regulations on Registration, Evaluation, Authorization and Restriction of Chemicals (REACH)	Compliant
SONY's Management Regulations For The Environment-Related Substances To Be Controlled Which Are Included In Parts And Materials	Compliant
European regulations on Packaging and Packaging Waste	Compliant
International Electrotechnical Commission International Standard for Material Declaration for Products of and for the Electrotechnical Industry (Halogen Free Requirements)	Compliant
EU Candidate List of Substances of Very High Concern (SVHC)	Compliant
EU Persistent Organic Pollutants Regulation (POPs)	Compliant
EU Directive on Waste Electrical and Electronic Equipment (WEEE)	- PSMC's products are not end products, therefore the regulation does not directly apply - After use by the final user, end-of-life product recycling is the responsibility of end-product manufacturers.

Potential Hazardous Substance Substitution Plan

In recent years, global attention has increased on the potential harm of perfluoroalkyl substances (PFASs) to human health and the environment. PSMC has fully completed the perfluorooctanoic acid (PFOA) and perfluorooctane sulfonic acid (PFOS) replacement project in 2019, resolutely no longer use related substances. In addition, regarding the chemical substance N-Methylpyrrolidone (NMP, 1-methyl-2-pyrrolidone), which is used in the cleaning of semiconductor parts and the dilution of photoresist, it has been confirmed to cause damage to human reproductive health. Since 2021, the company has been cooperating with suppliers to evaluate and develop alternatives to potential hazardous substances in process raw materials, and formulated a year-by-year replacement plan in response to the future of international regulatory. In 2024, PSMC achieved 97.75% of its original target for potential hazardous substance reduction through its hazardous substance replacement plan. The Company commit will continue working with suppliers to track the progress of material development for long-term hazardous substance replacement, and continue to invest in research and development resources to replace potential harmful substances in the future.

Future planning of potential hazardous substance substitution program

Material Name	NMP Concentration (%)	Process / Application	Substitution Status	Year of Substitution Plan	Target of Accumulated Potential Hazardous Substance Reduction Percentage (%)
Polyamide A	45-55	Protection	R&D in progress	2027	97.86
Polyamide B	50-60	Protection		2029	98.18
Polyamide C	1-3	Protection		2030	100.00

Column

Continuous Improvement Process (CIP) Competition



The CIP Competition was launched in 2019 with six competitions having been held so far. Ever year were held two competitions since 2023. The competitions are always actively responded by employees, and demonstrate remarkable achievements obtained through departments' teamwork. PSMC believes that the CIP Competition can not only help trigger employees' creativity and competitiveness, but can also improve the overall quality, efficiency, innovation, and service standard for the Company to provide better products and services to customers, and make more contributions to society.

Quality Improvement

Improvement projects for Wafer Yield, Wafer Defect, Wafer Acceptance Test (WAT), Process capability (Cpk), and Margin of error.

Efficiency Improvement

Projects for the efficiency of processes, systems or equipment, or for the timely management of online raw materials and parts inventory, automation, reduction, etc.

Innovative Thinking

Projects to achieve quality or workflow improvement through the introduction of new concepts, ideas or processes.

Service Improvement

Listen and understand the needs and feedback of internal and external customers, improve service quality and enthusiasm through improvement projects, and develop service processes that make customers more satisfied.

Stop & Fix

Detecting phenomena different from the usual baseline through visual, olfactory, auditory, tactile and other sensory alerts or through data analysis.

Equipment Improvement

Improvement projects to improve quality or production capacity through machine equipment maintenance and software and hardware optimization (for equipment/factory colleagues).

CIP Competition Status

There were 271 projects in total participating in the 2024 competitions, a slight decrease of 16 projects from 2023, primarily due to raised entry requirements to ensure project completeness. The participating units included 12-inch Production Operation Center, 8-inch Manufacturing Center, Quality Center, Advanced Technology R&D Center/Module Technology Center, Technology Development Center, Memory Development Center, etc.. Among them, the number of pieces with the theme of 12-inch production operation center was the largest.

Register and submission

- The total number of submissions in 2024 was 271

Document Review

- The factory/division chief will be the judge
- 45 projects selected for final-round competition

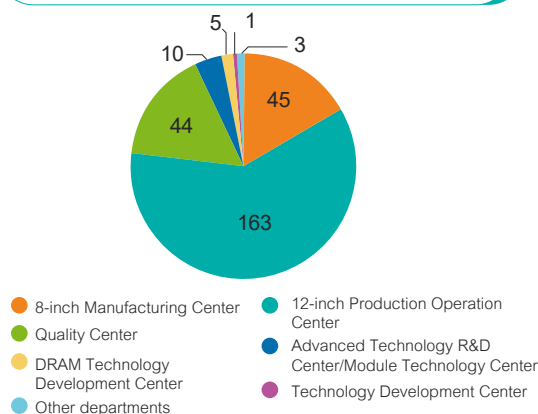
Finals Announcement

- The factory/division chief will be the judge
- Announcement of finalists by Quality Engineering Divisiong

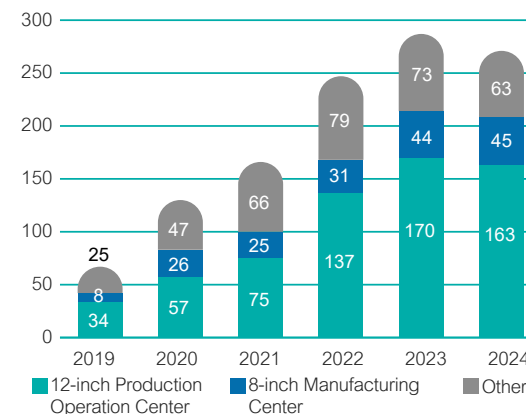
Awards Ceremony conducted in quarterly meetings

- Awards presented by the President

2024 Department Participation in CIP Competition



Number of CIP competition applications over the years



2024 CIP Competition Final-round Projects

Item	Wet Bench Automatic Optical Wafer Fragment Detection	Beyond Worldwide DNW Isolation Spacing (Deep N-Well)	Full Automation of WAT (Wafer Acceptance Test) Implementation
Project Description	Development of an optical inspection system for batch wet cleaning equipment (Wet Bench) used in wafer cleaning and etching processes, to improve detection of wafer fragments, missing wafers, and misaligned wafers. This addresses critical issues that lead to major wafer breakage	In driver ICs, circuits or IP blocks operating at different voltages are placed in separate DNW to prevent voltage noise and disturbances. However, due to excessive leakage current, DNW spacing could not be further reduced	Many WAT operations in the fab previously required manual intervention, including dispatching, transportation, operation, judgment, and record-keeping. Through this project, all operations have been fully automated, significantly reducing labor requirements
Overview of Project Content	- Develop strategies and improvement directions using the PDCA quality improvement methodology - Develop an innovative automated optical batch wafer inspection system with an automatic shutdown mechanism through collaboration project - Development, implementation, and validation of the wafer fragment detection model Image architecture setup: Utilize CCD (charge-coupled device) cameras and light source control systems to ensure high-resolution image output - Image capture: Integrate with machine signals to ensure synchronized image acquisition and equipment operation, enhancing real-time detection efficiency - Image processing: Apply region of interest (ROI) selection and multiple algorithms to accurately identify wafer fragments - Control system: When an anomaly is detected, the system automatically shuts down and sends a notification via the Line Equipment Management System (LEMS) to alert engineers for immediate action, minimizing operational impact	Through electrical characteristic analysis, key factors were identified, and an improvement experiment was designed, leading to the development of an innovative structure that significantly reduced leakage current and successfully minimized DNW spacing	Through analysis, four key improvement measures were proposed: - Transport System: Add tracks, transport vehicles, and equipment connections - Automated Dispatching: Obtain product-related information and assign it to equipment via the automated dispatching system - Programs: Automatically acquire probe pressure parameters, then select the corresponding product and program - Pattern Recognition: The first probe pressure pattern is detected through image recognition communication between the equipment and the big data server
Project Effectiveness	- Provide a real-time wafer integrity monitoring system for the process area - Prevent major wafer breakages in Wet Bench tanks. Between 2018 and 2024, incidents resulted in total material losses of NT\$15.85 million, 369 wafers scrapped, and 666 hours of downtime	This project utilized HV LDD P-type implant technology for high-voltage components, successfully achieving the following: - Leakage current reduction between DNWs (reduced from 1E-8A to 1E-12A, improving by four orders of magnitude) - Space scaling limit achieved (DNW spacing reduced from 5.5 μm to 2.7 μm) - Implemented Sharing Mask technology, enhancing process efficiency without increasing costs	- Labor cost savings - Cycle time (CT) optimization - Increased automation, reducing delays and variations caused by manual operations

1.3 Customer Relationship Management

Customer Communication

In order to deeply understand customers' needs, PSMC has established diversified communication channels to ensure that customers can receive immediate solutions.

Each channel has a dedicated person/line real-time communication and coordination window, and customer needs and feedback are obtained through customer meetings, customer satisfaction questionnaires and customer audits. The survey results will be discussed at internal business performance meetings to further plan accordingly. Improvement plans include building B2B product yield feedback data with customers and combining it with customer production information. A dedicated department is responsible for data analysis and product failure analysis, thereby improving production yield.

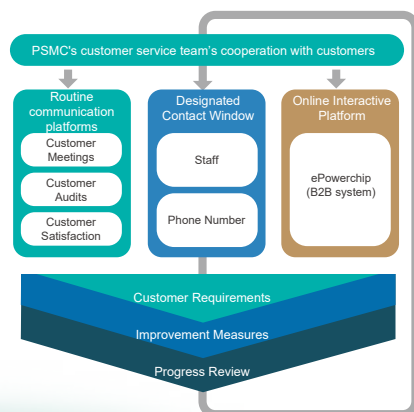
In addition, PSMC implements relevant improvement work through integrated analysis of quarterly product performance and regular discussions with customers to achieve the goal of improving product yield.

2024 Customer Communication Meeting Status

Meeting Type	Number of sessions held	Content of Discussion
Weekly / Monthly Meetings	>3,000	
Quarterly Meeting / Quarterly Operation Meetings	27	Product quality improvement, yield improvement, abnormal event countermeasures
Audit Meetings	50	Annual audits, ISO9001 /ATF16949 audits, Automotive Supply Chain Quality Control System (VDA6.3) audits

ePSMC Platform

The ePSMC platform provides customers with a channel for data upload and download. Various types of communication can be carried out on this platform. The communication content covers the distribution status of production lines, product research and development, process changes, engineering experiments, and customer satisfaction, etc. at each production stage.



Pteam Customer-Manufacturer Collaborative Communication Platform

To shorten the distance among customers, suppliers and employees, and at the same time to secure business secrets, PSMC has developed its own Pteam Business Collaboration Platform to carry out remote video conferencing, voice calls, message exchange, remote desktop connection, and augmented reality (AR) (collaboration with HoloLens) to improve communication efficiency, reduce communication costs, and effectively solve on-site problems such as equipment installation, borrowing-device testing, maintenance, and more.

The Three Benefits of Pteam's Customer-Manufacturer Collaborative Communication Platform

Improve work efficiency	Optimize cost effectiveness	Secure business secrets
178,000 online meetings were held through the Pteam Business Collaboration Platform in 2024, with a total of 1.517million participants engaged, and which including 3,000 of online meetings with business partners. - P1/2 Director Lin Yiliang gave feedback to Pteam that it has indeed improved the communication efficiency within Fab. Cross-department/cross-level communication becomes smoother and faster. - Pteam can provide of external business partners exclusive APPs starting in 2022, which can directly conduct collaborative work with employees such as text, voice, conferencing, remote desktop connections, etc., 2024, business partners sharing has been introduced total 44 stores with 210 users. - Customers can collaborate with remote experts through Pteam (approximately 131 people per month), and employees can cooperate with subsidiaries or customers/vendors within the group through Pteam to conduct product research and development. Under the principle of Pteam strictly guarding business secrets, ensures the quality of employees' work and the smooth progress of product development schedule. - To enhance communication efficiency between employees and international partners, PSMC introduced a large language model in 2024. Since April, the system has facilitated the translation of 3,300 documents. Since November, it has supported 350 online translation meetings, with a total of 3,400 participants. There were also 480 uses of voice translation pens.	- Self-development saves the cost of outsourced software such as instant messaging, remote platforms and online meetings, totaling approximately NT\$100 million, and the annual maintenance fees of instant messaging and remote platform and online meeting licensing fees are approximately NT\$8.05 million. - Reducing office and meeting room construction costs. - About 131 customers are provided with remote online borrowing-device testing services via the platform every month, reducing customers' transport costs. In addition, augmented reality (AR) (collaboration with HoloLens), allows customers/vendors can directly discuss through remote communication to effectively solve on-site problems. - This platform uses Open Source software for development, which greatly reduces development costs for the company, and uses it to introduce new software technologies and improve the technical capabilities of software developers. Before deciding open source software to adopt, the project team conducted a prudent and comprehensive technical assessment, including considering the functional requirements and business secret protection requirements of each unit of PSMC, and assessing whether it has continuous maintenance capabilities and must not have any concerns about information security vulnerabilities. - Pteam leads the industry and commercially available software, and can implement detailed authorization control in accordance with its internal, external or information security management policies within the company, maximizing the balance between individual work needs and information security.	Pteam provides security protection functions such as password secondary authentication and anti-photography. It can not only detect business secret documents that have been deliberately hidden, altered, and tampered with, but also design drawings, test programs, key intellectual property, etc. for similar business secrets, all can be detected with AI intelligence. To improve protection. Since its launch, Pteam has protected more than 111,000 business secret documents and approximately 63,000 design drawings, taking into account both efficiency and business secret protection.

Response to Customers' Needs

In response to customer complaints and product quality issues, PSMC used the electrical analysis results and related fault circuit location maps provided by customers to conduct product failure analysis to determine the abnormal site of the process, and provided an 8D improvement report. With the customer's consent, the customer complaint case can be close. In 2024, PSMC has set a customer complaint handling target of 1 case per quarter. In 2024, there are 1 customer complaints in the second quarter, and the remaining quarters met the target. A total of 1 customer complaints occurred in 2024, have obtained customer consent and been closed.

Impact Identification

When a customer receives reports of product anomalies from end-customers, if the customer's investigation is related to chip manufacturing, it may constitute a customer complaint event based on the level and quantity of the impact.

Response Mechanism

Customers can report customer complaints through business, engineering customers, or colleagues in the customer service department. Subsequently, the corresponding window of the customer service department will contact the customer and clarify the relevant follow-up events. If it is determined to be related to PSMC, the improved Eight Disciplines Problem Solving report will also be provided through the corresponding window of the customer service department and will be tracked until the customer agrees to close the case.

Handling Response

When dealing with customer complaints, we will conduct product fault analysis based on the electrical analysis results and related fault circuit location diagrams provided by the customer to determine the abnormal site in the process, and provide an Eight Disciplines Problem Solving improvement report, with the customer's consent, the customer complaint incident can be closed. If customers have dissatisfaction, in addition to promptly expressing their opinions through phone calls or emails, they can also be raised through various regular/irregular meetings. At the same time, the company has a dedicated window to contact customers and provide timely assistance.

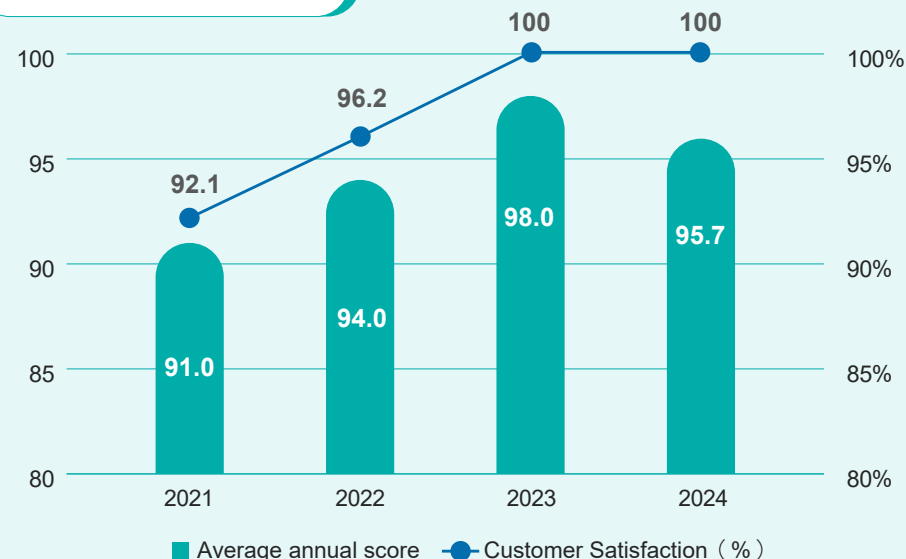
Review and Improvement

PSMC has established KPIs for handling customer complaints, which are discussed at monthly meetings held by the quality unit and semi-annual quality management review meetings, and strengthened management measures are formulated.

Customer Satisfaction Survey

At the beginning of each year, based on the customer's wafer production volume in the previous year, PSMC selects customers whose total wafer production volume is greater than 80% of the chip output as the objects of customer satisfaction survey. The survey coverage rate in 2024 reached 100%. Satisfaction evaluation will be conducted once a month, and the overall evaluation results will be provided at the end of the year and improvements will be discussed. The satisfaction survey covers five major aspects: quality, delivery time, technology, green products and others. PSMC set a customer satisfaction threshold of 88 points in 2024. A total of 28 customers participated in the annual satisfaction survey. Therefore, customer satisfaction in 2024 was 100%, with an average annual score of 95.7 points, reaching the 2024 customer satisfaction level. The annual goal of satisfaction is above 90%.

Customer Satisfaction



2 Synergized Sustainability

| Supply Chain |

2.1 Supply Chain Overview

2.2 Sustainable Supply Chain Management

2.3 Conflict Mineral Management



Synergized Sustainability

Supply Chain

PSMC actively promotes sustainable supply chain development in alignment with the core principles of the Responsible Business Alliance (RBA). In addition to establishing clear supply chain management processes and due diligence mechanisms, the Company has progressively expanded its management scope to all Tier 1 suppliers to ensure the stability of raw material and service supply. Furthermore, PSMC supports significant suppliers through audits and guidance, sustainability information sharing, and on-site greenhouse gas inventory projects, providing them with the necessary information and resources for sustainable transformation. The Company aims to scale operations alongside suppliers while minimizing negative impacts on environmental resources.



Corresponding United Nations Sustainable Development Goals (SDGs)



Corresponding Material Topics



Sustainable Supply Chain



Stakeholders who have priority reading

Suppliers / Contractors



Material Topics GRI Standards

GRI 204 : Procurement Practices

GRI 205 : Anti-corruption

GRI 308 : Supplier Environmental Assessment

GRI 408 : Child Labor

GRI 409 : Forced or Compulsory Labor

GRI 414 : Supplier Social Assessment



2024 Sustainability Performance Highlights

97%

Proportion of local procurement from
suppliers in Taiwan

100%

Supplier ESG Self-Assessment Questionnaire
response rate

100%

Proportion of significant suppliers who have
signed RBA Commitment Statement

100%

Free from sourcing of conflict minerals from
conflict-affected areas

2.1 Supply Chain Overview

PSMC, as a midstream manufacturer in the semiconductor industry chain, is specialized in wafer manufacturing and foundry services. In terms of procurement practices, PSMC divides our suppliers into three major categories: goods, equipment, and labor. All suppliers are required to comply with PSMC's Supplier Code of Conduct and risk assessment management mechanisms to ensure that they can meet the company's various sustainability standards. In addition, PSMC has focused our management resources on the suppliers with whom the annual procurement value has reached a certain level, and transactions are conducted on a continuous basis. Such suppliers are categorized as Tier 1 suppliers. Moreover, a tiered management system is implemented based on the supplier's degree of operational materiality to PSMC.

Raw Material Procurement

Focusing on the profound impact of the raw materials used in the manufacturing process on the environment and society, PSMC has formulated a "raw materials policy" to implement the spirit of sustainable raw material use in its procurement practices. Through the supplier risk management process, we evaluate whether the supplier's operation and production of raw materials have negative impacts or potential risks on social and environmental aspects such as labor safety, energy resource consumption, ethics, and quality, and prioritize the procurement of raw materials and use this process to confirm the traceability of raw material sources to suppliers.

To reduce the environmental and social impact of raw material production, PSMC requires suppliers to sign the Supplier Code of Conduct to ensure compliance with environmental and social regulations. Additionally, the Company actively communicates with suppliers via emails and meetings to encourage them to integrate biodiversity considerations into their operations and investment strategies. PSMC will in the future provide technical support and project cooperation with suppliers to reduce the negative external impacts of raw material production for making substantive contributions. In addition, in order to systematically implement the raw material management plan, PSMC has launched the development of in-plant recycling and remanufacturing technology for some raw materials and is gradually increasing the use of third-party verified sustainable raw materials and the proportion of procurement and use of recycled raw materials to continuously track and report on the progress of procurement targets. And through themed training courses, purchasing personnel can fully grasp the concepts and essentials of sustainable raw material purchasing, so as to achieve phased purchasing goals as scheduled.

Tier 1 Suppliers

Supplier Category		Asia (including Taiwan)	Europe	America	Subtotal
Goods Category (New Products & Raw Materials)	Number of Tier 1 suppliers	209	0	5	214
	Number of Tier 1 significant suppliers	73	0	1	74
	As proportion of Tier 1 significant suppliers (%)	14.34	0	0.20	14.54
	As proportion of Tier 1 significant suppliers' procurement value (%)	43.05	0	0.11	43.16
Equipment Category (Facilities & Processes)	Number of Tier 1 suppliers	101	3	5	109
	Number of Tier 1 significant suppliers	11	0	0	11
	As proportion of Tier 1 significant suppliers (%)	2.16	0	0	2.16
	As proportion of Tier 1 significant suppliers' procurement value (%)	0.60	0	0	0.60

Supplier Category		Asia (including Taiwan)	Europe	America	Subtotal
Labor Category (Contracting & Services)	Number of Tier 1 suppliers	184	0	2	186
	Number of Tier 1 significant suppliers	29	0	0	29
	As proportion of Tier 1 significant suppliers (%)	5.70	0	0	5.70
	As proportion of Tier 1 significant suppliers' procurement value (%)	4.24	0	0	4.24
Total	Number of Tier 1 suppliers	494	3	12	509
	Number of Tier 1 significant suppliers	113	0	1	114
	As proportion of Tier 1 significant suppliers (%)	22.20	0	0.20	22.40
	As proportion of Tier 1 significant suppliers' procurement value (%)	47.89	0	0.11	48.00

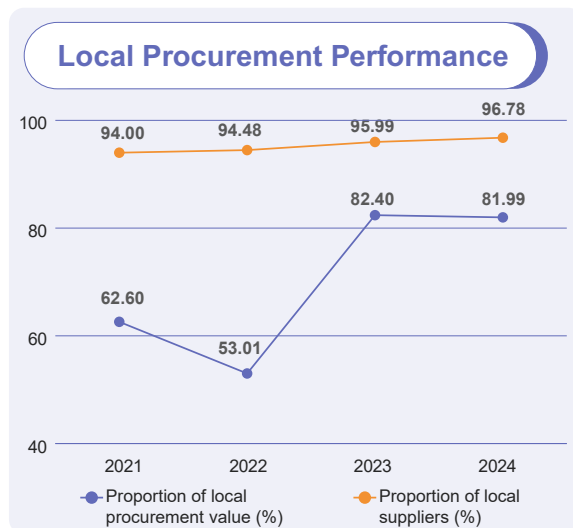
Note 1: Tier 1 suppliers refer to those whose annual transaction value with PSMC is greater than NTD\$10 million.

Note 2: There are three categories of significant suppliers, which are critical raw material suppliers (wafers used for fabrication), suppliers/contractors for large-quantity parts/components (whose semi-annual transaction value with PSMC is greater than NTD\$5 million), and irreplaceable suppliers or similar.

Note 3: Proportion of Tier 1 significant suppliers = (Number of Tier 1 significant suppliers in the specific category/region ÷ Total number of Tier 1 suppliers) * 100%.

Local Procurement

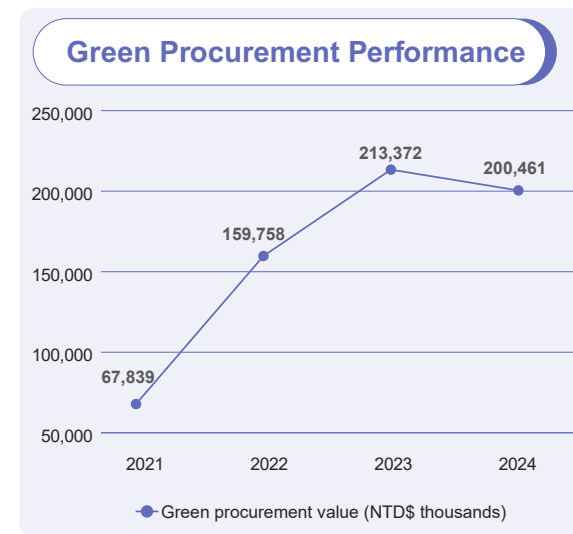
PSMC follows the principle of local procurement for our raw material sourcing operations. Local procurement not only helps to reduce greenhouse gas emissions from transportation, create local employment opportunities, and promote economic development, but also improves PSMC's ability to control product quality and lead times, while helping build a resilient and competitive supply chain. In 2024, local suppliers accounted for 96.78% of PSMC's suppliers, and their procurement accounted for 81.99% of total value. There was also a steady trend in these figures.



Note: PSMC's main operating base is located in Taiwan. The chart shows the proportion of purchases from local suppliers.

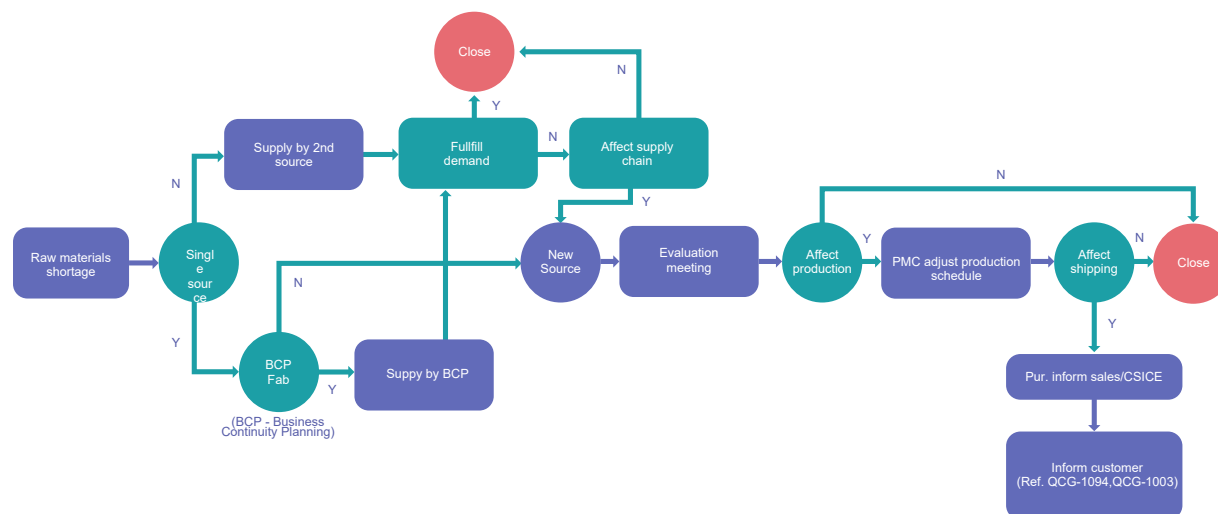
Green Procurement

At the supplier selection stage, PSMC gives priority to products and services that are Green Label certified, and recognized by governments and international organizations. The green product/service assessment criteria have been incorporated into the company's procurement practices and supplier assessment screening mechanisms. PSMC's total green procurement value in 2024 exceeded NT\$200 million. Despite weak consumer market demand in 2024, the Company maintained a procurement amount comparable to the previous year. The major procurement items were products that have passed Taiwan's Type I environmental labeling certification, and other certifications such as Energy Label, Carbon Label, Green Building Material Label, as well as internationally recognized ENERGY STAR Label, EPEAT (Electronic Product Environmental Assessment Tool) Registry Label, TCO Certified Label initiated by the Swedish Federation of Professional Employees, etc.



Business Continuity Plan – Supply Chain Risk Impact Management

The sustainability risks derived from changes in macro environments have urged PSMC to not only consider traditional risks that come from shortage of raw materials and parts in the global market, but also to incorporate emerging sustainability risks into the scope of risk management. By doing so, we establish more comprehensive supply chain response strategies and ESG plans, with the company's board of directors being responsible for supervising the implementation of supplier management affairs. To ensure the supply chain's continuous and stable development, and in accordance with PSMC's Measures for Emergency Response to Shortage of Production-purposed Raw Materials, when the quantity supplied by suppliers is lower than the company's demand quantity, the situation will be considered a significant supply shortage. When suppliers are unable to meet demand, in accordance with the Measures for Emergency Response to Shortage of Production-Purposed Raw Materials, PSMC classifies the situation as a significant supply shortage and immediately initiates cross-departmental evaluation meetings to discuss response measures. To control the potential impact of emerging sustainability risks on supply chain stability, PSMC's procurement-related units have carried out related advocacy, and implemented sustainable supply chain management measures to ensure suppliers' compliance with PSMC Supplier Code of Conduct, and avoid the potential impacts arising from conflicts with ESG index requirements. Moreover, PSMC has provided training courses focusing on the company's Supplier ESG Program for dedicated procurement personnel and internal stakeholders to raise the awareness of risk management among related personnel so that they can apply risk management principles in daily procurement operations.



2.2 Sustainable Supply Chain Management

Sustainable Supply Chain Development Strategy

Strategic Dimension	Guidelines for Strategy Implementation
Strengthening diversified strategic partnership	• Maintain at least two suppliers of the same raw materials to ensure a stable supply of raw materials. • Adhere to the principle of local procurement and strengthen the proportion of local suppliers. • Cooperate with suppliers on ESG projects to build suppliers' sustainable development capabilities.
Implementing risk management	• Establish supplier screening and risk assessment mechanisms to strengthen supply chain resilience. • Enhance the sustainability requirements for suppliers' economic, environmental and social aspects to ensure that PSMC can obtain the best quality of products and services.
Avoiding conflict minerals	• Implement the Conflict Mineral Due Diligence and management to ensure use of non-conflict minerals from qualified smelters.

Sustainable Supply Chain Management Process

PSMC has constructed a four-stage sustainable supply chain management process, using a dual-track parallel model through independent supplier assessment and active investigation by the company's procurement unit, and introduced an evaluation system covering ESG indicators to ensure that suppliers meet PSMC's various procurement standards and requirements.

Establishing and Following Sustainability Norms	ESG Risk Assessment and Management	Sustainability Auditing and Counseling improvement	Supplier Engagement and Development
• Formulate Raw Material Supplier Management Measures/ Contractor Evaluation and Management Rules to implement supplier management tasks • Require all suppliers to sign the "Supplier Code of Conduct Commitment" stipulated by PSMC	• Through the "Manufacturing Questionnaire", in addition to understanding the supplier's quality control, manufacturing and inspection technical capabilities, it also evaluates other ESG factors such as the characteristics of the products it supplies, industry attributes, operating conditions, and the region it belongs to • Only those who meet various evaluation standards and sign a letter of the "Supplier Code of Conduct Commitment" can become a qualified supplier	• Written evaluation: Based on the basic evaluation aspects such as quality, delivery, service, etc., combined with the supplier's "Sustainability Management Self-Assessment Questionnaire" and supporting documents provided by the supplier, assess sustainability risks on the Tier 1 significant suppliers written evaluation • On-site audit: Procurement, joint quality, engineering and other units conduct on-site audits on suppliers to understand the supplier's risk status • Provide improvement suggestions for deficiencies found in written and on-site audits, require suppliers to improve within a deadline and complete re-examinations	• Promote the latest sustainability information to suppliers from time to time, and provide necessary guidance and on-site technical support • Cooperate with suppliers to develop supplier building projects in order to equip suppliers with the ability to independently promote sustainable development

Establishing and Following the Sustainability Regulatory

PSMC requires suppliers to meet requirements for quality, cost, lead time, service, and other indicators, and to comply with local regulations. In addition, PSMC has formulated our Supplier Code of Conduct based on the Responsible Business Alliance (RBA)'s Code of Conduct, expanded to labor rights, health and safety, environmental protection, business ethics, management systems, and so on. The Code is also used as a guideline for the company to communicate and promote the concept of ethical management with/to all suppliers, who are then required to sign a Commitment Statement and abide by it. In addition, PSMC also conducts regular supplier risk assessments to ensure that the company's suppliers and contractors are not involved in any issues related to child labor or forced/compulsory labor. If any of the aforementioned violations is identified and the situation is of great concern, the business partnership will be terminated. Due to the high entry threshold of the semiconductor industry, the number of new suppliers in 2024 will be lower than in 2023, 100% of the company's qualified suppliers (with whom transactions have been conducted) and new suppliers completed signing of the Code of Conduct, and declared their commitment to comply with social and environmental standards, achieve the goal of setting 100% of suppliers signing the commitment letter by 2024. No supplier partnership was terminated in 2024 due to violation of the Code of Conduct.

Signing of Supplier Code of Conduct & Future Goals:

	Item	2023 Performance	2024 Performance	2025 Goal
Qualified suppliers with whom transactions have been conducted	Number of suppliers required for code-signing	975	1,387	-
	Proportion of suppliers completing code-signing (%)	100	100	100
New suppliers	Number of suppliers required for code-signing	288	7	-
	Proportion of suppliers completing code-signing (%)	100	100	100

Supplier Code of Conduct: <https://esg.powerchip.com/en-global/staticpage/supplier-and-contractor-management>

ESG Risk Assessment and Management

PSMC has established a proactive supplier screening process, incorporating factors such as industry and regional/country-specific risks, risks associated with supplied products, operational status, and other ESG assessment criteria as the selection standards for significant suppliers. Through preliminary investigation and assessment, the suppliers who cannot meet the minimum ESG requirements before specified deadlines will be weeded out. Moreover, PSMC will take suppliers' ESG performance into consideration when choosing suppliers or contractors. Priority will be given to the suppliers with better comprehensive ESG performance in the hope of reducing the negative impact of inferior suppliers on the overall supply chain.

Aspects of suppliers screening	Assessment Aspects
Environment	Greenhouse gas inventory, water resource use and management, carbon footprint survey, no hazardous substance control documents
Society	Human rights management, labor rights
Governance	Financial status, operational performance, conflict metals investigation, major violation incidents
Operating status	Basic information, major customers, industrial competitiveness, product quality and services, production and equipment status, investment and product continuity planning
Country-specific risk	Geopolitical relations, local policies/regulations and economic development status, natural and anthropogenic disasters
Sector-specific risk	RBA Supplier Code of Conduct Survey, industry-specific risks, labor composition structure
Commodity-specific risk	Procurement value, categories of products supplied, use of hazardous substances, Energy use

Sustainability Auditing and Counseling Improvement

Sustainability Risk Questionnaire (Written Assessment)

Based on our existing supplier assessment systems, PSMC has developed our ESG Management Self-Assessment Questionnaire (SAQ), which covers the five dimensions of Labor, Health and Safety, Environment, Ethics, and Management Systems, to carry out in-depth surveys on suppliers' implementation for various related issues. In addition to the questionnaire, suppliers are also required to provide related supporting documents such as management system certifications, etc. for the company's procurement unit to conduct analysis of survey results. Moreover, the unit will also examine suppliers' comprehensive performance in terms of quality, lead time, service, etc. to identify high-risk suppliers and formulate subsequent response measures. 100% of PSMC's Tier 1 significant suppliers took a written-format risk assessment survey in 2024. Of these, suppliers for five raw materials (chemical substances, special gases, photoresist, CMP slurry, and target materials) and contractors for related factory expansion operations under contracting companies, for a total of 50 companies, were further selected to complete the ESG Management Self-Assessment Questionnaire. The actual response rate was 100%.

2024 SAQ Results

Compliance rate	Risk Level	Improvement Actions	Number of Suppliers
Compliance Rate > 80%	Low Risk	-	42
60% < Compliance Rate < 79%	Moderate Risk	-	8
Compliance Rate < 60%	High Risk	For non-compliant items, suppliers were required to propose improvement plans and complete rechecks	0
Total			50

On-site Audits and Deficiency Improvement

With reference to the ESG Management Self-Assessment Questionnaire survey results and PSMC's Raw Material Supplier Management Measures/Contractor Evaluation and Management Rules, a sample of significant suppliers are selected as auditees for the current year's on-site audits based on indicators such as transaction frequency, procurement value, etc. and raw material suppliers' biannual evaluation results. The audits are carried out in forms including but not limited to on-site audits conducted by PSMC's procurement and related units, second-party audits (supplier audits/consultants), and third-party audits, or audits conducted based on the industry's accepted standards. For points of non-compliance, PSMC will analyze the deficiency causes with the suppliers, and propose suggestions for improvement. Suppliers are required to improve deficiencies and complete rechecks before specified deadlines. PSMC will also provide necessary guidance and on-site technical support in a timely manner. In terms of the written-format supplier assessments and on-site audits conducted in 2024, the goal achievement rate for auditees was 100%, with no major audit deficiencies or violations.

Item	2023	2024
Total number of suppliers audited by written evaluation and on-site	104	114
Percentage of significant suppliers evaluated (%)	100	100
Number of suppliers assessed to have significant actual/potential negative impact	3	0
Percentage of suppliers with significant actual/ potential negative impacts that have discussed corrective actions/improvement plans (%)	2.88	0
Number of suppliers with significant actual/potential negative impact and consequent termination of cooperation	0	0

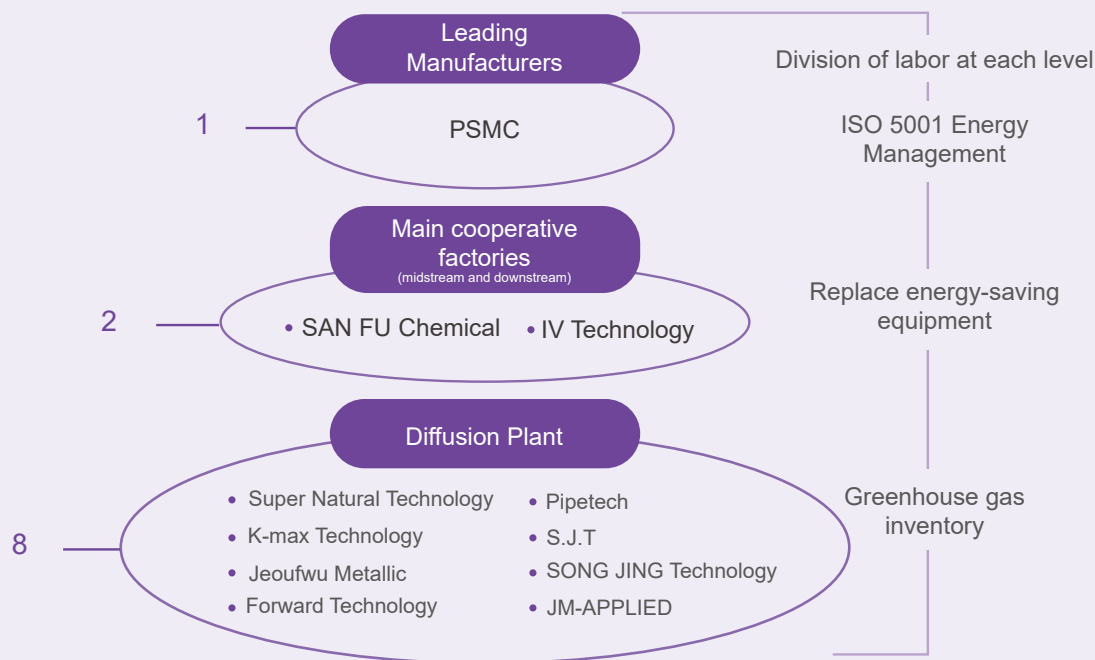
Supplier Engagement and Development

Based on our experiences in promoting sustainability (ESG) projects, PSMC always provides suppliers with suggestions on directions for ESG task implementation and required practical guidance. This helps them be aware of peer companies' ESG development status, so that they can continuously enhance their ability to achieve sustainable development. In addition, with such suggestions and guidance, suppliers can fully understand the plans and processes that PSMC has adopted for supplier ESG management. PSMC will continue to work with suppliers in the future to initiate different projects that will help build up suppliers' ability to facilitate ESG tasks, thereby reinforcing PSMC's connection with suppliers' business development.

Item	2023 Performance	2024 Performance	Target of 2024
Number of suppliers with building capacity	24	50	50
Percentage of significant suppliers (%)	23.08	43.86	-

"Leading by Example" – PSMC Joins Forces with Suppliers to Reduce Carbon Emissions

To assist suppliers in enhancing economic and social resilience while ensuring shared economic benefits for all, PSMC has responded to the Ministry of Economic Affairs' initiative to upgrade and transform industries and SMEs by applying for the "Leading by Example" Low-Carbon and Smart Manufacturing Upgrade Program. This initiative aims to strengthen industrial economic resilience and competitiveness. Starting in April 2024, PSMC has been assisting 10 suppliers in implementing a greenhouse gas inventory system and developing carbon reduction plans to enhance energy management and achieve effective supply chain carbon reduction. This project has been approved by the Ministry of Economic Affairs and is expected to be fully implemented by the end of October 2025.



2.3 Conflict Mineral Management

Tantalum, tin, tungsten, gold, and cobalt are essential minerals required for the manufacturing of electronic products. However, most of these minerals come from the Congo region and neighboring countries, where armed conflicts constantly occur, these minerals are thus called conflict minerals. To avoid using minerals from conflict-affected areas in product manufacturing processes, PSMC conducts precise supplier due diligence investigations, and also requires all suppliers for target materials and gases, with whom transactions have been conducted, to commit not to use materials or gases from upstream vendors/origin countries located in conflict-affected areas, in order to achieve the goal of 100% freedom from use of conflict minerals. PSMC also calls on all supplier partners who have business dealings with PSMC to declare that they will "not accept the use of metals from conflict-affected areas", as promoted by the Responsible Business Alliance (RBA). PSMC also requires raw material suppliers to comply with the following regulations to convey PSMC's No Conflict Mineral Use policy to their upstream suppliers and supply chain partners.

- Suppliers must fulfill social and environmental responsibilities, to mitigate the impact of business operations on local communities and the environment.
- Suppliers must ensure that their products do not use any minerals from upstream vendors/origin countries located in conflict-affected areas that have not been deemed qualified through third-party verification, nor minerals mined in poor working environment.
- Suppliers must disclose the sources of the metals/minerals they use, including tantalum, tin, tungsten, gold (3T1G), etc., and complete the investigation forms to implement PSMC's mineral origin investigations.

Conflict Minerals Management Measures

Types of Conflict Minerals	Management Measures
Tantalum, tin, tungsten, gold	Conduct Conflict Minerals Due Diligence Investigation and update related data on a yearly basis; irregularly review the Responsible Minerals Initiative (RMI) website for the latest information and require suppliers to complete the Conflict Minerals Reporting Template (CMRT) to ensure no use of conflict minerals.
Cobalt	The mineral has not yet been included in the scope of RMI's Conflict Minerals Reporting Template, PSMC continued to pay attention to updates to the Conflict Minerals Reporting Template. Once it is listed as a mineral for control and management, PSMC will firmly require our suppliers to implement control measures.

Conflict Minerals Due Diligence Process



PSMC follows conflict minerals reporting procedures in conducting our Conflict Minerals Due Diligence Investigations. We focus on suppliers whose materials contain tantalum, tin, tungsten, gold, or cobalt. In addition to assessing the necessity of such materials using each specific mineral, the investigations also check the origins of the suppliers' raw materials to ensure that they comply with the conflict minerals ethical and legal standards followed by PSMC. To effectively carry out due diligence, PSMC has completed registration on the RMI website, and adopts the Conflict Minerals investigations tool. Detailed information about suppliers' raw material origins and manufacturing processes collected through the investigations can not only be used as a basis for mineral origin country checks. Moreover, can be used for risk identification and guarantee of 100% of No Use of Conflict Minerals, at the same time, PSMC provides the results to customers as proof to confirm that PSMC's products do not use any minerals from upstream vendors or origin countries located in conflict-affected areas that have not been deemed qualified through third-party verification. The company also pays regular attention to updates on regulations and information released by RMI, to ensure the validity of our investigation procedures and their compliance with international standards.. In 2024, a due diligence investigation on conflict minerals was conducted on 70 major raw material suppliers. The investigation results were all in compliance with the requirements, confirming that 100% of PSMC's products do not use minerals from conflict areas.

Conflict Minerals Due Diligence Investigation



3 Commitment to Sustainability

| Environment |

- 3.1 Climate Strategy
- 3.2 Energy Management
- 3.3 Water Resources Management
- 3.4 Waste Management
- 3.5 Air Pollution Emissions Management
- 3.6 Biodiversity Management



Commitment to Sustainability

Environment

In response to Taiwan's increasingly stringent environmental regulations and policies, PSMC is committed to minimizing the environmental impact of its manufacturing processes by implementing renewable energy, adopting high-efficiency process equipment, optimizing water recycling management systems, and enhancing waste and air pollution treatment technologies. Looking ahead, we will continue to strengthen green collaboration across the supply chain, jointly develop low-carbon manufacturing models with partners, and deliver products with a competitive low-carbon advantage to our customers, fostering a resilient and sustainable ecosystem.



Corresponding United Nations Sustainable Development Goals (SDGs)



Material Topics GRI Standards

GRI 201: Economic Performance

GRI 305: Emissions

GRI 302: Energy

GRI 306: Waste



Corresponding Material Topics



**Energy
Management**



**Waste
Management**



**Air Pollution
Control**



**Climate
Strategy**



Stakeholders who have priority reading

Government agencies / Authorities, Employees, social / Local Communities



2024 Sustainability Performance Highlights



8,987.23 metric tons CO₂e

The overall carbon reduction in all fab was 10,117metric tons of CO₂e. More than 90 energy saving measures have been implemented, with energy consumption reduced by a total of 59,388.99 GJ.



1MW

The P5 fab completed installation of a 0.5 MW in-house solar photovoltaic system, bringing total installed capacity for the entire fab area to 1 MW, with an estimated annual generation of 1.25 GWh of green electricity.



88%

The overall average process water recovery rate in all fab reached 88%. Five water use efficiency improvement projects have been implemented.



97.71%

The average removal rate of VOCs exhaust reached 97.71%. Four air pollution improvement projects have been implemented.



42,405 metric tons CO₂e

All newly installed process equipment using fluorinated gases in the fab have been equipped with combustion-type exhaust gas treatment systems. In addition, existing process exhaust treatment equipment has been replaced and upgraded to expand the treatment scope. The total carbon reduction reached 42,405 metric tons of CO₂e, which is equivalent to the annual carbon absorption of approximately 110 Da'an Parks.



0

Number of violations of environmental regulations

3.1 Climate Strategy

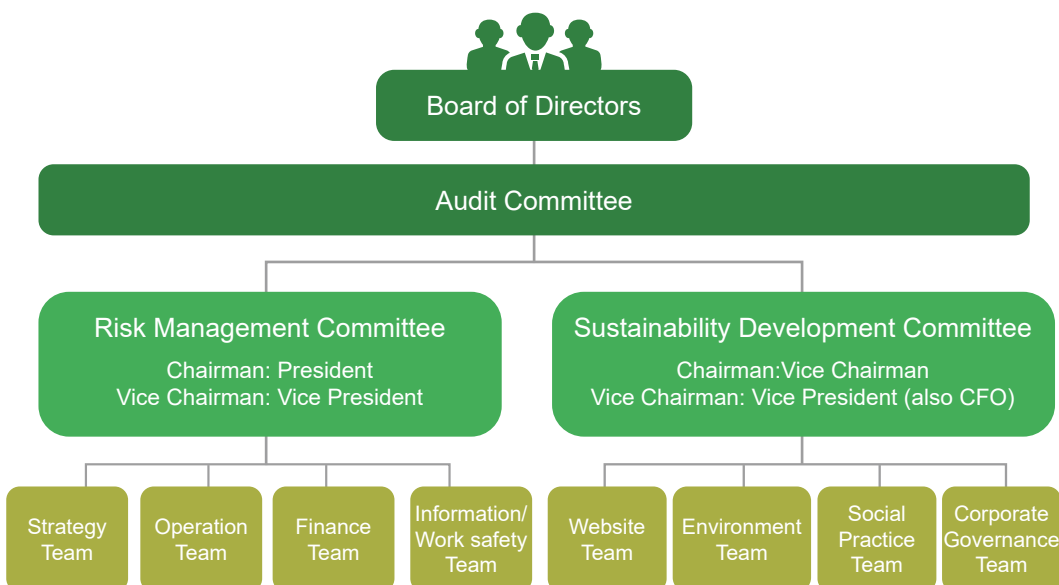
3.1.1 Climate Action

PSMC has introduced the Task Force on Climate-related Financial Disclosures (TCFD) framework to build a management structure to respond to climate risks and opportunities based on four major aspects: governance, strategy, risk management, indicators and goals, and identify potential risks and opportunities arising from climate change, and assess the actual impact and impact of climate issues on the company's business.

Climate Governance

The Board of Directors of PSMC serves as the highest climate governance unit, responsible for overseeing the achievement of climate change strategies and goals, and ensuring the consistency of the organization's operational development direction and goals. In order to improve the evaluation and implementation effectiveness of climate change strategies and plans, the Audit Committee is responsible for managing, integrating and organizing ESG issues and climate-related risks and opportunities, confirming the overall strategic direction and goals of the organization, and setting up the "Sustainability Development Committee" and the "Risk Management Committee" under its jurisdiction. The "Sustainability Development Committee" is responsible for formulating various climate change strategies. The four strategic implementation teams under the committee will assist each business unit in implementing the climate change strategy response plan and review the compliance and appropriateness of the response strategies and goals. The "Risk Management Committee" is responsible for identifying and analyzing the organization's climate-related risks, effectively controlling the potential risks faced by the organization on climate change, and ensuring that risk management can be implemented in the daily operations of each business unit through the four strategic implementation teams. The "Sustainability Development Committee" and the "Risk Management Committee" regularly report the strategic implementation of climate risks and opportunities of each department to the Audit Committee to facilitate it to track the development progress of each strategy and adjust the implementation plan in a timely manner.

TCFD Governance Framework



- Convened regularly every year
- Supervises climate-related strategies and goal achievement status

- Convened regularly every year
- Review the effectiveness of climate risk implementation and make recommendations for necessary improvement
- Report regularly to the Board of Directors on the effectiveness of the implementation

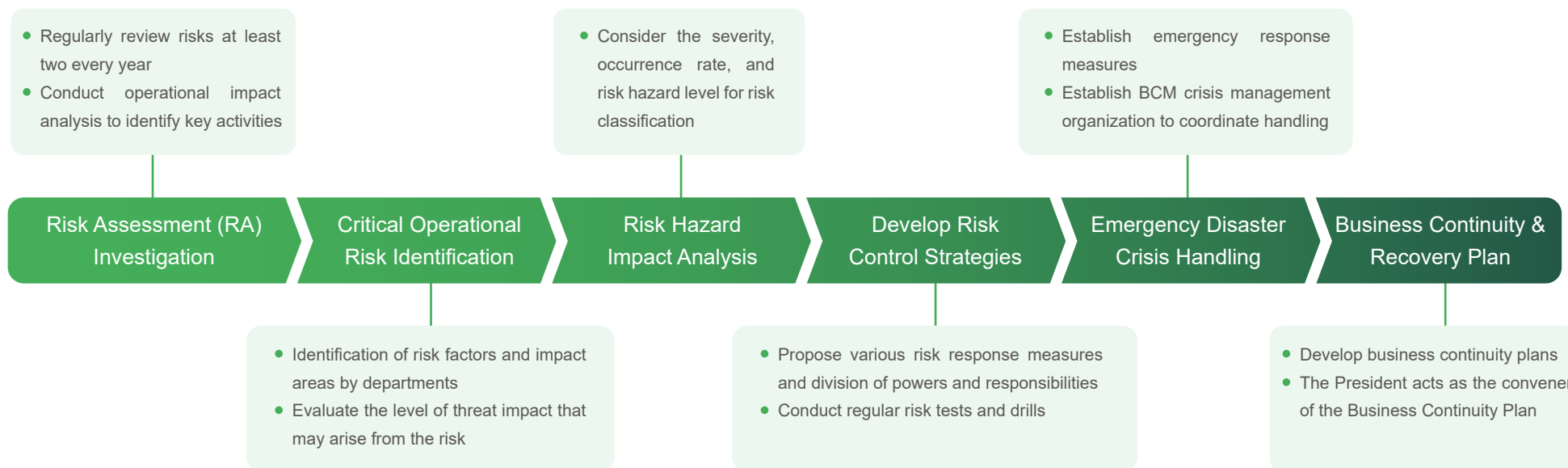
- Convened on a semi-year basis
- Responsible for identifying, analyzing, evaluating and responding to climate risks and opportunities within the business unit
- Implement Climate Risk Management
- Regularly report to the Audit Committee and the Board of Directors

- Convened on a quarterly basis
- Develops strategies and objectives related to climate-change-related
- Assists all departments in promoting water and energy saving, and emission and waste reduction measures, and ensures achievement of goals
- Regularly report to the Audit Committee and the Board of Directors

Climate Risk Management Process

PSMC uses a risk assessment, from key operating units of each department, based on the risk identification results, the three aspects of potential impact degree, vulnerability and probability of occurrence are considered, and key encountered risks at the occurrence stages are sorted. At the same time, PSMC corresponding risk control strategies are also developed, and must be reviewed at least every two years, and judge whether adjustment is needed according to the inspection results. In addition, through "establishing emergency response measures", when a risk disaster occurs, the heads of the responsible departments will carry out notification work in accordance with the factory disaster emergency response measures, and the crisis management team will coordinate the crisis handling and subsequent recovery work to maintain the company's continuous operations.

Risk Identification and Management Flow Chart



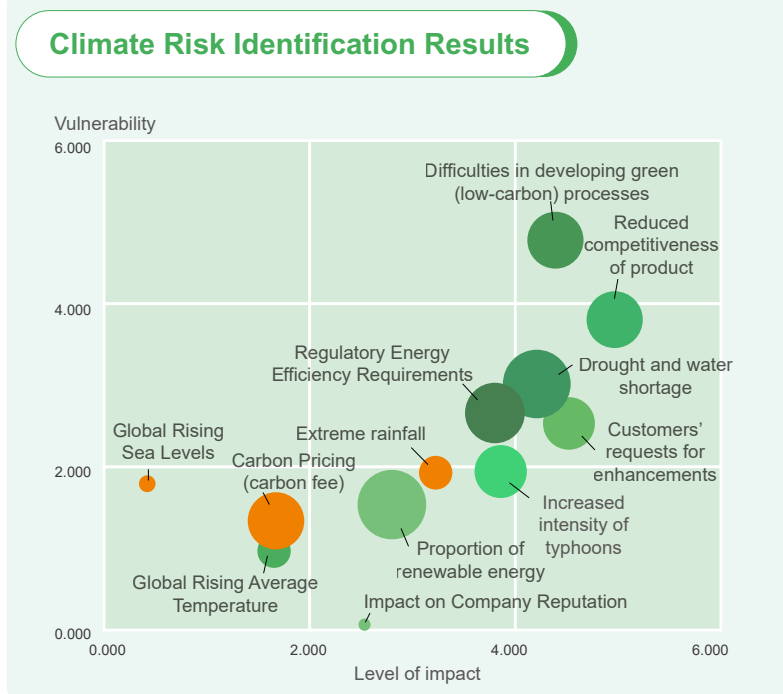
Climate Strategy

PSMC has identified 12 climate risks and 9 climate opportunities by considering the domestic and international climate change development trends and its own current operating conditions, and the relevant departments of authority have jointly evaluated the level of impact of climate related risks and opportunities. The evaluation covers the entire value chain of PSMC. Based on the level of impact, vulnerability, control, and probability of occurrence that each risk and opportunity, and the characteristics of the scale for short-middle-long term period has on PSMC, we identified the key issues that we should prioritize and address for climate change after an internal investigation, and formulated the strategies and targets in order to implement climate change management.

Climate Risk List

Type	Risk Items	Possible time of occurrence	Criticality	Risk assessment impact value chain scope		
				Upstream (Supply chain)	Operation	Downstream (Customer)
Physical Risk	Immediacy	Increased Intensity of Typhoons	Short Term	Medium	V	V
		Extreme Rainfall	Short Term	Low		V
	Long Term	Global Rising Average Temperature	Long Term	Low		V
		Drought and Water Scarcity	Medium Term	High	V	V
Transformation Risk	Regulations and Policies	Global Rising Sea Levels	Long Term	Low		V
		Regulatory Energy Efficiency Requirements (Including current and emerging regulations)	Short Term	High		V
		Proportion of Renewable Energy	Short Term	Medium		V
		Carbon Pricing (carbon fee)	Medium Term	Medium		V
	Market	Customers' Requests for Enhancements	Medium Term	Medium		V
		Reduced Competitiveness of Products	Medium Term	High	V	V
	Technology	Difficulties in Developing Green (low-carbon) Processes	Medium Term	High		V
		Impact on Company Reputation	Long Term	Low		V
	Reputation	Impact on Company Reputation	Long Term	Low	V	V
		Impact on Company Reputation	Long Term	Low	V	V

Note: Short Term: 1~3 years, Medium Term: 4~6 years, Long Term: 7 years or more

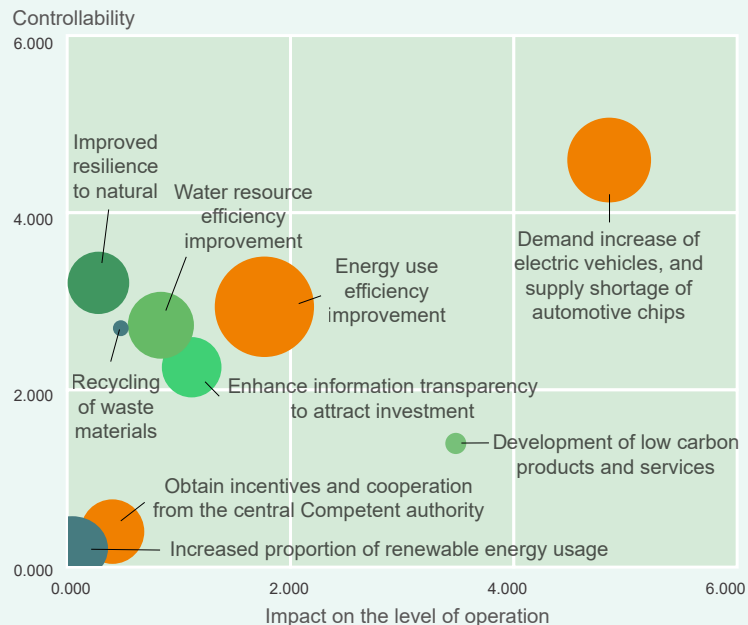


Note: Horizontal axis - degree of impact: if that particular risk were to occur, the extent to which it would most severely affect PSMC's revenue or operations; vertical axis - vulnerability: PSMC's readiness and ability to adapt to that risk if it were to occur; the size of the diameter of the circle - probability of occurrence: the probability of the occurrence of that risk factor affecting PSMC.

Key Climate Risks and Response Strategies

Risk Factor	Risk Description	Area of Impact (Service/strategy that will affect the company)	Response Strategy (Service/strategy that will affect the company)
Drought and water scarcity	Extreme weather causes uneven rainfall, long periods of no rainfall cause water scarcity, and water restriction policies cause disruptions in product production and reduced operational efficiency	• Increase of Operating Costs • Operational Disruption Risk	• The plant can withstand about a 20% of water restriction, and in the case of unbearable water restriction, water trucks will be activated to carry water. Based on the worst historical water shortage in 2021, where the water supply gap reached 17%, and assuming a water cost of NT\$550 per ton, the estimated cost for sourcing water via water trucks would be approximately NT\$14,000,000 • Plan for the use of recycled water. Based on the unit price of desalinated (recycled) water at NT\$31/m³ and tap water at NT\$12/m³, PSMC's 10% water demand would amount to 1,650 CMD. Therefore, the total additional investment cost is estimated at: 1,650 CMD x 365 days x (31-12) NT\$/m³ = approximately NT\$11,442,750 per year. (After the Hsinchu Seawater Desalination Plant begins operation, PSMC will purchase 1,650 CMD, accounting for about 10% of the Hsinchu fab's water usage.) • Emergency contingency plans for tap water shut-offs and water restrictions are in place to manage and control the water supply
Regulatory energy efficiency requirements	The Bureau of Energy of the Ministry of Economic Affairs (MOEA) has set the "Energy Conservation Targets and Implementation Plan for Energy Users", which stipulates that large industrial and commercial customers with electricity consumption of 800 kilowatts or more must conserve 1% of electricity each year and must continuously invest in energy saving cost improvements. In addition, if the electricity savings do not reach the 1% requirement, they will be punished by the central Competent authority	• Increase of Operating Costs	• By the implement of the ISO 50001 energy management standard, PSMC establish an energy baseline and control energy usage • Each plant promotes energy saving actions to achieve the goal of conserving more than 1% of electricity each year
Customers' requests for enhancements	Customers are advocating for international trends such as the Paris Agreement and the Carbon Disclosure Project (CDP), hoping that raw materials in the supply chain will also contribute to reducing carbon emissions; customers may demand a shift to renewable energy use (e.g. RE100), or a reduction in greenhouse gas emissions (e.g. commitment to SBT)	• Increase of Operating Costs • Increase of Investment Costs	• Establish the "Green Energy Project" team to carry out the planning of renewable energy installations and report the progress to the President on a monthly basis • Active participation in international ESG evaluations such as DJSI and CDP in response to customer requests
Reduced competitiveness of products	The increase in demand for low-carbon products and alternative raw materials from investors, customers and other stakeholders has caused the Company to lose competitiveness as its existing products are replaced by lower-carbon options, which in turn affects operating income	• Decrease of Revenue	• Continuously understand customers' preference to switch to low-carbon products • Gather information on low-carbon product markets or technologies to develop response strategies
Difficulties in developing green (low-carbon) processes	In order to achieve sustainable development, the company needs to move towards green (low-carbon) product development and green (low-carbon) process improvement, and the investment in human resources and research and development time will lead to increased operating costs	• Increase of Operating Costs • Increase of Investment Costs	• Continued development of higher-end, lower energy consumption process technologies and innovative applications (e.g., 3D packaging, power management ICs, low leakage oxide semiconductors, storage memory, and automotive electronics)

Climate Opportunity Identification Results



Note: Horizontal axis - degree of influence: if the opportunity occurs, how much revenue or cost savings will be brought to the company; Vertical axis - degree of control: if the opportunity occurs, the readiness with which the company is able to create a competitive advantage; Size of the diameter of the circle-probability of occurrence: the probability that the opportunity factor will bring benefits to the company.

Climate Opportunities List

Type	Opportunity Items	Possible time of occurrence	Criticality	Risk assessment impact value chain scope		
				Upstream (Supply Chain)	Operation	Downstream (Customer)
Resource efficiency	Water resource efficiency improvement	Short Term	High		V	
	Energy use efficiency improvement	Short Term	High		V	
	Recycling of waste materials	Medium Term	Low	V	V	
Energy sources	Increase the share of renewable energy	Medium Term	Low		V	
	Obtain incentives and cooperation from the central Competent authority	Short Term	Low		V	
Products and services ³	Development of low carbon products and services	Medium Term	Medium		V	V
Market	Enhance information transparency to attract investment	Long Term	Medium		V	V
	Demand for electric vehicles increases, and automotive chips are in short supply	Medium Term	High		V	V
Resilience	Improve resilience to natural disasters	Long Term	Medium		V	

Note: Short Term: 1~3 years, Medium Term: 4~6 years, Long Term: 7 years or more

Key Climate Opportunities and Response Strategies

Risk Factor	Risk Description	Area of Impact (Service/strategy that will affect the company)	Response Strategy (Service/strategy that will affect the company)
Water resource efficiency improvement	Make full use of water resources and improving water recycling and reuse will reduce reliance on natural water resources, and strengthen operational resilience in the face of extreme weather, while reducing the chance of impact and loss	Operating Cost Savings. Proactive water-saving measures across all fabs saved a total of NT\$25.748 million.	- Continuously improve water recovery rate and establish the following: process water recovery rate >85%, plant-wide water recovery rate >70% - The total investment cost for proactive water-saving measures across all fabs amounted to NT\$370,000
Energy use efficiency improvement	By examining the overall operational efficiency of our machinery, equipment and plants, we explore opportunities to improve energy efficiency and lower operating costs by reducing energy consumption while meeting regulatory and customer requirements for low carbon products	Operating Cost Savings	- Set a 1% of electricity savings usage reduction target each year, and continuously adjust the energy savings target on a rolling basis 13% of reduction in electricity consumption per wafer area compared to the base year
Demand for electric vehicles increases, and automotive chips are in short supply	The growing trend towards electric vehicles is obvious, and will drive the growing demand for automotive wafers. By increasing the volume of wafer tape-out for automotive wafer products and ensuring stable supply of products, we will be able to secure long-term cooperation with major automotive manufacturers	Increase revenue	- Customer planning for electric vehicle chip market technology products - Development of process technology for power management chips related to electric vehicles - Stabilize future revenue momentum by signing long-term prepaid automotive chips supply contracts with automotive manufacturers
Increased demand for IGBT end products in electric vehicles and inverter home appliances	Due to national policy requirements, the demand and market share for IGBT end products in electric vehicles and inverter home appliances are gradually increasing	Increase revenue	- Customer planning for electric vehicle market technology products - Development of IGBT-related power management chip manufacturing technologies - Stabilize future revenue momentum by signing long-term prepaid automotive chip supply contracts with automotive manufacturers

Note: The "Increased demand for IGBT end products in electric vehicles and inverter home appliances" opportunity factor was identified as a key climate-related opportunity that PSMC should address based on internal discussions in 2024, and has therefore been included in the response strategy management list.

Scenario Analysis

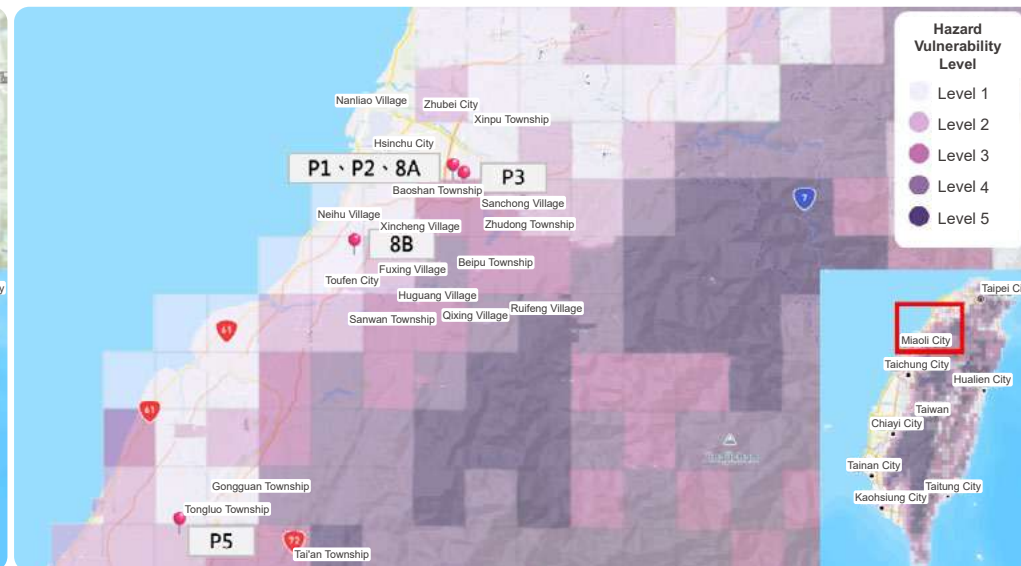
1. Physical Risk Scenario Analysis: Flooding & Landslide Disasters (Operational Locations)

In recent years, Taiwan has frequently experienced short-duration intense rainfall events, which may overwhelm municipal drainage systems and pose flooding risks to fab facilities. Extreme rainfall or uneven rainfall distribution could also lead to natural disasters such as landslides and debris flows. As of the end of 2024, PSMC operates four 12-inch and two 8-inch wafer fabs in Taiwan, located in Hsinchu City and Miaoli County. According to simulations from the National Science and Technology Center for Disaster Reduction's Disaster Risk Adaptation platform under the SSP5-8.5 scenario for the mid-century period (2041–2060), all wafer fabs are situated in areas classified as the lowest flood risk level (Level 1), based on a spatial grid resolution of 40 meters. However, under the SSP5-8.5 scenario and using the rockfall, debris slide, rockslide, and dip slope areas interpreted by the Geological Survey and Mining Management Agency, all wafer fabs are located within Level 1 to Level 3 slope disaster risk zones (with a spatial resolution of 5 kilometers).

SSP 5-8.5 Scenario-Flooding Risk



SSP 5-8.5 Scenario-Slope Disaster Risk

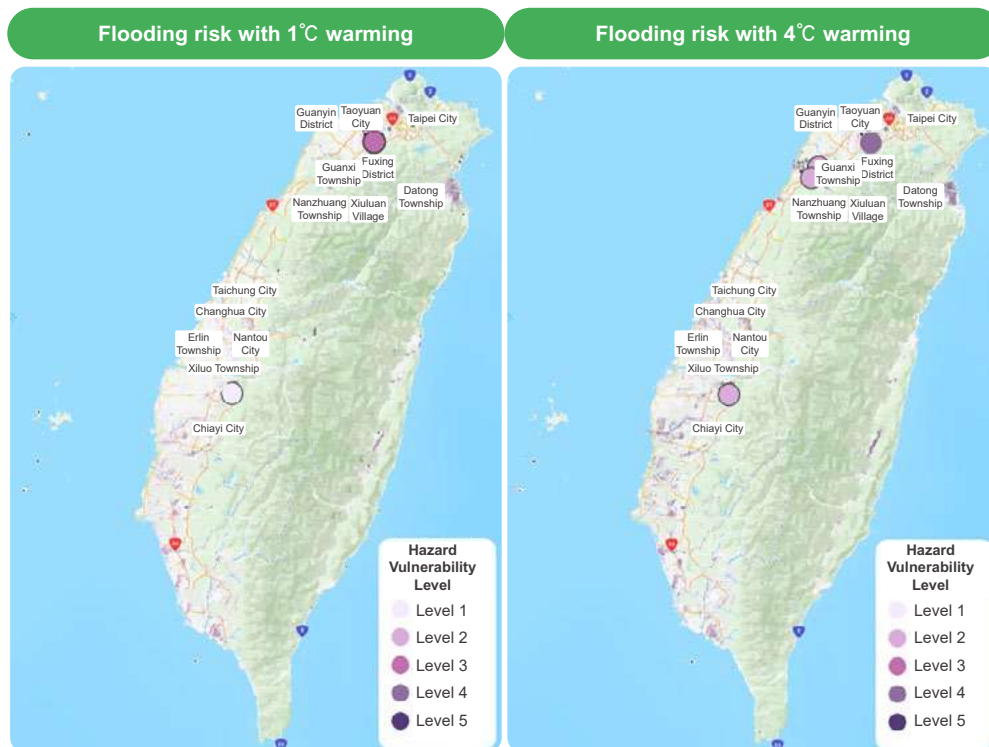


PSMC has considered the risks of extreme-climate-caused heavy rainfall and slope-land disasters as selecting sites for fab construction. The selected sites are all relatively high above areas subject to potential slope-land disasters, that is, at 100.8 meters above sea level; buildings' entrance floors have also all been elevated 1.2 meters above the surrounding road surface. Rainwater interception ditches have been set around buildings. There are also sump pits and pumps under the fab structures. When necessary, the lowest underground floor can be closed and used as a temporary water reservoir or flood detention layer to cope with flash floods. In addition, PSMC has formulated corresponding crisis event management plans and regulations to respond to a variety of possible emergencies, and carries out drills every year to review the effectiveness of the adopted countermeasures. An Emergency Response SOP has also been established, based on which emergency response taskforces will be activated in the event of disasters, to reduce impacts caused.

2. Physical Risk Scenario Analysis: Flood Disasters (Suppliers' Operating Sites)

The natural risks faced at suppliers' locations could impact the stability of the supply chain. Therefore, PSMC selected the top five suppliers of critical components (including photoresist, wafers, chemicals, and gases) in Taiwan and used the National Science and Technology Center for Disaster Reduction's Disaster Risk Adaptation platform to simulate flood risk levels at suppliers' locations under global warming scenarios of a 1°C and 4°C temperature rise. Based on procurement value, under the warming scenario of a 1°C temperature rise, two suppliers were located in Level 1 and Level 3 flood risk areas, with an exposure level of 4.09%. Under the warming scenario of a 4°C temperature rise, three suppliers were located in Level 2 areas and one supplier was located in a Level 4 area, with an exposure level of 12.87%. However, no suppliers were located in the highest flood risk level (Level 5) areas.

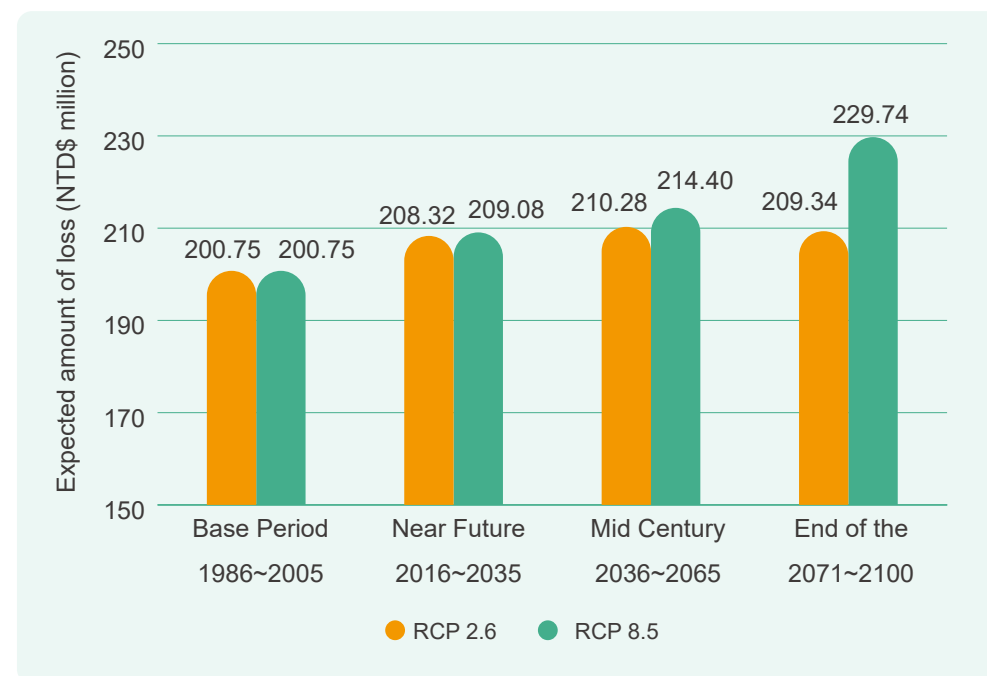
PSMC has estimated the possible risk exposures under a variety of scenarios, based on the degree of hazards, vulnerability, and possible impact when floods occur. With the aim of reducing the risk, PSMC has introduced a substitute supplier system for critical suppliers to reduce the possibility of supply interruption due to flooding.



3. Physical Risk Scenario Analysis: Drought and Water Scarcity

Extreme climate events have led to increasingly polarized geographical and seasonal distribution of rainfall, with some regions or periods experiencing excessive rainfall while others face severe water shortages. Taiwan is not only facing prolonged periods without rainfall causing water resource scarcity, but also faces significant impacts on PSMC's process stability and delivery schedules due to government-imposed water restrictions. PSMC has compiled statistics on the maximum water shortage volumes during past water restriction events, factoring in parameters such as water consumption per unit product, production volume, and unit product pricing at each fab. Based on simulations from the National Science and Technology Council's Taiwan Climate Change Projection Information and Adaptation Knowledge Platform (TCCIP) project under RCP 2.6 and RCP 8.5 scenarios, covering near-future, mid-century, and end-century continuous non-rainy day rate changes, PSMC has estimated potential production capacity reductions and anticipated financial losses under a variety of future water scarcity risk scenarios.

To mitigate financial losses caused by water shortage risks, each PSMC fab is actively promoting a variety of water conservation projects, to improve process water recycling rates, optimize wastewater treatment and reuse systems, and thereby reduce reliance on municipal water supplies. For more details on water resource management measures and performance, please refer to Section 3.3, [Water Resources Management](#).

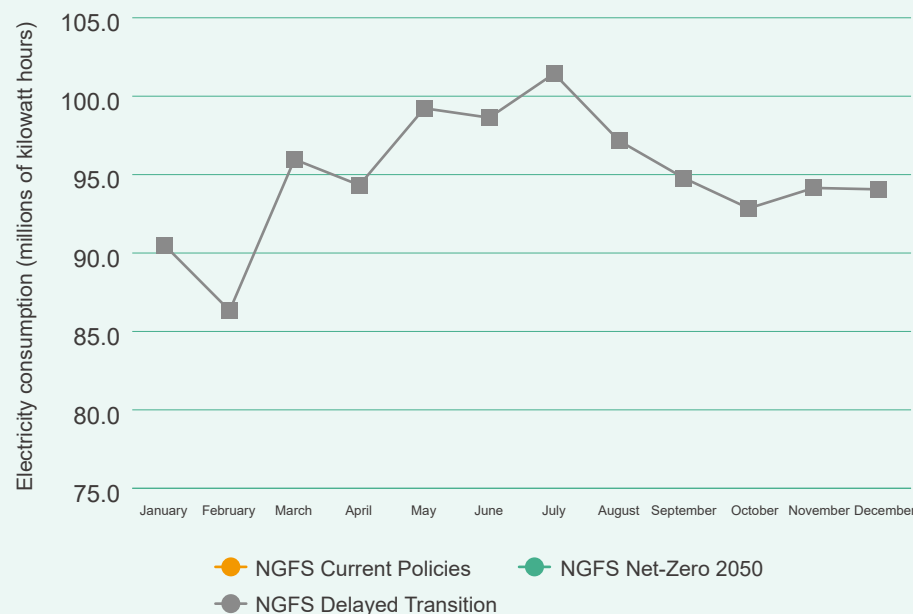


4. Physical Risk Scenario Analysis: Rising Global Temperatures

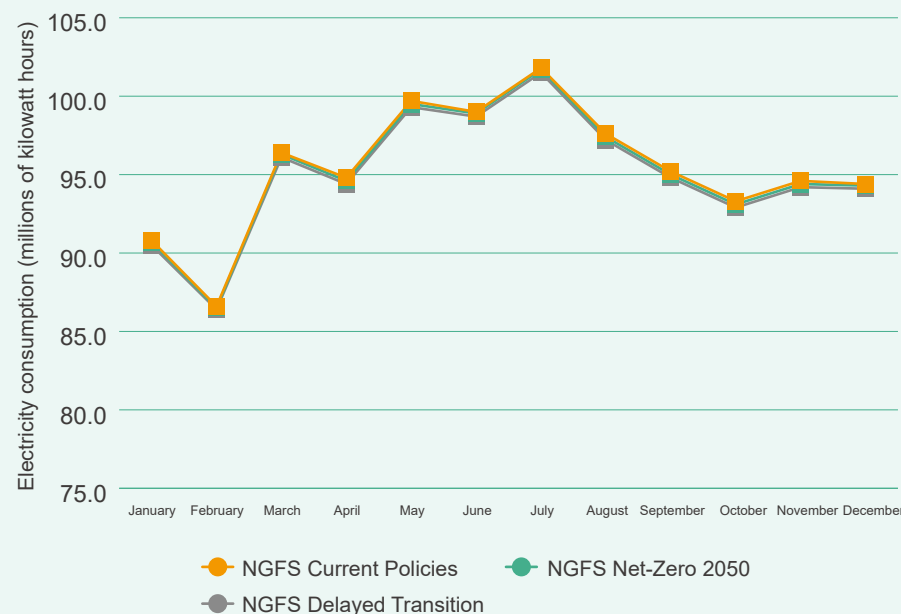
In 2024, PSMC's energy use is mainly electricity, accounting for about 88.13% of the total energy use. Further analysis of the electricity consumption data found that direct air conditioning electricity consumption accounts for more than 30% of the entire plant's electricity consumption, and other electricity consumption is for manufacturing processes, fixed electricity consumption and the other static factors. To address this, PSMC has focused on improving the energy efficiency of air conditioning and process equipment, and continues to promote a variety of electricity-saving initiatives.

PSMC selected the climate scenarios of the Network of Central Banks and Supervisors for Greening the Financial System (NGFS) to evaluate the three scenarios: Current Policies, Delayed Transition, and Net Zero 2050 for Taiwan's greenhouse gas emission path and future average temperature rise in each month. In addition, PSMC according to the data on the total power consumption of the plant in recent years, to model simulation to evaluate the correlation between the increase in the current average outside air temperature and power consumption, and selected the direct correlation dynamic variable outside air average temperature and production quantity. After using the multiple regression equation, the simulated factory power consumption change equation is obtained as the average outside air temperature increases. Combined with the NGFS scenario, the increase in electricity consumption in 2030 and 2050 compared to 2020 in each month under different scenarios is estimated. The estimated electricity consumption of the entire plant in 2030 and 2050 is as follows.

NGFS scenario estimates the monthly electricity consumption of the entire factory in 2030

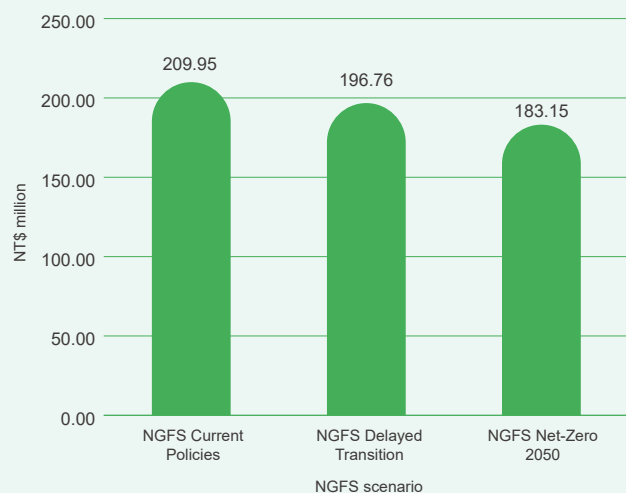


NGFS scenario estimates the monthly electricity consumption of the entire factory in 2050



Under the most severe Current Policies scenario of global warming, PSMC's future electricity expenditures are projected to increase by approximately NT\$25.34 million in 2030 compared to the 2020 baseline electricity costs, and the electricity costs are expected to rise to about NT\$210 million by 2050.

Increase in electricity costs in various scenarios in 2050



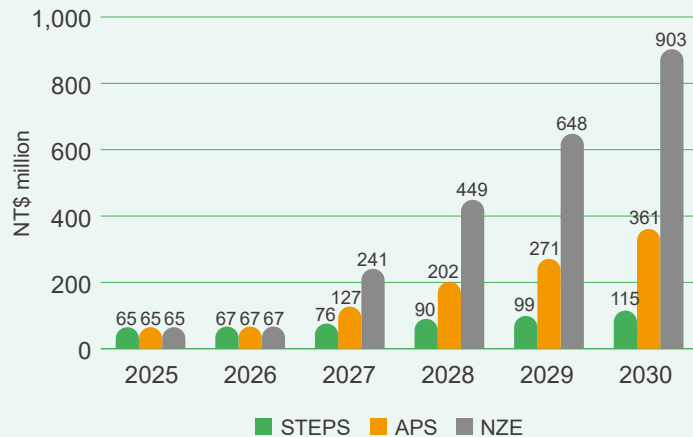
5. Transformation Risk Scenario Analysis: Carbon Fee

Taiwan will start levying carbon fees in 2026. To assess the financial impact of carbon fees, in order to assess the impact of carbon fee collection on financial, PSMC under the IEA World Energy Model the Stated Policy Scenario (STEPS), the Commitment Target Scenario (APS) and the 2050 Net Zero Emissions Scenario (NZE), PSMC calculate the carbon fee and evaluate the carbon reduction path in each scenario to understand the changes in carbon costs before setting carbon reduction targets and after setting company targets (see the figure below). Among them, the company goals set by PSMC include the emission intensity of greenhouse gas Scope 1 and 2, the company's production targets and the expected introduction period of green electricity.

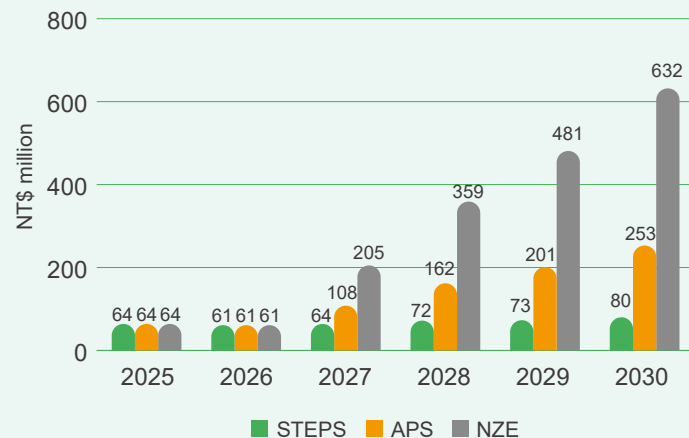
Scenario	Scenario I - STEPS	Scenario II - APS	Scenario III - NZE
Scenario Description	Development under established policies, global average temperature in 2100 will be 2.6°C higher than pre-industrial.	All the commitments made by various countries were fulfilled as scheduled, and the global average temperature in 2100 will be about 2.1°C higher than pre-industrial.	Assess global progress towards achieving the 1.5°C goal and other energy-related sustainable development goals.
Carbon Fees	2030: NT\$100/ton (Applicable preferential rates) 2050: US\$140/ton (Applicable preferential rates)	2030: US\$40/ton 2050: US\$160/ton	2030: US\$100/ton 2050: US\$200/ton
Carbon Reduction Path	2030 compared to the base year: 1.17% 2050 compared to the base year: 12.72%	2030 compared to the base year: 14.00% 2050 compared to the base year: 66.16%	2030 compared to the base year: 37.64% 2050 compared to the base year: 100.00%

PSMC's greenhouse gas emissions are mainly Scope 2 indirect energy emissions. In terms of green energy adoption, the goal is to achieve RE30 by 2030, reducing emissions from indirect energy through the introduction of green electricity and progressively increasing the proportion of renewable energy use each year. Under the highest NZE (Net Zero Emissions) carbon fee collection scenario, it is projected that by setting carbon reduction targets, the Company will reduce its carbon fee payment by approximately NT\$271 million by 2030 as compared to the scenario without reduction targets, a significant decrease from the NT\$797 million estimated the previous year. The main reason for this decline is that PSMC submitted a program that met the competent authority's conditions and therefore qualified for a preferential carbon fee rate, reduced from NT\$300 per ton to NT\$100 per ton. Nevertheless, this analysis still highlights the importance of setting carbon reduction targets and actively implementing reduction actions. PSMC will continue to plan more aggressive carbon reduction strategies to minimize the material financial impacts of carbon fee collection.

Carbon fees in different scenarios before the company sets carbon reduction targets



Carbon fees in different scenarios after the company sets carbon reduction targets



Indicators and Objectives

PSMC has developed a variety of climate management indicators, and consults trends and policies/regulations around the world to make adjustments on a rolling basis. Moreover, by regularly checking goal achievement status, the potential impacts of climate change on the organization's operations are continuously reduced, and the organization's climate resilience is improved.

TCFD Disclosure Content	Promoted Strategy
Indicators used by the organization to assess climate related risks and opportunities in accordance with the strategy and risk management process	Establish greenhouse gas emissions, electricity consumption, renewable energy use, water recovery rate, etc. as indicators to measure climate-related risks and opportunities
Scope 1, Scope 2 and Scope 3 (if applicable) greenhouse gas emissions and associated risks	The plants of PSMC calculate the emissions and disclose the related data in accordance with ISO 14064-1, refer to Chapter 3.1.2 for details
The objectives used by the organization to manage climate-related risks and opportunities, and the performance in achieving those objectives.	Climate related objectives are set, covering water resources, electricity, renewable energy, greenhouse gases, and waste. For related data and objectives results, refer to Chapter 3.2, Chapter 3.3 and Chapter 3.4 for details.

Climate Related Aspects	Indicators	Objectives
Resources Usage	- Manufacturing process water recycling rate - Full plant water recycling rate	- Manufacturing process water recycling rate per year >85% - Full plant recycling rate per year >70%
Electricity Usage	- Total power consumption - Power consumption per wafer area	- Save 1% of electricity per year 13% (short-term), 15% (medium-term), 20% (long-term) reduction in electricity consumption per wafer area compared to the base year (2020)
Renewable Energy Usage	Proportion of renewable energy usage	- Procure 26,260 renewable energy certificates, equivalent to 26.26 GWh of green electricity. (short-term) - Including self-generated solar power, the total annual renewable energy usage reaches approximately 27.51 GWh. (short-term) - Total installed renewable energy capacity reaches 52.5 MW, with annual renewable energy consumption totaling approximately 65.6 GWh. (medium-term) - Achievement of the RE30 initiative objective: 30% of renewable energy use (480 GWh) (long-term)
Greenhouse Gas Emissions	Greenhouse gas emissions	Promote energy conservation/carbon reduction solutions, average annual reduction 1% of greenhouse gases
Waste Production	- Waste recycling rate - Number of waste reduction cases	- Waste recycling rate ≥ 92% (short-term); ≥ 93% (medium-term); ≥ 94% (long-term) - Waste reduction projects: 40 projects/year (short-term); 45 projects/year (medium-term); 50 projects/year (long-term)

Note: Short Term - 2025, Medium Term - 2027, Long Term - 2030

3.1.2 Greenhouse Gas Emissions and Management

Greenhouse Gas Inventory and Reduction

PSMC conducts greenhouse gas inventory every year in accordance with the ISO 14064-1 procedure, all operational locations had been verified by a third-party impartial inspection unit to fully understand the overall greenhouse gas emission sources and emissions, and set various reduction targets and plan. Each department has also formulated related greenhouse gas reduction measures and plans. Statistics show that PSMC's energy-saving and carbon-reduction programs achieved a reduction performance of 1.34% in 2024, meeting the annual reduction target of 1%. In 2024, total greenhouse gas emissions from Scope 1 and Scope 2 combined increased by more than 20% compared to the previous year. The main reasons were the commencement of operations at the newly built P5 fab (thus included in the inventory), increased production capacity at fab sites, and the use of the newly-announced Greenhouse Gas Emission Factors as stipulated by the Ministry of Environment's Climate Change Administration. From the results of the 2024 annual greenhouse gas inventory, it can be seen that the overall emission hot spots of PSMC focus on the fluorine-containing greenhouse gases used in the scope 1 process and the use of purchased electricity in scope 2. The aforementioned two scopes accounted for 95% of the total emissions. Therefore, PSMC has listed the reduction of fluorine-containing greenhouse gases and the reduction of purchased energy as key projects in its greenhouse gas reduction plan.

Greenhouse gas emissions

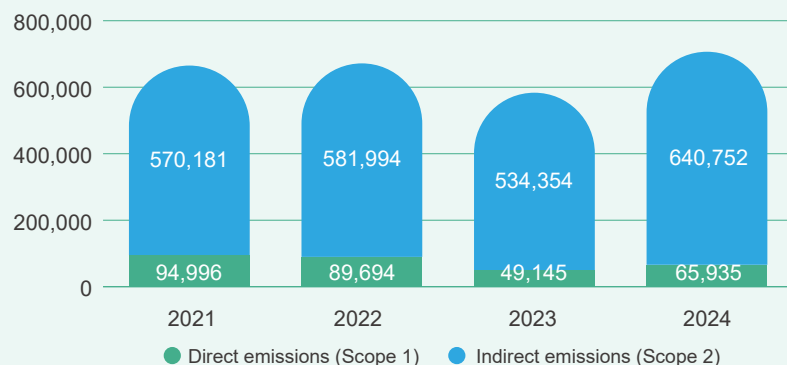
Unit: metric tons CO₂e

Scope	Year	2021	2022	2023	2024
Scope1		94,996	89,694	49,145	65,935
Scope2	Regional Coefficient Method	570,181	581,994	540,850	647,227
	Market Coefficient Method	570,181	581,994	534,354	640,752
Total	Regional Coefficient Method	665,177	671,688	589,995	713,162
	Market Coefficient Method	665,177	671,688	583,499	706,687

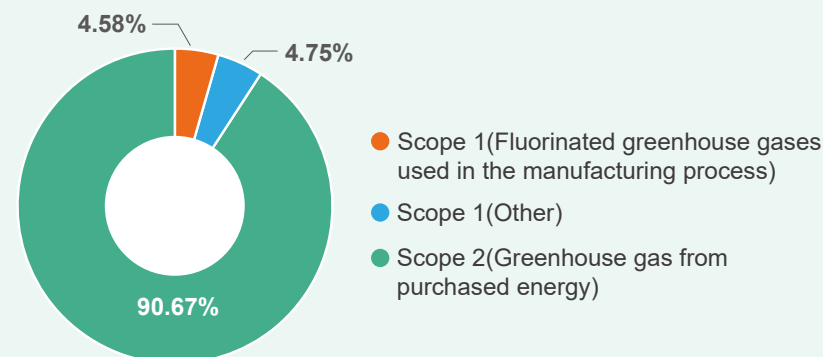
Note: The greenhouse gas emissions in 2023 and 2024 cover PSMC and its consolidated financial reporting subsidiaries.

Scope 1 and Scope 2 greenhouse gas emissions

Emission Unit: Metric tons of CO₂e



Proportion of Scope 1 and 2 greenhouse gas emissions



Note 1: The greenhouse gas emission coefficient mainly refers to the Ministry of Environment announced the greenhouse gas emission coefficient on February 5, 2014. The electricity carbon emission coefficient is calculated using 0.494kg CO₂e/kWh as Scope 2 (this calculation method is based on the Market Coefficient Method for statistics).

Note 2: The source of the Global Warming Potential (GWP) used in this table is the "IPCC Fifth Assessment Report (2013)" edition.

Note 3: Greenhouse gas types include nitrous oxide (N₂O), methane (CH₄), carbon dioxide (CO₂), hydrofluorocarbons (HFCs), perfluorocarbons (PFCs), sulfur hexafluoride (SF₆), and nitrogen trifluoride (NF₃).

Note 4: Method used for aggregating greenhouse gas volumes: Operational controlling rights method.

Note 5: Standards, methodologies, assumptions and/or calculation tools used: The energy consumption is derived from the measured values of the chargeable receipts, and the checklist does not have any estimated values.

Note 6: The emission coverage of Scope 1 and Scope 2 is 100%.

PSMC also follows the procedures of ISO 14064-1:2018 to conduct Scope 3 inventory and obtain third-party external verification. The total emissions of greenhouse gas Scope 3 in 2024 was 214,549 metric tons of CO₂e, increased compared to the previous year, not achieving the annual reduction target of 1%. The main reason is that the P5 plant was completed and started operation and was included in the inventory and the plant's production capacity increased. The inventory results show that in Scope 3, fuel and energy-related activities and purchased goods and services are the main emission hotspots. PSMC will continue to negotiate with value chain partners to expand the influence of its climate strategy and create opportunities for cooperation.

Indirect greenhouse gas emissions in Scope 3 for 2024

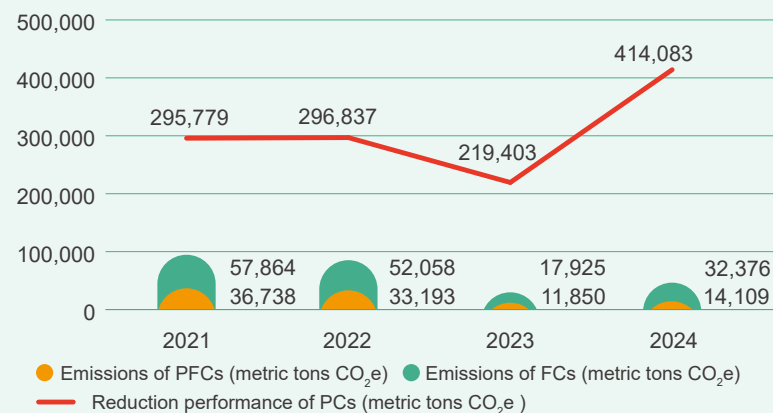
Unit: metric tons CO₂e

Item	Source of calculation	Greenhouse gas emissions
Upstream Transportation Distribution	Carbon emissions from the transportation of raw materials from the suppliers to the plant	754
Downstream Transportation Distribution	Carbon emissions generated from the transportation of produced wafers from the plant to the customer's designated location	8
Staff Commute	Carbon emissions from company transportation vehicles	565
Business Travel	Carbon emissions from staff travel by public transportation	128
Purchased Goods and Services	Carbon emissions from purchased primary raw materials and auxiliary raw materials in the manufacturing process	76,152
Fuel and energy-related activities (not included in Scope 1 or Scope 2)	Carbon emissions from the use of fuels and energy sources at the plant calculated by subtracting the usage from the entire life cycle	135,535
Waste generated from business operations	Carbon emissions from waste treatment and transportation at the plant	1,407
Total		214,549

Note: Coefficients are quoted from Carbon Footprint Information Platform of the Ministry of Environmental, and business travel are additionally quoted from the High Speed Rail Transit Service Carbon Footprint.

In order to reduce the fluorine-containing greenhouse gases used in the manufacturing process, in addition to continuous source management and evaluation of material replacement, we are also committed to improving the processing efficiency of end-of-pipe pest control devices or replacing old equipment, hoping to achieve the reduction of perfluorinated compound emissions in the manufacturing process with a rate of 90% by 2030 for the long-term goal. In 2024, all units will actively promote the replacement of old process exhaust gas treatment equipment with new ones, increase the treatment range of process exhaust gas, and install high-efficiency process exhaust gas treatment equipment in new machines and other actual reduction plans. The carbon reduction from fluorinated gases (FCs) is 414,083 metric tons of CO₂e, an increase of 194,680 metric tons of CO₂e, compared to 219,403 metric tons of CO₂e, in 2023, and the reduction performance has increased by about 88.7%.

Emissions and reduction performance of Perfluorinated (PFCs) and fluorinated (FCs)



Note 1: Reduced gas emission types include hydrofluorocarbons, perfluorocarbons, SF₆, and NF₃.

Note 2: The reduction is in the scope of direct greenhouse gas emissions (Scope 1).

Note 3: Calculated using the Intergovernmental Panel on Climate Change (IPCC) Tier 2c emissions calculation method.

Item	2021	2022	2023	2024
PFCs emissions (metric ton PFC)	5.00	4.54	1.81	1.97
PFCs emissions (metric ton CO₂e)	36,738	33,193	11,850	14,109

2024 Fluorinated Greenhouse Gas Reduction Effectiveness Statement

Unit: metric tons CO₂e

Program Name	Implementation Method	Reduction Effectiveness
Replacement of on-site process exhaust gas treatment equipment	Evaluate and replace process exhaust gas treatment equipment for etching production machines.	330
Increase the treatment range of process exhaust gas	Connect adsorption-type process exhaust gas treatment equipment to the PFC process exhaust gas treatment system to improve PFC exhaust gas treatment efficiency.	520
Properly handle fluorine-containing gases emitted during the process	The factory has newly installed fluorine-containing gas processing equipment, and all of them are equipped with combustion process tail gas treatment equipment	41,555
Total		42,405

3.1.3 Environmental Return on Investment

In order to measure the investment benefits of environmental protection, PSMC calculates the return on environmental investment every year and sets a target of more than 6%. In recent years, the return on environmental investment has remained stable at above 6%. However, due to the significant capital investment in new equipment at the P5 Tongluo fab in 2023, the return on environmental investment dropped to 1.43%. Although capital investment in the new fab decreased significantly in 2024, additional investments were made to address energy conservation, carbon reduction, and increasingly stringent environmental regulations. Consequently, capital investment, operational expenses, and cost-saving initiatives all increased compared to previous years, leading to a recovery of the return on environmental investment to 5.55% in 2024.

Unit: NT\$,%

Item	2021	2022	2023	2024
Capital Investment	206,513,454	229,373,276	4,178,491,273	682,507,214
Operation Expenses	496,423,956	637,849,901	661,360,764	1,184,529,181
Total Expenses(=Capital Investment+ Operating Expenses)	702,937,410	867,223,177	4,839,852,037	1,867,036,395
Cost Savings 、 Cost Avoidance 、 Income 、 Tax Incentives, etc.	53,945,581	52,385,332	69,355,194	103,617,976
ROI	7.67	6.04	1.43	5.55
Operation Site Coverage Rate	100	100	100	100

3.2 Energy Management

3.2.1 Energy Policy and Management

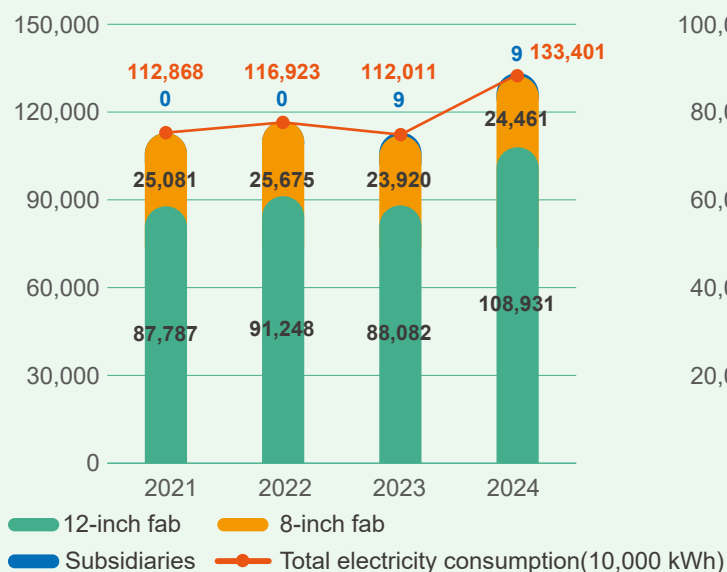
To analyze energy consumption and improve efficiency, PSMC has fully implemented the ISO 50001 Energy Management System and engages third-party verification bodies to conduct regular audits and re-certifications to ensure the effectiveness of the management system. In addition, PSMC has set an annual energy saving target of 1% or more, and provides incentives through the proposed improvement system and CIP competition to encourage employees to take the initiative to propose recommendations around energy savings, climate response measures, productivity improvements, and work efficiency. At the same time, PSMC is committed to continuously carry out and improve the following tasks and promote management plans based on this to achieve the mission of environmental sustainability:

- Provide energy improvement resources and enforce energy laws
- Enhance equipment energy management and reduce energy consumption in order to achieve energy saving targets
- Continue to carry out energy reduction measures to optimize energy use efficiency
- Increase the procurement of equipment with energy-saving labels and introduce energy-saving design products and processes
- Establish appropriate energy strategies and engage all employees to promote sustainable business operations
- Provide employees with education and training related to energy systems and energy efficiency to enhance their awareness of energy conservation and equip them with the ability to discover and continuously improve energy efficiency

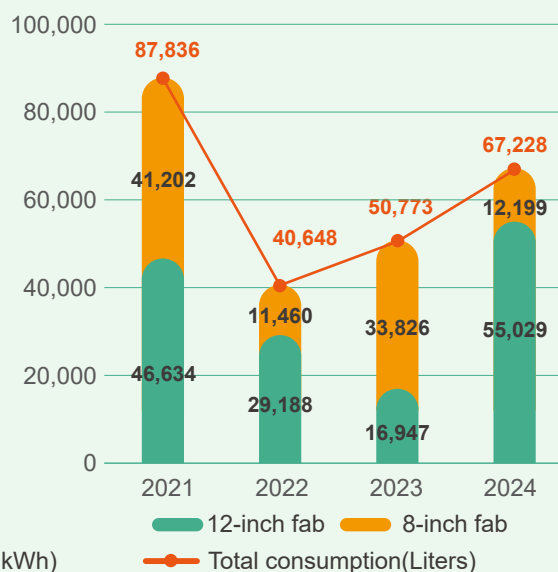
Energy Resources Consumption

PSMC's energy usage is mainly composed of purchased electricity, accounting for approximately 88.13% of the overall energy consumption. The usage statistics of various energy resources are as follows:

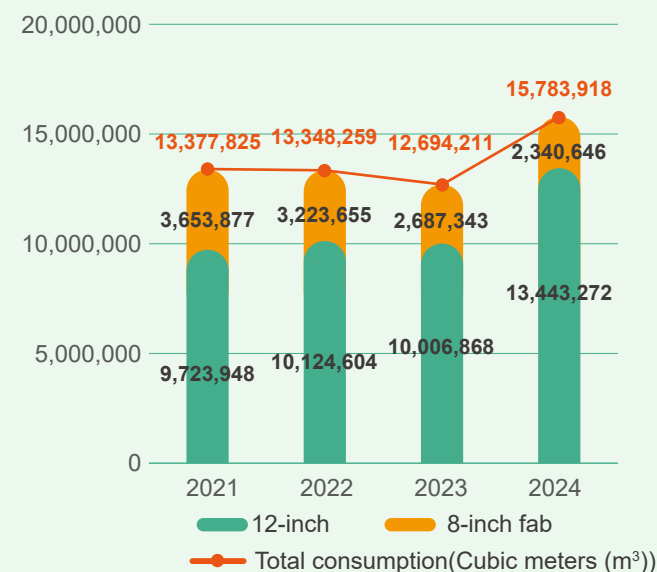
Purchased electricity consumption



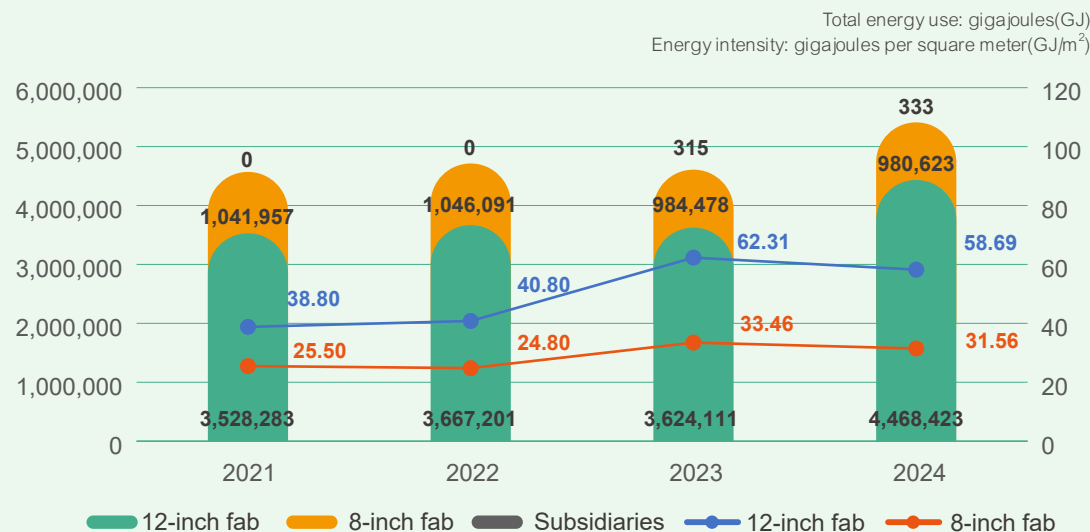
Diesel consumption



Fuel consumption



Total Energy Consumption and Energy Intensity



Energy Consumption in 2024

Unit: Gigajoules (GJ)

Item		Volume of use
Electricity	Purchased electricity	4,802,454
	Purchased renewable electricity	47,185
	Self-generated renewable electricity	2,734
Natural Gas		594,643
Diesel		2,364
Total Energy Consumption		5,449,380

Note 1: Energy heating value: The heating value of purchased electricity: 3,600 gigajoules/million degrees; converted using the coefficients in the "Greenhouse Gas Emission Coefficient Management Table Version 6.0.4" of the Climate Change Administration of the Ministry of Environment. The coefficients used for natural gas and diesel are 9,000 kcal/cubic meter and 8,400 kcal/liter respectively.

Note 2: Calculated as 4.186 kilojoules per kilocalorie.

Note 3: The scope of coverage is PSMC and its consolidated reporting subsidiaries, and the subsidiaries only use purchased power for energy.

Note 4: Renewable electricity is solar photovoltaic.

Annual Energy Consumption Statistics

Unit: Million Watt hours (MWh);%

Item		2021	2022	2023	2024
Non-Renewable Energy Consumption	Non-renewable fuels	140,831	140,088	133,341	165,835
	Purchased non-renewable energy (electricity)	1,128,680	1,169,228	1,118,104	1,334,015
	Total	1,269,511	1,309,316	1,251,445	1,499,850
Renewable Energy Consumption	Total	0	0	13,679	13,866
Data coverage rate		100	100	100	100

Note: Data statistics covers PSMC and its consolidated reporting subsidiaries.

3.2.2 Energy Saving Measures

Energy Saving Goals and Key Solutions

In accordance with the Ministry of Economic Affairs' "Energy Conservation Targets and Implementation Plan for Energy Users", PSMC has set a target of 1% annual energy savings. 2024 saw the implementation of about 93 energy saving measures, reducing energy consumption by 33,803.82 gigajoules (including electricity, diesel and natural gas) and 8,987.23 metric tons of CO₂e compared to the 2023. The energy saving rate of the entire plant is 1.65%, achieving the set target. The energy savings programs are summarized in the table below:

Project Category	Project Details	Number of Projects	Investment Amount (NT\$10,000)	Power Savings (kWh/year)	Heat Reduction (GJ/year)	Cost Savings (NT\$10,000/year)	Carbon Reduction (t CO ₂ e/year)
Lighting system	- Smart control - LED lighting replacement	18	605.44	2,898,791.57	3,081.34	995.87	1,460.07
Motors/drivers	- Inefficient motor replacement - Frequency converter installation	5	521.00	430,714.00	1,127.94	137.79	212.99
Pure/wastewater system	- Equipment maintenance - System equipment replacement	4	1,299.00	3,697,170.00	13,290.59	1,183.10	1,826.40
Equipment replacement/upgrading	- Process equipment replacement/upgrading - Old parts and materials replacement	31	11,333.80	6,615,655.90	11,657.78	2,203.95	3,263.69
Process efficiency improvement	- Parameter optimization - Workflow refinement	35	1,539.00	4,289,071.73	4,646.17	1,440.73	2,224.08
Total		93	15,298.24	17,931,403.20	33,803.82	5,961.44	8,987.23

Invest in innovation , reduce energy consumption

Chilled Water System Energy-Efficient Operation Management

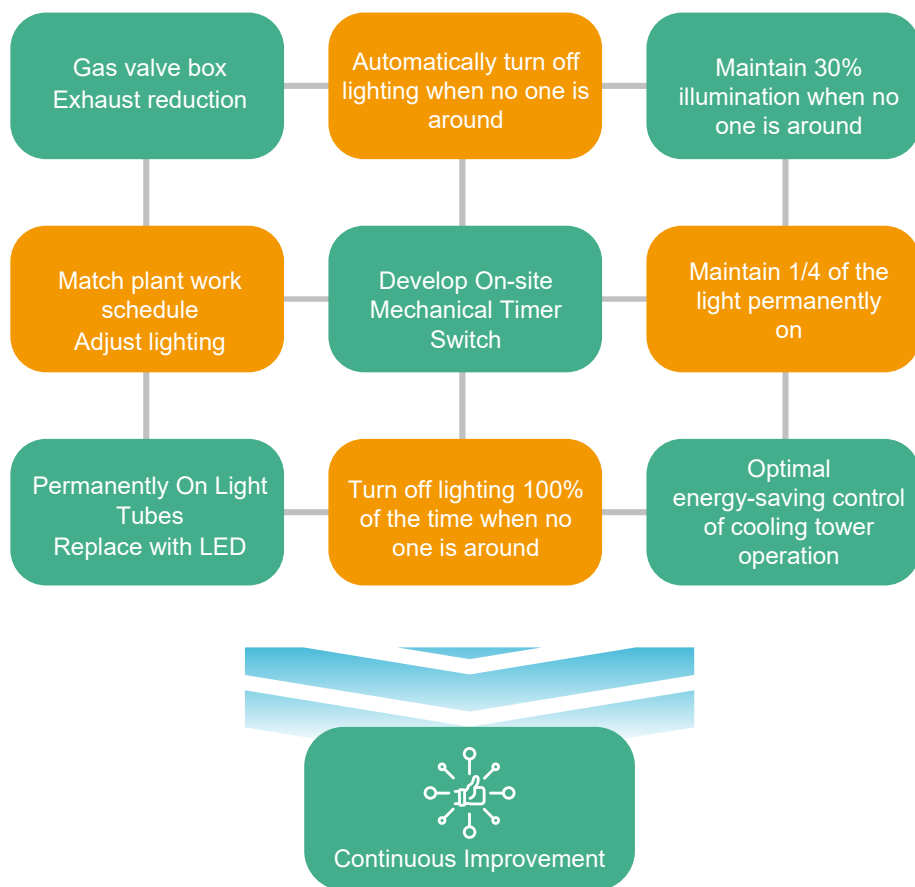
PSMC is actively expanding smart energy management. For example, at the P2 fab, an energy-saving system was officially introduced on November 21, 2024. This system utilizes outdoor wet-bulb temperature as a baseline for data clustering and compares the coefficient of performance (COP) efficiency of the chilled water system before and after improvements under the same outdoor wet-bulb temperature conditions. The original data is sourced from the facility monitoring historical database, with average values recorded every 10 minutes. The extracted data is then analyzed to evaluate system performance. By integrating intelligent management with renewable energy, PSMC is committed to providing customers with low-carbon manufacturing services.

Intelligent Air Compressor Operation Management

PSMC has also developed and introduced AI technology in an industry-university cooperation model with Professor Zhang Guohao of Tsinghua University starting in 2022, creating an advanced intelligent control system for air compressor operations. This system not only monitors and predicts the health status of air compressors, enabling more precise maintenance to prevent abnormal energy consumption caused by inefficiencies, but also flexibly manages multiple air compressor units, constructing an optimized model to reduce overall energy consumption.

Basic Energy Saving Solutions for Factory Area

PSMC has basic energy-saving solutions, such as automatically turning off lighting when no one is around, adjusting lighting to accommodate the factory's work schedule, and replacing frequently used fluorescent tubes with LEDs. We look forward to working with our employees to save energy and maintain a sustainable environment through basic energy-saving solutions.



3.2.3 Renewable Energy

In 2022, PSMC established a Green Energy Team responsible for planning renewable energy procurement and implementing solar power generation systems to enhance energy self-sufficiency and reduce greenhouse gas emissions. In 2024, the P5 fab completed the installation of a 0.5MW in-house solar photovoltaic system, bringing PSMC's total installed capacity to 1MW (with an estimated annual green power generation of 1.25 GWh). Additionally, PSMC purchased 13,798 renewable energy certificates, equivalent to 13.798 GWh of green electricity.

Renewable Energy Goals

Short Term Goals (2025)

- Procure 26,260 renewable energy certificates, equivalent to 26.26 GWh of green electricity
- Including self-generated solar power, the total annual renewable energy usage reaches approximately 27.51 GWh

Medium Term Goals (2027)

- Total installed renewable energy capacity reaches 52.5 MW, with annual renewable energy consumption totaling approximately 65.6 GWh

Long Term Goals (2030)

- Achievement of the RE30 initiative objective: 30% of renewable energy use (480 GWh)

3.3 Water Resources Management

3.3.1 Water Resources Policy and Management

The water used at all of PSMC's fab is tap water, mainly from the Hsinchu Baoshan Reservoir No. 1 and No. 2 and the Miaoli Yonghe Mountain Reservoir, and the water resource stress in the Hsinchu Science Park is Low Medium according to the World Resources Institute (WRI) water resource stress map. In 2024, the total water withdrawal across all fabs reached 7,397.62 million liters. The primary water consumption activities include process water, air conditioning water for cleanroom operations, scrubber water for process exhaust gas treatment, and domestic water usage within the facility. Of these, process water accounts for the largest share, making up 85% of total water consumption. PSMC has set a cumulative water-saving target of 216 million liters by 2030. For more details on how PSMC enhances water recycling rates and process water efficiency, please refer to section, [3.3.2 Water Reclamation Management](#).

Due to the special needs of the semiconductor manufacturing process, tap water or process recycled water must be obtained through appropriate treatment procedures to meet usage standards. The ultrapure water (UPW) usage in 2024 reached 9,695,383 cubic meters, with a reduction rate of pure water usage per unit product area at -18.01%, falling short of the annual target reduction of 1%. The primary reason for this increase in UPW consumption was the start of operations at the P5 fab and the rise in production capacity. The UPW usage data covers 100% of all fabs' UPW consumption. Since the P5 fab officially went into production in 2024, it has used 600 cubic meters of reclaimed water per day.

In order to improve the efficiency of water resource use and ensure the stability of water intake, PSMC has promoted various management plans with the following work items as the core, and deepened the management results year by year:

- The Manufacturers Association held a water resources communication meeting with the Water Resources Department to ensure the consistency of management objectives between management authorities and enterprises
- During the low-water period, the manufacturers and the Water Resources Agency increased their cooperation on water resources allocation
- Participated in the water conservation counseling program for science park manufacturers
- During the low water period, water trucks were sent to the Hsinchu City Keya Water Resources Center to carry reclaimed water
- Continued to participate in the Hsinchu regional water usage plan and support the construction of the Hsinchu seawater desalination plant project
- Provide employees with water resource efficiency improvement and other water resource-related education and training to enhance employees' awareness of water conservation and equip them with the ability to optimize process water efficiency

Water Resource Usage

Unit: Million liters

Item		2021	2022	2023	2024
Water Intake Volume	Taken from a third party - Taiwan Water Corporation	6,030.55	6,359.99	6,582.31	7,397.62
Discharge Volume	Discharged to a third party	4,040.84	4,359.01	4,399.70	5,054.42
Total Water Consumption		1,989.71	2,000.98	2,182.61	2,343.20
Water Withdrawal per Unit Product Area (Million Liters/m ² of Wafer)		0.05	0.05	0.08	0.07

Note1: Total water intake = tap water + recycled water (condensate, rainwater), tap water intake is based on the water bill.

Note2: The discharge volume is the actual volume of water measured by the flow meter in the plant.

Note3: Data covers PSMC and its subsidiaries included in the consolidated financial statements.

Note4: The water resources pressure in the covered areas are all Low-Medium, and the water quality is fresh water ($\leq 1,000$ mg/L total dissolved solids).

Ultrapure Water Usage Statistics

Unit: Cubic meter; %

Ultrapure Water	2021	2022	2023	2024
Ultrapure Water Usage	8,765,869	8,810,525	8,215,560	9,695,383
Data Coverage Rate	100	100	100	100

3.3.2 Water Reclamation Management

To address the increasingly uneven rainfall patterns in Taiwan in recent years, PSMC has optimized processes and invested in equipment to enhance water recovery rates and process water efficiency. In 1996, it became the first semiconductor company in the Hsinchu Science Park to commit to a process water recovery rate of more than 85%. The process water recovery rate in 2024 has reached an average of 88%, which has reached the set water recovery rate target of 85%, and the water volume of the 12-inch factory is controlled below 95% of the permitted water consumption.

Average Annual Water Recovery Rate for the Manufacturing Process Recovery Program

Unit: million liters; %, number

Fab	Item	2021	2022	2023	2024
12 inch	Annual manufacturing process water recovery volume	7,856.44	8,039.70	10,107.04	13,653.74
	Average annual manufacturing water process recovery rate	87	88	89	89
8 inch	Annual manufacturing process water recovery volume	2,949.50	3,845.32	3,819.94	4,275.72
	Average annual manufacturing water process recovery rate	87	87	87	88
Total	Annual manufacturing process water recovery volume	10,805.95	11,885.01	13,926.98	17,929.46
	Average annual manufacturing process water recovery rate	87	89	88	88
	Number of times each drop of water used	3.53	3.41	3.67	3.72

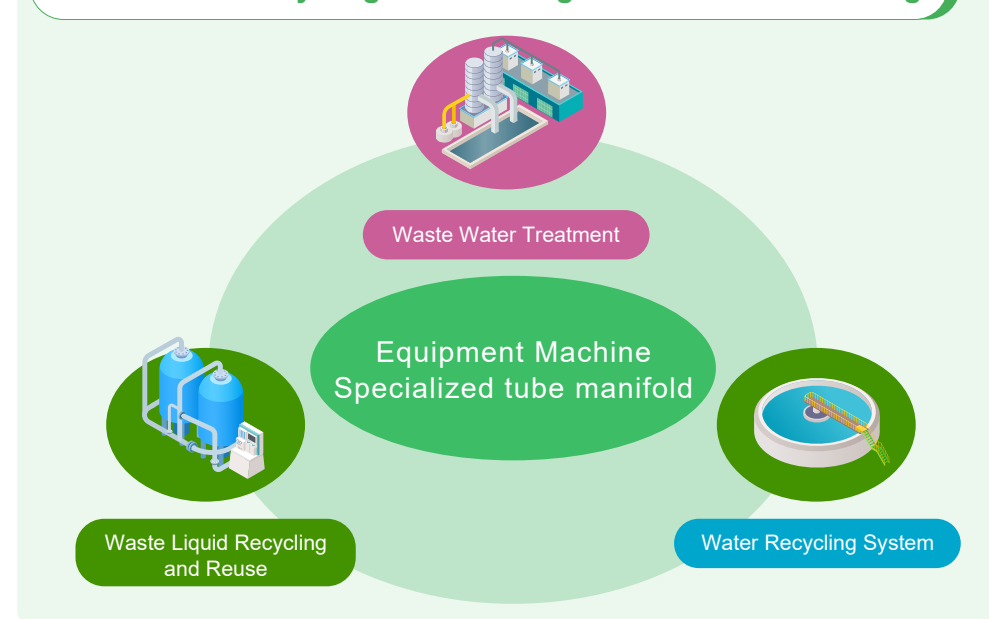
Water Efficiency in 2024

Fab	Project Details	Implementation Results
P1/2	Optimization of air pollution L/S water daily recovery volume	Reduced water consumption by 16,790m ³ per year
P3	Enhancing UPW/WWR backwash water recovery	Reduced water consumption by 8,030 m ³ per year
P5	Recycling and reusing LSR RO concentrated water	Reduced water consumption by 75,920 m ³ per year
8A	Upgrading ammonia-nitrogen system with high recovery RO membranes	Reduced water consumption by 6,132 m ³ per year
8B	Extending 2B3T water extraction time to reduce use of recycled tap water	Reduced water consumption by 8,119 m ³ per year

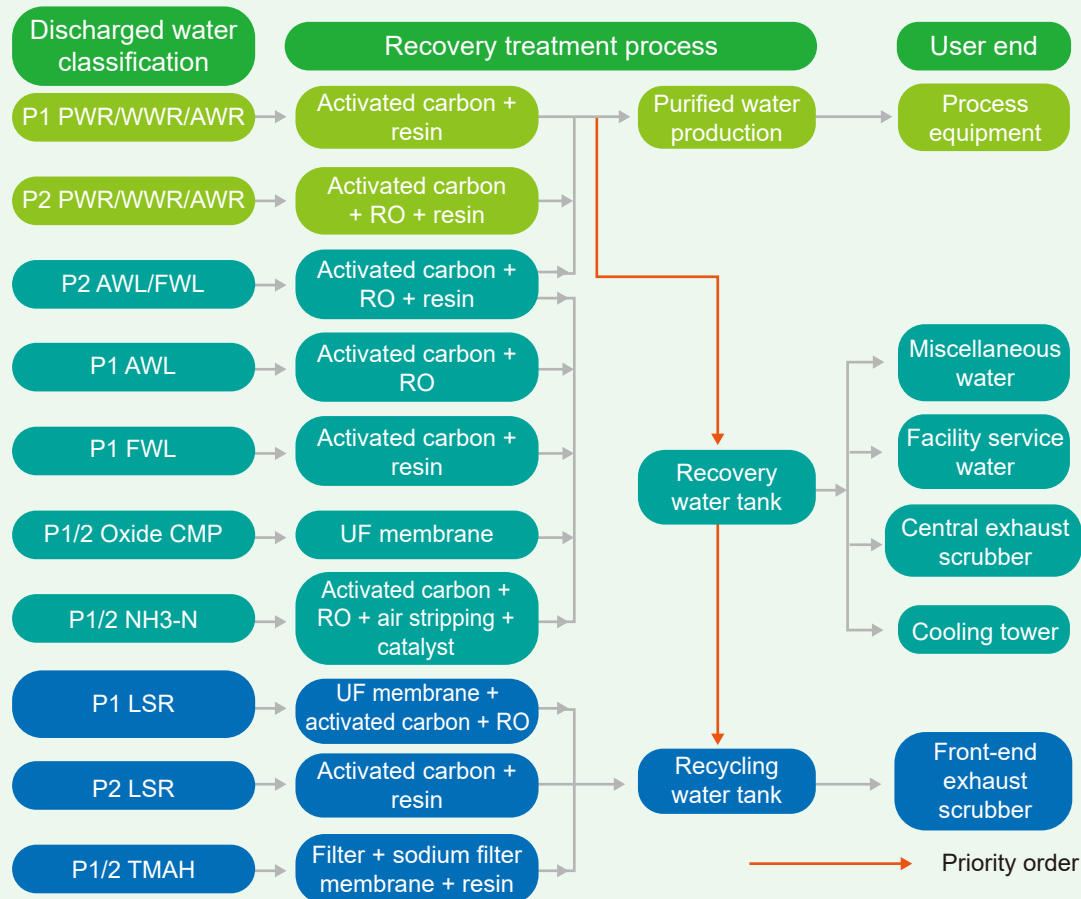
3.3.3 Wastewater Management

The manufacturing process wastewater from all of our plants is treated in our wastewater treatment plant before being discharged into the wastewater treatment plant in the Science Park for subsequent treatment and discharge. In order to grasp the effectiveness of the treatment of different components, a total of 30 different water systems, including recoverable and non-recoverable, are diverted for treatment according to the different characteristics of the discharge liquid at the manufacturing end, such as type, concentration, and conductivity. Some waste acids (phosphoric acid/copper sulfate/sulfuric acid/ammonium hydro fluoride/ammonium fluoride) and organic waste liquids (isopropyl alcohol/photoresist/de-photoresist/photographic solution) still have economic value for recovery. Also having separate diversions can not only reduce the amount of dosing in the wastewater plant, but can also reduce the difficulty of waste treatment at the backend and the burden on the environment. PSMC have pH and Fion testing equipment installed in our effluent process, and the Hsinchu Science Park Administration conducts sampling tests on effluent quality twice a month to strictly control the effluent quality of our plants, and to prioritize the management of effluent quality in accordance with plant regulations. In recent years, the annual average test values of the effluents from each plant have been far lower than with the Science Park's standard of control, and there have been no incidents of non-compliance with the effluent value limits.

Waste Water Recycling and Discharge Classification Planning



Treatment Process of Reclaimed Water System in P5 Fab



Effluent Water Quality Monitoring

Item	Unit	2021	2022	2023	2024
Ammonia Nitrogen (NH ₃ -N)	mg/L	11.1	8.3	5.9	6.0
Chemical Oxygen Demand (COD)	mg/L	73.9	67.4	45.5	62.2
Total Suspended Solids (TSS)	mg/L	33.8	28.6	21.3	30.2

Note 1: The wastewater discharge is the actual amount of effluent measured by the flow meter in the plant.

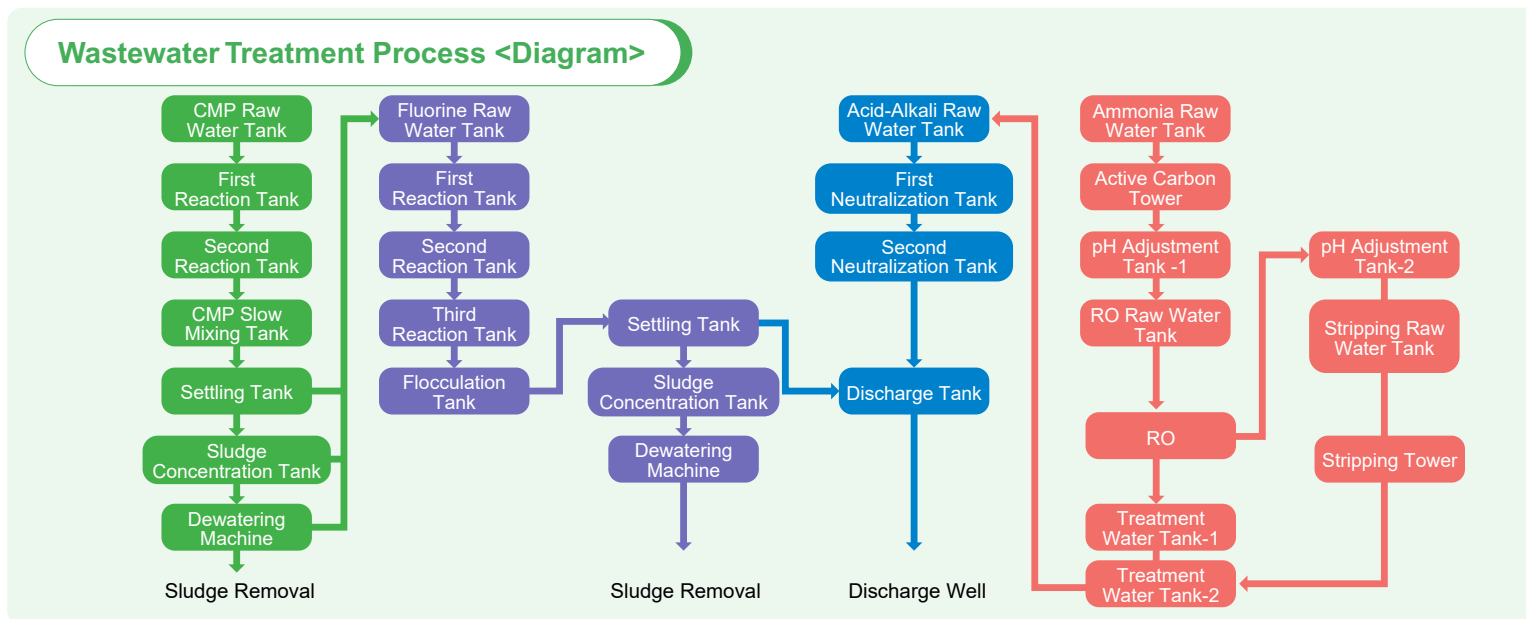
Note 2: The measurement value of each pollution indicator is the annual average of the two monthly measurements by the Hsinchu Science Park Administration.

Note 3: The ammonia nitrogen (NH₃-N) values are not listed in the Science Park where Fab P5 and 8B are located. Therefore, the data is sourced from the SGS self-inspection reports obtained from the fab.

Note 4: PSMC's drainage water is remitted to the wastewater treatment plant in the Science Park and total dissolved solids have not been tested.

Wastewater Treatment Process

The recycled water from the manufacturing process is treated by the appropriate recycled water system and then returned to the pure water system for reuse. For high concentrations of unrecoverable wastewater, it is discharged to the wastewater treatment system (including acid-alkali, fluorine acid, grinding, and ammonia nitrogen wastewater treatment system) and then piped into the wastewater treatment plant at the Science Park for subsequent treatment and discharge, so our effluent is not directly discharged into rivers and bodies of water, and has no direct impact on bodies of water and related habitats. There has also been no serious leakage of wastewater from any of the PSMC's plants over the years, and also there have been no major fines.



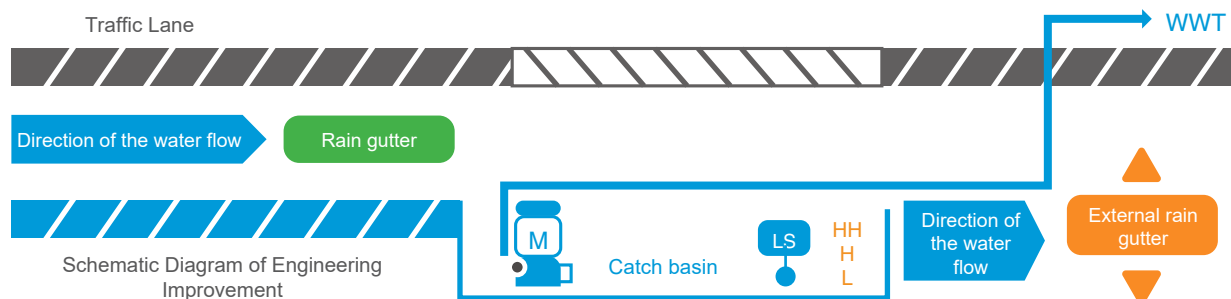
Wastewater Ammonia Nitrogen Reduction and Treatment Technology

Item	12 inch Fab			8 inch Fab	
	P1/2 Fab	P3 Fab	P5 Fab	8A Fab	8B Fab
Wastewater ammonia nitrogen reduction (%)	>90	>95	>95	>90	>85
Treatment technology	Air stripping + Cat-alyst	Air stripping and catalytic conversion	Air stripping + Catalytic	Air stripping + sulfuric acid washing	Air stripping + sulfuric acid wet scrubber
Final product	None	Nitrogen	Nitrogen	Ammonium sul-fate	Ammonium sulfate
Wastewater ammonia nitrogen reduction benefits	Reduce sewage charges by NT\$230 million per year	Reduce sewage charges by NT\$110 million per year	-	Reduce sewage charges by NT\$44.7 million per year	Reduce sew-age charges by NT\$40.55 million per year

Storm Water System

PSMC has installed overflow prevention dikes and flood pumps at the perimeter of each plant, and flood pumps at the ground floor of each building. In the event of an abnormal leak or heavy rainfall, the flood pumps will pump the water to the wastewater treatment plant and then discharge it to the sewage drain in the Science Park. In addition, chemical tanker filling areas are equipped with overflow prevention ditches to prevent chemicals from leaking and polluting the environment. In the storm water drains of the high-risk areas, abnormal event drainage pumps are also installed to pump abnormal sewage back to the wastewater treatment plant for treatment in order to prevent pollution incidents, all relevant measures are in accordance with the Company's internal storm water drainage practices.

Schematic Diagram of the Rainwater System



Water Resource Sharing

PSMC also actively participates in vendors' cooperation and provides water-saving guidance and technology sharing through its employees as representatives of industry associations. As of the end of 2024, PSMC has participated in a total of 40 water-saving coaching projects for vendors, with cumulative water-saving results reaching 1.3862 million metric tons. The vendors receiving counseling in 2024 are expected to save 382,000 metric tons of water per year, while the 10 vendors participating in the water-saving counseling are tracking to save approximately 341,000 tons of water in 2023, with a water saving achievement rate of 83.7%. In 2024, two water consumption fee briefing sessions, two water-saving education & training sessions and one P3 LSR engineering highlight training session, with a total of 129 participants. By sharing experience, PSMC not only helps suppliers adaptively reduce their dependence on water resources but also promotes communication and collaboration among science park members.

3.4 Waste Management

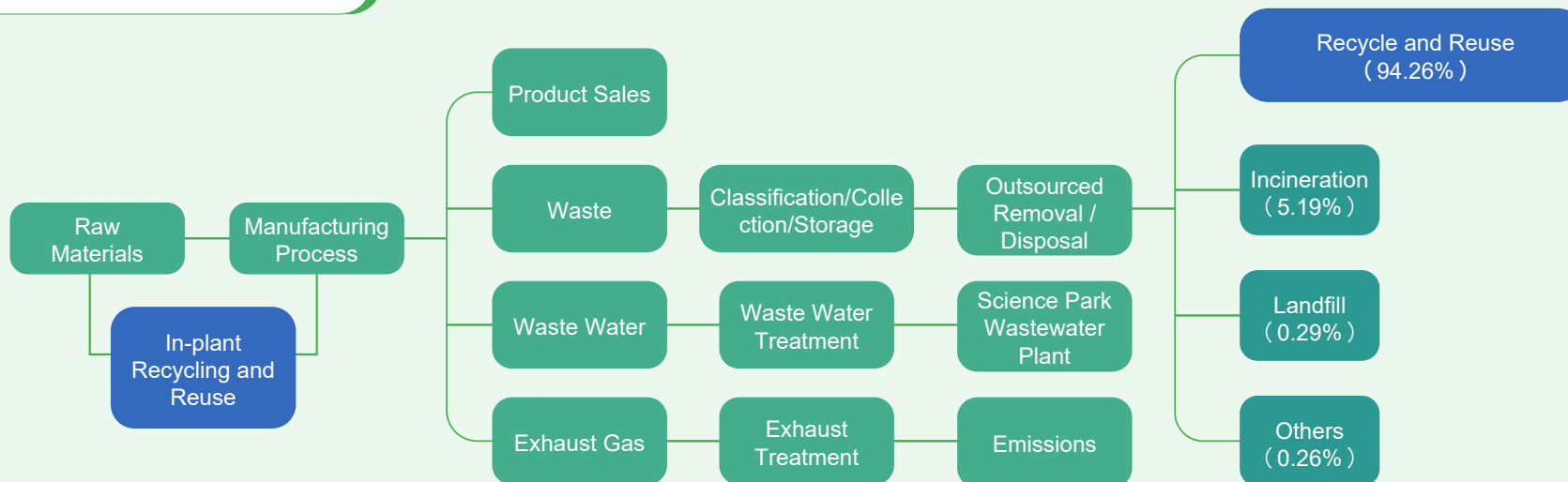
3.4.1 Waste Policy and Management

PSMC has designated personnel at each fab to manage waste disposal, and in 2024, the Company continued to reduce waste generation through monitoring, verification, and equipment upgrades, such as replacing outdated machines and pipeline valves. In 2024, the waste disposal cost was approximately NT\$132 million, while the total waste generated reached 18,896 metric tons, representing a 44.56% increase compared to 2023. The primary reason for this increase was the initiation of mass production at the P5 fab. The overall waste recycling rate was 94.26%, successfully achieving the 2024 target of $\geq 91\%$. Meanwhile, the waste landfill rate was 0.30%, also meeting the 2024 goal of $\leq 0.5\%$.

Short Term Goals (2025)	Medium Term Goals (2027)	Long Term Goals (2030)
- Waste recovery rate $\geq 92\%$ - Waste landfill rate $\leq 0.5\%$ - Proportion of on-site audit and counseling for waste vendors ^(Note) 100% - Provide guidance to waste vendors to obtain ISO14001 certification $\geq 60\%$ - Waste reduction cases of 40 cases/year	- Waste recovery rate $\geq 93\%$ - Waste landfill rate $\leq 0.4\%$ - Proportion of on-site audit and counseling for waste vendors ^(Note) 100% - Provide guidance to waste vendors to obtain ISO14001 certification $\geq 70\%$ - Waste reduction cases of 45 cases/year	- Waste recovery rate $\geq 94\%$ - Waste landfill rate $\leq 0.3\%$ - Proportion of on-site audit and counseling for waste vendors ^(Note) 100% 80% Provide guidance to waste vendors to obtain ISO14001 certification $\geq 80\%$ - Waste reduction cases of 50 cases/year

Note: The definition of key waste disposal vendors is as follows: 1. Vendors handling waste disposal projects that comply with waste management regulations, including hazardous industrial waste, sludge, waste plastics, waste electrical wires, and cables; 2. vendors that are the sole partner responsible for a specific waste disposal project for the Company.

Flow Chart of Waste Generation



Waste Generation

Unit: Metric tons;%

Category			2021	2022	2023	2024
Output	Waste from general business		11,333.55	10,862.04	8,399.15	11,027.35
	Waste from hazardous business		6,454.36	6,204.18	4,672.03	7,868.64
	Total Output		17,787.91	17,066.22	13,071.18	18,895.99
Transfer (Recycle/ Reuse)	Waste from general business		10,710.90	10,335.53	8,135.98	10,563.62
	Waste from hazardous business		5,516.50	5,190.71	4,034.40	7,246.88
	Total Transfer		16,227.40	15,526.24	12,170.38	17,810.50
Direct Disposal	Waste from general business	Incineration	448.50	382.51	218.99	358.92
		Energy Recovery	0.00	0.00	0.00	0.00
		Landfill	127.42	94.31	43.64	56.14
		Other methods	46.73	49.69	0.54	48.67
	Waste from hazardous business	Incineration	880.09	1,004.66	632.13	621.64
		Energy Recovery	0.00	0.00	0.00	0.00
		Landfill	0.00	0.00	0.00	0.00
		Other methods	57.77	8.81	5.50	0.12
	Total Direct Disposal		1,560.51	1,539.98	900.80	1,085.49
Recycling and Reuse Rate for General Industrial Waste			94.51	95.15	96.87	95.79
Recycling and Reuse Rate for Hazardous Industrial Waste			85.47	83.66	86.35	92.10
Data coverage rate			100	100	100	100

Note 1: "Transfer (reuse)" is done by adopting the reuse flow of the Climate Change Administration Ministry of Environment or entrust and jointly process the flow to adopt a reuse method.

Note 2: Incineration is the waste treated by incineration (Z05) and does not include low calorific value organic waste liquids.

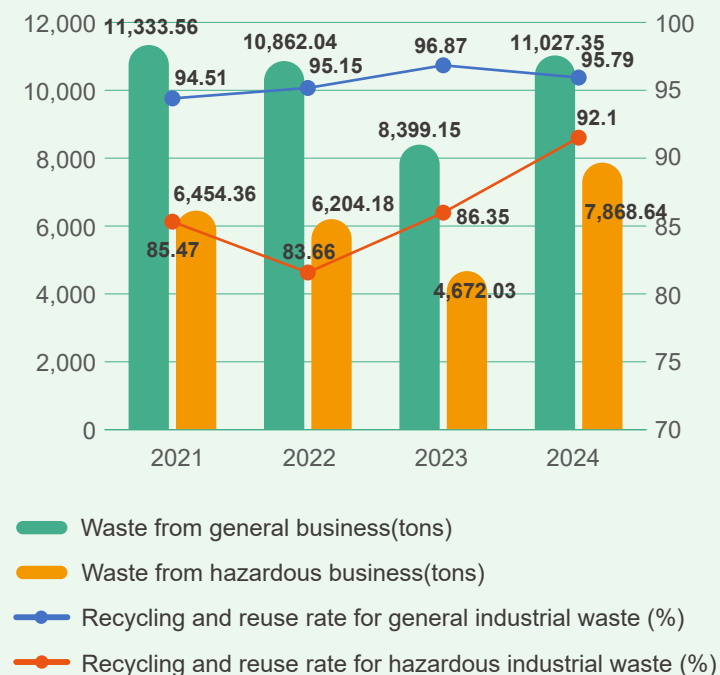
Note 3: Landfill is the waste disposed of by burying in the landfill (X01).

Note 4: Other wastes that are not treated by the above treatment methods (solidification treatment, offshore treatment or intermediate treatment of wastes that does not change the shape of the wastes).

Note 5: The quantity of waste disposal is declared in accordance with the relevant regulations, and all of them are off-site disposal.

Note 6: Data covers PSMC and its subsidiaries included in the consolidated financial statements.

Waste Generation and Recycling Rate



3.4.2 Waste Reduction Measures

PSMC adopts reduction, recycling, and reuse as its core waste management principles. In addition to seeking waste reduction opportunities through process improvements, PSMC also provides employees with education and training on waste reduction to ensure that employees can correctly handle the waste produced in the factory, reducing waste generation and reducing environmental risks. The plant's safety, health, and environmental committee compares the reasonableness of waste production (raw material volume vs. waste) every quarter, and completes internal management review procedures in accordance with the ISO 14001 environmental management system, and conducts PDCA effectiveness evaluations on emission management. In line with the four major goals set by PSMC: "Source reduction, End of process waste reduction, Enhancement of waste reuse value, Proper waste disposal and waste flow tracking", we plan and implement various waste reduction plans.

Objective 1. Source Reduction

Item	Description
Reducing isopropyl alcohol (IPA) usage	At the 8B fab, the chuck cleaning process was modified by replacing IPA with DIW (deionized water) as the cleaning medium. This allowed complete substitution of IPA with water, significantly reducing IPA consumption. - Phase 1: Adjusted chuck clean settings, reducing IPA usage from 9L/min to 7L/min. - Phase 2: Fully replaced IPA with pure water, achieving 100% IPA reduction. The monthly IPA consumption was reduced from nearly 40,000L/month at its peak to an average of 8,000L/month.
Reusing concentrated isopropyl alcohol (IPA)	To reduce pipe-end waste solvent disposal costs and raw material consumption, one waste IPA online recycling system and one waste liquid classification & collection system have been installed at Fab 8A. Through these, higher-concentration waste IPA is first classified, then transferred to the online recycling system for re-purification, and finally transferred back to and reused in the process IPA supply system. In 2024, 147,251 liters of IPA raw material was saved, helping reduce raw material consumption and waste solvent disposal as well.
Reducing photoresist usage	At the 8A fab, two optimization projects were implemented on photoresist coating equipment, aimed at reducing raw material consumption while ensuring quality. Type A equipment: Photoresist usage per wafer was reduced from 2c.c. to 1c.c., achieving a 50% reduction. Type B equipment: Photoresist usage per wafer was reduced from 1.5c.c. to 1c.c., achieving a 33.3% reduction. Annual savings: Approximately 374 liters of photoresist, reducing costs by NT\$808,000. At the P1/2 fabs, the equipment department implemented a photoresist reduction plan on six coating machines. Without affecting production volume or quality, the dummy wafer photoresist usage was reduced from 1.4c.c. to 1c.c. Annual savings: Approximately 639 bottles of photoresist, reducing costs by NT\$14.78 million. At the P3 fab, a big data system was used to analyze photoresist usage and production volume, leading to the removal of idle photoresist. Annual savings: Approximately 27 bottles of photoresist (one bottle is ~3,785c.c.), reducing costs by NT\$3.2 million.

Objective 2. End of Process Waste Reduction

Item	Description
Ammonia nitrogen wastewater treatment	PSMC's 12-inch fab prioritize eco-friendly approaches in assessing and planning construction of ammonia nitrogen wastewater treatment systems, and incorporate derivative waste disposal into the assessments. For those reasons, we hot air stripping plus catalytic oxidation disposal, which produces no derivative waste, to reduce overall waste generation. As a result, the ammonia nitrogen in water can be reduced by more than 85%. This disposal approach has been recognized by the HSP Bureau with an Award for Outstanding Achievement in Ammonia Nitrogen Wastewater Reduction. In addition, PSMC has worked with the HSP Bureau and other agencies as well as industry peers in HSP to share experience in using this wastewater treatment system, providing reference information for a variety of industries to develop treatment systems that fit their specific scales. In 2024, the system continued to operate without generating any derivative waste. Through regular system maintenance and inspections, the replacement cycle of MD film and RO front filter in Fab 8A has been extended; and this has helped reduce waste consumables generation by 117.6 kg per year.
Sludge reduction	Due to processing machine characteristics, the hydrofluoric acid waste liquid generated by 537.7 tons from Fab P3, and the concentration reaches 20% or more. In order to reduce environmental pollution resulting from the treatment of high-concentration waste liquid, and to avoid wasting usable resources, the fab held discussions with onsite departments then decided to established a dedicated recycling system to collect waste fluid for separate control and treatment. Legal disposal service providers have also been commissioned to recycle the high-concentration hydrofluoric acid waste liquid generated at the fab, instead reusing it to make fluoride substitutes that can be used as fluxing agents. In addition to improving the fab's waste reuse rate, this approach can also help reduce the amount of waste sludge.
Adjusting chemical mechanical polishing/planarization (CMP) operation mode	Fab P1/2 has adjusted its chemical mechanical polishing/planarization (CMP) operation mode, increasing treatment volume through the ceramic membrane filtration (CMF) system. This helps reduce the sludge system's treatment volume, and lower the frequency of system operation and backwashing. It also reduces waste generation, improves treatment efficiency, and reduces chemical consumption. We estimate that these adjustments can decrease CMP sludge generation by 18 tons per month, and decrease waste sludge generation by about 216 tons per year, while also reducing clearance costs by about NT\$2.27 million.
Landfill treatment for waste reduction	Starting from 2022, PSMC conducted a company-wide feasibility assessment of waste reuse for landfill treatment. After evaluation and review, more qualified reuse vendors have been added to reuse and dispose wastes such as waste quartz, waste thermal bricks, waste zeolite, sand and gravel that were originally treated by landfill. A total of 57.62 tons has been completed in 2024. The relevant cooperative vendors are all removal and treatment agencies qualified by the competent authorities, and some vendors have passed IOSO 14001 certification.

Objective 3. Enhancement of Waste Reuse Value

Previously, chemical mechanical polishing (CMP) sludge was reused as a low-strength cement additive in the cement industry. However, starting in June 2024, the P1/2 fabs partnered with a new supplier utilizing patented technology to implement a silicon recycling system. Through this system, impurities are removed from CMP sludge, and silicon dioxide is extracted and refined to manufacture high-value-added products, such as rubber and construction material fillers, coatings, and conductive material catalysts. This initiative further supports circular economy goals. Since 2024, a total of 60.7 metric tons of CMP sludge has been recycled and reused.

Objective 4. Proper Waste Disposal and Waste Flow Tracking

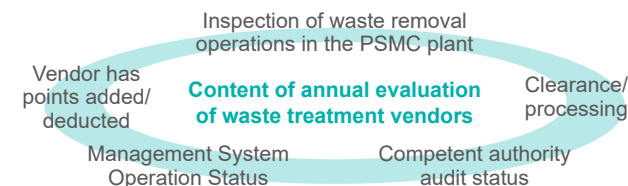
The waste generated at PSMC is accurately classified according to categories and disposal methods, and is entrusted to qualified waste vendors for cleaning and reuse. It is strictly required that waste cleaning should comply with the provisions of the Waste Cleaning Law and international regulations.

2024 Waste Recycling Rate and Reuse Methods

Types of Waste	Recycling rate (%)	Fab Area	Reuse Method
Empty barrel containers	100	P1/2、P3、P5、8A、8B	Recycled and reused by suppliers, cleaned and crushed by manufacturers and made into plastic and glass materials
Sulfuric acid waste	100	P1/2、P3、P5	Purified and reprocessed into industrial grade phosphoric acid by the vendors
Copper sulfate waste	100	P1/2、P5	Refined into copper sulfate powder by the vendors
Activated carbon waste	100	P1/2、P3、8B	Regenerated by the vendors for secondary use after desorption
Inorganic sludge	100	P1/2、P3、P5、8A、8B	Recycled by vendors and made into artificial stone granules, alternative raw materials for cement, artificial fluorite
Phosphoric acid waste	100	P1/2、P3、8B	Purified and reprocessed into industrial grade phosphoric acid by the vendors
Slurry residue	100	P1/2、P3	Potassium silicate raw material is produced by the vendors after dosing.
Tetramethylammonium sulfate waste	100	P1/2、P3、P5	Recycled by the vendors into tetramethylammonium hydroxide (TMAH) for use in panel factories
Diluents waste	100	P1/2、P3、P5、8A、8B	Purified and remanufactured into optoelectronic cleaning fluid (EBR)/paint material by the vendors
Ammonium sulfate waste	100	8A、8B	Purified and reconstituted into solid ammonium sulfate by the vendors
Waste mercury lamps	100	P1/2、P3、P5、8A、8B	Dismantled and refined for reuse by the vendors
Hydrofluoric acid waste	100	P3、P5	Remanufactured into fluorite for use by the vendors
Ammonium fluoride	100	P1/2、8B	Remanufactured into fluorite for use by the vendors

3.4.3 Waste Treatment Vendors Management

PSMC develops a waste manufacturer audit plan every year, cooperates with industry associations and its own factory audit schedules, regularly visits industrial waste removal and treatment vendors, and waste treatment compliance as the primary audit items. For deficiencies identified during audits, PSMC utilizes its electronic auditing system (E-Auditing) to issue audit reports and track improvements. All corrective actions are completed within the specified deadline, and vendors are required to strengthen the implementation and management of environmental, safety and health/fire protection regulations to avoid the risk of illegal cooperation by cooperative vendors. In addition, the annual audit results are provided to the procurement department as a reference for the following year's purchasing allocation decisions.



Unit: Vendors; %

Year		2022	2023	2024
Item				
Waste vendors collaborating		47	43	49
Written and on-site audits conducted ^(Note)		47	43	49
Waste vendor audit percentage		100	100	100
Audit Results	Excellent (85+ points)	19	20	30
	Qualified (70-85 points)	25	23	19
	Needs Improvement/Non-Compliant (<70 points)	1	0	0
	Deficiencies identified & improvement status	60 deficiencies, all rectified	34 deficiencies, all rectified	36 deficiencies, all rectified

Note: On-site audits conducted >85%.

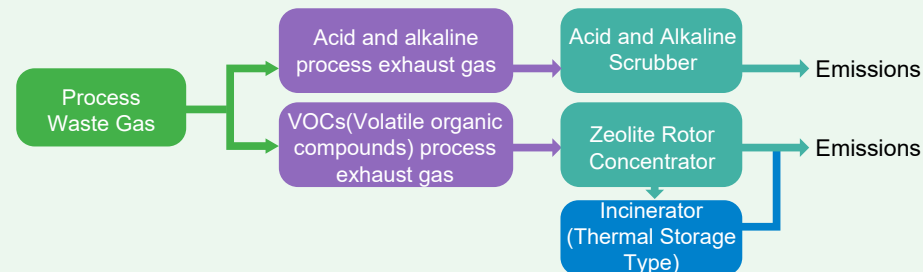
PSMC has signed the "High-tech Industry Waste Cleanup and Self-Discipline Convention" initiated by the Taiwan Semiconductor Association since 2017, and continues to cooperate with the association in implementing the "Waste Processing Agents" "Counseling, Assessment and Promotion Plan" to implement conduct audits of waste vendors to confirm that management and operations comply with the requirements of the self-discipline convention. In recent years, PSMC have not been punished by relevant authority units due to environmental pollution incidents and have not suffered any losses, and there have been no pollution disputes since its establishment.

3.5 Air Pollution Emissions Management

3.5.1 Air Pollution Prevention and Control Policy

Air pollutants produced by the semiconductor manufacturing industry are divided into two categories: acid/alkaline gases, and VOCs gases. Acid/alkaline waste gases are treated through wet scrubbers and then released. VOCs gases, after being adsorbed and concentrated through a zeolite rotor concentrator, are thermally incinerated and then discharged into the atmosphere. The Company adopts the best technologies to prevent pollution, in order to ensure that discharged pollutant content will meet or exceed regulatory requirements.

Flow chart of air pollutants treatment



Treatment of Acid and alkaline exhaust gas

The first stage: the local scrubber equipment is set up at the end of the machine to treat the toxic, flammable and fluorinated greenhouse gases (FG).
The second stage: the gases are then discharged to the central scrubber system on the back end and then discharged into the atmosphere through the discharge pipe.



Acid and alkaline exhaust gas treatment facility

Treatment of Volatile organic compounds exhaust gas

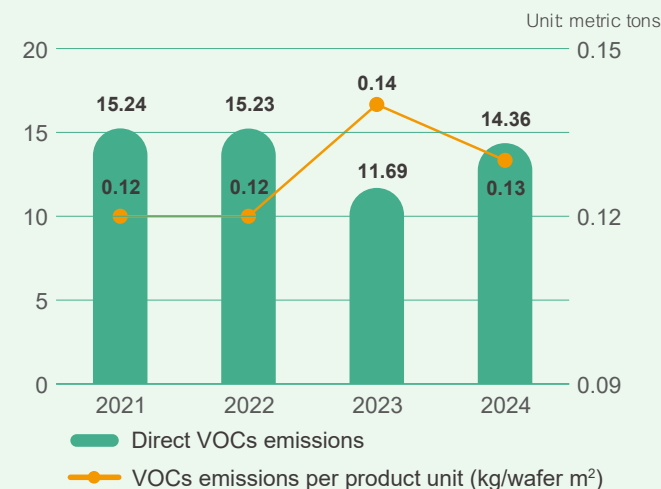
Volatile organic compounds exhaust (Solvent Exhaust) is discharged based on "low-temperature adsorption and high-temperature desorption" through the zeolite rotor equipment of volatile organic exhaust gas.



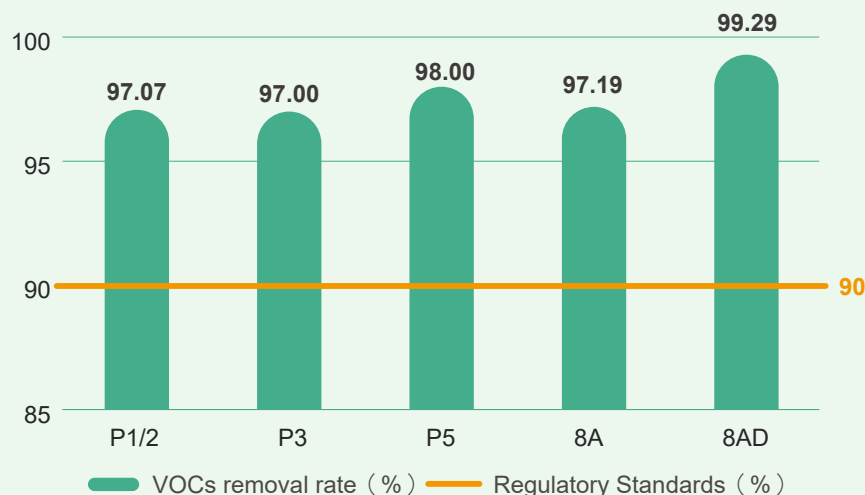
Volatile organic compounds exhaust gas treatment facility

The volatile organic waste gas systems in all factories of PSMC are equipped with zeolite runner treatment equipment. Fab P1/2,P3,P5 and 8A also each have a set of volatile organic compound treatment and control equipment, so that the removal rate can be more effectively controlled. In 2024, the overall average volatile organic waste gas (VOCs) removal rate of PSMC has reached 97.71%, which is better than the best control technology conditions for semiconductor manufacturing processes in the "Best Available Control Technology for Stationary Pollution Sources" (≥ 92%). The Zeolite concentration conversion wheel system reduction rate has also increased year by year.

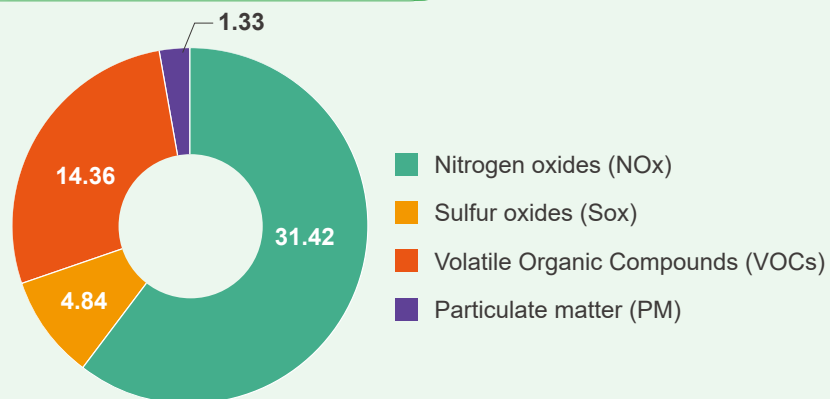
Emitted volatile organic compounds (VOCs) emissions



Volatile Organic Compounds (VOCs) removal rate of each plant



Air pollutant emissions (tons)



Note: Air pollutant emission volume is calculated and reported based on categories specified by the competent authorities; therefore, persistent organic pollutants (POPs) and hazardous air pollutants (HAPs) are not required to be included within the scope of monitoring.

3.5.2 Air Pollution Management Measures

In addition to taking measures to strengthen in-factory air pollution control equipment against possible polluting species, PSMC also rolling assessment whether to add or improve control equipment for the manufacturing process, including such as Local Scrubber (L/S) operation optimization and treatment efficiency verification, Central Scrubber (C/S) treatment efficiency verification, efficiency improvement, addition or replacement of control equipment, and white smoke improvement projects, etc. The air pollution improvement projects and results in 2024 are as follows:

Fab	Project details	Implementation description/effectiveness
P1/2	Thermal storage material replacement	Replaced thermal storage material to address decreased pollutant treatment efficiency
P3	VOCs#1 rotor replacement to improve VOCs removal efficiency	Replaced VOCs#1 rotor to enhance VOCs removal efficiency. Expected to increase air pollution control equipment efficiency by 5% and reduce total pollutant emissions by 2,426.52 kg
P5	Additional organic exhaust gas treatment equipment installation	- Increased total organic exhaust treatment capacity - Improved reliability of the organic exhaust system
8A	ATWP ammonia exhaust system project implementation	Improved equipment availability through diversion treatment



3.6 Biodiversity Management

In the course of its industrial development, PSMC continues to make efforts to protect and maintain biodiversity. In response to international concerns about biodiversity and forest conservation, PSMC have formulated the "PSMC Biodiversity Policy" to demonstrate our commitment to the conservation of the ecological environment through the monitoring and promotion of corporate sustainable governance.

Biodiversity and Non-Deforestation Commitment

PSMC is committed to maintaining biodiversity in accordance with SDG15, with ecosystem protection as the starting point. The Company is gradually implementing biodiversity risk assessments and actively engaging customers, suppliers, and industry associations to jointly support biodiversity preservation and forest conservation. This initiative aims to promote the well-being of mankind and the planet in response to the environmental and ecological challenges posed by climate change.

Biodiversity and Non-Deforestation Policy

- Avoid constructing manufacturing facilities or engaging operations adjacent to globally or nationally protected biodiversity areas.
- Avoid relative operating activities to destroy protected forests and creatures.
- If operating activities touch on biodiversity and ecological conservation areas, actions must be taken to eliminate, mitigate, offset and restore the richness and variability of biodiversity.
- According to the characteristics of the operating site and the nearby ecosystem, priority is given to measures that can help mitigate or adapt to climate change, and the introduction of biodiversity enhancement practices, as well as attention to ecological indicators monitoring and maintenance.
- Comply with international and local forest-related laws or specific regulations, and follow the principle of Zero Net Deforestation, so as to avoid deforestation.
- Actively pursue conservation and sustainable use in response to the Convention on Biological Diversity signed by the United Nations Environment Programme.
- Bringing together the upstream and downstream value chain and working with customers, suppliers and public association to support and practice biodiversity conservation.
- Actively promote the concept of biodiversity, enhance public awareness of ecological conservation, and encourage ecological conservation actions.

Biodiversity Risk Assessment

In response to the Taskforce on Nature-related Financial Disclosures (TNFD) initiative, PSMC has adopted the WWF Biodiversity Risk Filter starting in Q4 2024. This tool helps identify biodiversity risks that the semiconductor manufacturing industry may face. By supporting and implementing biodiversity conservation goals, PSMC aims to promote biodiversity awareness, enhance ecological conservation efforts, and reduce the negative environmental impact of operational pollution.

The biodiversity risk assessment tool covers eight major categories and 33 indicators. Since PSMC's operational sites are all located within science parks governed by the Hsinchu Science Park Bureau (HSPB) under the National Science and Technology Council, and there are no globally or nationally designated key biodiversity areas within a 2-kilometer radius, certain categories—including natural resources, disasters, and cultural services—were excluded from the risk assessment. The evaluation focused on two main categories: pressures on biodiversity, and environmental factors. A total of nine indicators were assessed. The risk assessment results indicate that all operational sites are classified as having a medium-to-low physical risk level. PSMC will continue to collaborate with the HSPB on a variety of ecological monitoring, conservation, and species protection programs. Additionally, the Company plans to follow the TNFD-LEAP methodology for a more comprehensive and in-depth assessment and management of biodiversity risks. PSMC prioritizes avoidance and minimization measures for potential biodiversity risks, focusing on the protection of endangered species and mitigating ecological impacts.

Biodiversity Conservation Initiative

To uphold its commitment to biodiversity conservation and ecological balance, PSMC proactively inquired with the Hsinchu Science Park Bureau (HSPB) regarding the ecological monitoring efforts around Tongluo Science Park. The HSPB conducts quarterly environmental quality monitoring (terrestrial ecology) at Tongluo Science Park and has installed 10 infrared cameras for wildlife observation. In 2024, the monitoring recorded many different species, including birds, mammals, reptiles, amphibians, and butterflies. Summary of monitoring results are as follows:

- Birds: 38 to 47 species were recorded, with population size influenced by habitat conditions and migratory patterns.
- Mammals: 5 to 9 species were detected, with the Japanese house bat being the dominant species.
- Reptiles: As ectothermic animals with reclusive behavior, their detection was opportunistic.
- Amphibians: Population numbers spiked during the rainy season.
- Butterflies: Fewer were observed in winter and early spring, while populations increased in summer and early autumn, mainly influenced by the blooming periods of nectar plants.
- Infrared camera monitoring recorded 35 species from 12 orders and 24 families, including bats, squirrels, muntjacs, laughingthrushes, and sparrows.

Regarding the conservation efforts at Tongluo Science Park, in 2024, the Hsinchu Science Park Bureau completed the construction of a Leopard Cat Ecological Corridor around PSMC's Tongluo fab (P5 fab) in the designated conservation and park areas. The corridor connects habitats on both sides of the leopard cat's range through aboveground, ground-level, or underground passages. Additionally, it integrates with existing drainage systems by incorporating openings, mesh installations, and climbing ramps to create a more wildlife-friendly environment, minimizing disruption to native species and ecosystems. New road signs and speed reduction markings have also been installed within the park to remind drivers to slow down, helping reduce the risk of wildlife roadkill incidents.

Leopard Cat Ecological Corridor



Added a new sign



Speed reduction line



In addition to tracking external environmental monitoring results, PSMC has incorporated multi-layered greenbelt plantings along the perimeter fence of the Tongluo fab to balance greening efforts with ecological conservation. In 2024, PSMC organized several biodiversity management initiatives, including: Removal of invasive species, specifically mile-a-minute weed (*Mikania micrantha*); Nature Exploration Team Let's Go lectures; and a Mountain Cleanup event at the Tongluo Skywalk. These activities aim to raise awareness of environmental conservation, encourage PSMC employees to adopt an ecological mindset, and promote the protection of native plants, trees, and biodiversity. Through these ongoing efforts, PSMC continues to foster a harmonious coexistence between its operations and the surrounding ecosystem.

Mountain Cleanup event at the Tongluo Skywalk



Nature Exploration Team Let's Go lectures



Mile-a-minute weed (*Mikania micrantha*) invasive species removal



Multi-layered greenbelt planting



4 Collaborative Sustainability

| Talent |

- 4.1 Talent Attraction and Retention
- 4.2 Talent Nurturing and Development
- 4.3 Human Rights
- 4.4 Occupational Health and Safety



Collaborative Sustainability

Talent

PSMC values people-centric management and is committed to creating a supportive work environment for employees. The Company offers competitive salaries and comprehensive training programs, working hand in hand with employees to maximize individual and team potential, and fostering a creative workplace. PSMC also implements occupational safety and health risk identification and employee health care systems. The Company has established clear performance evaluation mechanisms, labor-management communication systems, and technology talent collaboration programs with universities. By ensuring workplace safety, PSMC continues to create shared value for employees and stakeholders.



Corresponding United Nations Sustainable Development Goals (SDGs)



Stakeholders who have priority reading

Government / Authorities, Employees, Suppliers / Contractors, Social / Local Communities



2024 Sustainability Performance Highlights



Material Topics GRI Standards

GRI 401 : Employment

GRI 402 : Labor/Management Relations

GRI 403 : Occupational Health and Safety

GRI 404 : Training and Education

GRI 405 : Diversity and Equal Opportunity



Corresponding Material Topics



Occupational
Health and Safety



Talent Attraction and
Retention



Human
Resource
Development



Silver Award

1111 Job Bank Happy
Business Awards –
Technology and Energy
R&D Category



Subsidies of NT\$1.25 million

Supported 6 employees to
continue their studies and
obtain higher academic
qualifications, and 688
employees to obtain
certificates or licenses
required for their jobs.



Scholarships of NT\$3.24 million

52 collaborative research
projects implemented
with four universities'
semiconductor academies



27 students

"Pre-recruitment program
for current students"
provides fresh graduates
with pre-recruitment
qualifications



100%

Completion rate of
employees' human-rights-
related training



35,849 participants achieved

Conducted 254 EHS educational trainings



No cases of occupational diseases

44 special health examinations held, with
2,433 participants

4.1 Talent Attraction and Retention

PSMC is committed to protecting employee rights, personal privacy, prohibiting inappropriate discrimination and sexual harassment, and providing an equal employment opportunity and environment. The recruitment policy is diverse, equal, and inclusive, and the selection of personnel is based solely on their professional ability and suitability for the position, with no differential treatment based on gender, age, race, nationality, marital status, religion, physical or mental disability, political affiliation, or gender orientation. We have clear rules and regulations for promotion, training, employee benefits and performance appraisal, to ensure that every employee enjoys equal opportunities and treatment.

To strengthen talent management and protect labor rights, PSMC has formulated the PSMC Labor and Human Rights Policy. This clearly outlines the international guidelines and domestic regulations that the Company follows. The Company also implements a variety of practical measures to comprehensively safeguard employee rights. For example, employee salaries are adjusted based on the annual consumer price index (CPI), industry salary standards, and the Company's operating conditions, ensuring that fixed salaries comply with the minimum wage standards set by the Labor Standards Act and are recognized as a living wage internationally. PSMC reasonably manages working hours to prevent excessive overtime, ensuring that regular weekly working hours do not exceed 40 hours. The Company also promotes gender equality by guaranteeing that there are no salary disparities in hiring based on gender. Furthermore, in the event of major workforce adjustments, the Company is required by law to notify the competent authorities and affected employees at least 60 days in advance, thereby safeguarding employees' rights to work and change jobs.

PSMC adopts diversified employee recruitment channels, including participating in large-scale recruitment activities such as campus recruitment and employment fairs, and publishing recruitment information through social media. Personnel interviews are flexibly conducted in both physical and video formats to ensure the annual recruitment goals effectively achieved. In recent years, it has actively participated in the semiconductor college program of top universities, expanded and established close cooperative relations with educational institutions, and strived to fill the talent gap in the semiconductor industry.

4.1.1 Diversified Workforce Composition

As of the end of 2024, PSMC had a total of 8,387 full-time permanent (formal) employees, with no part-time or non-guaranteed hours employees. The average employee age was 39 years old, and the average tenure was 10.9 years. Due to the completion and operational launch of the P5 fab, the total number of employees increased by 2.73% compared to the previous year. For non-permanent (temporary) employees, PSMC continued its pre-employment student recruitment and internship programs in 2024, hiring 29 new interns to help graduates transition smoothly into the workforce. In addition, to meet operational needs, PSMC outsourced services including corporate catering, environmental cleaning, security, equipment maintenance, and facility repair/expansion to external contractors. The total number of outsourced personnel reached 18,975, a significant increase compared to the previous year, primarily due to the launch of the P5 fab and its related operational demands.

To help create a diverse, equal, and inclusive workplace, PSMC has hired 17 employees of Indigenous descent and 55 employees with disabilities, meeting the government-mandated employment ratio for persons with disabilities. As part of its global expansion strategy, PSMC actively participated in international talent recruitment events organized by regulatory authorities and universities in 2024. The Company continues to increase the proportion of foreign employees, aiming to foster cross-cultural and professional exchanges that enable international employees to maximize their strengths and influence.

Workforce Composition Statistics (by Employment Type)

Employment Type	Category	Number of People	Percentage of total employees (%)
Formal Employees	Female	3,174	37.70
	Male	5,213	61.90
	Total	8,387	99.60
Temporary Staff	Female	17	0.20
	Male	17	0.20
	Total	34	0.40

Note: Temporary employees are those who have signed a fixed-term contract.

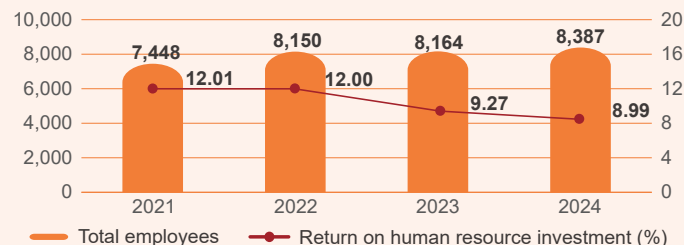
Workforce Composition Statistics (by Formal Employees Category)

Formal Employees Category	Category	Number of People	People Percentage of total employees (%)
Gender	Female	3,174	37.84
	Male	5,213	62.16
Management Level	Non-managerial general employees	7,044	83.99
	Junior Management Level	755	9.00
	Middle Management Level	565	6.74
	Senior Management Level	23	0.27
Nationality	Taiwan	7,871	93.85
	The Philippines	446	5.32
	Japan	45	0.54
	Others	25	0.29
Age	Under 29 years old	1,609	19.18
	30~49 years old	5,859	69.86
	older than 50 years old	919	10.96

Note 1: Junior management level: Production line deputy leader, production line leader, production line supervisor, team leaders or supervisors (excluding technical/project leaders/supervisors), middle management level: department and division level heads/supervisors, senior management level: assistant vice president/director level or above.

Note 2: The percentages of formal employees of a variety of nationalities in managerial positions are as follows: Taiwan 98.36%, Japan 1.19%, and others 0.45%.

Human Resource Investment Performance



Note 1: Return on human resource investment = [Operating revenue - (Operating expenses - Employee salaries and benefits)] / Employee salaries and benefits.

Note 2: In 2024, both operating revenue and employee-related expenses increased over 2023. However, due to a significant rise in operating expenses, the return on human resource investment experienced a slight decline.

Distribution of Female Workers

PSMC attaches great importance to the promotion and development of female employees. However, due to the characteristics of the semiconductor industry, the majority of management and professional engineering personnel are male employees, while the majority of technical personnel and administrative personnel are female employees. In order to increase the proportion of female employees participating in decision-making, PSMC promotes communication among female employees from the perspective of career planning and workplace mentality through education, training and themed lectures, comprehensively supports female employees and creates promotion opportunities to promote the innovative value of inclusion. In 2023, we set a goal of reaching 30% of female managerial personnel (including junior-level, middle-level and senior management positions) by 2025.

Category	Female		Male	
	Number of People	As a percentage of this category (%)	Number of People	As a percentage of this category (%)
Management Level	376	28.00	967	72.00
Junior Management Level	301	39.87	454	60.13
Middle Management Level	74	13.10	491	86.90
Senior Management Level	1	4.35	22	95.65
Revenue-Creation-Related unit Managerial	354	27.63	927	72.37
Number of employees in science, technology, engineering, and mathematics (STEM) related positions	1,191	20.18	4,710	79.82

Note 1: Junior management level: Production line deputy leader, production line leader, production line supervisor, team leaders or supervisors (excluding technical/project leaders/ supervisors), middle management level: department and division level heads/supervisors, senior management level: assistant vice president/director level or above.

Note 2: The managerial positions in revenue-creation-related units are defined as those in units other than the supporting units (IT, legal affairs, human resources, finance /accounting).

4.1.2 New and Departed Employees

New Hire Overview

916 new employees were recruited in 2024, focusing on talents related to equipment, processes, facilities, R&D, and other related fields, and including younger workers, foreign blue-collar workers, and white-collar workers.

Type	Fiscal Year	2021	2022	2023	2024
Number of new employees		974	1,397	516	916
New Hire Rate (%)		13.08	17.14	6.35	10.92
Average Cost of Recruitment (NT\$)		7,317	6,853	14,026	7,606

Note 1: New hire rate = Number of new employees / Total number of employees at the end of the year.

Note 2: Average Cost of Recruitment = Cost of Hiring New Personnel for the Year / Number of New Employees

Statistics of New Formal Employees

Category	Type	2021		2022		2023		2024	
		Number of People	Percentage (%)	Number of People	Percentage (%)	Number of People	Percentage (%)	Number of People	Percentage (%)
Gender	Female	324	33.26	503	36.01	196	37.98	303	33.08
	Male	650	66.74	894	63.99	320	62.02	613	66.92
Age	Under 29 Years Old	588	60.37	830	59.41	360	69.77	616	67.25
	30~49 Years Old	375	38.50	560	40.09	151	29.26	282	30.79
	Older than 50 Years Old	11	1.13	7	0.50	5	0.97	18	1.96
Management Level	Non-managerial Level	955	98.05	1,383	99.00	505	97.87	870	94.98
	Junior Management Level	18	1.85	5	0.36	5	0.97	29	3.16
	Middle Management Level	1	0.10	9	0.64	6	1.16	14	1.53
	Senior Management Level	0	0.00	0	0.00	0	0.00	3	0.33
Nationality	Taiwan	937	96.20	1,235	88.40	432	83.72	783	85.48
	The Philippines	35	3.60	150	10.74	78	15.12	121	13.21
	Japan	1	0.10	0	0.00	0	0.00	3	0.33
	Others	1	0.10	12	0.86	6	1.16	9	0.98

Employee Departure and Internal Rotation

PSMC continuously tracks employee turnover trends. In 2024, the overall turnover rate was 8.26%, while the voluntary turnover rate was 8.11%, both showing slight increases over 2023. Through turnover rate analysis, PSMC has identified that employee departures are primarily influenced by salary and benefits, work environment, and personal career development considerations. To enhance employee retention, the Human Resources Department conducts exit reason analyses for departing employees. Based on their skills and experience, PSMC offers job adjustments, including changes in job responsibilities, department transfers, or relocation opportunities, to encourage internal rotation and career development.

In terms of the internal rotation mechanism, basically, the staff-requesting unit must submit an internal recruitment request, which shall then be released by the HR unit and be open for current employees' application. After the preliminary review is completed and the qualified candidate is determined, the HR unit shall send the application form to the applicant's home unit and the staff-requesting unit for the form to be signed off. After the consent of the two units are obtained and the form is signed off by the two units, the reassignment can be carried out.

Category \ Fiscal Year	2021	2022	2023	2024
Number of Departed Employees	700	695	537	693
Turnover Rate (%)	9.40	8.53	6.58	8.26
Voluntary Turnover Rate (%)	9.30	8.42	6.51	8.11
Internal Rotation Rate (%)	42.50	38.40	60.20	46.90

Note 1: The number of departed employees include those who left voluntarily or by dismissal, as well as those who retired and stayed without pay.

Note 2: Turnover rate = Number of employees who left the company / Total number of employees at the end of the year.

Statistics on Departed Formal Employees

Category	Type	2021		2022		2023		2024	
		Number of People	Percentage (%)	Number of People	Percentage (%)	Number of People	Percentage (%)	Number of People	Percentage (%)
Gender	Female	237	33.86	263	37.84	211	39.29	262	37.81
	Male	463	66.14	432	62.16	326	60.71	431	62.19
Age	Under 29 Years Old	336	48.00	256	36.83	189	35.20	249	35.93
	30~49 Years Old	345	49.29	413	59.43	309	57.54	389	56.13
	Older than 50 Years Old	19	2.71	26	3.74	39	7.26	55	7.94
Management Level	Non-managerial Level	682	97.43	669	96.26	482	89.75	631	91.05
	Junior Management Level	5	0.71	3	0.43	22	4.10	27	3.90
	Middle Management Level	10	1.43	20	2.88	32	5.96	30	4.33
	Senior Management Level	3	0.43	3	0.43	1	0.19	5	0.72
Nationality	Taiwan	672	96.00	640	92.09	452	84.17	628	90.62
	The Philippines	27	3.86	53	7.63	75	13.97	63	9.09
	Japan	1	0.14	1	0.14	5	0.93	2	0.29
	Others	0	0.00	1	0.14	5	0.93	0	0.00

Column

Pre-employment Internship Program



Talent Recruitment Outpost - Pre-employment Internship Program for Current Students

Since 2022, PSMC has been actively promoting the Pre-Employment Internship Program for Current Students, aimed at recruiting outstanding students to gain early exposure to the workplace. This program allows students to familiarize themselves with the Company's environment and business operations, while also developing professionalism, teamwork, and other soft skills necessary for their future careers. Through this internship, participants can gain insights into the corporate culture, while PSMC can identify and attract top talent to join the Company. The program serves as a talent pool, preparing a future workforce to support the Company's long-term development.

In 2024, a total of 521 students applied for the intern engineer position. After a two-stage interview process, 29 interns were selected. Each intern was assigned a full-time employee as a mentor to help them become familiar with the work environment. After completing a 2- to 3-month internship, 27 students received pre-employment offers and expressed their willingness to join PSMC upon graduation. Since the program's launch in 2022, a total of 20 students have graduated and become full-time employees at PSMC, demonstrating the program's effectiveness in talent attraction. The high conversion rate highlights that most interns recognize and appreciate PSMC's friendly work environment, pragmatic corporate culture, and strong team dynamics, reinforcing the success of the pre-employment internship program.

Furthermore, in 2024, PSMC held a total of ten 2024 Pre-Employment Internship Information Sessions at different universities. These sessions provided students with a comprehensive understanding of the internship program, and encouraged them to jump at the chance to apply for corporate internships. A total of 590 students attended the sessions. This year, online information sessions were also continued to reach a wider audience, helping more students to learn about and join the pre-employment internship program.

Performance Category	2023	2024
Number of Participants in the Program	21	29
Percentage of People That Qualified for Formal Positions (%)	80.95	93.10
Actual Job Conversion Rate (%)	66.67	42.86
Amount of Scholarship Awarded (NT\$)	120,000	120,000

Yuan Ze University Pre-Employment Internship Information Session



National United University Pre-Employment Internship Information Session



Column

Semiconductor Academy Program

Top Talent Incubator - Semiconductor Academy Program

In order to continue to expand Taiwan's semiconductor technology leadership, the Executive Yuan passed the "Innovation Regulations on Industry-Academic Cooperation and Talent Cultivation in Key National Areas" and the "Special Law on Innovation Regulations for Higher Education Sandboxes", spending 9.6 billion over 12 years to establish "semiconductor colleges" at National Taiwan University, National Cheng Kung University, Tsinghua University, National Yang-Ming National Chiao Tung University, and National Sun Yat-sen University, training an additional 400 master's and doctoral students every year. Chairman Huang Chongren believes that the key to maintaining its leading position lies in the cultivation and self-sufficiency of talents. The reason why PSMC invested in the Semiconductor Academy is to train more Taiwanese engineers, and also become to one of the few sponsors of the four major semiconductor colleges in China that fully participates and donates funds. It estimate to invest NT\$100 million each year to fund the four university semiconductor colleges and hopes to cultivate more than 5,000 semiconductor professionals in Taiwan within 12 years. At the same time, PSMC recruit more talents to join PSMC and contribute to Taiwan's semiconductor industry.

2024 Semiconductor Academies Research Project Achievements:

- Held a presentation conference on the research project results of the School of Industry-Academic Innovation Research of Yang-Ming Jiaotong University and National Taiwan University of Science and Technology, and invited 9 professors from Yang-Ming Jiaotong University to PSMC to explain and exchange the results of the research project, and collaborate with Yang-Ming Jiaotong University on research projects. One of the projects has passed the company's internal review and is in the process of applying for a patent. Two other projects have passed intellectual property unit reviews and have been submitted for

journal publication. This year, the Graduate School of Advanced Technology (GSAT) at National Taiwan University (NTU) presented its research findings in the form of a poster presentation. There, graduate students explained their research posters to PSMC employees, showcasing 19 collaborative research projects. Of these, five research projects are currently applying for patents. Four projects have been submitted for journal publication, and 16 papers have been submitted to conferences for presentation.

- Collaborated with National Tsing Hua University on the Self-Powered AlGaIn/GaN HEMT-Based Sensor Integrated with Rotational Triboelectric Nanogenerator for Comprehensive Water Quality Analysis research project. This has been submitted to the Nano Energy journal and is currently awaiting publication.
- Three research projects in collaboration with National Yang Ming Chiao Tung University (NYCU) have been submitted for patent applications, while two others have been submitted for journal publication and have passed the internal intellectual property review.
- Phase one of the research projects at National Cheng Kung University (NCKU)'s Academy of Innovative Semiconductor and Sustainable Manufacturing was completed by the end of 2024, covering 12 projects in total. Six projects focused on automated visual monitoring (AVM), to enhance product quality prediction and control in production lines. Six projects applied intelligent predictive maintenance (IPM) technology, enabling PSMC equipment engineers to monitor machine conditions and key components, and lowering the risk of unexpected failures. The insights gained from industry-university collaboration have also facilitated internal development of additional related projects within PSMC.

In addition, PSMC also conducts exchanges and discussions with the semiconductor colleges of the four universities from time to time, and arranges for supervisors to give book discussions or lectures at the semiconductor colleges.

Performance Category	2022	2023	2024
Number of collaborative technical R&D projects	46	49	52
Project investment cost (NT\$)	over 40,000,000	over 40,000,000	59,100,000
Total amount allocated for scholarships (NT\$)	7,000,000	3,160,000	3,240,000

NTU Graduate School of Advanced Technology Achievement Presentation



4.1.3 Compensation System

PSMC's compensation system is based on the principles of fairness and reasonableness, and provides competitive compensation packages, including base salary, allowances, employee cash bonuses, and employee rewards, as well as an incentive bonus plan based on performance and job responsibilities, regardless of race, management level, language, religion, political affiliation, marital status, nationality, place of birth, gender, sexual orientation, age, or group affiliation. PSMC's compensation and benefits system is formulated by the Compensation Committee, taking into account factors such as the Company's operational performance, industry development trends, the overall economy, and sustainable corporate development. The compensation plan is reviewed and adjusted annually on a rolling basis, and distribution recommendations are prepared and submitted to the Board of Directors for consideration to ensure that the Company's overall compensation structure remains competitive in the industry.

In addition to fixed remuneration such as base salary and allowances, PSMC also provides employee incentive bonuses, retention bonuses and employee stock option certificates and other variable remuneration. In 2023, PSMC passed the employee stock subscription plan approved by the supervisory authority. The plan will be released in 2024. The amount of variable remuneration is determined based on individual performance, the company's profit for the year and unit target achievement rate, and does not differ based on gender. PSMC rewards employees for their excellent performance and shares operating results, and encourages colleagues to actively strive for performance and create healthy competition of working environment. In 2024, the average salary of full-time employees who are not in supervisory positions was NT\$1.245 million, and the median salary of full-time employees who are not in supervisory positions was NT\$1.181 million. The ratio of the annual salary of the highest individual to the median annual salary of all employees (excluding the individual with the highest annual salary) is 23.81, and the percentage increase of the annual salary of the highest individual to that of all employees (excluding the individual with the highest annual salary) annual median salary increase percentage rate is 0.

Average salary, median salary and salary gap of full-time non-supervisory employees

Item	Fiscal Year	2022	2023	2024	Difference between 2024 and the previous year
Average Salary of Full-Time Employee (NT\$10,000)		167.7	109.7	124.5	14.8
Median Salary of Full-Time Employee (NT\$10,000)		145.3	104.5	118.1	13.6

Note: The data related to non-supervisory employees was calculated by consulting the definition of "supervisory employees" given by TWSE

PSMC adheres to the principle of exceeding regulatory requirements in hiring employees, and ensures that starting salaries and overall compensation are not influenced by gender. In 2024, the differences in compensation between female and male employees were primarily due to a combination of factors that include job category, job responsibilities, length of service, and individual performance. Overall, PSMC continues to monitor and address gender pay disparities and is committed to providing a market-competitive compensation and benefits system to help employees achieve a healthy balance between career development and family life.

Gender Remuneration Indicators

Item	Fiscal Year	2023	2024
Difference in the average salary by gender (%)		2024	13.80
Difference in the median salary by gender (%)		12.20	11.50
Difference in the average bonus amount by gender (%)		21.40	19.50
Difference in the median bonus amount by gender (%)		23.30	20.40

Note 1: The scope covers indirect employees other than managers.

Note 2: Salary gender difference = (male salary - female salary) / male salary; Bonus gender difference = (male bonus - female bonus) / male bonus.

Note 3: Equal pay for equal work means "granting the same compensation to employees whose job content and performance are equivalent", and pertains to individual-level pay equity. In contrast, gender pay disparity is an overall indicator, which may vary due to differences in employee distribution, job level structure, and statistical methods. Therefore, it cannot be directly used as a basis for determining whether equal pay for equal work has been implemented.

Retirement Life Security

In accordance with the Labor Standards Act and the Labor Pension Act, there is a retirement plan for all regular employees, and those who have worked for PSMC for at least 10 years and are at least 45 years old, and with the sum of the first two items reaching 60, are eligible to apply for retirement. Employees subjected to the Labor Standards Act (also known as the old system) are required to contribute 2% of their monthly salary to the Bank of Taiwan's Labor Retirement Fund Supervisory Committee's special account. If the balance of the special account is not sufficient to pay for workers who are expected to meet the retirement requirements in the following year before the end of the year, and the balance must be replenished by the end of March of the following year. Employees subjected to the Labor Pension Act (also known as the New Labor Retirement System) are required to contribute 6% of their monthly wages to the personal pension account established by the Labor Insurance Bureau. For more information on pension liabilities, please refer to PSMC's annual report of 2024.

Item	Details	Employees' Participating in the Retirement Plan
Labor Standards Act Pension (Old System)	Employer: Contribute 2% of monthly salary to workers' retirement reserve fund	100%
Labor Pension Act (New System)	Employer: 6% of employee's monthly salary Workers: 0~6% of monthly salary	100%

4.1.4 Employee Benefits and Support Programs

Benefits and Employee Rights

The Human Resources Division and Employee Welfare Committee (below, the Welfare Committee) has developed comprehensive employee benefits systems to provide employees with diverse welfare facilities and activities for employees to enjoy a lifestyle full of creativity and energy both at work and in leisure time. In addition to the basic benefits of gift vouchers for three major festivals and birthday, the Company also provides marriage & baby allowances, hospitalization subsidies, children's scholarships, and other thoughtful subsidies/allowances of different amounts. Moreover, to promote work-life balance, there are diversified employee clubs and a variety of courses for employees to choose, through which employees can expand their social networks, cultivate hobbies, enhance physical fitness, and build connections with more people. Besides, all-embracing activities such as theater appreciation, arts and cultural activities, Family Day, brisk walking, festive (e.g. Christmas) activities, etc. are also frequently held to extend the Company's care to employees' family, thereby creating a sense of belonging among members of the whole PSMC family.

Health Care and Life Care

- In accordance with the Labor Standards Act, employees are given 3-30 days of paid leave depending on their employment period.
- In accordance with the government's maternity policy, in addition to the 8 weeks regulated by the law, an additional 2 weeks of postnatal care leave with half pay is granted.
- Regular planning of health screenings and health improvement activities to protect and track the physical and mental health of employees, as well as implement workplace stress management.
- In addition to providing employees with labor insurance and universal health insurance in accordance with the law, we also provide free and self-funded group insurance (including disability, medical, and life) to benefit their spouses and children, so that both employees and their families can be protected.
- We provide hospitalization, funeral and burial assistance, and emergency loans to help our colleagues cope with the difficult times.
- If employees need family care leave, they can apply for a flexible work system to help employee more easily achieve a balance between work and life
 - The supervisor of the unit controls the work hours of the employees.
 - With the approval of the unit supervisor, one can apply to work from home.
 - We provide 7 additional days of paid flexible leave each year. This exceeds legal requirements, and the days can be used for personal or family care needs.
 - Volunteer leave for participating in volunteer activities
- Provide part-time job options: hire part-time visually impaired employees and provide free shoulder and neck stress relief massage services to full-time employees
- We establish the welfare measures such as wedding red envelope, maternity leave, maternity examination leave, maternity subsidy, child care leave without pay, breastfeeding room, and scholarship for children, etc. and encourage colleagues to have childbirth with a subsidy of NT\$20,000 for each newborn baby
- We have signed special deals with several contracted childcare providers to offer special packages to our colleagues.
- We cooperate with many banks to provide preferential loan or deposit programs to meet the capital needs of our employees.
- Employee stock purchase plan: Individual members can allocate a fixed monthly amount of NT\$2,000 from their salary (self-withdrawal), and the company simultaneously allocates NT\$2,000 as a subsidy. A total of NT\$4,000 is entrusted to a financial institution to purchase the stocks of PSMC which will be transferred to personal CHEP account on the 15th of the following month.

Festivals and Annual Events

- Organize family days and trekking activities to provide employees with outdoor corporate activities with their families, so that employees can relieve stress and maintain physical and mental health, and effectively unite colleagues and strengthen family identification.
- Organize arts and cultural activities, such as parent-child theater, celebrity lectures, and concerts. On the one hand, PSMC sponsor arts and cultural groups, and on the other hand, we use arts and cultural activities to enhance colleagues' participation in art activities, further optimize the quality of life of employees, and promote parent-child interaction to enable employees to achieve an appropriate balance between work and leisure life
- The welfare committee subsidizes the operating funds of clubs to encourage employees to participate in and establish clubs, and to participate in external competitions, so that employees can develop interests and interpersonal relationships outside of work.
- Give vouchers for three major festivals and birthday, and gifts for Labor Day, Mid-Autumn Festival, Christmas, and other important festive events. An e-platform was introduced in 2023 to provide employees with more diverse and flexible electronic vouchers options.
- Issue gift certificates for three holidays, birthday gift certificates, and important holiday gifts such as Labor Day, Mid-Autumn Festival, and Christmas. From 2023, an E-based platform has been introduced to provide employees with more diversified and flexible electronic ticket choices.

Complete Work Environment

- On-site staff restaurants, convenience stores, and cafes.
- Free transportation to Hsinchu and Miaoli areas, and staff dormitory shuttles
- Recreational facilities such as libraries, gymnasiums and other sports halls



Arts and Cultural Events

Family Theatre Appreciation and Celebrity Special Lectures

Each year, PSMC organizes a variety of arts and cultural activities, including family theater performances and lectures by celebrities, to bring the arts closer to the lives of employees and cultivate a humanistic atmosphere, and to promote parent-child interaction and learning, and optimize the work-life balance of employees. In 2024, PSMC hosted two theater appreciation events. First was My Spectral Wedding, by Spring River Performing Arts Troupe, attended by 982 people; second was We're No Angels, by Godot Theatre, attended by 964 people. In total, nearly 2,000 participants enjoyed the performances. The events provided employees and their families with warm, joyful experiences, creating cherished moments in a lively and engaging atmosphere.

In addition to theatre performances, PSMC also hosted six arts and cultural lectures in 2024. These covered a wide range of topics: Nutrition lecture by Ming-Hua Sung, "Smart Choices for a Healthy Life"; Financial management lecture by Chi-Fen Lin, "Easy Financial Planning for a Prosperous Life"; Oral health lecture by Yu-Hsin Chou, "A Comprehensive Look at Adult Dental and Jaw Health"; Parenting lecture by Kuang-Tsai Hao, "Unlocking Children's Potential and Giving Them Wings to Fly"; Health lecture by Teacher Vicky, "Spinal Care for Better Health"; and a drama appreciation lecture by Chih-Min Liang, "We're No Angels and an Introduction to Godot Theatre's Well-Known Productions". A total of 833 participants attended these lectures. These renowned speakers delivered insightful and engaging sessions, sharing their life experiences and expertise. Their talks not only broadened employees' understanding of different fields, but also encouraged knowledge-sharing and personal growth.

Nutrition lecture by Ming-Hua Sung

Smart Choices for a
Healthy Life



Parenting lecture by Kuang-Tsai Hao

Unlocking Children's
Potential and Giving
Them Wings to Fly



Health lecture by Teacher Vicky

Spinal Care for Better Health

PSMC Family Day

Powerchip Family Day is an event exclusively for PSMC members' family members. In 2024, the event was held at Leofoo Village Theme Park on July 20 and August 3. A total of 5,667 employees registered to participate in the event, and on the event days, 12,899 employees and their family members turned out, for an attendance rate of 87.01%. Employees together with their family totaling nearly 13,000 people participated in the event, which not only facilitated husband-wife and parent-child interactions, but also encouraged more extensive interpersonal communication among participants. A variety of lovely food, fun facilities, and interesting stage performances were presented in the event for participants to enjoy the cheerful atmosphere together to effectively gather the centripetal force of employees and strengthen the identification of family members.

Powerchip Family Day Photos



Baoshan Second Reservoir Hiking Event

In 2024, PSMC organized two sessions of a family-friendly hiking event at Baoshan Second Reservoir. A total of 1,374 employees registered for the event, with attendance rates of 70.7% and 66.8% at the two sessions. In total, 942 employees brought their family members along to participate. Beyond the health benefits of hiking and exercise, the event also featured a variety of food stalls, interactive game challenges, stage performances, and a lucky draw, creating a fun and engaging experience for the whole family while promoting health and wellness.

Baoshan Second Reservoir Hiking Event photos



Diverse Clubs and Sports Events

Club Activities and Interdepartmental Competitions

With PSMC's active encouragement, participation in club activities has flourished in recent years. 2024, the Company hosts 12 clubs, in several different categories: Ball sports; fitness and recreational activities; public welfare; music and arts; and general learning clubs. The Employee Welfare Committee provides financial subsidies to support club operations. Ball sports clubs, in particular, are responsible for organizing the PSMC Cup sports tournaments, where prize money is awarded based on the number of participating teams. Additionally, members receive subsidies for jerseys and sports equipment to further encourage engagement. In addition, if a club represents the Company to participate in an external competition and wins an award, it will be granted an additional incentive subsidy of NT\$32,000. The total subsidy funds for all ball game clubs have reached NT\$1.5 million, and a subsidy of NT\$20,000 has been provided for each general club and study-oriented club to support their activities.

As stipulated in the Regulations Governing Employee Welfare Committee's Club Management, clubs must complete the next year's budget itemization in October every year, and club activities must be carried out by following the planned budget. The operating procedures are as follows:

1. Itemize the next year's budget in October every year, which shall be submitted to the Welfare Committee for approval in the quarterly meetings.
2. Clubs are required to fill out an Activity Application Form before the implementation of club activities. Club activity information and regulations must be announced on the Company's Welfare Bulletin after the application is reviewed and approved by the Welfare Committee.
3. Submit a case-closing report within 15 days after the activity is completed to apply for expense reimbursement.

Club sports competitions enhance employees' sense of belonging to the Company and further strengthen teamwork and cooperation. In 2024, in addition to continuing to host a variety of PSMC Cup sports tournaments, clubs also actively represented the Company in external competitions, bringing recognition and honor to PSMC.

Club Name	PSMC Cup		In 2024 PSMC Team Achievement
	Number of teams	Number of members	
Softball Club	19	340	Second place in the Mayor's Cup Slow-Pitch Softball Invitational (Recreational Division) Fourth place in the Science Park Cup Slow Pitch Softball Invitational Tournament (Healthy Group)
Tennis Club	23	182	-
Basketball Club	37	567	-
Table Tennis Club	40	385	Third place in the Zhunan Township National Table Tennis Championship Group Category Fourth place in the Science Park Cup Table Tennis Tournament Women's Group
Badminton Club	52	465	Second place in the Hsinchu Mayor Cup Badminton Championship Social Group Category Fourth place in the Science Park Cup Badminton Tournament
Volleyball Club	34	369	-
Bowling Club	104	709	-



Child Care Leave without Pay and Maternity Benefits

PSMC has established a complete unpaid paternity leave system and environment based on the rights that workers are entitled to in accordance with the Act of Gender Equality in Employment. This includes parental leave without pay, as well as paid use of lactation rooms when necessary. In addition, in response to the government's maternity policy, PSMC has introduced a new "Maternity Allowance" in July 2022, which provides a subsidy of NT\$20,000 for each newborn baby; a total of 86 applicants and a sum of NT\$3.04 million were paid out as of the end of December 2024. A new "Postnatal Care Leave" was introduced in September 2022 that is superior than the regulatory requirements, which allows female employees to apply for a two-week postnatal care leave (half pay) after the expiration of the statutory eight-week maternity leave, which makes a total of 10 weeks of paid parental leave, in order to provide more recuperation and care for female employees.

In 2024, the number of employees who were entitled to parental leave was 633, and the actual number of applicants for parental leave was 49. The job reinstatement rate was 73.91%, and the retention rate was 81.82%, with both the reinstatement and retention rates showing a stable status. All of our employees who have not returned to work after their leave of absence without pay, and all of them left voluntarily due to family care needs and personal career development.

Child Care Leave without Pay Statistics

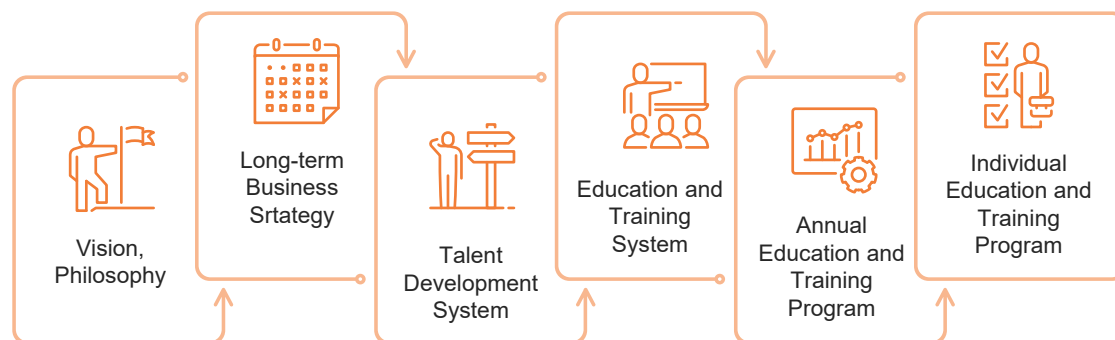
Item	Female	Male	Total
Number of employees entitled to child care leave in 2024	174	459	633
Actual number of employees applying for child care leave in 2024	33	16	49
Number of employees reinstated in 2024	30	16	46
Actual number of employees reinstated in 2024	26	8	34
Actual number of employees reinstated in 2023	23	10	33
Number of employees who are still working 12 months after their reinstatement in 2023	18	9	27
Reinstatement rate (%)	86.67	50.00	73.91
Retention rate (%)	78.26	90.00	81.82

Note 1: Reinstatement rate (%) = (Actual number of staff reinstated in FY2024 / Number of staff to be reinstated in FY2024) * 100%.

Note 2: Retention rate (%) = (Number of employees still working 12 months after reinstatement in FY 2023 / Actual number of employees reinstated in FY 2023) * 100%.

4.2 Talent Nurturing and Development

PSMC spares no effort in creating a comprehensive learning environment for employees by providing a variety of training courses and multiple learning channels as well as systematic training programs to help employees acquire the professional knowledge and skills needed for their work. At the same time, the Company also encourages employees to maintain a habit of self-directed learning after work by establishing online learning resources, providing subsidies for statutory professional license examinations and on-the-job training courses, and other support measures to create a work environment for continuous learning and progress and to inspire employees' potential.



4.2.1 Educational Training Framework

PSMC follow the Procedures for Employee Education & Training in which the PDCA methodology has been introduced. By following the PDCA principles, managerial personnel must develop next year's education and training plan by the end of each year based on comprehensive consideration of the Company's vision and business philosophy, medium-/long-term business strategies, and individual unit's training needs. The Human Resources Division responsible for organizing a variety of in-person and online education and training activities, keeping track of employees' training participation status, and recording feedback from course participants to continuously improve educational training programs' contents..



The education and training system is centered around the three core areas of "New Employee Orientation Training", "Functional Training", and "Knowledge Inspiration", to design suitable with two major categories of training contents for "new associates" and "on-the-job associates". To shorten the adaptation period for new employees, PSMC has also arranged a mentorship program for newcomers, with training content focused on building fundamental knowledge related to semiconductors. Further, through on-the-job training (OJT), new associates receive support to quickly integrate into the Company's operational system and culture. In order to meet the company's core philosophy of continuous learning for employees, the vocational training area is divided into three categories: professional, management and general education. Based on the employee's academic background, work style and professional functions, the supervisor and the employee jointly plan and formulate course combinations for different career stages to provide the most timely and appropriate training assistance programs to inspire the employee's long-term goal of lifelong learning and self-development.

Three Cores of Education and Training System



Educational Training Content

Target	Category	Course Content
New associates	Onboarding Training	Assist employees to understand the company's quality, environmental safety, information security and other regulations and policies, as well as company regulations, benefits and rights that are relevant to them.
	Professional Basics	All colleagues are trained to have basic knowledge of semiconductor basic technology, intellectual property rights and relevant business knowledge.
	On-the-Job Training (OJT)	The head of each department will design the training contents for the new employees in line with their duties, and the departments will evaluate the training to ensure that the employees can quickly grasp the key points of the business.
	Engineering Program	Provides essential business skills required for personnel in engineering and technical positions.
On-the-Job Associates	Professionals	Provide professional courses in operation technology and administration to improve the professional skills of our staff.
	Management	Provide stage by stage management skills refinement for management/succession to nurture excellent managers and successors.
	General Competency	Provide courses including presentation skills, problem-solving skills, time management, etc. to cultivate soft skills

PSMC encourages employees to take professional license or qualification examinations. For employees who need to obtain statutory licenses, the Company will provide full subsidies for exam registration fees and license fees. Employees who need to obtain higher academic qualifications can submit study applications in accordance with the Management Directions for On-the-job Study. 6 employees obtained higher academic qualifications in 2024, among whom 3 employees obtained a master's degree. Additionally, in 2024, PSMC organized four in-house certification courses, assisting 370 employees in obtaining certifications for organic solvents, supervisors of operations involving specific chemicals, and radiation equipment operation. Furthermore, 318 employees participated in external training programs to acquire work-related certifications. Total subsidies provided to employees obtaining certifications in 2024 exceeded NT\$1.25 million.

Employee Development Program

In addition to education and training programs, PSMC plans long-term training programs for all employees(include contractual and part time employees) based on the principle of continuous improvement of core functions.

Program Category	Program Name	Program Benefits
Leadership development	Succession Planning	- Ensures smooth personnel transitions and continuity in critical positions, to reduce operational risks. - Appoints backup candidates for key positions within each department, to mitigate the impact of personnel changes. - Identifies potential successors and key talents, to support future company growth and development. - Provides understanding of potential successors' competency gaps, then provides them with appropriate development planning.
Multicultural education	- Cross-Cultural Course Exchange (India) - Colleagues Share India Travel Experiences	In collaboration with Tata Electronics to build a new fab, PSMC invited Indian teachers based in Taiwan and employees with experience in India to share insights on Indian culture. This helped employees gain an initial understanding of Indian customs and traditions, thereby reducing the impacts of cross-cultural differences. Related seminars and participant numbers are as follows: "Understanding India" (in-person + livestream) "Discovering India's Diverse Religions, Cultures, and Ethnicities Through Movies" (in-person + livestream) "Ganges, Saris, and Marigolds: Incredible India" Travel Sharing Session – Total of 612 participants.
Employee transition	Internal Job Rotation and On-the-Job Training (OJT)	To enhance cross-functional collaboration, strengthen departmental cooperation, and promote innovation, PSMC has established an internal job rotation system. During internal rotations, employees are required to complete designated standard courses sequentially; the new departments also provide on-the-job training to assist employees in adapting to their new roles. PSMC's internal rotation rate is 46.9%.
Digital transition	AI Training and Re-Skilling Courses	- In response to the AI digital trend, PSMC invited professors from National Yang Ming Chiao Tung University to deliver courses on deep learning and machine learning. By understanding fundamental theories and engaging in case-based practice, employees strengthened their AI-related knowledge. - Two sessions of the "AI Deep Learning and Machine Learning" course were held, each lasting 36 hours, with a total of 68 participants.

Employee's On-the-job training and License Subsidy



Flexible and Diverse Learning Formats

In order to improve learning willingness and effectiveness, PSMC integrates internal and external training resources of the company and plans a dedicated training website, so that colleagues can master the most up-to-date professional knowledge, regulatory changes and the company's latest decisions through more flexible learning models, multiple channels and other information, etc.. PSMC also uses the online learning platform to grasp the learning status of colleagues in real time and provide timely guidance and support.



In 2024, the total training hours of employees reached 345,012 hours, and the average training hours per employee was approximately 41.14 hours; annual training-related expenses reached NT\$6,252,942, and the average employee education and training expenditure was approximately NT\$745. Through the company's internally developed in-factory communication software Pteam, various online training courses are continuously arranged. The achievement rate of the training plan for 2024 (including the E-Learning system) was 100%. A total of 36 physical standard courses were offered, with an average rate of the satisfaction survey was 94.7%.

Education and Training Hours Statistics

Category	2021	2022	2023	2024
Total training hours (hours)	225,420	315,974	277,928	345,012
Training hours per capita (hours)	30.27	38.77	34.04	41.14

Average training hours by job level

Category		Total training hours (hours)	Number of employees	Training hours per capita (hours)
Non-managerial employees	Female	81,956	2,798	29.29
	Male	224,742	4,246	52.93
	Total	306,698	7,044	43.54
Junior Management Level	Female	7,630	301	25.35
	Male	16,405	454	36.13
	Total	24,035	755	31.83
Middle Management Level	Female	1,701	74	22.99
	Male	12,428	491	25.31
	Total	14,129	565	25.01
Senior Management Level	Female	5	1	5.00
	Male	145	22	6.59
	Total	150	23	6.52

Note 1: Management positions refer to supervisors at or above the team or section level (excluding technical/project section chiefs and technical/project assistant managers).

Note 2: The training course hours statistics include the e-Learning system.

Note 3: At the non-management level, the total number of hours of education and training is higher because the majority of engineers are male.

4.2.2 Performance Evaluation System

PSMC has designed a fair and equal performance appraisal system. Supervisors at all levels will review, track and review goals and key performance indicator achievement status and matters to be improved based on the daily work performance of employees affiliated to their departments. In accordance with the Company's policies, all employees are required to undergo performance evaluations after completing three months of employment. In 2024, 100% of employees, regardless of position level or gender, participated in biannual performance appraisals. The performance appraisal results will be used as an important basis for employee promotions and rewards. For employees with unsatisfactory performance, supervisors will provide coaching and necessary resources according to the situation. If necessary, their job content will be adjusted based on the employee's situation to ensure that their performance output can be improved.

Type of Performance Evaluation

Type of Assessment	Applicable Target	Frequency	Execution Method
Management by objectives	All Employees	Once every half-year	At the end of each year, the Company establishes its operational policies and goals for the following year. After each unit has set departmental goals based on the Company's goals, the departmental employees and their direct supervisors jointly set quantitative and specific performance targets as individual performance goals at the beginning of the year, which are then tracked every six months and evaluated by the direct supervisors at the end of the year according to the status of individual performance.
Team-based performance appraisal	All Employees	Once every quarter	In addition to the employee's personal evaluation, the achievement of team goals will also affect the overall employee evaluation results, so as to motivate employees to collaborate with each other and maximize individual and team performance.
Multidimensional performance appraisal	Unit Supervisors	Once a year	In addition to the employee's personal evaluation, the achievement of team goals and the contribution to the Company's overall goals and other factors, the annual assessment is based on the results of questionnaires for the Employee upward appraisal.
Conversation & Feedback & Recognition (CFR)	All Employees	Once every quarter/ immediately	The president and employees have quarterly conversations, and department supervisors communicate with employees about their work performance in order to review and track the progress of employee performance and work status. We also provide appropriate feedback to employees on work-related problems, and give praise and encouragement to employees when they perform well or make significant progress. If the employee is not in good working condition, the supervisor will also provide necessary counseling and support immediately.

4.2.3 Labor Relations and Communication

Open Communication Channels

PSMC to protect the individual and group negotiation rights of employees, actively develop various one-way advocacy and two-way communication channels, also hold through quarterly statistics labor-management meetings to discuss issues related to the rights and interests of our employees, such as working hours, leave and benefits, or any opinions and concerns related to the company's operations, and the results of the agreement apply to all workers.

In order to respond to employees' needs and direct them back to the right track in a timely manner, an Employee Relations Department has been established under the Human Resources Division to provide employees with the most appropriate and prompt assistance and consultation/referral services in the shortest possible time. Five free-of-charge psychological consultations are offered each year, and information on mental health, emotional stress management, communication skills and other related information from time to time are also provided to strengthen mental health promotion.

Since the establishment of PSMC, labor relations have been harmonious, thus there has been no labor union established, also no collective agreement signed with employees. However, the company is committed to safeguarding the rights of workers and ensuring that employees can express their opinions and receive fair treatment by holding regular labor-management meetings, setting up complaint channels and employee care mechanisms. In 2024, there were also no incidents of labor disputes or violations of freedom of association or group consultation that resulted in penalties imposed by the competent authorities. In addition, in the event that employees are dismissed or discharged as a result of Article 11, 13, or 16 of the Labor Standards Act, the minimum notice period required by law is applied at each plant and the Company will follow up on these matters.

Online Communication Platform

Through the "Partner's Word" communication platform, employees are provided with a channel to respond to comment on the company's system, management practices or specific issues, and are referred to the relevant responsible departments to respond to their questions and suggestions. 603 responses were received from employees through the "Feedback Corner" communication platform in 2024, with a 100% response rate.

Workplace Abuse and Workplace Sexual Harassment Complaint Mechanism

In addition to the various complaints and reports, which are handled according to the procedures established by each responsible unit, the Employee Relations Department has set up an "Employee Care Site" to create a friendly workplace with dignity and gender equality, and receives complaints in accordance with the "Regulations on the Prevention of Unlawful Acts of Assault in the Performance of Duties", "Procedures for Notification of Unlawful Acts of Assault in the Performance of Duties", and "Measures to Prevent, Report a Complaint and Punish Sexual Harassment in the Workplace", and convenes an investigation committee to make decisions and impose sanctions to protect the rights and interests of employees.

Labor & Management Meetings

In accordance with the provisions of the Labor Standards Act, each plant completely holds regular quarterly labor-management meetings in 2024 and additionally holds the extraordinary meeting by the requirement of issues to coordinate various labor-management issues and promote harmonious labor-management relations and the results of the agreement apply to all workers

Employee Feedback Survey

Employee opinion surveys are conducted through questionnaires and interview plans of high/middle supervisors are executed, the survey results and suggestions will be used as an important reference for the Company to develop/revise its operating guidelines and formulate various policies.

Employee Engagement Survey

In order to understand employees' perspectives and suggestions on the long-term development of PSMC, we periodically conducted surveys through questionnaires every year to understand employees' concerns regarding the operational aspects and potential problems of the organization, and to plan refined countermeasures to address key aspects and issues, with the goal of building a more resilient operational model that supports the well-being of our employees.

Employee Engagement Survey design considerations for questionnaire aspects

Aspect	Question
Work Vigor	I really enjoy my daily work routine and feel happy most of the time while working.
Work Dedication	I am willing to take on additional duties and responsibilities at work and can accept an appropriate level of work-related pressure.
Work Focus	The Company's work environment enables me to maximize my work efficiency, and my job has a clear purpose.
Intrinsic Satisfaction	Relative to what I put into the Company, I am satisfied with my job, and my compensation is reasonable.
Extrinsic Satisfaction	The Company provides sufficient training and learning resources.

Survey Results

Number of respondents: 6,391	Questionnaire survey coverage rate: 100%	Questionnaire response rate: 76.2%
--	--	--

The average score of the 2024 Employee Work Engagement Survey was 3.9 points (on a 5-point scale). The percentage of actively engaged employees (i.e. those obtained an overall score of 4 or above) was 63%, and the average score for Recommend Joining PSMC was 6.7 points (on a 10-point scale). The results showed that males' average score was slightly higher than that of females, and managerial personnel's average score was higher than that of non-managerial personnel, suggesting that the higher an employee's position rank is, the higher his or her overall engagement score will be.

Employee Engagement Survey results (by aspect)

Aspect	Average score for female employees	Average score for male employees
Work Vigor	3.5	3.5
Work Dedication	3.5	3.7
Work Focus	3.6	3.6
Intrinsic Satisfaction	3.6	3.7
Extrinsic Satisfaction	3.7	3.9

Employee Engagement Survey results (by category)

Category		2022	2023	2024
Gender	Female	4.5	3.9	3.8
	Male	4.5	3.9	3.9
Managerial personnel	General employees (non-managerial)	4.5	3.8	3.8
	Operative level managerial personnel	4.6	4.0	4.0
	Middle level managerial personnel	4.7	4.0	4.2
	Higher level managerial personnel	4.7	4.1	4.3
Tenure	Less than 2 years	4.5	3.9	3.9
	3-5 years	4.4	4.2	3.7
	6-10 years	4.5	3.8	3.8
	Over 10 years	4.5	3.9	3.9
Total		4.5	4.0	3.9

Employee Engagement Survey results and targets

Fiscal Year	Questionnaire survey coverage rate (%)	Engagement Index (%)	Target (%)
2022	100	86	85
2023	100	70	80
2024	100	80	63

Note1: In the Employee Engagement Survey, management levels are classified as follows: Entry-Level Management: Assistant Supervisors, Supervisors, Production Line Supervisors, and Section-Level Managers (excluding technical/project roles); Mid-Level Management: Department-Level and Division-Level Managers; Senior Management: Assistant Vice Presidents and above.

Note2: The implementation target depends on the survey aspects and content of the questionnaire design of the current year. Due to the continuous optimization of the questionnaire in recent years, the target differences between previous years have occurred.

Post-Survey Improvement Measures

- The survey analysis results indicate that scores were relatively low in the areas of Employee Vitality, Employee Focus, and Salary Satisfaction. In response, the Company has initiated a series of talent retention programs and launched projects to boost employee morale, aiming to improve the workplace atmosphere and enhance work efficiency.
- For issues with weak results, improvement action plans will be prepared by the corresponding departments and reviewed on a quarterly basis in order to manage the direction of improvement of the company's operational development.

4.3 Human Rights

4.3.1 Human Rights Policy Communication

Based on a human-oriented business philosophy, PSMC strictly follows local laws and regulations in employee recruitment, and will never discriminate anyone on account of the individual's race, gender, age, marital status, religious belief, political affiliation, disability status, or other situations, to comply with the international trend that aims to achieve a balance among environmental, social and corporate governance development, and to protect the basic human rights of employees, suppliers, business partners and other stakeholders. PSMC, by following the RBA Code of Conduct, UN Guiding Principles on Business and Human Rights, Declaration of Fundamental Principles and Rights at Work, UN Universal Declaration of Human Rights, and other regulations, formulated its Labor and Human Rights Policy in 2022, which has been signed by the President and announced on the Company's official website. The Company also implements the human rights policy through education/training, human rights due diligence, and RBA Self-Assessment Questionnaire (SAQ), and requires suppliers and business partners to cooperate in completing questionnaire surveys on their compliance with the RBA Code of Conduct, and on-site audits conducted on an irregular basis to build a friendly workplace of diversity and inclusion.

Labor and Human Rights Policy: <https://esg.powerchip.com/en-global/staticpage/respect-human-right>



PSMC is committed to strictly prohibiting the employment of child labor, over the age of 15 but under the age of 16, and will not arrange workers under the age of 18 (youth workers) to do work that may endanger health, safety or ethics, including overtime or night work. The employment of foreign workers is in compliance with the age requirement for workers of the exporting countries. The Company does not recruit workers by means of rape, coercion, detention, debt collection, human trafficking or other illegal methods, and prohibits forced labor or related coercive measures, including but not limited to physical punishment, intimidation, verbal violence, withholding of workers' finances, identity documents or other unlawful forced labor to ensure that the basic human rights of foreign workers are not unreasonably restricted. If there is a need to extend the working hours, the employee must apply for the extension after agreeing to do so, and be paid overtime according to the number of overtime hours shown in the system. If overtime hours exceed the standard limit, the system will display a warning message, and the pre-overtime application cannot be submitted. In 2024, there were no forced or compulsory labor incidents, nor were there any incidents involving the use of child labor or infringement of the rights of indigenous peoples. Therefore, no penalties were imposed by the competent authorities for this reason.

In order to implement the human rights policy, PSMC provides human rights-related training to all employees and security personnel, PSMC conducts five human rights protection related training courses, including Corporate Social Responsibility and Business Ethics and Labor Laws and Regulations, for a total of 11.5 hours, with a total of 17,695 participants and a 100% employee training attendance rate, and achieved a 100% of completion training rate.

PSMC has also established the Workplace Sexual Harassment Prevention Measures, Complaint, and Disciplinary Procedures, and requires employees to complete a mandatory online training course on workplace discrimination and harassment prevention. This initiative aims to protect employees from discrimination and harassment in the workplace. The total duration of the course is three hours, with 8,387 participants.

4.3.2 Human Rights Issue Management

PSMC adopted a standardized risk assessment template (Self-Assessment Questionnaire, SAQ), designed by RBA, every year identify its human-rights-related risks by PSMC (with the scope covering PSMC's own operations, the value chain, and new business partnerships such as merger, acquisition, joint venture, etc.), in two year implement external verification. In addition to focusing on discrimination, harassment, forced labor, child labor, human trafficking, freedom of association, and other human rights-related issues, emphasis is also placed on the rights of employees, children, women, indigenous people, immigrants, and other related groups. The P1/2 fabs successfully passed the RBA Validated Audit Process (VAP) verification in late 2021 and obtained the Gold-level Certificate in Q1 of 2022. The P3, 8A, and 8B fabs completed the RBA SAQ in 2022 and underwent the RBA VAP verification in November 2023. The P1/2 and 8B fabs were awarded the Gold-level Certificate, while fabs P3 and 8A received the Platinum-level Certificate. In the same year, the P3, 8A, and 8B fabs conducted an independent RBA self-assessment. In early October 2024, PSMC completed an internal RBA audit across all facilities and held the RBA Corporate Social Responsibility Management Review Meeting at the end of the month. No business ethics violations were identified in the audit. A third-party verification for all facilities is scheduled to be conducted by the end of 2025.

In addition, through a systematic investigation process conducted regularly on potential human rights risk issues, PSMC can promptly identify major human rights-related risks for the year and formulate mitigation, remediation measures, and response strategies accordingly. Based on the results of the 2024 Human Rights Risk Assessment, mitigation and remedial measures will be established for specific target audiences to reduce the likelihood of risk occurrence in order to avoid the occurrence of human rights abuse.

Target Item	All Employees	Foreign, female employees	Suppliers	Contractors
Risk Issues	- Working Hours - Sexual Harassment	- Forced Labor - Anti-Discrimination	- Forced Labor - Working Hours - Conflict Minerals	- Forced labor - Work hours
Mitigation Measures	- Recruit sufficient staff to fill the labor gap and avoid overtime work due to lack of manpower - Ensure that human rights-related training and advocacy are carried out in the plant, and establish a sexual harassment complaint and handling process.	Review and evaluation through various complaint channels/internal audits/internal management procedures such as "prohibition of forced labor" and "prohibition of discrimination and harassment"	- Conduct RBA compliance survey - On-site field audits	- Questionnaire surveys on compliance with the RBA Code of Conduct - On-site audits
Remedial Measures	- Control the number of overtime hours through the work hour management system and provide alerts to remind supervisors of the situation when the maximum number of legal hours is reached - Follow up on complaint cases to ensure effective implementation of counseling measures / disciplinary actions as a basis for future improvement and adjustment	Units where forced labor or discrimination has occurred will be placed on the priority list for future audits.	- Conduct RBA compliance survey - On-site field audits	- Questionnaire surveys on compliance with the RBA Code of Conduct - On-site audits

4.4 Occupational Safety and Health

4.4.1 Occupational Safety and Health Management

Environmental Safety and Health Management System

PSMC has established an in-house ESH Management System in accordance with the ISO 14001 Environmental Management System and the ISO 45001 Occupational Safety and Health Management System, covering all of the Group's plants, which includes all 8,387 employees and 18,975 non-employee workers, with a coverage rate of 100% (30.65% and 69.35% of total workers for employees and non-employee workers respectively).

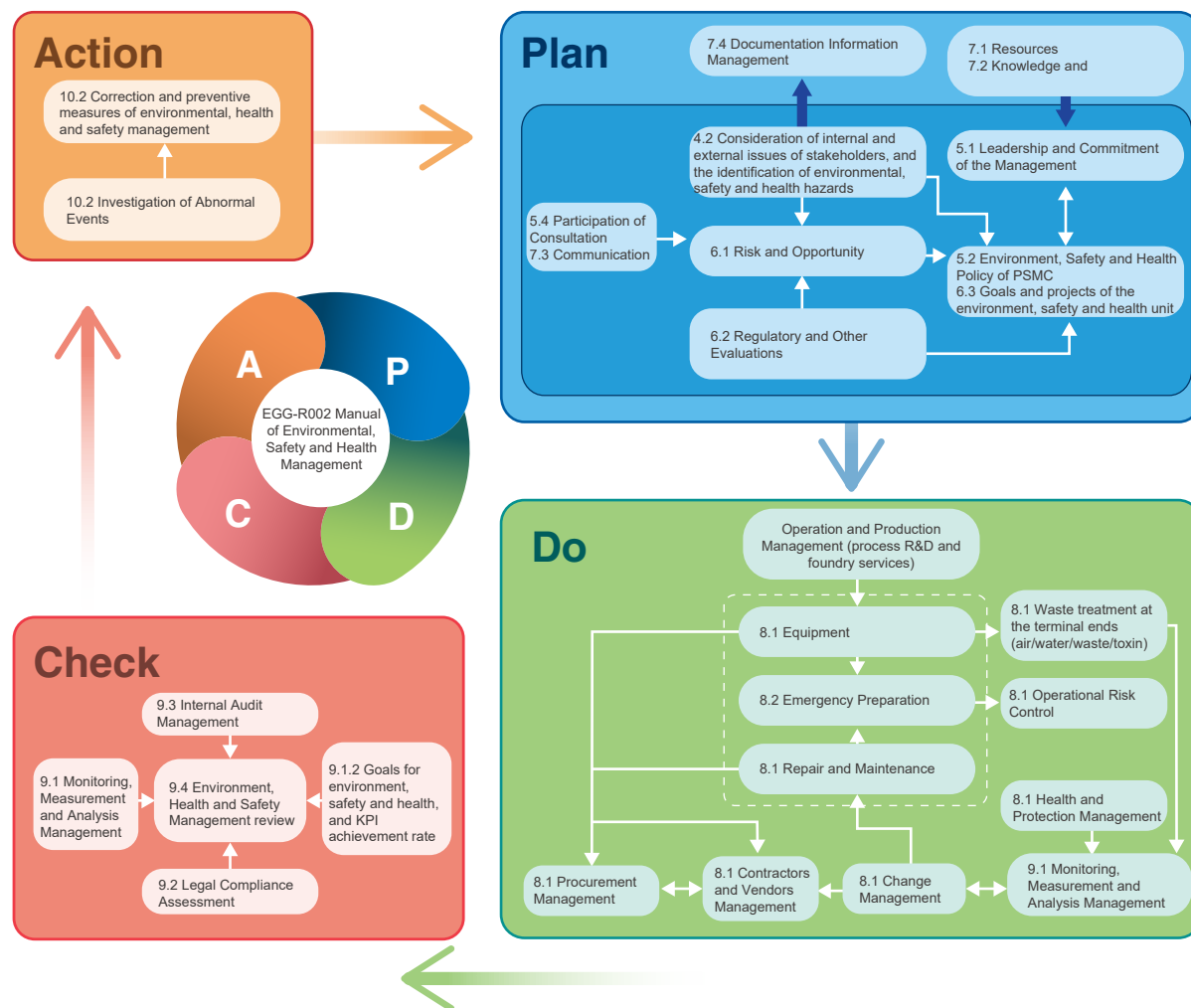
PSMC takes the Environmental, Safety, and Health Policy signed by the President as the highest guiding principle for environmental, safety, and health (ESH) management, and has established a Safety, Health, and Environmental Protection Committee, led by the President, as the highest decision-making body for the ESH management system. The environmental safety and health audit team composed of internal environmental, safety and health auditors in each factory conducts internal environmental, safety and health audits every six months with reference to the "Environmental, Safety and Health Internal Audit Operation Management Measures" and "Internal Environmental, Safety and Health Auditor Recognition and Registration Rules" and other procedures. activities, and at the same time entrust an external Lloyd's Register Verification Company (LRQA) to verify the overall environmental, safety and health management system operation and execution records in compliance with the provisions of the validity, and convene the Safety, Health and Environmental Protection Committee every quarter to review, amend the management plan, and evaluate prevention / Reduce health and track the achievement of safety and health hazard risks for the purpose of continuous improvement. In August 2023, the three-year re-verification of the environmental and occupational safety and health management system has been completed to continue the effectiveness of the management system.

Environmental Health and Safety Policy:

<https://esg.powerchip.com/en-global/staticpage/safe-working-environment>



Process Flow of ESH Management System



Safety and Health Organization

The Risk Management Department and the Risk and Environmental Safety Department of each plant at PSMC are in charge of the overall environmental safety and health related business, and have established the head office and the Safety, Health and Environmental Protection Committee at each plant respectively, which are responsible for planning and promoting the auditing, verification, risk assessment and education and training related to environmental safety and health. On a quarterly regular safety, health and environmental management review meetings are held, and labor representatives are elected to participate in the meetings in accordance with the law, so that employees can fully understand the company's safety, environmental safety and health operation model and provide a formal channel for discussion and debate.



Plant	HQ	P1/2	P3	P5	8A	8B
Number of labor representatives (persons)	26	11	7	8	11	9
Total number of committee members (persons)	68	28	20	23	27	20
Percentage (%)	38.24	39.29	35.00	34.78	40.74	45.00

Note: Percentage = (number of labor representatives / total number of committee members) * 100%

Employee Consultation and Communication

Information related to the environmental safety and health management system, such as environmental safety and health policies, environmental safety and health objectives, laws and regulations, stakeholder requirements, and changes to the environmental safety and health management system, will be communicated to the responsible units through internal safety, health and environmental committee meetings, environmental safety and health representative consultation and communication meetings, and safety, health and environmental newsletters in accordance with the communication procedures of the "Environmental Safety and Health Management Consultation and Communication Regulations". Employees can also raise concerns, provide feedback, and submit complaints regarding environmental, safety and health issues through the internal Employees' Voice communication channel. If the complaints and communications received by the Human Resources Department are confirmed to be directly related to the company's operations, they will be forwarded to the responsible departments for risk and opportunity assessment through the intranet, the Environmental Safety and Health Law Management System, and the DCMS Document Management System to ensure that questions, comments, or complaints raised by employees are appropriately responded to and handled.

In 2024, the Company held 24 Occupational Safety and Health Committee meetings, 40 contractor consultation and communication meetings, and 17 EHS representative consultation and communication meetings. Related workers and stakeholders were invited to participate in the meetings and join the consultations regarding the Environmental & Occupational Safety and Health Management System's establishment.

4.4.2 Occupational Safety and Health Risk Assessment

Environmental Safety Risk Identification and Improvement Measures

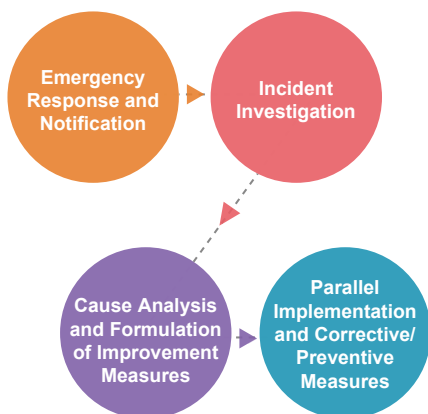
In order to control the possible environmental, safety and health risks existing in business activities or product production processes, through process/activity hazard identification and risk assessment, PSMC has identified related environmental safety and health risks and formulated countermeasures in advance to effectively reduce the risks caused by a variety of operations or activities. Individual departments must examine the contents of their operations or information about the activities they are engaged in based on the scope and procedures of related processes/activities, including activities of contractors and visitors in plant areas, and EHS issues arising in the process when products are delivered to customers. Next, the departments must take inventory of the raw materials, parts, cleaning supplies, and personal protective equipment they use, summarize the information about the wastewater/exhaust/waste, etc. they produce, and fill out the Department Process/Activity Step List, which shall serve as a basis for subsequent EHS risk assessments.

In the implementation of environmental safety and health risk assessment, various assessment techniques such as "machine safety inspection, change management, hazardous workplace process safety assessment, SEMI standard, Hazard and Operability Analysis (HAZOP), Failure Mode and Effects Analysis (FMEA), Fault Tree Analysis (FTA) and SWOT analysis" can be applied depending on the needs of the process activity. In addition, environmental safety and health audit activities are conducted every six months, where we conduct internal/external audits on the effectiveness of our environmental safety and health management system, covering all employees and non-employee workers such as contractors with qualified work permits.

In accordance with the Operational Guidelines for EHS Risk Assessment applicable for in-plant areas, risk identification shall focus on the general chemical, physical, biological and ergonomic hazards. After summarizing worker consultation results, each department must identify its environment, safety and health hazards and risks based on its input and output processes, and confirm potential EHS hazard/risk items. Through the filling-out of the Department Risk Assessment Form, the inherent risk level and control risk level shall be assessed. Three indicators must be adopted for calculation when conducting risk assessment, which are a hazard's probability of occurrence (P), severity (S), and risk weight (W). The most severe level risk (when all soft-service/hard-service protections have failed) must be considered when determining the level of the inherent risk. When determining or changing existing risk control measures, the order of consideration shall be elimination, substitution, engineering controls, redeployment, labeling/warning/administrative controls, and appropriate personal protective equipment. Next, determine the level of control risk after implementing the existing control measures such as operation controls, safety devices/protective measures, emergency response measures, etc., based on which corresponding improvement measures and goals shall be formulated. The implementation progress of the improvement measures shall be reviewed in EHS management & review meetings.

Therefore, for employees who are at risk of being exposed to significant environmental safety and health risks, they have completed relevant education and training before taking up their duties to achieve hazard awareness and the ability to avoid immediate risks. If there is a risk of immediate danger during the performance of their duties, they can stop their work and evacuate to a safe place without endangering the safety of other operators, and will not be subjected to any adverse punishment, and should immediately report to their immediate supervisor. In addition to providing a safe and healthy working environment, PSMC regularly identifies internal and external violations in accordance with the "Regulations for the Prevention of Unlawful Acts Against the Performance of Duties" to put an end to the occurrence of inappropriate incidents such as workplace violence or bullying..

In the event of environmental pollution, occupational disaster/injury and health impact incidents, the incident investigator will analyze the cause of the incident and propose corrective and improvement measures in accordance with the standard operating procedures of the "Abnormal Incident Investigation Procedures" in conjunction with relevant personnel, and make improvements to address potential or existing non-conformities in order to reduce the impact of environmental safety and health incidents. In addition, parallel investigations are conducted across all fabs to ensure the formulation and implementation of corrective and preventive measures.



Formulating Environmental Safety and Health Strategy Process and Target

PSMC through the SWOT analysis to the specific risks and opportunities identified. The management objectives and implementation plans are discussed through the environmental safety and health representatives at each plant, and then implemented after the results are reported upward by the management audit committee.

Environmental Safety and Health Policy / Target Strategy

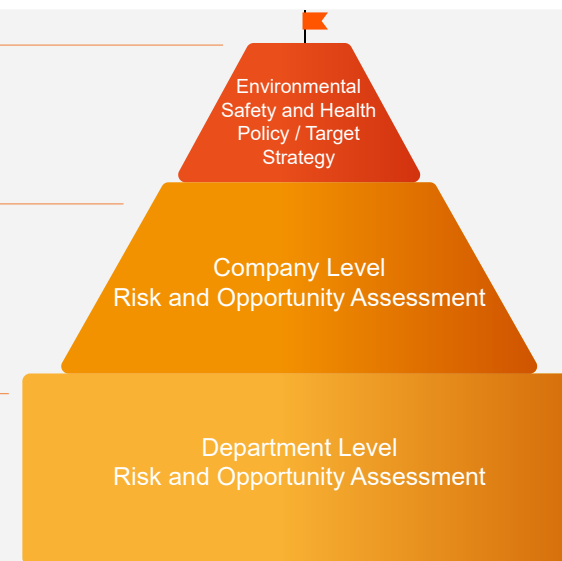
- Innovate for sustainable development and fulfill corporate social responsibility
- Enhance the effectiveness of safety and health management
- Strengthen environmental safety and emergency response capabilities
- Reduce operational losses caused by internal and external abnormal events

Company Level Risk and Opportunity Assessment

Through the sustainable major issues identification process, each plant identifies major issues related to environmental safety and health to find "risks" to set environmental safety and health goals and "opportunities" to improve the performance of the management system, and presents the analysis results at the management review meeting, as well as to discuss environmental safety and health policies and target.

Department Level Risk and Opportunity Assessment

Each responsible unit reviews and updates the manufacturing process activity step-by-step checklist (including process and SWOT analysis) and risk assessment form, and reviews departmental risks and opportunities with assessment items such as "occurrence of major abnormal events, unacceptable risks, stakeholder concerns, workplace injuries, major nonconformities deficiencies, and regulatory impacts".



Contractor Management Mechanism

PSMC has established the Contractor Safety, Health, and Environmental Protection Regulations to ensure that contractors working within the fab premises implement proper safety management mechanisms, and thus achieve the goal of zero workplace injuries. 2024 all fabs follow the "One Permit, One Card" principle to control contractor activities within the premises. Contractors must pass required training and obtain a special work permit and access control card, along with a work permit approved by PSMC's responsible department before entering the designated work areas. Each business department receives regular monthly contractor management reports, which are analyzed to identify areas for enhanced management and to supervise contractors with frequent violations, encouraging self-management. Furthermore, contractor performance is reviewed in the quarterly Safety, Health, and Environmental Protection Committee meetings. To further optimize communication efficiency, PSMC plans to launch a contractor communication platform in 2025, which will include training materials accessible to contractors at any time.

Contractor evaluation and auditing are also key aspects of the management system. Occupational safety and health-related requirements and regulations are incorporated into procurement contracts to ensure that suppliers, subcontractors, and contractors are fully aware of PSMC's operational guidelines and ESH legal requirements. Each fab's risk and environmental safety department, in cooperation with the procurement department, conducts regular contractor evaluations and has established a standardized contractor evaluation system. Contractors are required to respond with corrective actions and undergo re-evaluation if deficiencies are found. The evaluation results also serve as a reference for future contract renewals. In 2024, PSMC selected 190 contractors for evaluation, all of whom passed the assessment and were deemed eligible to continue the partnership.

4.4.3 Workplace Safety Protection

In terms of implementation, follow "Environmental Safety and Health Management Manual", supervisors at all levels provide adequate resources according to their duties, and encourage departmental colleagues to actively participate in proposals for improvement and training discussions to enhance the vigilance of business execution and the safety of the working environment, and to strengthen the overall environmental safety and health quality. In terms of supervision and management, we regularly review the level of compliance of the operation of the environmental safety and health management system in accordance with the Environmental Safety and Health Management Review Regulations and Abnormal Incident Investigation Practice on a quarterly basis, evaluate the progress and results of achieving the goals of preventing and reducing risks of health and safety hazards and deliver the audit results to the management and stakeholders as a point of reference for the improvement and adjustment of the management system. In addition, PSMC has established abnormal event investigation procedures, which include investigation of occupational disasters (including traffic accidents) and abnormal events, hazard identification, safety inspections, and on-site audits to confirm the causes of accidents and propose measures to prevent accidents and correct and prevent their recurrence.

Operating Environment Monitoring and Protection

For hazards and risks that have occurred or have the potential to occur in the working environment, the Operating Environment Monitoring Plan must be used to determine monitoring locations and sampling plans. The results of the operating environment monitoring cannot exceed the requirements of the Allowable Exposure Standards for Labor Workplaces as announced by the Ministry of Labor, and if they do, improvement measures must be immediately formulated in accordance with the control measures of the Environmental Safety and Health Risk Assessment Practice, and the effectiveness will be evaluated again and records kept.

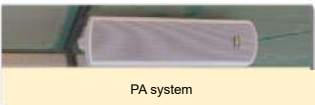
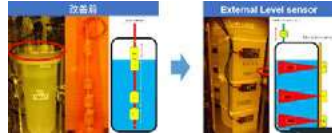
In order to grasp the actual working environment of employees and evaluate the exposure of hazardous factors, PSMC performs semi-annual monitoring of items required by law, including chemical hazards (including organic solvents and specific chemical substances), and physical hazards, in accordance with the Regulations on Implementation of Labor Environment Monitoring. It also commissions an operating environment monitoring organization approved by the Ministry of Labor to perform various environmental testing tasks, review the data of operating environment monitoring results, quantify the variation characteristics of the data distribution, and provide feedback to the next sampling strategy planning for reference, and conduct follow up operating environment monitoring plans and notification of monitoring results.

The operating environment monitoring results and improvement measures are all described in the Safety, Health and Environment Committee meetings to labor representatives and meeting participants for related employees to be well aware of relevant issues and build up good safety and health protection concepts. The 2024 monitoring results all met regulatory requirements.

Monitoring Items	Monitoring Details	Monitoring pass rates for first/ second half of 2024 (%)
Chemical Factors	Organic solvents and specific chemical substances (including regional measurements and personal sampling)	100
Physical Factors	Illumination and noise measurement (including area measurement and individual sampling)	

In terms of reducing employee workplace risks, the responsible unit evaluates and improves the potential risks of human-induced hazards and occupational health in each unit based on the operating environment monitoring results and environmental safety and health risk assessment practices. 60 applications for potential risks evaluations and improvement proposals were approved by the Risk Safety and Environment Department of each plant in 2024, exceeding the annual target of more than 55 proposed improvements.



Item	Operating Environment Optimization Project				
	Establishing a real-time transformer thermal imaging recognition system	Electronic information board and voice broadcast system	High-noise and gas leakage monitoring technology guidance project	Establishing an automated SMS notification system for chemical leak detectors	TMAH buffer tank level sensor modification
Project Description	To continuously improve and reduce occupational safety and health risks in the fab, a real-time transformer thermal imaging recognition system has been implemented. By integrating smart monitoring software, hardware technologies, and image recognition functions, the system enables real-time monitoring of equipment operations and collects operational data (such as temperature). Through the SPC abnormal warning mechanism, the system instantly detects potential anomalies in equipment operations, and provides early warnings to engineers, notifying them to perform immediate troubleshooting.	Occupational injuries due to traffic accidents have long been the primary cause of lost work hours. To reduce the occurrence of traffic accidents and enhance employees' vehicle driving awareness, electronic information boards and voice broadcast speaker systems have been installed. Dedicated control software has also been developed to automatically play scheduled announcements and broadcast traffic safety reminders through both text displays and voice alerts.	In collaboration with the Science Park Administration, PSMC participated in the Thermal Imaging Acoustic Instrument Monitoring Technology Promotion as a pilot site, monitoring four designated areas within the fab. High-noise monitoring and gas leakage detection were conducted to identify pump areas with excessive noise levels and to detect potential gas leaks.	PSMC has developed an intelligent chemical leak detection and notification system, integrating high-sensitivity sensing technology with a real-time automated SMS communication function. By leveraging the smart IoT technology platform within the fab, alarm notifications are immediately sent to both machine managers' mobile devices and on-duty personnel's phones, significantly reducing emergency response time, minimizing property damage, and mitigating environmental safety risks. This initiative aims to enable early leak detection and instant active alerts.	In 8A fab's photolithography PX machines, the original TMAH buffer tank level sensor was installed inside the tank by the manufacturer. If the sensor malfunctioned, it required opening the tank lid for replacement, which posed chemical splash and exposure risks. To address this issue, collaboration with the Maintenance Department developed an external level sensor, which was later installed independently by the 8A fab's Photolithography Engineering Department. With this improvement, sensor repairs and replacements can now be performed without opening the tank lid.
Project Results	1. Installation of IP cameras to capture temperature data of the power system, with image recognition technology uploading the readings to the big data (IRCA) platform for monitoring and control. 2. Upgrading from single-point monitoring to comprehensive real-time monitoring. IR detection frequency upgrade: Previously conducted once per quarter, now upgraded to real-time monitoring. 	According to an employee survey, 95% of employees expressed that displaying traffic safety messages and playing voice reminders during commuting hours is helpful in raising awareness, reducing accident rates, and fostering a culture of traffic safety.  Electronic information board at storage and transport dock entrance/exit  PA system	1. High-noise monitoring: According to the operating environment monitoring results for the first half of 2024, 14 locations within the monitored areas were identified as high-noise work zones (>85dBA) and were prioritized for improvement. Follow-up monitoring in the second half of 2024 showed a decreasing noise trend (94dBA → 87dBA). Future monitoring will continue regularly to ensure no upward trend in noise levels. 2. Gas leakage monitoring: No abnormalities were detected in special gas pipelines or natural gas systems during the monitoring process. However, 18 locations were found to have inert gas (N₂) leaks from pipelines and valve components. The monitoring results have been provided to responsible departments for further evaluation and potential improvements.	By integrating chemical leak detectors with the fab's smart IoT platform, this innovative solution not only enhances safety within the fab but also optimizes response efficiency, demonstrating outstanding technological performance in risk management.	1. Successfully installed across all ten PX machines, with an investment of NT\$980,000. 2. If the external level sensor malfunctions, it can be replaced or repaired from outside the tank, eliminating the need for direct TMAH contact. 

4.4.4 Occupational Safety and Health Education and Training

In accordance with the Environmental Safety and Health Education and Training Regulations and the Employee Education and Training Procedures, PSMC has designed occupational safety and health general education courses and departmental/plant-organized courses to train all employees to understand the operation model of the PSMC's environmental safety and health system and to possess the necessary knowledge and skills for workplace safety and disaster prevention. To effectively enhance the professionalism of our personnel and their ability.

General Occupational Safety Training

PSMC's environmental safety and health education and training is divided into three major categories, including "environmental safety and health management system, legal license, and operational control". Each department conducts the suitability assessment of personnel who may affect the performance of environmental safety and health and compliance obligations in order to plan the required education and training categories, and formulates the "departmental environmental safety and health education and training plan". In 2024, a total of 168 training plans were completed and incorporated into departmental goals, with quarterly progress checks to ensure a 100% achievement rate. Additionally, an environmental safety and health education and training system was established to help departments update and track safety and health training results. For instance, in waste reduction-related training, a total of 97 training sessions were conducted, with 4,152 participants. The training results were recorded in the electronic system, improving overall training management and audit efficiency.

In 2024, a total of 254 education and training sessions were conducted, with 35,849 participants. This included 28 sessions for new employee training, attended by 903 participants. In particular, hazard identification/risk assessment and control courses were introduced to enhance the ability of risk assessment personnel in different departments to identify hazards during risk assessment operations. A total of 455 employees completed the training. As of the end of 2024, a total of 1,025 employees held a variety of ESH-related positions, all possessing the certifications required by regulations. In addition, PSMC provided workplace hazard notification, as well as safety and health training for new contractor personnel, completing training for 3,538 individuals.

Key Environmental, Safety and Health Training Courses and Their Training Results

Target Audience	Topics	Training Results
Plant engineering, equipment engineering and manufacturing department staff	- Points of Note for High-Risk Operations Control - Developing Incident Investigation Skills	A total of 11,241 people completed the training sessions
Departmental staff in administrative categories	General Occupational Safety and Health Training	A total of 2,144 people completed the training sessions
Foreign personnel	Fall Prevention and Rolling Object Hazard Mitigation	A total of 409 people completed the training sessions
All Personnel	- Ecological Conservation Promotion (Nature Exploration Team Let's Go) Lecture - Impact of Climate Change on Water Resources and Adaptation Strategies lecture	A total of 555 participants

Emergency Response and Fire Safety/First Aid Training and Drills

The fire safety training and practical drill programs held in 2024 were divided into ERT (Emergency Response Team) training and ERT drill. Training sessions related to how to wear personal protective equipment and how to use a variety of response equipment are organized every year to help personnel skillfully, correctly and effectively use response equipment in emergencies and avoid ERT personnel injuries during accident rescue. PSMC conducted 125 ERT training sessions in 2024, with 3,040 qualified ERT rescue team members being trained, and 122 members completing the ERT first aid team training (online CPR and AED courses). In addition, PSMC has formulated the Plant Area Disaster Emergency Response Regulations to ensure that in the event of emergencies, the personal injuries, property losses and environmental damage caused by accidents and the degree of impact on the environment can be mitigated through correct, effective, and systematic response procedures. 295 ERT drills were conducted in 2024, including cleanroom evacuation drills, department's self-directed response drills, interdepartmental joint drills, etc.



Contractor Education Training

PSMC continued to implement operations related to contractor work permits in 2024 by providing contractors with course materials focusing on safety, health, and environmental regulations, with a total of 4,766 contractors completing self-training. Since there were no major engineering projects in 2024, the number of contractor training participants was limited to those required for routine operational activities at the specific fab. As a result, the number of self-training participants decreased compared to 2023. In addition, contractors are required to undergo construction capability certification before plant-entry. 1,222 contractors have been certified by a variety of consultative organizations, and the long-term in-plant service providing contractor personnel such as staff canteen workers, security personnel, cleaners, cylinder-replacing operators, etc. are required to receive at least 1 hour of safety-health-related educational training every year. There were 402 contractor plant-entry training sessions conducted in 2024 with 3,750 participants.

4.4.5 Occupational Injuries and Illnesses

In the event of occupational accidents, the accident investigation procedures must be followed to give feedback to the management system, through which the accident source can be controlled and improvement measures can be parallelly implemented to avoid the recurrence of same problems. The corresponding preventive improvement measures have been parallelly implemented in all plant areas. In addition, PSMC has taken into consideration the recommendations of experts and scholars to consolidate the business operations of the plants and to specify the implementation of occupational illness prevention and management measures at the company to achieve the ultimate goal of "zero illness at work and a healthy life" through three approaches: prevention, return to work, and compensation (PRC). Moreover, screening of potentially hazardous operations and near miss improvement and tracking have also been conducted, and the List of Potentially Hazardous Operations has been updated and incorporated into in-plant monitoring and management systems.

2024 Injury Index

Item	Category	PSMC	The Subsidiary	Total
Total hours worked (hours)		17,717,160	56,000	17,773,160
Frequency rate (FR) of disabling injuries		0.33	0	0.33
Severity Rate (SR) of disabling injuries		10	0	10
Sum of Injury Index (FSI)		0.05	0	0.05

Note 1: Calculated in accordance with the Occupational Safety and Health Administration of the Ministry of Labor's Occupational Accident Statistics Format.

Note 2: Injury incident statistics do not include employee commuting traffic accidents.

Occupational Injury Statistics

Item	Fiscal Year	2022	2023	2024
Total hours worked (hours)		16,900,136	17,420,608	17,773,160
Total number of occupational injuries (recordable occupational injuries)		1	6	6
Work-days lost (Day)		10	187	182
Work Injury Rate (IR)		0.06	0.34	0.33
Work-Days Lost Ratio (LDR)		0.59	10.73	10.27

Note 1: Serious occupational injury refers to workers who have lost more than six months of work.

Note 2: Work Injury Rate (IR) = (number of occupational injuries / total hours worked) * 1,000,000.

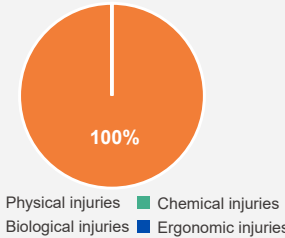
Note 3: Work-Days Lost Ratio (LDR) = (number of work-days lost / total hours worked) * 1,000,000.

Note 4: In 2024, there were no fatalities or serious occupational injuries caused by occupational injuries.

Note 5: In 2024, no occupational injuries occurred among non-employees.

Note 6: Occupational injury incident statistics do not include employee commuting traffic accidents.

Occupational injury types for 2024



Occupational injury causes and improvement measures

Physical injuries

Causes:

- (1) Falls/slips
- (2) Abrasion

Improvement measures:

- (1) Posting warning signs
- (2) Enhancing internal employee safety training and awareness
- (3) Establishing SOPs for key work areas

Occupational disease statistics

Item	Fiscal Year	2024	
		Employee	Non-Employee
Recorded instances of occupational disease		1	0
Number of deaths caused by occupational disease		0	0
Mortality rate due to occupational disease		0	0
Description of occupational disease types		Human hazards - musculoskeletal	N/A

Near Miss Statistics

Near miss category	2024	Improvement Measures
Chemical use hazards	10	Immediately restore the unsafe environment to normal, analyze incident causes, formulate preventive and corrective measures, evaluate the measures' effectiveness, and keep track of improvement status.
Fire hazards	24	
Operating environment hazards	408	
Moving objects	135	
Machine/equipment/system anomalies	55	

Note: A "near miss" is defined as an accident that has a minor impact and does not cause casualties, property losses, or process interruption, but comes as a shock to people onsite.

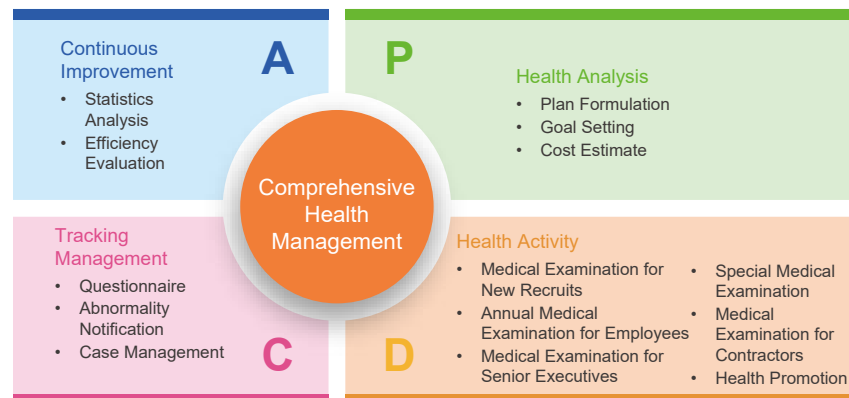
Absence Rate Statistics

Item	Fiscal Year	2021	2022	2023	2024	2024Target
Absentee rate (%)		0.35	0.71	0.63	0.51	0.75
Percentage of total employees covered (%)		100	100	100	100	100

4.4.6 Comprehensive Health Care

Building a Healthy Workplace

PSMC integrates the results of environmental safety and health risk assessment with the concept of a Three-Sections and Five-Levels prevention, and adheres to the implementation model of PDCA and the spirit of continuous improvement to plan complete health checkups and health promotion activities, and actively participates in the health promotional activities organized by the Ministry of Health and Welfare, the Public Health Bureau of Hsinchu City, the Ministry of Labor, and the Hsinchu Science Park Administration to build a healthy and friendly workplace environment with a systematic health management model and diversified resources.



PSMC followed the Operating Rules for Contractor Work Permit Application to compile statistics on the contractor personnel such as outsourced security personnel, cleaners, staff canteen workers, etc. who performed in-plant duties for 3 or more months in the past year and shared the same work environment or had regular work contact with PSMC employees. For such in-plant personnel, contractors have been required to provide their health check reports or a health declaration issued by the contracted companies on a yearly basis so that the Company can keep track of the in-plant personnel's health status. Those who do not provide health check reports or explain reasons shall be forbidden from entering PSMC plants. The control achievement rate was 100%.

Based on a human-oriented business philosophy, PSMC has established the Three-Sections and Five-Levels Prevention Strategy to develop a full range of health check campaigns and conduct a series of health promotion activities, for employees to receive the most timely and appropriate improvement treatment. PSMC's health management system is carried out through "advanced physical examination, general physical examination, physical examination for special operations personnel, new employee physical examination, foreign worker physical examination, and contractor physical examination", which is then reinforced through different levels of health management measures and anomaly tracking and improvement mechanisms for the Company to stay on top of employees' health status. Case tracking shall be conducted for personnel of high-risk groups such as workers engaged in special operations, etc. The cases shall be required to receive rechecks, and will be assisted in visiting Powerchip Clinic or onsite occupational medicine specialists to receive consultations.

A Three-Sections and Five-Levels Prevention Strategy

Initial Section of Prevention		Secondary Section of Prevention	Final Section of Prevention	
Promoting Health	Special Protection	Early Diagnosis and Appropriate Treatment	Limiting Disability	Rehabilitation
- Annual health check-ups - Organized health education - Provided in-plant sports field and gym	- Prevention of Infectious Diseases - Avoidance of occupational disease hazards -occupational physicians to visit operational plant site	- Case Screening - Special Medical Screening - Organized activities for high-risk groups - Regular sending disease hygiene Education - Occupational Medicine Interview Service - Medical services provided by Powerchip Clinic	- Avoiding exacerbating the disease - Referral for medical treatment - Evaluation of work assigned	Referral to appropriate medical institutions

Health Promotion Activities

Based on the results of the employees' annual health checkups, PSMC assess the types of health promotion needs, the specific coverages are as follows:

- Evaluate health examination data such as employee health examinations and medical examinations for new employees, analyze items with high abnormality rates, and plan them as key tasks in the health promotion plan.
- PSMC use the Employee Health Needs Questionnaire to understand common health problems and implement various health promotion activities such as stress management, physical fitness activities and dietary guidance.
- In accordance with laws, seasonal and employee needs, PSMC post new health information (divided into three categories: safety and health newsletters, general medical knowledge and health education for chronic disease groups) to enhance employees' general health knowledge.
- Through medical checkups of staff and newcomer, those with chronic diseases are screened and health education pamphlets are regularly sent every three months to keep employees up-to-date with the latest health education knowledge.

Health Promotion Activities

Activity Topics	Total number of events held	Total number of Participants
Special Health Screening (including abdominal ultrasound, thyroid ultrasound, breast ultrasound, gynecological ultrasound, prostate ultrasound, Pap smear, bone density test and eye pressure test, etc.)	44	2,433
Health Promotion Lectures	10	805
Blood donation	8	442
Self-paid flu shot	2	280
Low Calorie Healthy Meals	52 weeks(Every working day)	1,070
Noise Awareness Seminar	5	293
Weight Loss Program	1	785

Powerchip Clinic

In order to meet the medical service needs of employees and contractors at work, PSMC has set up the Powerchip Clinic within the plant to provide health management services such as outpatient consultation, health counseling, and preventive vaccination. Relevant health information and activities are available on the company's intranet "Health Management System" for inquiries and appointments. In addition, the clinic has also appointed on-site occupational medicine specialists to provide health education, guidance and planning, and assist in employees' fitness-for-work arrangement. In view of shift work type (four shifts and two rotations), in addition to the full-time nurse practitioner at Powerchip Clinic, there is another nurse practitioner on duty 24 hours a day in the plants to provide professional health consultation, health promotion activities and emergency care services to caring for the health of our colleagues.

5 Integrating Sustainability

| Society|

5.1 Societal Influence

5.2 Social Co-Prosperity Development



Integrating Sustainability

Society

In collaboration with the Powerchip Cultural Foundation, PSMC has long been committed to addressing social needs and development opportunities. Guided by three core pillars – environmental conservation, public welfare investment, and arts & culture festivals – PSMC strongly invests in environmental protection, supports social resources in rural areas, and promotes the development of domestic sports and arts performances. The Company is dedicated to transforming intangible resources into concrete actions to foster positive societal impacts. PSMC firmly believes that participating in public welfare and social services not only strengthens employee cohesion, but also amplifies our corporate influence to raise public awareness of social issues. Through concrete actions, PSMC seeks solutions and strives to create a corporate development environment that does well while also doing good for society.



Corresponding United Nations Sustainable Development Goals (SDGs)



Stakeholders who have priority reading

Society / Local Community



2024 Sustainability Performance Highlights



452 volunteers; 1,511 hours

Volunteers and hours committed to social co-prosperity activities



90,719 persons benefitted

Environmental conservation activities with 1,500 persons participating; 1,700 persons benefitted from the Company's public welfare commitment; and 87,519 participants in arts & culture promotion activities



1,000 trees

Tree-planting and forest protection effectiveness



700 kg

The waste removed via mountain and beach cleaning activities

5.1 Social Influence

Based on the management philosophy of "committing to energy-saving, and carbon-reducing environmentally friendly measures and fulfilling corporate responsibilities," PSMC has established three main strategic principles of social co-prosperity: environmental conservation, public welfare investment, and arts & cultural festivals. PSMC has extended along these principles to provide the necessary assistance and support to the environment, society, and specific ethnic groups in the public welfare activities' areas of ecological maintenance, care for the disadvantaged, support for arts and cultural activities, and community building, in order to solve society's long-term structural problems. To this end, PSMC has integrated internal and external resources such as the Group's Public Welfare Committee and the Powerchip Culture Foundation through systematic social investment planning, and has joined hands with employees, suppliers/contractors, competent authorities, and other stakeholders jointly in social welfare activities and demonstrate its corporate values through practical actions.

Social Co-Prosperity Strategy Principles and Achievement

Social Co-Prosperity Strategy Principles	Strategy Principles SDGs Alignment	Business Drivers	Action Plans	Business Efficiency and Key Performance Indicator	Social Impact	Social/Environmental Benefits and Key Performance Indicators
Environmental Conservation	 	There is a global consensus that the impact of climate change must be mitigated, and that this is relevant to the sustainability of business operations. PSMC committed to making a tangible contribution to environmental conservation, cultivating a philosophy of environmental conservation for stakeholders, and expanding the impact and continuity of social investment.	Animal Care, Environmental Education, Planting trees and protecting forests	Expanded staff participation and enhanced staff dedication - Recruited 398 employees to participate in environmental conservation educational activities Enhance social recognition and brand image through community interaction - Conducted Leopard cat conservation and promotion activities, inviting approximately 1,500 local people to participate, strengthening connections with local communities	- Raised awareness of environmental conservation among employees and the public - Reduced the impact of human activities on the environment	Number of environmental conservation activities - In 2024, a total of 8 projects were launched and 8 events were held Employee Volunteer Service Hours - A total of 1,091 accumulated service hours Total amount of waste removal - Cumulative 700 kg of general waste was removed Area of park cleaning - Park adoption of 7,088.49 square meters Tree-planting & forest protection 1,000 trees planted
Public Welfare Investment	 	PSMC's sustainable management philosophy is to create a positive cycle between corporate operations and society. Through long-term investment in rural care, community development, and the promotion of education for rural students, PSMC is committed to bridging the gap between urban and rural areas and strengthening the link between social development and corporate sustainability.	Community Engagement and Rural Assistance	Expanded staff participation and enhanced staff dedication - A total of 52 employees participated in volunteer efforts - A total of 404 hours that volunteers put in	- Provided necessary resources and assistance to rural communities, disadvantaged communities and students - Supported and promoted the development of Taiwan's sports culture - Promoted industry-academia cooperation to cultivate outstanding talents in semiconductor innovation	Established a positive corporate image - A total of 1,700 individuals benefited from the community involvement and rural care investment Number of social welfare investment beneficiaries - More than 7 social welfare organizations, institutions, schools, farmers, and other organizations and individuals benefited
Arts & Culture Festivals		The diversified sponsorship programs demonstrate the possibility of combining corporate management with arts and cultural activities. PSMC aims to promote arts and cultural activities and devotes resources to support the arts and cultural industry through donations, sponsorships, and strategic cooperation to drive the overall cultural development of the country.	Powerchip's Arts and Cultural Festival, Music Festival Education Series Activities	Enhanced social recognition and brand image through arts and cultural events series A total of 87,519 people benefited from the events	- Supported the development of Taiwan's arts and cultural industries and enhanced social arts culture - Created a stage for outstanding Taiwanese musicians to perform	Number of social welfare investment beneficiaries In 2024, a total of 49concerts, 2 weeks music festival (a total of 80) and 16 days of art and cultural exhibitions sponsored

Social Engagement Impact Assessment

PSMC had reference to the B4SI (Business for Societal Impact) Community Investment Measurement Framework. We measure the benefits and impact of each public welfare activity and resource investment, and adjust investment and resource allocation plans in a timely manner, so that social investment programs can be closely connected with societal issues. In 2024, PSMC had more than 452 volunteers volunteering for a total of 1,511 hours. We hope that in addition to investment in human, material, and other physical resources, PSMC can also continue our invisible influence on long-term social issues such as the promotion of cultural education, establishment of positive values for school children, elimination of resource shortages in rural area communities, etc. By doing this, we extend our strengths to create positive cycles in public welfare commitment.

5.2 Social Co-Prosperity Development

5.2.1 Environmental Conservation

By responding to the Hsinchu Science Park Administration's "Industrial Safety and Environmental Protection Month" plan, we promote a series of activities with main themes including "environmental protection" and "environmental education", calling on colleagues, family members and the general public to contribute to environmental protection, and cultivate the environmental awareness of colleagues and the public in order to demonstrate the determination to protect the terrestrial and marine ecological environment.

Moreover, PSMC engages in and sponsors research, discussions, and related activities in the field of environmental conservation in order to promote environmental protection education and the application of environmental protection at home and abroad. PSMC is actively applying the results of these programs to individuals, society, and enterprises, expecting to put environmental conservation actions into practice in everyday life.

Environmental Protection

Keep the Sea Clean

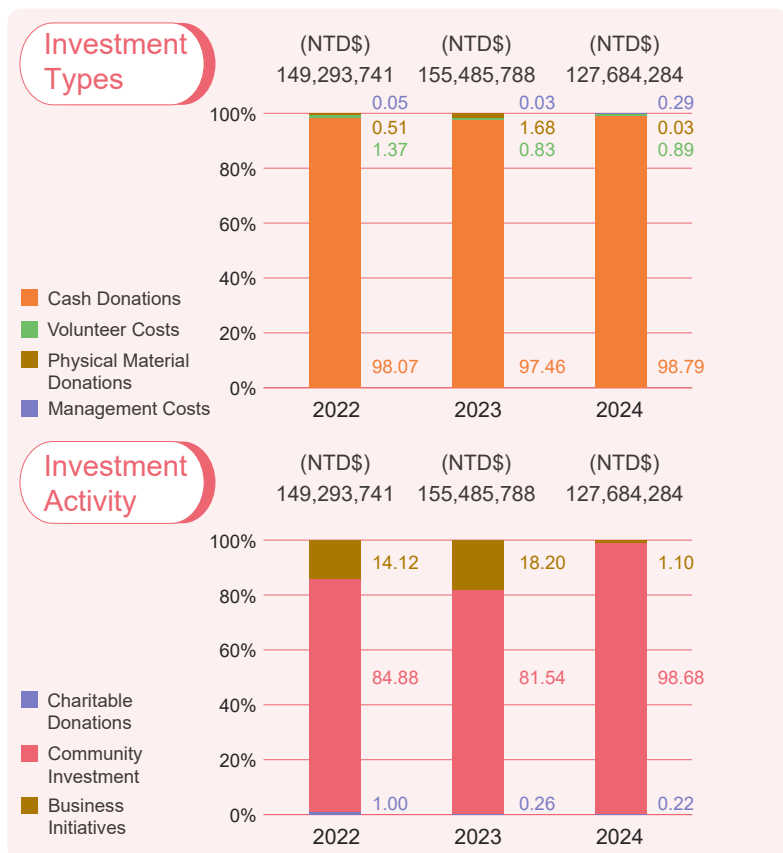


PSMC gathered 149 employees and their families to participate in a beach cleanup activity on the north shore of the Guanyin Temple at Nanliao Fishing Harbor, Hsinchu City, Taiwan, and removed non-marine litter. The beach cleanup activity was also documented in the International Coastal Cleanup (ICC) log sheet, which monitors marine debris. According to the item-by-item records of the garbage picked up by the group, the total area of the beach cleaned on that day was about 500 meters, and a total of 454.8 kg of various types of waste were removed.

Mountain Cleanup initiative



100 participants, including PSMC employees, partners from TATA, and contractors, gathered at the Tonglue Skywalk to take part in a mountain cleanup event. Through cleaning the environment near the fab, the event aimed to enhance environmental awareness among participants and demonstrate PSMC's commitment to corporate social responsibility. The cleanup route was c. 2.6 km long, and 245.2 kg of waste was collected during the event.



Nature Exploration Team Let's Go



PSMC continues to expand our focus and resource investment in local environmental conservation. Hosted in-person lectures at the P5 fab and live-streaming sessions on Pteam have had 266 participants. The program incorporates the concept of citizen science, engaging participants through visual, auditory, and olfactory senses to explore lowland forest ecosystems. The initiative aims to cultivate scientific literacy and environmental conservation awareness, while also helping participants develop basic observation skills and learn to record and explore nature's mysteries through scientific methods.

Adopting Parks, Public Restrooms, and Parking Lots



Nanliao Ecological Park



public restrooms at Zhunan Science Park community park



parking lots near Tongluo Skywalk



public restrooms at Tongluo Skywalk

To uphold the spirit of local community service, PSMC has partnered with the Hsinchu Science Park Administration for two consecutive years to organize clean-up and maintenance adoption activities for nearby parks. In collaboration with the Occupational Safety and General Affairs Department, we arranged for cleaning personnel to perform public restroom cleaning, grass cutting, sidewalk plant trimming, and dead branch removal. In 2024, PSMC adopted public restrooms at Zhunan Science Park community park, and Tongluo Skywalk surrounding parking lots, as well as Nanliao Ecological Park. These efforts help strengthen relationships with government agencies and local residents.

Flea Market



Employees donated unused household items for a charity sale during the event, promoting resource reuse and environmental sustainability. Proceeds from the event, which totaled NT\$168,218, were entirely donated to charitable causes.

Environmental Education



Supporting Sustainability with Creativity – Eco-Friendly DIY



Eco-Education Tour– Explore and Learn

EHS Communication

1. Sending 50 on a variety of topics EHS messages to enhance employees' sensitivity of pollution prevention, climate change, safety & health, health promotion, disaster prevention and safety awareness, and other EHS-related issues.
2. Supporting Sustainability with Creativity – Eco-Friendly DIY Workshop: This event encouraged employees to create beautiful potted plants using recycled waste materials, promoting the concept of "circular economy" from an aesthetic perspective.
3. The Impact of Climate Change on Water Resources and Adaptation Seminar: A professional scholar delivered an insightful lecture to help employees understand environmental protection and the challenges posed by climate change. The course enhanced awareness of water resources, treatment, and reuse. In 2024, a total of 255 employees participated in this seminar.
4. Eco-Education Tour – Explore and Learn Activity: Employees participated in environmental education site visits, where they gained knowledge, values, attitudes, and skills related to sustainability, and shared their insights with colleagues.

5.2.2 Social Welfare Investment

Following PSMC Public Welfare Committee's planning focused on community building and charitable donations, the Company takes employees to rural areas, campuses, and communities to promote and participate in local public affairs. Through concrete volunteering commitment and resource donations, the Company strives to mitigate urban-rural gaps, support local agricultural products/special products, promote equal educational opportunities for young people, and encourage sports activities. All of this is aimed at using our corporate strengths to achieve collective prosperity, collective benefits, and sustainable development with society.

Community Building

Promoting Badminton and Supporting Local Badminton Varsity Teams



With the aim of promoting Hsinchu's local sports development, PSMC has sponsored the Zhudong Junior High School/Tongtex Secondary High School badminton team for three years in a row; the sponsorship fund totals NT\$648,000. The Company was also invited to participate in the opening ceremony for Hsinchu County Badminton Tournament. By providing the team with more complete training resources and funds required for competition participation, we hope that an excellent environment can be created to cultivate young badminton talents in Taiwan.

Public Welfare & Art Corridor



PSMC's Public Welfare & Art Corridor event showcased digitally reproduced paintings with authorized production, promoting aesthetics and integrating art into the workplace and daily life. Through concrete actions, PSMC supports local artists and advances art and cultural public welfare initiatives.

Supporting Disadvantaged Groups



PSMC has been supporting a charitable store run by Mr. Lin, a survivor of severe burn injury, to provide meals for and donate foods to disadvantaged groups on a long-term basis. The Company also helps pay part of the store rental every month to encourage public welfare actions.

Hsinchu Nanliao Beach Cleanup Activity



At the beach near the Nanliao Stairway, a free blood pressure measurement booth was set up to serve employees and the public participating in the cleanup. In addition, conservation-themed transfer stickers featuring the leopard cat and other species were distributed to promote conservation awareness.

Nanliao Ecological Park – Corporate Adoption and Maintenance



PSMC has adopted Nanliao Ecological Park and conducts environmental maintenance twice a year, once in the first half and once in the second half. The maintenance activities include weeding, trimming sidewalk plants, clearing fallen branches and leaves, and garbage removal.

World Vision Taiwan – Sponsoring Overseas Performances for Jinping Elementary School Choir



Jinping Elementary School Choir in Jianshi Township is dedicated to training students in traditional Atayal songs, helping them develop a sense of cultural identity through learning and practice. By participating in a variety of performances, students build confidence and courage. PSMC Public Welfare Committee sponsored the choir's trip to perform in Malaysia, continuing our support for education, character development, and the promotion of Indigenous arts and culture for underprivileged children.

Christmas Charity Market



PSMC invited multiple disadvantaged groups, such as the Syn-Lu Social Welfare Foundation, Yu'an Children's Home, World Peace Association, and Down Syndrome Foundation, to set up charity stalls at PSMC in two rounds, free of charge. This initiative not only provided these organizations with opportunities to increase their income but also helped raise employees' awareness of and care for disadvantaged communities.

Public welfare booth placement



PSMC participated in the 2024 Occupational Safety and Environmental Protection Month fair organized by the Hsinchu Science Park Bureau at Zhudong Ecological Riverside Park. During the event, PSMC conducted a public blood pressure measurement activity, fulfilling corporate social responsibility and promoting community engagement.

Tree Planting and Forest Conservation Activity



PSMC, in collaboration with the Tse-Xin Organic Agriculture Foundation, continues to support the Taiwan Green Great Wall Afforestation Initiative. In this effort, 500 trees were planted in the following locations: Mao'erding Section in Zhubei, Hsinchu, and Jixin/Jibao sections in Wuji Township, Yilan, to protect the nearby coastline and prevent land erosion.

Folk Sports Development Program



PSMC has been a long-term sponsor of the Diabolo Program under the Folk Sports Development Program at Houlong Elementary School in Miaoli County, providing NT\$100,000 in funding for international competitions. In 2024, this sponsorship supported students in participating in the Osaka International Diabolo Competition (OIDC) in Japan, helping them achieve outstanding results on the global stage.

Leopard Cat Carnival



The Leopard Cat Festival event at PSMC's Tongluo fab was organized in collaboration with the Tongluo Hakka Cultural Center, the Leopard Cat Association of Taiwan, and the Tongluo Farmer's Association. The festival featured local farmers' and vendors' stalls, parent-child DIY activities incorporating leopard cat conservation themes, and screenings of authorized conservation-related films to promote awareness of environmental and wildlife conservation. The event attracted approximately 1,500 participants.



Charitable Donations

Old Shoe Donation campaign



Employees were invited to participate in the Old Shoe Donation initiative organized by Step30 International Ministries, donating unused shoes from their homes to be sent to underprivileged communities in East Africa. This initiative aimed to help protect the health of African children. A total of 1,379 pairs of shoes were collected.



Donating Used Computers to Charity

After conducting a donation assessment, PSMC sent hundreds of replaced computers to the ASUS Foundation to support the Recycled Computer Project of Hope. Through this, the donated computers were delivered to disadvantaged organizations in need. Two computer donation sessions were carried out by PSMC in 2024.



Sending Love to Rural Elementary Schools and Tribes During the Holiday Season

In order to share the festive atmosphere with students in rural areas, PSMC gathered its employees to visit Chien Shih Elementary School and the Chien Shih Hou Shan Tribe in Hsinchu County on the eve of the Mid-Autumn Festival. PSMC's employees brought mooncakes, pomelos, and other seasonal delicacies, giving students an opportunity to learn about and participate in the traditional festival culture, fulfilling the Company's mission to educate and care for rural communities.

5.2.3 Art & Cultural Festivals

PSMC aims to "help arts & culture education take root, and create stages for performing artists". As part of this, PSMC made a cash donation of NT\$25 million to the Powerchip Cultural Foundation in 2024. In addition, driven by the goals of Encouraging Music Education, and Promoting Arts & Cultural Performances, the Company has collaborated with Powerchip Cultural Foundation for 20 years and counting. Through this partnership, we promote development of arts and cultural activities, and support musicians and symphony orchestras with great potential in Taiwan through hosting/co-organizing cultural events, performances, competitions, and similar activities. PSMC launched new arts and cultural activities in 2024 including the Powerchip 2024 Classic Series, the Music Festival Education Series, the Arts & Cultural Exhibition, etc., and sponsored other related activities that help achieve the Foundation's goals. By doing so, we improve the environment for arts and cultural development in Taiwan, and enhance the value of the performing arts industry.

Promoting Music Culture



Powerchip Arts and Cultural Festival 2024 Series Events

The Powerchip Cultural Foundation's flagship program for sponsoring symphonic performances and arts and cultural exhibitions is the Powerchip Arts and Cultural Festival. We have long focused on outstanding musicians and renowned symphony orchestras from domestic and foreign countries, and have exclusively sponsored musicians and orchestras to perform in Taiwan. This also helped create a platform for the development of outstanding musicians in Taiwan by allowing Taiwanese music fans to experience an international music feast in person in Taiwan, and providing them with an opportunity to further exchange musical knowledge with international masters. In 2024, we sponsored a total of 49 concerts, attracting more than 65,478 Taiwanese music fans to experience these musical festivities.



Music Festival Education Series

Rooted in Music Education – 2024 Taipei Music Academy & Festival

The Taipei Masters Music Academy & Festival is the most international classical music festival and educational music camp in Taiwan. The festival entered its sixth year in 2024, with over 14,841 participants in a series of promotions, lectures, and guided music education activities a total of 80. The festival, jointly facilitated by the Ministry of Culture and community organizations, brings together a faculty of leading musicians from all sections of the world's top orchestras. In addition to solid individual courses and chamber music instruction, the festival also offers master classes and professional lectures, providing students with the opportunity to learn from the masters, and creating more musical possibilities through bilateral exchanges between teachers and students, expanding a broad international perspective.

Promoting Arts and Cultural Performance

"Hope, Peace, Love" Immersive Performance Exhibition



In 2024, the Powerchip Cultural Foundation sponsored the "Hope, Peace, Love" Immersive Performance Exhibition, led by Chairperson Chia-Hui Lu of the Egret Cultural and Educational Foundation. This event marked Taiwan's first-ever 5G+4K fully immersive digital exhibition, integrating immersive art, performing arts, and interactive art. Through the synergy of technology and art, and the fusion of virtual and physical experiences, the exhibition helped viewers to immerse themselves in a boundless artistic ocean, personally experiencing a spiritual journey of love and peace, and setting sail on a voyage of hope.

6 Efficient Sustainability

| Corporate Governance |

- 6.1 Corporate Governance
- 6.2 Integrity Management
- 6.3 Regulatory Compliance
- 6.4 Responsible Taxation
- 6.5 Risk Management
- 6.6 Information Security



Efficient Sustainability

Corporate Governance

While strengthening the functions of the Board of Directors, PSMC is committed to fostering a corporate culture and environment of integrity and accountability. The Company operates with the highest standards to ensure the rights and interests of stakeholders and uphold the core values of integrity management. In addition, to enhance risk control and adaptability, PSMC has established a comprehensive risk management and audit system, with the Board of Directors overseeing risk supervision and control. This ensures that all business operations comply with legal and regulatory requirements while also equipping the Company and its employees with the ability to respond to potential risk impacts.



Corresponding United Nations Sustainable Development Goals (SDGs)



Stakeholders who have priority reading

Government/Authorities, Shareholders/Investors, Customers



Material Topics GRI Standards

GRI 201 : Economic Performance

GRI 205 : Anti-corruption

GRI 206 : Anti-competitive behavior

GRI 405 : Diversity and Equal Opportunity

GRI 418 : Customer privacy



Corresponding Material Topics



Economic
Performance



Regulatory
Compliance



Information
Security



Integrity
Management



Corporate
Governance



2024 Sustainability Performance Highlights

Top 5%

TWSE Corporate Governance
Evaluation

>50%

Percentage of independent
director seats

9.3 hours

Directors' average educational
training hours

100%

Completion rate of all employees'
training on information security
management/control measures

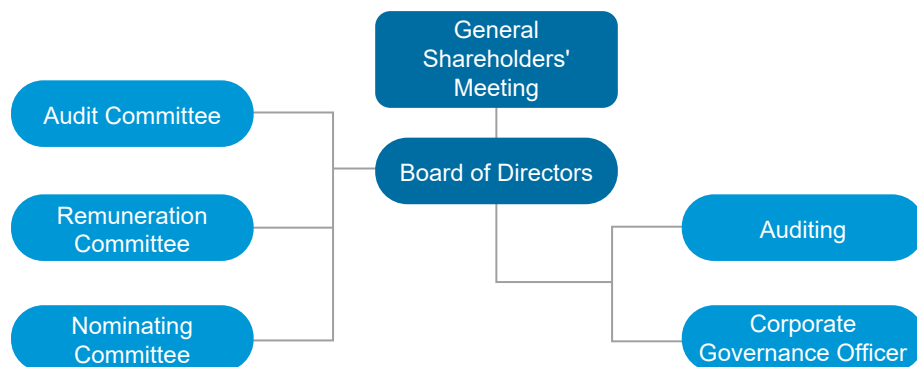
0

Information breach incidents

6.1 Corporate Governance

6.1.1 Corporate Governance Framework

The PSMC Shareholders' Meeting is responsible for reviewing major matters and decisions related to the Company's operations. Under the Shareholders' Meeting, the Board of Directors is established as the Company's highest governing body. In accordance with the Company's director election procedures, board members are elected by shareholders through a voting process and are responsible for overseeing the Company's operations, development, and supervision. The duties of the Board of Directors include setting the Company's vision, operational strategies, and budget plans, as well as planning the Company's mid- to long-term development direction. In addition, the Board supervises the implementation of operational plans. All board members possess experience and expertise in risk oversight and management, guiding PSMC in expanding our international market presence amid an evolving economic landscape.



Functional Committees

Name of Committee	Duty & Authority	2024 Achievements
Salary and Compensation Committee	Responsible for assist the board of directors in formulating and reviewing measures for director and manager performance evaluation, developing compensation policy, system, standards and structure, and regularly evaluating and determining the aforementioned personnel's compensation to ensure its competitiveness and reasonableness.	The Committee held 4 meetings in 2024. The committee members' actual attendance rate was 100%.
Audit Committee	Responsible for supervising the fair representation of the Company's financial statements, the appointment or dismissal of attesting CPAs, the evaluation of CPAs' suitability, independence, and performance, the effective implementation of the Company's internal control, the Company's compliance with relevant laws and regulations, and the control and management of the Company's existing or latent risks to serve as a countervailing mechanism for corporate governance.	The Committee held 8 meetings in 2024. The committee members' actual attendance rate was 93%.
Nomination Committee	Responsible for formulating and reviewing the composition, qualification criteria, and selection standards for directors and senior executives. It also assesses the independence of independent directors and evaluates potential conflicts of interest among governance unit members, while developing succession plans.	The Committee held one meeting in 2024. The committee members' actual attendance rate was 92%.

Note : For more information on the composition, authority, and resolutions of the functional committees, please refer to the Company's 2024 Annual Report.

6.1.2 Operation of the Board of Directors

As required by relevant regulations, the Board was formed based on a single-track system, and is currently composed of 4 directors and 5 independent directors, for a total of nine directors, who serve a 3-year term of office (As of the end of 2024, the average service tenure of all directors was 4.9 years). The proportion of independent directors is 55.56%, which exceeds the preset requirement that independent director seats should account for more than 50% of the entire director seats, and no independent director is concurrently serving as an independent director in more than three publicly listed companies. To meet the Company's organizational development needs, Chairman Frank Huang concurrently serves as the Company's CEO to ensure the smoothness of communication channels with all directors. Eight board meetings were held in 2024, and the average actual attendance rate was 93.06%, with a 100% attendance rate including proxies, exceeding the minimum 80% attendance rate required by the Company's board performance evaluation standards.

As stated in Article 193 of the Company Act, directors shall perform their duties in accordance with laws, the Articles of Incorporation, and shareholders' meeting resolutions. If their violation of such causes damage to the company, they shall be liable to compensate the company for damages. Such liability is a mandatory legal provision and cannot be restricted or exempted through internal company rules or the Articles of Incorporation. Therefore, PSMC does not have any provisions or agreements that limit directors' liability. To strengthen corporate governance and enhance Board of Directors functions, the Company also complies with the policies promoted by the regulatory authorities by maintaining directors' liability insurance throughout directors' terms of office. Furthermore, as stated in Article 277 of the Company Act, any amendment to the Articles of Incorporation must "be adopted by a majority of the shareholders present [at the meeting of shareholders] who represent two-thirds or more of the total number of its outstanding shares".

Policy on Nomination and Election of Directors

The Board of Directors has established the "Regulations Governing the Election of Directors" as the criteria for the eligibility and selection of director candidates, and has adopted a strict candidate nomination procedure and a cumulative voting system; moreover, it has encouraged shareholders to participate in the nomination and election process. The verification of candidate eligibility and the determination of whether there has been any violation of the provisions of Article 30 of the Company Act are conducted and announced in accordance with the laws, and this process remains autonomous. The method of nomination and election of candidates, the professional qualifications of independent directors, the restrictions on concurrent employment, and any other matters that should be complied with are handled in accordance with the Company Act, the Securities and Exchange Act, and other relevant laws and regulations. The nomination process not only takes into account the professional background and industry experience of each candidate, but their integrity and leadership ability are also critical points of consideration. The results of the Board of Directors' performance evaluation, which is conducted on an annual basis, are also used as a reference for the nomination of directors for reappointment.

Director Independence and Conflict of Interest Management

The directors and independent directors are independent individuals who exercise their duties and responsibilities independently. Based on the "Operating Procedures and Conduct Guidelines for Integrity Management" and "Regulations for Board of Directors Meetings", which has clear regulations on the recusal of conflict of interest, and the also have provisions on the recusal of interest, whereby those who have a stake in the matters of the meetings, whether personally or in the capacity of a legal representative, are not allowed to join the discussion or vote, and the relevant recusal is stated in the minutes of the meetings. As of the end of December 2024, no material conflict of interest has occurred at the Company. Please refer to pages 17 of the Company's 2024 Annual Report for the implementation of the recusal of directors from motions of interest.

Diversity of Board Members

In view of the diversity policy and in an effort to strengthen corporate governance and to promote the development of sound board composition and structure, PSMC has formulated a "Code of Corporate Governance Practices" which clearly states that the composition of the Board of Directors should take into account diversity and the complementary nature of the directors' professional capabilities, and that appropriate diversity policies should be formulated with respect to PSMC's own operations, business model, and development needs, including but not limited to the following:

- (a) Basic qualifications and values: Gender, age, nationality, culture, race or ethnicity.
- (b) Professional knowledge and skills: Operational judgment ability, accounting and financial analysis ability, managerial competencies, crisis handling ability, industrial knowledge, global market views, leadership, and decision-making skills.

The current board of directors consists of 9 members who are equipped with diverse backgrounds, professional capabilities and practical experiences, and can provide different professional advice to facilitate PSMC's corporate governance. Through directors' rich management experiences, leadership/decision-making abilities, industrial knowledge, risk management capabilities, and other competencies, the Board's functions can be significantly strengthened. Among the directors, the non-executive directors all possess backgrounds related to the financial sector or electronics industries, whose profound experiences in global investment risk assessment and relevant expertise will help PSMC enhance the decision-making quality when carrying out its global deployment in the future. In 2024, PSMC achieved the goals for board diversity management that the proportion of elderly board members should be less than one-third, and the service of all independent directors should be controlled within three terms.



Implementation Status of Board Member Diversity Policy

Diversity Core Values Name	Basic Components			Professional Background				Professional Knowledge and Skills						GICS Level 1 ^(Note1)
	Nationality	Title	External Independence ^(Note 2)	Accounting	Industry	Finance	Technology	Operational Judgment	Operational Management	Leadership & Decision Making	Crisis Management	Industry Knowledge	International Market Insights	
Huang Chung-Jen*	Republic of China (Taiwan)	Chairman	Executive Director		√		√	√	√	√	√	√	√	Industry, Healthcare, Finance, Information Technology, Communication Services
Hsieh Tsai-Chu		Director	Executive Director		√		√	√	√	√	√	√	√	Information Technology
Chu Hsien-Kuo*		Director	Executive Director		√		√	√	√	√	√	√	√	Information Technology
Hsu Ching-Hsiang		Director	Non-Executive Director Independent Director		√		√	√	√	√	√	√	√	Information Technology
Wu Chung-Yu		Independent Director	Non-Executive Director Independent Director		√		√	√	√	√	√	√	√	Healthcare, Information Technology
Chang Chia-Lin		Independent Director	Non-Executive Director Independent Director		√	√	√	√	√	√	√	√	√	Finance, Information Technology, Communication Services
Yeh Shu		Independent Director	Non-Executive Director Independent Director	√	√	√	√	√	√	√	√	√	√	Finance, Information Technology, Communication Services
Chen Chun-Sheng		Independent Director	Non-Executive Director Independent Director		√		√	√	√	√	√	√	√	Finance, Information Technology, Communication Services
Tsao Shih-Lun		Independent Director	Non-Executive Director Independent Director		√		√	√	√	√	√	√	√	Industry

Note 1: The director's industry experience assessment was conducted in accordance with the Global Industry Classification Standard (GICS) Level 1.

Note 2: Directors meet the external independence assessment criteria and adopt the following criteria for independent directors. Directors must be non-executive directors and must meet at least 4 of the following 9 indicators, of which the first 3 must meet at least 2:

- (1) During the past one year, the director has not held any senior executive within the Company.
- (2) The director and their family members did not receive more than US\$60,000 from the Company, its parent, or its subsidiaries this year, except as permitted under Rule 4200 of the US SEC.
- (3) No family member of a director worked as a senior executive of the Company, its parent, or its subsidiaries this year.
- (4) The director is not an advisor to the Company or the management team and has no vested interest in the Company's advisors.
- (5) The director has no vested interest in the Company's major customers or suppliers.
- (6) There is no service contract relationship between the director and the Company or the management team.
- (7) The director has no vested interest in a nonprofit organization that is a major recipient of corporate contributions.
- (8) During the past one year, the director has not served as an external auditor or as a partner of the Company.
- (9) There is no conflict of interest between the director and the independent operation of the Board.

In addition, pursuant to Article 4 of the Regulations Governing Appointment of Independent Directors and Compliance Matters for Public Companies, the independent directors of the Company shall not serve as independent directors of more than three other companies.

Note 3: The percentage of directors aged 51 or above is 100%, all of whom are male.

Note 4: To ensure the inclusiveness of stakeholders, the board members are representative of various stakeholders such as senior citizens, veterans, government workers, investors, and nonprofit organizations.

Note 5: Please refer to the Company's 2023 Annual Report for a brief biography of each director.

Note 6: Those names marked with an * are: Director Huang Chung-Jen, who is also the CEO of PSMC; Director Chu Hsien-Kuo, who is also the President of PSMC.

Professional Development for Board Members

In consideration of the diversified approaches, such as operational patterns and development needs, while at the same time responding to the advancement of domestic and international corporate governance and sustainability issues, PSMC, based on the characteristics of the industry and taking into account the professional background of the directors, assists in drawing up a plan for professional development and arranges for courses of at least 6 hours per year in accordance with the law. The total length of directors' continuing education achieved in 2024 were 84 hours, and the average training hours for each director was 9.3 hrs. By reference to international development trends and feedback given through directors' self-assessment, sustainable development issues such as ethical management, global economy, ESG risks supervision, etc. were incorporated into the topics for directors' continuing education in 2024. Additionally, it was arranged for directors to participate in training courses related to risk management topics, enabling them to stay informed about different risk impacts in a timely manner. For detailed director continuing education information, please refer to page 23-25 of the Company's 2024 Annual Report.

Performance Evaluation and Compensation for Board Members

PSMC has formulated the "Measures for Board of Directors Performance Evaluation" in accordance with "Corporate Governance Best Practice Principles for TWSE/TPEX Listed Companies" to carry out the performance evaluation for board of directors and functional committees, and adjust evaluation focuses by reference to international trends and industrial development tracks to more comprehensively measure governance units' operation and supervision performance. PSMC follows stipulated evaluation procedures and relevant metrics to implement the board's internal self-assessment, board members' appraisal/self-assessment, and functional committees' self-assessment on a yearly basis, and commission's external independent professional institutions or expert-scholar teams to conduct evaluation at least every three years. The evaluation procedures shall be completed before the end of the first quarter of the following year. The evaluation report shall be reviewed by the Board, and improvement and upgrade measures shall be formulated accordingly. PSMC submitted the 2024 evaluation results to the Board of Directors and each functional committee on February 25, 2025, for more information about Board of Directors Performance Evaluation, please refer to page 18 of the Company's 2024 Annual Report.

PSMC adopts the results of evaluation conducted in accordance with "Measures for Board of Directors Performance Evaluation" and "Regulations on Manager Performance Evaluation and Remuneration" as a basis to assess the compensation and remuneration for directors and managers. The performance evaluation and compensation policies are set by the Salary and Compensation Committee. At least 2 meetings are held each year to review the implementation outcomes to ensure the competitiveness and reasonableness of the personnel's compensation. The Company does not commission any external consultants to assist in its compensation operations for the 2024 year.

The annual payment of directors and managers is mainly based on the company's future operational development and operational risks, while evaluating their individual performance in order to strike a balance between sustainable management and risk control. The compensation of the directors is reviewed by the Salary and Compensation Committee, submitted to the Board of Directors for approval, and proposed to the shareholders meeting for reporting. The managers' compensation includes fixed salary, bonuses, and employee compensation. Each manager's compensation is based on the manager's participation in and contribution to the Company's operations, is paid based on the annual salary survey report with reference to industry salary standards, and is submitted to the Board of Directors for approval after review by the Salary and Compensation Committee. On the other hand, in order to incentivize senior executives such as the Chair, President, vice presidents, and assistant vice presidents to focus on long-term overall performance and achieve the goal of the Company's sustainable operations, PSMC has incorporated sustainability performance indicators related to their respective duties into the compensation policies for these senior executives.

Index	Weight Function	Description
Operational Performance Indicators	30~50%	Such as POIC、EPS achieve rate
Individual Annual KPI Achievement Rate	40-60%	Annually set 1 to 4 items according to individual's job functions
Sustainable Performance Indicators	5-25%	Such as renewable energy, energy conservation and carbon reduction, diverse workplace, participation in committees, etc.
Personal Growth and Learning	5%	Cross-disciplinary learning and self-improvement

- The compensation of the CEO is based on the achievement of financial and non-financial targets related to the Company's operations, in addition to the individual's operating performance. PSMC has issued employee stock option certificates and uses the company's financial metrics (such as consolidated revenue, EPS, ROIC, etc. achievement rate) to set performance goals with reference to peers' relative financial metrics (such as total shareholder return, etc.). Then we establish long-term performance goals for a period of 4 years, and also use the performance of non-financial metrics related to sustainable governance (such as energy conservation and carbon reduction goals, green energy use and water conservation goals, etc.) as variable compensation evaluation items to align operational goals with sustainable results.
- In order to ensure and enhance the ethics and conduct of senior management and all employees, PSMC has established the Code of Conduct for Integrity and the Procedures and Guidelines for Integrity Management. In the event of a violation of the Code and the Guidelines, PSMC reserves the right to claim the profits from the improper conduct and the variable compensation paid to the employees to the extent permitted by applicable law, and to impose penalties, including salary reduction, withholding of bonuses and bonuses, demotion, dismissal or legal action, depending on the severity of the situation or the negative impact on the Company's reputation, in order to protect the interests of the Company and the shareholders as a whole, as well as to safeguard the long-term development of the Company by requiring employees to strictly adhere to ethical standards.
- In order to align the long-term interests of managers with those of shareholders, PSMC encourages managers to hold a certain amount of stock for a long period of time during their employment.

6.2 Integrity Management

PSMC is committed to "Integrity, Service, Quality, and Innovation" as its business beliefs, and strict requirements for its directors, managers, and employees to comply with the anti-corruption and anti-bribery laws and regulations applicable in the countries in which they operate. PSMC also strives to enhance the work ethics and professional competence of its internal staff through the three main pillars of "Establishing a Integrity Management Regulation", "Implementing Education and Training", and "Establishing a Whistleblower System" for integrity management and communication.

Integrity Management Operation Status

PSMC "Ethical Corporate Management Best Practice Principles" and "Operating Procedures and Conduct Guidelines for Ethical Management" have been formulated after being approved by the Audit Committee and passed by the board of directors, and have been implemented after being reported to the Shareholder Meeting. In addition, "Procedures for Whistleblowing, Complaints, Suggestions, and Employees' Participation in Feedback Management" have also been added on the basis of the aforementioned regulations for the Company to communicate PSMC's policy commitments to employees, business partners, suppliers, and other stakeholders through the Company's websites, orders / invoices, major engineering equipment trading contracts, internal websites, and other possible channels. A zero-tolerance policy is upheld when dealing with employees' violations of business ethics and related behaviors. For any violations of the Code of Conduct, punitive measures such as disciplinary admonitions, demerits, etc. shall be taken against the violators depending on the severity of violations. For serious violations, the involved personnel shall be dismissed.

To ensure employees' behavioral compliance with the various requirements, employees' behavioral performance shall be linked up with their remuneration through the Company's rewarding and disciplinary systems, and Code-of-Conduct-related rewarding / disciplinary records shall also be considered in employees' annual performance evaluation. In addition to regulating relevant internal personnel, the policies are also applicable to suppliers, customers, or other business counterparties. It is necessary to consider the legality of the aforementioned parties, and check if they are involved in any unethical acts before business relationships are established. The Company also requires new suppliers to complete the "Supplier Survey", and current suppliers and plant-construction contractors to sign the "RBA Commitment Statement", or adds ethical management provisions to contract contents so that business counterparties can jointly practice business ethics and ethical management policies.

In operation, PSMC's Professional Ethics Advancement Committee, chaired by the Director of Legal Affairs, is composed of the heads of the Legal Affairs Office, Finance Division, Accounting Division, Human Resources Division, and Risk Management Division, with the Director of Legal Affairs serving as the Chairperson. The Professional Ethics Advancement Committee is responsible for promoting the development and implementation of integrity management policies and preventive programs. Through annual self-assessments, all departments and plants are should to conduct self-assessments of their operations, including assessments of compliance with laws and regulations and potential risks related to corruption, which serve as a reference for adjusting the design and implementation of internal control systems. The implementation status is also reported to the Audit Committee and the Board of Directors on a regular basis each year in order to carry out self-monitoring and track the status of the improvements..

PSMC conducts corruption-related risk assessments by department every year in accordance with the operating rules for integrity management risk assessment and management. A total of "infringement of business secrets, trademark rights, patent rights, potential corruption and integrity risks such as copyrights and intellectual property rights will be

strictly monitored through the internal risk management mechanism. In 2024, there were no incidents that PSMC violated the integrity management such as corruption or bribery, conflicts of interest, money laundering or insider trading, etc.

Advocacy and Educational Training

In addition to irregularly announcing the variations of laws and regulations to all directors and employees through email or other means, PSMC also advocates the concepts of anti-corruption and emphasis on business ethics to all employees through educational training and online tests, at the same time, ensure that employees clearly understand the operating procedures of the reporting mechanism. The completion rates of the courses related to "material information, insider trading prevention, ESG development, ethical management, internal control & risk management training", and "social responsibility & business ethics", etc. organized by the Company for entire employees in 2024 all reached 100%, and the nine directors of this term all completed the educational training on anti-corruption policies.

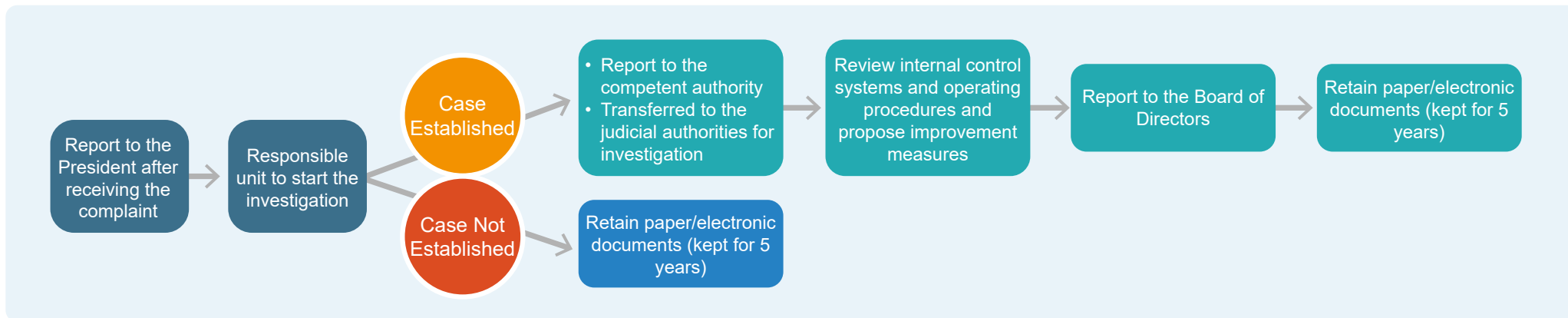
Complaint and Reporting Mechanism

PSMC has set up the "Business Ethics Maintaining Whistleblowing System" and "whistleblowing mailbox", through which external whistleblowers can report unethical cases in an anonymous or a named way to the Company. In addition, whistleblowing channel information is conveyed to external stakeholders through the Company's ESG website, orders and invoices for stakeholders to make whistleblowing reports or file complaints if any violations are identified.

Reports, Complaints and Suggestions Incidents	Reporting, Complaint and Suggestion Channels
Business Ethics Related Issues	- Report email : ethic@powerchip.com - Professional Ethics Incident Reporting System (for internal use only)



PSMC's Professional Ethics Advancement Committee is responsible for handling complaints and will follow the following procedures:



When the Investigation Team to immediately create a file upon receipt of a reported case and notify the actual informant, complainant, or advocate in writing within 14 working days to present the facts in person, and the Investigation Team may conduct an investigation based on the facts. Additionally, the complaint investigator shall declare in writing that the identity of the "whistleblower" and the content of the report shall be kept confidential, and establish "Whistleblower Protection and Anti-Retaliation Management Procedures" to protect employees, suppliers, and other internal or external parties from exercising their whistleblowing rights in accordance with the law, and provide them with due protection and immunity from retaliation. In case of anonymous reports, the investigation team shall take into consideration the improvement of the case. If there is any intentional planting of evidence, smearing, defamation, or damage to the character of others, the case shall be handled in accordance with the relevant laws and regulations in order to strengthen the efficiency and effectiveness of the operation of the complaint and reporting mechanism.

Corporate Donations and Related Expenses

The technical innovation in domestic semiconductor industry is a key factor for Taiwan to gain a foothold in the global market. Through industry-government-academia collaboration and investment, a more completed semiconductor ecosystem shall be established. PSMC has been investing resources in supporting domestic semiconductor industry development for a long time by focusing on issues including technical innovation, environmental sustainability, talent cultivation and development, supply chain management, etc. to continuously provide energy for industrial innovation and breakthrough.

2024 Issues of Focus and Investments

Issue	Description of the position/level of involvement	Investment Capital (NTD\$)
Talent Nurturing	Based on the "Act of National Key Fields Industry-University Cooperation and Skilled Personnel Training" passed by the Legislative Yuan in May 2022, the Ministry of Education approved the establishment of semiconductor academies at NTU, NYCU, NTHU, and NCKU to create momentum for continuous semiconductor technological advancement and the reinforcement of Taiwan's semiconductor industry's next-generation competitiveness. PSMC supports the operations of the domestic semiconductor academies with its practical actions to promote industry-academia collaboration and uphold the evolution of public higher education system policies. In addition, the Company has made monetary donations to National Tsing Hua University for the university to construct the Building of Technology and Engineering to optimize the university's academic environment.	225,000,000
Environmental Sustainability	In response to the global push for net-zero emissions, the 2050 net-zero emissions target, and the trend toward carbon border adjustment mechanisms in Europe and the United States, PSMC actively participates in domestic and international industry associations such as the Taiwan Semiconductor Industry Association (TSIA) and the Allied Association for Science Park Industries (ASIP). Through these platforms, PSMC engages in exchanges with industry peers to share insights and experiences in implementing environmental sustainability initiatives and advocates for sustainability-related policy needs and initiatives to the government. At the same time, PSMC collaborates with supply chain partners to continuously focus on climate change-related measures and regulations, including energy management, water management, waste management, and air pollution prevention. In 2024, Assistant Vice President Li-Wen Ting serves as the Chief Coordinator of the Environmental Protection Committee within ASIP, and also participates in the TSIA's Environmental, Safety and Health Committee. PSMC remains committed to working with industry partners to contribute to environmental sustainability.	1,619,025

2024 Issues of Focus Association Participation and Investment

Name of Association	Main Activities	Investment Capital (NTD\$)
Taiwan Semiconductor Industry Association (TSIA)	The mission of the Taiwan Semiconductor Industry Association (TSIA) is to promote mutual cooperation among its member companies and within Taiwan's semiconductor industry, enhancing the industry's overall development. TSIA represents the industry in international negotiations, participates in the formulation of international standards, and engages in activities related to the semiconductor sector. Additionally, TSIA conveys industry opinions and needs to the government, providing a reference for policy-making related to the semiconductor industry. By ensuring that the government understands the association's direction and requirements, TSIA seeks governmental support to strengthen and further develop Taiwan's semiconductor industry. As a standing director of the association, PSMC actively participates in discussions and advocacy initiatives promoted by TSIA, including: Jointly issuing the "Taiwan Semiconductor Industry Association Net-Zero Emissions Voluntary Reduction Goals"	1,205,025
Allied Association for Science Park Industries	The Allied Association for Science Park Industries (ASIP) encompasses six major industries: Integrated circuits; computer peripherals; communications; optoelectronics; biotechnology; and precision machinery. ASIP operates through cluster synergy, vertical integration, and specialized division of labor to construct an effective operational mechanism. A variety of specialized committees are established to address and advance professional issues relevant to member companies. In 2024, as a standing director of the association, PSMC actively participated in discussions and initiatives through the ASIP, including: - Hsinchu Science Park Occupational Safety and Environmental Protection Month events - Joint Audit Program for Waste Disposal and Treatment Vendors - Establishing the 2050 Low-Carbon Science Park Promotion Platform - Carbon Neutral Alliance Consultation Meeting of the Chinese National Federation of Industries - Recommended amendments to the Carbon Fee Draft Regulations - Water and Energy Conservation and Carbon Reduction Assistance Program for Hsinchu Science Park Companies (2023–2024) In addition, PSMC's Assistant Vice President for Environment, Safety, and Health (EHS), Li-Wen Ting, serves as the Chief Coordinator of the Ministry of Environmental, representing the association at various workshops and discussions held by the Ministry of Environment (MOENV). He also collaborates with the MOENV in policy advocacy, establishing an industry dialogue platform to communicate current policy developments, regulatory updates, and industry trends. This initiative helps corporate executives stay informed about environmental policies and legal requirements while fostering consensus on environmental protection issues, contributing valuable insights for the government's environmental policy planning.	414,000
Global Semiconductor Alliance	The GSA is a platform that brings together enterprises to establish a highly efficient, profitable, and sustainable global semiconductor and high-tech ecosystem, encompassing semiconductors, software, solutions, systems, and services. The GSA provides a unique, neutral collaboration platform where global senior executives can interact and exchange ideas with industry peers, partners, and customers to drive innovation. This not only accelerates industry development but also maximizes returns on investment and intellectual capital. The GSA has also established the Women's Leadership Initiative (WLI), which aims to increase female participation in boardrooms and leadership positions. Through the influence of female leaders in the semiconductor industry, the WLI advocates for greater public awareness of healthcare, the eradication of extreme poverty, the importance of clean water, climate change mitigation, and equal access to education opportunities.	207,285

Public Affairs Participation and Investment

Unit: NTD\$

Category	Investment Capital			
	2021	2022	2023	2024
Interest Group Lobbying	0	0	0	0
Donations to political organizations, candidates, etc.	0	300,000	0	0
Participation in industry unions, trade organizations, think tanks, or non-profit organizations	1,904,703	2,593,871	2,879,909	3,011,640
Other (e.g. election bills, referendum-related expenses, etc.)	0	0	0	0

Note: All donations are in cash.

Column

Implement RBA Core Spirits and Fulfill Ethical Management Culture



Based on the regulations related to labor, environmental health and safety, and business ethics stipulated by the Responsible Business Alliance (RBA), PSMC has corresponding formulated preventive measures. The Company conducts systematic examination every year on the compliance of the Company's systems concerning workers' human rights, health and safety, environment, business ethics and management systems' risk assessments, feasible measures, and actual operations. In addition, a standardized risk assessment template (Self-Assessment Questionnaire, SAQ) designed by RBA is used to identify the risks related to the actions and management systems adopted for five major aspects of business operations, which are labor, health and safety, environment, ethics, and management systems. Dedicated units have been assigned to be responsible and implement each aspect's relevant affairs. The 2024 SAQ self-assessment scores achieved by the Headquarters and all fabs all fell within the range of "Low Risk". In response to the RBA 8.0 revision and our new P5 fab going into mass production in 2024, modifications were made to comply with the updated requirements. The scope of internal audits was also expanded to include interviews with fab operators and labor service-based contractors to better understand the issues that employees prioritize. The goal is for all fabs to pass the RBA VAP verification by the end of 2025.

Item \ Fab	P1/2	P3	8A	8B
SAQ Score	85.5	90.0	91.4	86.3
SAQ self-assessment score: Low risk(≥ 85), Moderate risk($\geq 65 \& < 85$), High risk(< 65)				

With the Responsible Business Alliance (RBA) pledge, PSMC demonstrates that it is working together with all of its employees, suppliers and third parties to fulfill and comply with the requirements of the international community and its customers, and fulfilling commitments towards social responsibility as a world-class company.

Please refer to the Company's official website for the RBA pledge and certificate : <https://esg.powerchip.com/en-global/news/content/rba>



6.3 Regulatory Compliance

6.3.1 Regulatory Compliance Management

PSMC has established a regulatory compliance system to help personnel in all departments clearly understand and comply with the laws and regulations related to their business activities, and to avoid financial and reputational losses resulting from legal and regulatory violations. This system includes regulatory tracking and assessment, identifying potential compliance risks in business operations, formulating internal compliance policies, and regular education and training programs. The Board of Directors and the Legal Affairs Office are further responsible for overseeing the compliance status of each department.

PSMC adjusts its annual legal compliance focus in accordance with its business development priorities, and proceeds to identify, manage, and publicize laws and regulations related to product import and export, labor safety, environmental protection, and securities trading. In accordance with domestic laws and regulations, the Company has also established the principle that a single criminal law case or administrative law case with a penalty of NT\$1 million or more, or a major incident that seriously affects the Company's operations, must be disclosed, and major losses and improvements must also be publicly disclosed. In addition, we will review our operating procedures and strengthen our auditing mechanism to prevent the recurrence of such incidents. As of the end of 2024, no major violations of environmental or social laws and regulations, or anti-competitive, antitrust or monopolistic practices have resulted in financial penalties at any of our locations.

2024 Regulatory Compliance Status

Human Rights Laws and Regulations

- Compliance with Labor Standards Act
- No use of child labor
- No incident involving discrimination or harassment
- No infringement of indigenous rights
- No infringement of freedom of association and assembly
- No forced labor

Anti-competitive Behavior

- No anti-competitive behavior involved
- No Antitrust Incidents
- No monopolistic incidents

Relevant Company Operating Laws and Regulations

- No violation of the Company Act
- No violation of Commercial Law
- No violation of securities and financial laws and regulations
- No incident involving corruption or bribery, conflict of interest, money laundering, or insider trading
- No incident involving customer privacy data leakage

2024 Legal Compliance Educational Training

Training Format	Course Topics	Frequency	Target Participants	Completion rate (%)
Online	• Training on PSMC's social responsibilities and business ethics • Material information, insider trading prevention, ESG development, ethical management, internal control and risk management	Regular	All employees	100
Online and in-person	• Information concerning the export of finished strategic high-tech commodities	Regular	Customs & business related units, legal affairs & product technology units, system management units, and production planning units	100

6.3.2 Internal Control and Internal Audit

The auditing office under the Board of Directors of PSMC is staffed with the full-time auditors. The audit office is responsible for coordinating internal audit operations, planning annual audit projects and conducting audit assignments with reference to the regulations set forth by the competent domestic authorities and risk assessment results, as well as performing project based audits from time to time as needed, in order to assist the Board of Directors and managers in checking and evaluating the effectiveness of the internal control system duly, so that PSMC can reasonably ensure the achievement of the following objectives under the operation of the internal control system:

- Effectiveness and efficiency of operations (including profitability, performance and safeguarding of assets)
- The report has reliability, timeliness, transparency and is in compliance with relevant regulations
- Compliance with relevant laws and regulations

PSMC's internal control system includes a self-monitoring mechanism. Whenever deficiencies are identified or potential risks are detected, immediate corrective actions must be taken to prevent the deficiencies from expanding their impact. In addition, the Audit Office supervises each unit's and subsidiary's internal control system implementation, and conducts annual reviews of the self-assessment results. It also adjusts internal control design and implementation procedures in a timely manner to facilitate the Company's self-supervision mechanisms and ensure the quality of relevant implementation. For deficiencies indicated in the aforementioned self-assessment results or identified during Audit Office's audits, and the improvement status of anomalous items, audit reports are regularly issued and then reviewed by the Audit Committee as a principal basis for PSMC's board of directors and president to issue the Statement of Internal Control System. The 2024 statement that confirmed the effectiveness of PSMC's internal control system design and implementation was approved by the board of directors on February 25, 2025. With precise internal control system implementation, no major internal control deficiency was identified in 2024.

In accordance with the 2024 revision to the Regulations Governing Establishment of Internal Control Systems by Public Companies, listed companies are required to incorporate sustainability information management into their internal control systems. In compliance with these regulations, PSMC's Board of Directors approved the revision of the internal control system in the second half of 2024. Starting in 2025, sustainability information will be included within the annual audit plan, and beginning in 2026, the Company will report the results of the annual audit on sustainability information management.

6.4 Responsible Taxation

Tax Governance Policy

To match up with global tax governance trends, PSMC formulated its "Tax Governance Policy" aimed at supporting enterprise innovation and promoting economic growth in 2022 in accordance with tax laws and regulations applicable in the places of business operations, and the "OECD Transfer Pricing Guidelines for Multinational Enterprises and Tax Administrations". The policy has been implemented after being resolved by the board of directors to ensure the effective control of tax risks to protect shareholder value and equity.

Tax Governance Policy :

https://esg.powerchip.com/upload/media/sustainability_development/Tax_Governance_Policy.pdf



Tax Governance Policy Highlights

- **Tax management** : Regulatory Compliance, Honest Taxation
- **Tax Relations** : Healthy, Effective, Transparent and Mutual Respect
- **Prudent assessment of tax risk and impact** : conduct taxation assessment to manage risk and create shareholder value
- **Transparency of tax information** : Tax information is disclosed in accordance with relevant regulations and standards, and is regularly disclosed to stakeholders through public channels

Tax Governance Unit

The Office of Accounting is PSMC's tax administration unit, for which personnel with professional knowledge and practical experiences are appointed to carry out general tax administration and management affairs. In addition, departmental meetings are held on a monthly basis to review and convey contents of laws and regulations to relevant personnel. In response to the expansion of PSMC Group's operating bases, the Company has planned to incorporate external professional institutions' tax-related advice into its tax management system to ensure the tax policy's compliance with the latest laws / regulations and global transformation trends, and the effectiveness of the Company's tax administration operations. To avoid the financial losses caused by tax risks, relevant tax pre-assessments shall be conducted before major operating decisions and transactions are made to thoroughly consider the tax laws and regulations applicable in the places of business operations and relevant international tax guidelines so that the outcome of double taxation can be avoided under Audit Committee's supervision, and PSMC's compliance with each operating base's tax laws and regulations can be guaranteed. In addition, PSMC promises that its management framework and transaction arrangement are consistent with substantive economic activities, and shall not carry out any tax planning for the purpose of tax avoidance, nor shall it transfer corporate profits to low-tax countries or regions, demonstrating the Company's commitment to establish a cooperative relationship with local tax authorities based on mutual respect, transparency, and trust.

Tax Information

Country-specific tax information

Unit: NT\$ in thousands

Names of all the resident entities	Primary Business	Number of employees	Operating Revenue	(Loss) Income before Income Tax	Income Tax (Income) Expense	Income Tax paid
Taiwan	Wafer Foundry OEM	8,532	44,725,710	(6,716,046)	(58,536)	267,573
Japan	Integrated Circuit Research and Development	35	0	(115,703)	4,086	4,012

Effective Tax Rate and Cash Tax Rate

Unit: NTD\$ in thousands

Item	2023	2024	Average
(Loss) Income before Income Tax	(2,086,038)	(6,831,749)	(4,458,894)
Income Tax (Income) Expense	(442,497)	(54,450)	(248,474)
Effective Tax Rate (%)	21.21	0.80	5.57
Income Tax paid	3,940,481	271,585	2,106,033
Cash Tax Rate (%)	(188.90)	(3.98)	(47.23)

PSMC's 2024 effective tax rate was 0.80%, which was lower than the average effective tax rate of 13.96% for the global "semiconductors and semiconductor equipment" industry announced by CSA Handbook 2025. The main reason for this is that losses for 2 consecutive years. However, the income tax paid in 2024 was calculated based on income before income tax of the previous year, and a "net loss before income tax" was presented in 2024, so the 2024 cash tax rate was (3.98) %, which is lower than the current year's effective tax rate of 0.80% and the average cash rate of 13.82% for the global "semiconductor and semiconductor equipment" industry announced by CSA Handbook 2025.

6.5 Risk Management

6.5.1 Risk Management Culture

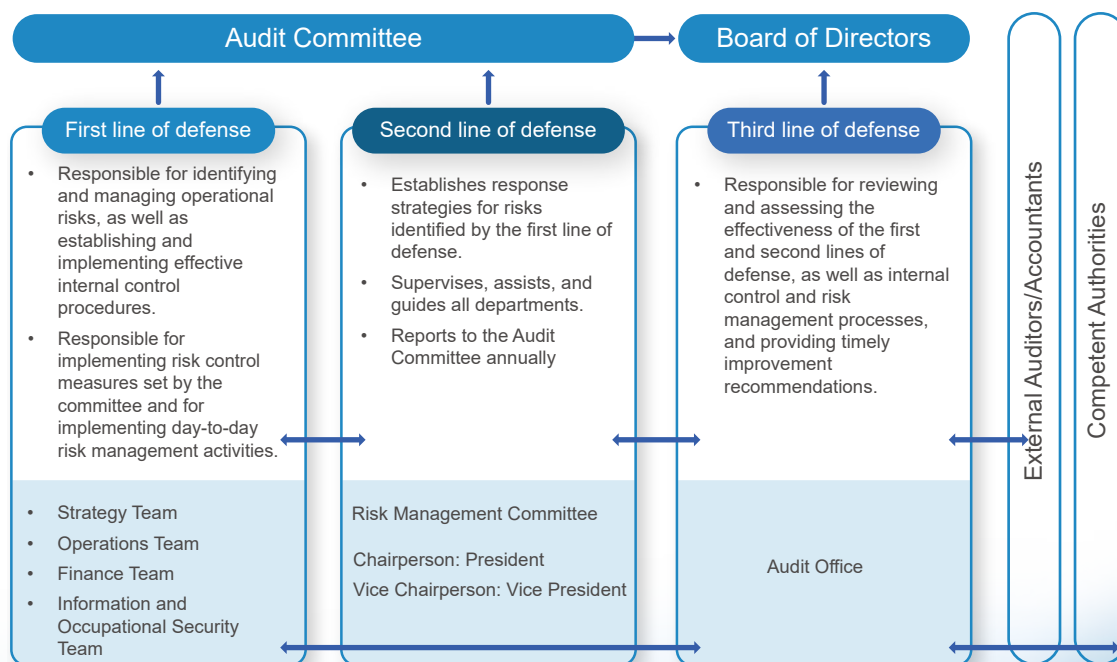
Risk Management Framework

The Board of Directors serves as PSMC's top governance unit for risk management. PSMC formulated "Risk Management Policy and Procedures" in 2023, which has been approved by the Board of Directors and shall serve as the highest guiding principles for risk management. The Audit Committee is responsible for strategic guidance and supervising the implementation of risk management measures to ensure risk management effectiveness. The Risk Management Committee under its jurisdiction is responsible for planning, executing and supervising risk management and operational compliance and other related matters to ensure that management goals can be achieved. And the Risk Committee appoints a front-line business implementation team to be responsible for implementing the risk control measures set by the committee and daily risk management activities. In addition, risk management outcomes are also included as audit items in the annual audit plan. Through internal audit operations, PSMC ensures that all risk categories involved in business operations are effectively controlled, providing an objective and independent assurance of the Company's risk management system and its effectiveness. With three lines of defense in risk management, PSMC aligns risk management with the Company's strategies and goals, defines key risk areas, and enhances the comprehensiveness, foresight, and completeness of risk identification results.

Shaping the Risk Culture

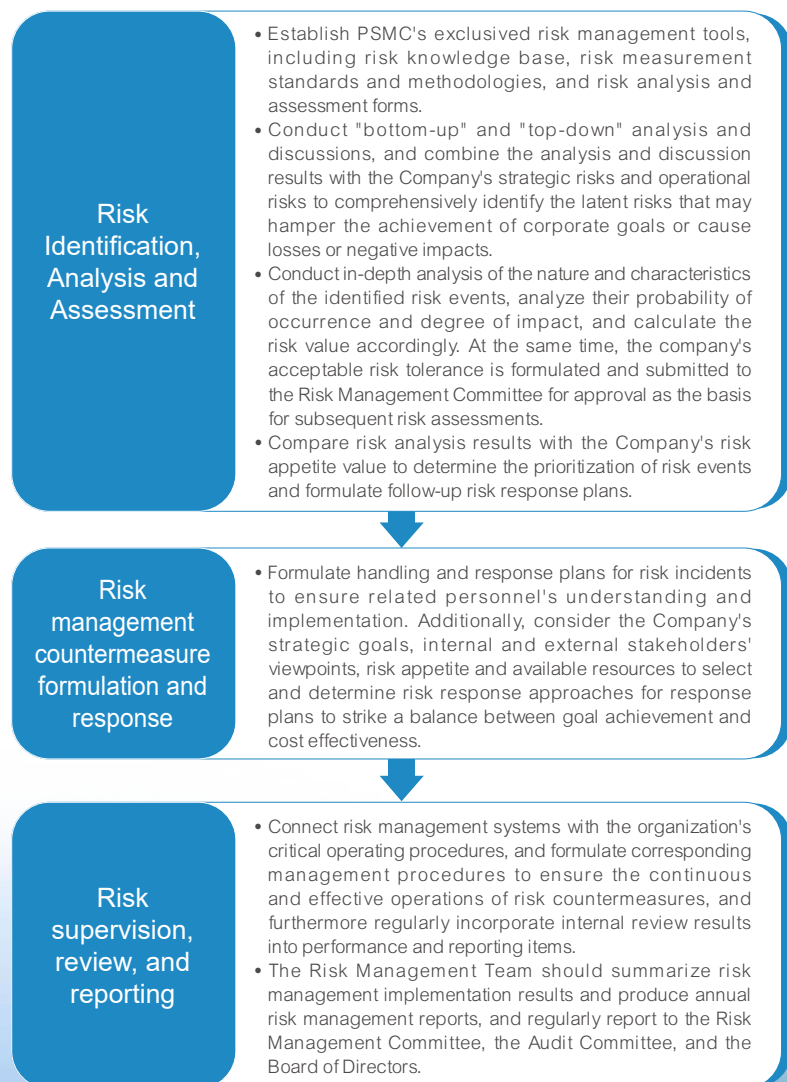
To enhance the risk awareness of all employees and establish a corporate culture of implementing risk management on a daily basis are core philosophy of risk management at PSMC. PSMC has included operational risk items such as climate and occupational safety and health in the performance evaluations of senior executives and business department heads, and has developed various financial incentives to encourage management to refine their risk management capabilities over the long term. In addition, PSMC also includes "Risk Management and Crisis Awareness" in the employee performance evaluation, so that all employees clearly understand the potential risks in daily business. PSMC not only provides regular risk management training for non-executive directors responsible for supervising operations, but also conducts a total of 6 company-wide risk management on line education and training courses for front-line implementation business colleagues in 2024, sharing risk practice cases, covering risk aspects such as human resources, marketing communication, compliance, reporting, information, corporate governance, sustainability, finance, law, supply chain, etc., to integrate risk management awareness to daily decision-making and operational activities.

PSMC has also established measures and channels for individual employees to proactively identify and report potential risks to the entire organization, including a whistleblower mechanism such as the "Feedback Corner" online platform, an extension line and email, and encourages employees to participate in a structured feedback process to continuously refine the risk monitoring and management system. In addition, PSMC incorporates risk assessment standards into the raw material procurement, product development or approval stages. And according to the internal innovation management methods and proposal improvement system, when employees propose specific and feasible risk management or other improvement suggestions, they will be rewarded according to the degree, and the risk management indicators will be strengthened to be linked to financial incentives. We will continue to introduce various innovative risk culture activities that are conducive to risk assessment and management, and shape a comprehensive corporate risk management culture.



6.5.2 Risk Identification Process

PSMC regularly conducts risk identification, assessment, and internal audits every year, and conducts rolling reviews in a timely manner to ensure the effectiveness of risk management procedures and related control measures, and to continuously improve.



Risk Management and Corresponding Strategy

With the operations of systematic risk management mechanisms, the effectiveness of management measures and response strategies can be significantly improved, and the impact arising from potential risks shall be minimized. The potential risks and response strategies related to information security, climate change, human rights, etc. shall be described in separate chapters. For the remaining major risks identified by PSMC in 2024, the analysis results and response strategies are listed in the table below :

Strategy

Risks from the improvement of global semiconductor technology strategies, which will hamper the Company's growth in the future

Risk Description	Potential Impacts
If the stages of determining new product / technology platform, developing management and mass-production transition management measures, and enhancing mass production yield are not comprehensively completed, there will be adverse effects on the Company's overall reasonable resource deployment and business expansion. This trend will pose severer competition on the Company in the future.	- Failure to use new technologies shall lead to increased costs, lowered product or service quality, and decreased productivity. - Failure to complete the development and launch of new technologies in time shall make the Company miss out on market opportunities. - Prolonged product development timeline and the introduction of new technologies that affect product development timeline and product quality shall lead to increased costs, lowered product or service quality, and decreased productivity.
Risk Tolerance Level	Corresponding Strategy
- Reputation: Moderate-level risk: Delayed technical development timeline shall affect specific customers' product plans and become a gossip topic among peer companies, by which enterprise reputation shall be impacted. - Marketing timing's lower hit rate shall lead to delayed development timeline, which will reduce customers' confidence in the Company, and is classified as a high-level risk.	Enhance technical capabilities, including: - Implement and strengthen development procedures standardization and self-directed management. - Implement and strengthen project review committees, business development meetings and vertically integrated manufacturer mechanisms. - Establish a dedicated organization to enhance the consistency and accuracy of design rule specifications during the development phase. - All platform management related to IP must be formally discussed and resolved in the IP Committee meetings. - List handover items for different modules to reduce product defects caused by unaddressed improvements during handovers. - Continue discussions with customers and design companies on technology platform development and product planning. - Utilize photomask and test revisions to enhance product performance and reduce design flaws.

Operation

Global semiconductor companies compete for talent, leading to risk of future talent shortage

Risk Description

- In order to meet market demand, the world's major wafer manufacturers have significantly increased capital expenditure and expanded recruitment scales. However, being affected by increasingly intensified birthrate decline in Taiwan, the manpower is obviously falling short of the level needed to support the development of the industry and the Company, making the industry faced with difficulties in talent recruitment expansion.
- The Company's core manpower loss shall lead to decreased competitiveness, and the failure to develop suitable talent / employee retention policies and the lack of or the incomplete succession personnel cultivation plans shall also cause core manpower loss and other related problems, and affect the organization's operations and production.

Potential Impacts

- Talent shortage: Foundries may be faced with difficulties in recruiting and retaining competent technical talents. Moreover, with the worldwide increasing demand for semiconductor industry talents, foundries also need to cope with intensified competition for talents among peer companies.
- Decreased production efficiency: Being affected by core personnel gap caused by the lack of key talents or complete succession personnel cultivation plans, foundries' production efficiency may decrease and their market competitiveness and profitability may be affected.
- Decreased technological innovation capabilities / business competitiveness: Failure to attract talents and retain competent technical talents who possess market foresight and comprehensive market views shall limit foundries' technological innovation capabilities and affect their positions in the market.

Risk Tolerance Level

- Increased costs and difficulties in recruiting new employees: Moderate-level risk: With the keen competition prevailing in Taiwan's talent market, increased salary may cause specific impacts to operating costs.
- Inadequate technology R&D talents: High-level risk: The lack of technology R&D talents shall affect the Company's new process development and long-term technical competitiveness.
- Talent gap: Moderate to high level risk: Incomplete succession personnel cultivation plans shall lead to core manpower loss.

Corresponding Strategy

- Talent succession planning, including:
- Investment in training and development: Invest resources in talent training and development to enhance employees' skill and knowledge standards, and expand the Company's talent base.
 - Build up a culture that will attract talents: Provide flexible work arrangement and diverse benefits and compensation programs to build up a corporate culture that will attract and retain competent technical talents.
 - Reinforce industry-academia collaboration: The Company has collaborated with top universities in establishing semiconductor academies to reinforce industry-academia collaboration and carry out joint research and development projects to attract competent technical talents. PSMC will continue to place emphasis on talent management by formulating effective talent management strategies to attract and retain competent technical talents to ensure the Company's market competitiveness and long-term development.



Emerging Risks

Emerging Risk	Risk Category	Risk Description	Potential Impact	Response Measures
U.S. CHIPS and Science Act	Geopolitics	The U.S. chip sanctions on China's semiconductors might work to a certain extent, yet they have forced Chinese government to expand investment in the semiconductor industry. As a result, the local momentum in China has increased rapidly, particularly the mature processes above 28nm, which have already posed a threat to Taiwan's second-line wafer fab.	Product lines' total demand amount is shared by competitors, which will affect the long-term growth of the businesses between the Company and customers. If the development timeline cannot be properly managed, it will impact the Company's mid-term order intake and wafer production.	1. Continue to develop customized processes based on current technology level and customers' needs to enhance customer stickiness. 2. Plan the development of advanced process platforms to widen the competitive gap. 3. Pay regular visits to customers (and customers' customers) to understand market status and stay on top of customers' needs in a timely manner, which will help the Company stay one jump ahead of competitors to launch new products and enjoy the bonuses for first movers. 4. Optimize processes based on existing platforms to meet customers' needs and optimize cost competitiveness. 5. Establish a dedicated organization to strengthen progress tracking for engineering horizontal deployment. 6. Provide indicators regarding evaluation project to assist senior management in decision-making and focus development resources.
Climate change	Environment	Failure to monitor and reduce greenhouse gas emissions, and lack of a specific, effective carbon reduction plan.	If the Company fails to formulate response strategies in accordance with government-promoted policies (such as internal carbon pricing, carbon fees, and net-zero pathways), subsequent input costs will gradually increase.	1. Formulate and implement corporate responsibility and sustainable development policy regulations. 2. Continuously monitor legal regulations and promptly update corporate policies and planning. 3. Establish an ESG project team for improvement initiatives. 4. Incorporate greenhouse gas considerations into technology development to minimize emissions. 5. Enhance awareness of greenhouse gases, such as the EU's emission reduction timeline.
Artificial Intelligence	Information Technology	When introducing data, inadequate analysis and risk assessment are conducted, leading to the direct adoption of information without proper evaluation.	Bias in machine learning affects data prediction results, leading to misinformed decision-making.	1. Privacy – Commit to safe and reliable artificial intelligence (AI) while ensuring compliance with relevant laws and fulfilling corporate, social, and ethical responsibilities. 2. Accountability and governance – Apply principles that prioritize social and ethical responsibilities in AI applications to prevent any human rights violations. 3. Apply review mechanism to continuously verify and supervise AI applications, implementing strong security controls to prevent accidental harm and malicious threats. 4. Ensure human oversight of AI. Help users understand AI functions and provide a transparent data process. 4.1 Data identification – Machine learning outcomes, derived from historical data, will be regularly reviewed and verified with users. 4.2 Identifying incorrect data and models – System personnel will make necessary corrections. 4.3 Updating incorrect data and models – Re-run predictions after corrections to reduce biases generated by the model.
Talents and Succession Plan	Social	When faced with Chinese national industries' talent attraction policies and the active talent recruitment strategies adopted by leading companies in the industry, if the Company fails to attract and retain employees, manpower loss will occur and the organization's operations and production will be affected.	1. Key talent drain or gap 2. The lack of required talents or skills shall impact the Company's operations. 3. Decreased business competitiveness 4. Failure to attract and retain talents shall affect the implementation of business plans.	1. Higher level managerial personnel nominate succession candidates in writing on a yearly basis based on candidates' work performance and personal characteristics. HR units provide planned training to strengthen the candidates' strengths for them to meet succession requirements. 2. Activate headhunting projects, and optimize recruitment plans to recruit key talents from external human resources.

Sensitivity Analysis and Stress Testing

Financial Risk

Risk category: Interest Rate Changes

Unit: NTD\$ in thousands; %

Item	Year	2023	2024
Consolidated interest expense		626,568	1,534,307
Consolidated net operating revenues		44,022,552	44,725,710
Consolidated interest expense as a percentage of net operating revenues		1.42	3.43

Corresponding Strategy

Interest expense as a percentage of net operating income for 2023 and 2024 is 1.42% and 3.43%, respectively, which is a small percentage for each period. This is mainly due to the interest incurred by the Company in financing its borrowings from financial institutions for operating needs and capital expenditures, and although market interest rates fluctuate, they do not yet have a significant adverse effect on all revenue and profitability. PSMC will monitor and analyze in a timely manner the cash flow impact generated from interest rate changes in the financial market on all of the Company's interest-bearing liabilities, maintain good relationships with banks, and conduct timely assessment of the potential interest rate risks which the interest-bearing liabilities are subject to, thereby reducing the impact of interest rate changes on the year's profit / loss.

Risk category: Fluctuations in the exchange rate

Unit: NTD\$ in thousands; %

Item	Year	2023	2024
Consolidated net foreign currency exchange gain (loss)		(82,894)	1,226,022
Consolidated net operating revenues		44,022,552	44,725,710
Consolidated net foreign currency exchange gain (loss) as a percentage of net operating revenues		(0.19)	2.74

Corresponding Strategy

PSMC's net foreign currency exchange gain (loss) in 2023 and 2024 accounted for (0.19) % and 2.74 % of the year's net operating revenue respectively. Since the Company's capital expenditure and manufacturing costs are mainly paid in US dollars or Japanese yen, and most revenue comes from US-dollar income, its revenue performance is significantly influenced by exchange rate fluctuations. In response to this risk, by considering exchange rate changes in the market and the Company's actual positions and funding status, PSMC adopts natural hedging strategies and uses spot foreign exchange transactions and forward foreign exchange contracts to hedge exchange rate risks within the scope permitted by the Company's policies.

Risk category: Inflation

With the interference by factors such as inflation, destocking, etc. in the industries, the global market demand became slacked in 2024. In response to this situation, PSMC keeps monitoring price fluctuations in the market, and maintaining good interactions and relationships with suppliers and customers to actively adapt to the slack-off impact caused by inflation to the Company's overall profit performance.

Non-Financial Risks

Risk category: Potential Risks of Capacity Expansion

The 12-inch wafer fab constructed in Tongluo Science Park, Miaoli officially commenced mass production in 2024. However, demand forecasts may undergo significant changes due to fluctuations in the market environment. In the event of reduced demand, the purchased equipment and additional hired personnel may impact PSMC's profitability.

Corresponding Strategy

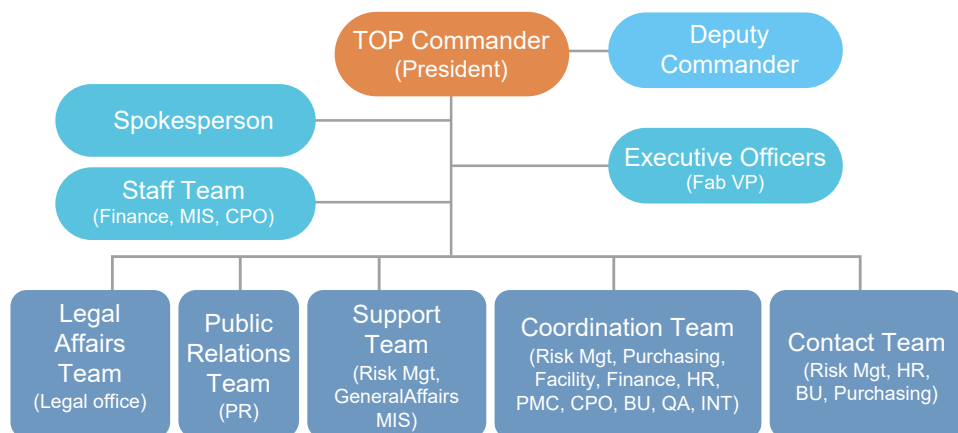
In the medium to long term, foundry capacity is still a key resource for customers' growth, and customers will actively acquire more capacity. However, in 2024, there is significant pressure in the mature semiconductor process market, due to weak demand in the end-consumer electronics market and the aggressive capacity expansion of mature processes in China. PSMC will continue to observe market changes and closely collaborate with customers. In the events when market demand keeps falling short of the expected level, the Company shall adjust production capacity plans on a rolling basis. In principle, plant and production capacity expansion projects shall be carried out throughout several phases or stages within the Company's affordable budget and the minimum risk level to concurrently achieve stable corporate growth and meet customers' production capacity needs.

Business Continuity Management System (BCMs)

With the aim of building up various units' ability to respond to and handle crisis incidents, the Business Continuous Plan (BCP) and themed drills are carried out has been formulated for specific risk scenarios such as prolonged power supply interruptions, water outages, fire accidents, earthquakes, climate change, raw materials supply interruptions, labor shortages, information system hacking, critical equipment failure, etc. to ensure that in the events of major accidents or crisis incidents, the normal operations can be resumed as soon as possible.

BCM Crisis Management Organizational Framework

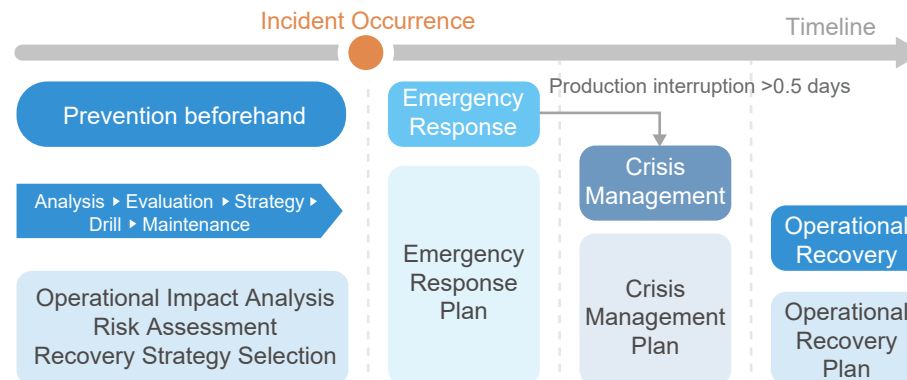
The President serves as the top commander, with the spokesperson, staff team (encompassing units of finance, information technology, and strategic planning) and fab executive officers serving as assistants to promote relevant affairs. Several functional teams have been set for different emergencies and incidents, including legal affairs team, public relations team, support team, coordination team, and contact team.



Implementation Results of Business Continuity Management (BCM) Plan

The annual drills focus on the pre-emptive measures for handling operational shocks, conducting operational shock analysis, risk assessment, and selecting recovery strategies. The internal departments of PSMC prepare Standard Operating Procedures (SOPs) for emergency events and initiate emergency response plans immediately upon the occurrence of the event. If the emergency is so severe that production is interrupted for more than half a day, a crisis management plan will be conducted to eliminate the crisis and eventually resume operations in the shortest time possible.

1. Plant Accidents 2. Plant Supply System 3. Information System 4. External Supply Chain



Year	Fab	Type	Scenario	Implementation Result
2021	8B	Electricity	Construction of LMOC(Land Mark Optoelectronics Corporation) dug out power pipelines, which caused a whole plant power outage, one-day production interruption and machine damage.	Personnel of various units worked together to activate emergency response measures, and production line operations were resumed within 90 minutes in all scenarios.
2022	P1/2	Cyber hacking	Ransomware disrupted and caused factory production interruption.	
2023	8A	Gas	The CPC main gas pipeline was accidentally cut during digging activities, which caused a major gas leak, and gas supply needed to be cut off for emergency repair. This accident made 8A Plant suffer one-day gas outage, resulting in cleanroom temperature / humidity deviation, VOC exhaust and process local scrubber equipment shutdown, and cessation of all machines.	
2024	P3	Earthquake	The magnitude 6 earthquake on April 3, 2024, caused a disruption in fab production lines.	

6.6 Information Security

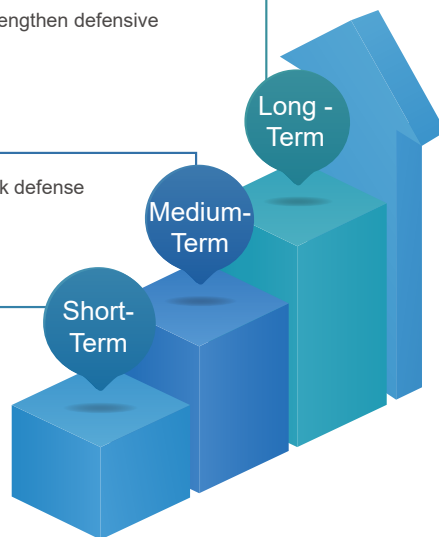
Information Security Management Objectives

With PSMC's continuous progress in technological development, information security and confidential information protection has become more and more important. PSMC has passed the ISO 27001 Information Security Management System Certification, and successfully completed recertification in March 2024. Through clear short-, medium-, and long-term information security objectives, the Company ensures that technological R&D achievements and trade secrets are properly protected.

- Cultivate the Red Drills Team to continuously strengthen defensive capabilities
- Develop a business secret registration system

- Introduce red team drills to improve hacker attack defense capabilities
- Strengthen business secret management

- Continuously strengthen employee education and training to enhance information security awareness
- Strengthen drills to shorten the recovery time
- Accurate identification and verification of business secrets



6.6.1 Information Security Management Framework

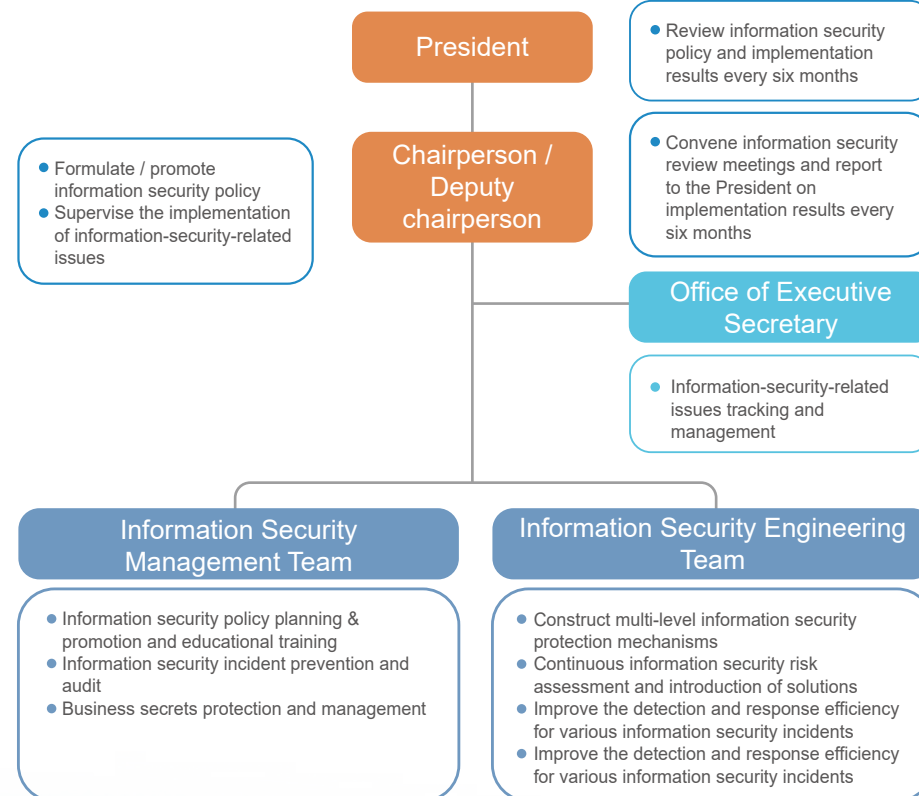
PSMC's "Information Security Committee" formed by representatives designated by related departments, and lead by president, follows the "Information Security Policy" and the "Information Security Management Measures" to formulate and implement information security-related strategies, plans, and risk management. The committee oversees the operation of information security management across departments and evaluates the implementation results of security plans. It also discusses and makes decisions on information security-related issues, covering aspects such as human resources, physical security, and information security. Under the Information Security Committee, there are two subordinate teams: the Information Security Management Team and the Information Security Engineering Team. These two teams are respectively responsible for information

security policy planning, and for education and training, and regularly report to the President. To ensure alignment between information security management and the Company's operational goals, the Information Security Committee also regularly reports to the Sustainable Development Committee, which in turn presents relevant matters to the Audit Committee and the Board of Directors to strengthen the information security management mechanism.

In the event of a major change or an information security incident, an emergency meeting will be convened to discuss and address the situation. Educational training and internal announcements are adopted every year to continuously share and communicate the latest information security news to employees to maintain employees' information security protection consciousness.

Information Security Committee Framework

Organization effective date: May 20, 2022



6.6.2 Implementation of Information Security Protection

Business Secrets Protection

Management policies and actions

- Formulate "Business Secrets Management Regulations", "Documents and Files Management Regulations", and other related regulations to ensure the information security and effective management of the Company's business secrets
- Introduce AI-based business secret detection and protection system (AIRS) to carry out real-time monitoring of printed document contents to check if any confidential information is contained with the aid of AI technology

Employees

- Employees are required to participate in the course of "Information Security Management / Control Measures" every year. All PSMC employees participated in the course in 2024, achieving a completion rate of 100%.
- Summarize recent information security incidents on a quarterly basis, and make information security rules known to all employees.
- Business secret documents should be labeled; supervisors' approval should be obtained for duplication or printing
- Strengthen access control and monitoring, manage information system access authorization, and retain and review access records

Business Continuity Management

PSMC performs regular server vulnerability scanning and system patching, and instantly fixes major deficiencies to ensure that the Company's operations are not affected by any threats. In addition, joint drills are regularly conducted by relevant units to ensure that PSMC can rapidly and effectively respond in the events of crises. PSMC conducted a ransomware attack disaster recovery drill in 2023, and completed to carry out the Web Application Penetration Testing in 2024. The drill result was a successful defense with no intrusion incidents. The recommendation report will serve as a reference for strengthening information security protection, ensuring the safeguarding of corporate assets and stakeholders' interests.

Information Security Incident Reporting and Handling Procedures

The incident reporting and handling process and information security risk levels are clearly documented in the Information Security Management Measures. All potential security incidents must be restored or have damage control completed within a specified timeframe after confirmation. The number of information breach incidents in 2024 is 0.



Information Security Incident Levels and Time Limit for Handling

Incident Level	Impact Level	Incident Level Description	Time Limit for Handling	Reporting Level
Level 1	Low	The incident caused temporary threats and hazards to information systems or services, and had little impact on data's confidentiality, integrity and availability. The response measures can be implemented through daily operating procedures.	Within 24 hours	Incident occurring unit's section-level and department-level supervisors
Level 2	Moderate	Operations of applications or systems are interrupted due to anomalous processing or external intrusion. The interruption timeline can still be controlled and preliminarily judged by the personnel in charge of the operations or systems, or the signs of attacks can be detected and the threats of possible attack damage can be estimated by information security personnel.	Within 12 hours	Incident occurring unit's director; and all PSMC employees should be notified by email or through public announcement
Level 3	High	- The incident may have a long-term impact on the Company, or cause hazards to important system operations. - The incident has caused significant impact or harm to information systems and services under the information security management system, and emergency response measures must be taken immediately to handle the situation.	Within 6 hours	Incident occurring unit's VP; and all PSMC employees should be notified by email or through public announcement

Privacy and Personal Information Protection

PSMC has formulated the Privacy Policy in accordance with Taiwan's Personal Information Protection Act and other related regulations, with the policy scope covering all employees, business partners, and users. The Company has also incorporated privacy-related issues into the scope of compliance management to identify relevant legal and regulatory compliance on a quarterly basis. In addition, internal audits, risk assessments, customer audits, and supply chain security audits are conducted every year, and the RBA External Audit Certification is conducted every two years to ensure that there is a just and reasonable connection between the information collection act and its purpose and also commit not to use customer information for a second time. Stakeholders' autonomy over private information can be enhanced. In 2024, there were no substantiated complaints concerning breaches of customer privacy and losses of customer data, or other violations of laws and regulations at PSMC, and the proportion of secondary use of customer information was 0%, demonstrating the company's personal data management and privacy commitment and effectiveness in safeguarding human rights.

Appendix

About this Report

Sustainability Information Disclosure Guidelines

Independent Third Party Verification Statement



About this Report

The Powerchip Semiconductor Manufacturing Corporation 2024 Sustainability Report is PSMC's seventh sustainability report, the contents of which were provided and compiled by various units of the Company. It is intended as a communication channel to address stakeholders' concerns, and aims to enhance transparency regarding the Company's sustainability plans, implementation history, and the status of its performance achievement, with the hope of driving more positive changes for the society.

Scope of Reporting

The reporting period is from January 1, 2024 to December 31, 2024. The scope covered in this Report includes the plant areas in Taiwan (the Headquarters and all fabs in Taiwan), and the subsidiaries PSMC Japan Corp. and Maxram Inc. in Japan. The reporting period and scope are consistent with those of the Consolidated Financial Report. The Report contains the issues of stakeholders' concern related to PSMC's concrete practices in economy and governance, environment, people and human rights, etc. The financial figures were cited from the Company's CPA audited annual financial reports, and are presented in New Taiwan Dollars (NTD). Some information includes data from the past three years to highlight long-term performance achievements. Any adjustments to previous data within any chapter is specifically noted.

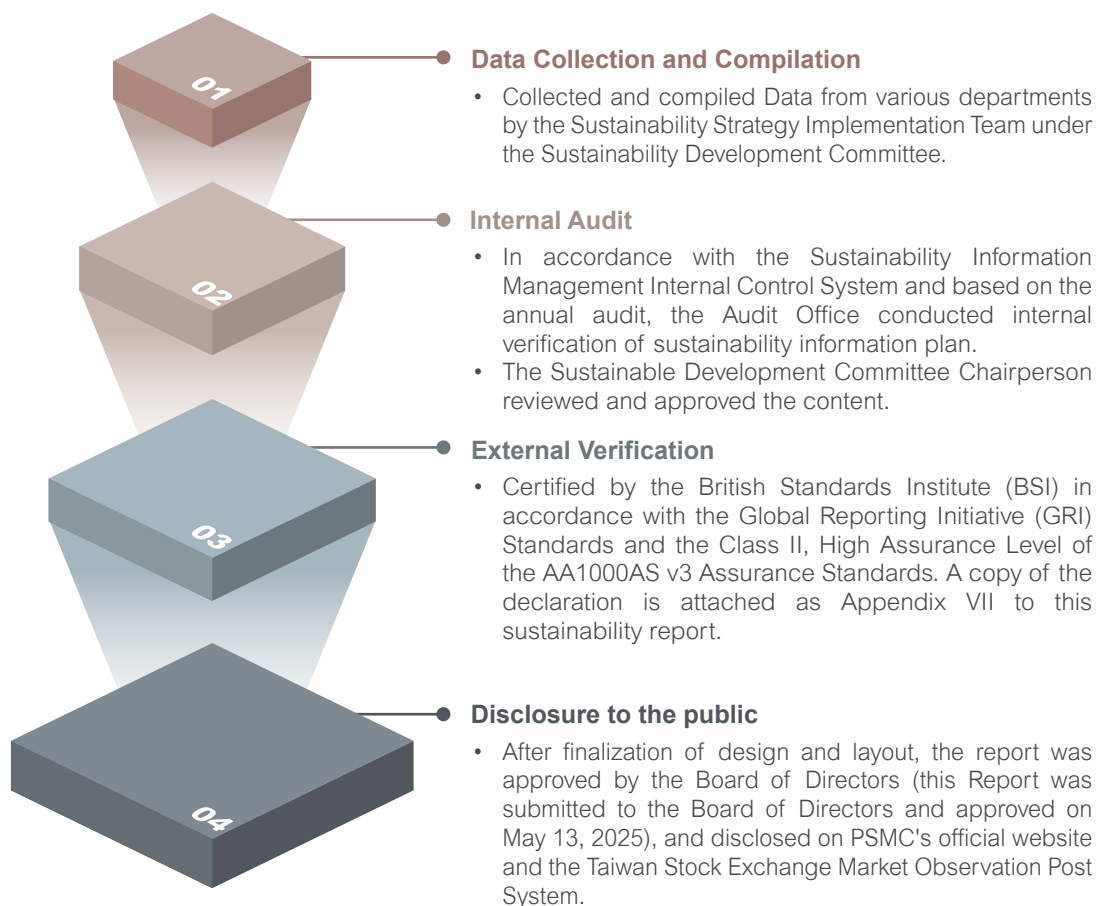
Standard of Reporting

The criteria followed and referred to in this report are listed below：

Standards Issuing Organization	Standards Guidelines
The Global Reporting Initiative (GRI)	GRI Universal Standards 2021
Sustainability Accounting Standards Board (SASB)	Industry Standards - Semiconductor Industry
Financial Stability Board (FSB)	Task Force on Climate-related Financial Disclosures (TCFD)
International Sustainability Standards Board(ISSB)	Sustainability Disclosure Standards IFRS S1 General Requirements for Disclosure of Sustainability-related Financial Information IFRS S2 Climate-related Disclosures
Taiwan Stock Exchange (TWSE)	Code of Practice for Sustainable Development of Public Traded Companies
	Procedures for the Preparation and Filing of Sustainability Reports by Publicly Traded Companies
United Nations (UN)	Sustainable Development Goals (SDGs)
	United Nations Global Compact
International Organization for Standardization(ISO)	ISO 26000: 2010 Guidance on Social Responsibility

Management Process

In accordance with the Sustainable Development Best Practice Principles, this Report is prepared, and internal and external verification matters are planned. The process is explained below, and the final version of the Report is approved by the Board of Directors before finally being publicly released.



Release Date

- Previous Year's (2023) Sustainability Report Issue Date: June 2024
- Current Year's (2024) Sustainability Report Issue Date: June 2025
- Next Year's (2025) Sustainability Report is Scheduled for Release: June 2025

Contact Information

- Capital Planning Department, Powerchip Semiconductor Manufacturing Corporation
- Address: No. 18, Li-Hsing 1st. Road, Hsinchu Science Park, Hsinchu City, Taiwan 300, R.O.C.
- Telephone: +886-3-5795000
- Fax: +886-3-5792040
- E-mail: csr@powerchip.com
- Website: <https://esg.powerchip.com/en-global>



PSMC Official Website



PSMC Official Website -
Sustainability Section

Sustainability Information Disclosure Guidelines

I : GRI Sustainability Reporting Standards Content Index Table

Terms of Use	Powerchip Semiconductor Manufacturing Corporation has reported the content for the period of 2024 (January 1, 2024 to December 31, 2024) in accordance with GRI standards.
GRI 1 used	GRI 1: Foundation 2021
Applicable GRI Industry Standards	No applicable GRI industry standards were published at the time of this sustainability report

GRI Standards Categories / Topics	Number	GRI Standards Disclosure Content	Corresponding Chapters	Page No.	Omitted/ Remarks
General Disclosure					
GRI 2: General Disclosure 2021	2-1	Organizational Details	Company Profile	4	
	2-2	Entities included in the organization's sustainability reporting	About this Report	139	
	2-3	Reporting frequency, period, and contact point	About this Report	139	
	2-4	Restatements of information	About this Report	139	
	2-5	External assurance/ reliability	Sustainability Information Disclosure Guidelines	154	
	2-6	Events, Value Chains, and Other Business Relationships	Company Profile 2.1 Supply Chain Overview	4 44	
	2-7	Employees	4.1 Talent Attraction and Retention	85	
	2-8	Workers who are not employees	4.1 Talent Attraction and Retention	85	
	2-9	Governance Structure and Composition	6.1 Corporate Governance	119	
	2-10	Nomination and selection of the highest governance body	6.1 Corporate Governance	119	
	2-11	Chair of the highest governance body	6.1 Corporate Governance	119	
	2-12	Role of the highest governance body in overseeing the management of impacts	Organizational Structure for Sustainable Development	10	

GRI Standards Categories / Topics	Number	GRI Standards Disclosure Content	Corresponding Chapters	Page No.	Omitted/ Remarks
GRI 2: General Disclosure 2021	2-13	Delegation of responsibility for managing impacts	Sustainable Development Strategy	10	
	2-14	Role of the highest governance body in sustainability reporting	Sustainable Development Strategy About this Report	10 139	
	2-15	Conflicts of interest	6.1 Corporate Governance	119	
	2-16	Communication of critical concerns	Organizational Structure for Sustainable Development	10	For more information, please refer to II. Corporate Governance Report-3. Corporate Governance Practices of the Company's 2024 Annual Report pages 16
	2-17	Collective knowledge of the highest governance body	6.1 Corporate Governance	119	
	2-18	Evaluation of the performance of the highest governance body	6.1 Corporate Governance	119	
	2-19	Remuneration policies	4.1 Talent Attraction and Retention 6.1 Corporate Governance	85 119	
	2-20	Process to determine remuneration	6.1 Corporate Governance	119	
	2-21	Annual total compensation ratio	4.1 Talent Attraction and Retention	85	

GRI Standards Categories / Topics	Number	GRI Standards Disclosure Content	Corresponding Chapters	Page No.	Omitted/ Remarks
GRI 2: General Disclosure 2021	2-22	Statement on sustainable development strategy	Message from the Chairman and Sustainable Development Committee Chairman	3	
	2-23	Policy Commitments	4.3 Human Rights 6.2 Integrity Management	99 123	
	2-24	Embedded policy commitments	4.3 Human Rights 6.2 Integrity Management	99 123	
	2-25	Processes to remediate negative impacts	Materiality Analysis 1.3 Customer Relationship Management 6.2 Integrity Management	13 40 123	
	2-26	Mechanisms for seeking advice and raising concerns	Stakeholder Communication 6.2 Integrity Management	21 123	
	2-27	Compliance with laws and regulations	6.3 Regulatory Compliance	127	
	2-28	Membership associations	Company Profile	4	
	2-29	Approach to stakeholder engagement	Stakeholder Communication	21	
	2-30	Collective bargaining agreements	4.2 Talent Nurturing and Development	94	
Material Topics					
GRI 3: Material Topics 2021	3-1	Process to determine material topics	Materiality Analysis	13	
	3-2	List of material topics	Materiality Analysis	13	
	3-3	Management of material topics	Materiality Analysis	13	
Economic and Governance Aspects					
Material Topic — Economic Performance					
GRI 3: Material Topics 2021	3-3	Management of material topics	Economic Performance Materiality Analysis	7 13	
GRI 201 Economic Performance 2016	201-1	Direct economic value generated and distributed	Economic Performance	7	

GRI Standards Categories / Topics	Number	GRI Standards Disclosure Content	Corresponding Chapters	Page No.	Omitted/ Remarks
GRI 201 Economic Performance 2016	201-4	Financial assistance received from government	-	-	The Company received a total of NT \$ 1,055,198 thousand from the government in 2024.
Material Topic — Product Liability and Quality					
GRI 3: Material Topics 2021	3-3	Management of material topics	Materiality Analysis 1.2 Product Liability and Quality	13 33	
GRI 416 Customer Health and Safety 2016	416-1	Assessment of the health and safety impacts of product and service categories	-	-	All products and services provided to customers have no significant impact on health and safety.
	416-2	Incidents of non-compliance concerning the health and safety impacts of products and services	-	-	There were no such incidents in 2024.
Material Topic — Compliance with laws and regulations					
GRI 3 : Material Topics 2021	3-3	Management of material topics	Materiality Analysis 6.3 Regulatory Compliance	13 127	
GRI 205 Anti-corruption 2016	205-3	Confirmed incidents of corruption and actions taken	6.3 Regulatory Compliance	127	
GRI 206 Anti-competitive behavior Topic 2016	206-1	Legal actions for anti-competitive behavior, anti-trust, and monopoly practices	6.3 Regulatory Compliance	127	

GRI Standards Categories / Topics	Number	GRI Standards Disclosure Content	Corresponding Chapters	Page No.	Omitted/ Remarks
Material Topics — Information Security					
GRI 3 : Material Topics 2021	3-3	Management of material topics	Materiality Analysis 6.6 Information Security	13 135	
GRI 418 Customer Privacy 2016	418-1	Substantiated complaints concerning breaches of customer privacy and	6.6 Information Security	135	
Material Topic — Integrity Management					
GRI 3: Material Topics 2021	3-3	Management of material topics	Materiality Analysis 6.2 Integrity Management	13 123	
GRI 205 Anti-corruption 2016	205-1	Operations assessed for risks related to corruption	6.2 Integrity Management	123	
	205-2	Communication and training about anti-corruption policies and procedures	2. 2 Sustainable Supply Chain Management 6.2 Integrity Management	46 123	
Material Topics — Corporate Governance					
GRI 3: Material Topics 2021	3-3	Management of material topics	Materiality Analysis 6.1 Corporate Governance	13 119	
GRI 405 Diversity and Equal Opportunity 2016	405-1	Diversity of governance bodies and employees	4.1 Talent Attraction and Retention 6.1 Corporate Governance	85 119	
Material Topics — Customer Relationship Management					
GRI 3: Material Topics 2021	3-3	Management of material topics	Materiality Analysis 1.3 Customer Relationship Management	13 40	
Material Topic — Sustainable Supply Chain					
GRI 3: Material Topics 2021	3-3	Management of material topics	Materiality Analysis Synergized Sustainability Supply Chain	13 46	
GRI 204 Procurement Practices 2016	204-1	Proportion of spending on local suppliers	2.1 Supply Chain Overview	44	

GRI Standards Categories / Topics	Number	GRI Standards Disclosure Content	Corresponding Chapters	Page No.	Omitted/ Remarks
GRI 205 Anti-corruption 2016	205-2	Communication and training about anti-corruption policies and procedures	2.1 Supply Chain Overview 6.2 Integrity Management	44 123	
GRI 308 Supplier Environmental Assessment 2016	308-1	New suppliers that were screened using environmental criteria	2.2 Sustainable Supply Chain Management	46	
	308-2	Negative environmental impacts in the supply chain and actions taken	2.2 Sustainable Supply Chain Management	46	
GRI 408 Child Labor 2016	408-1	Operations and suppliers at significant risk for incidents of child labor	2.1 Supply Chain Overview 4.3 Human Rights	44 99	
GRI 409 Forced or Compulsory Labor 2016	409-1	Operations and suppliers at significant risk for incidents of forced or compulsory labor	2.1 Supply Chain Overview 4.3 Human Rights	44 99	
GRI 414 Supplier Social Assessment 2016	414-1	New suppliers that were screened using social criteria	2.2 Sustainable Supply Chain Management	46	
	414-2	Negative social impacts in the supply chain and actions taken	2.2 Sustainable Supply Chain Management	46	
Other Disclosures					
GRI 207 Tax 2019	207-1	Approach to taxes	6.4 Responsible Taxation	128	
	207-2	Tax governance, control, and risk management	6.4 Responsible Taxation	128	
	207-3	Stakeholder engagement and management of concerns related to taxes	6.4 Responsible Taxation	128	
Environmental Aspect					
Material Topic - Energy Management					
GRI 3: Material Topics 2021	3-3	Management of material topics	Materiality Analysis 3.2 Energy Management	13 66	

GRI Standards Categories / Topics	Number	GRI Standards Disclosure Content	Corresponding Chapters	Page No.	Omitted/ Remarks
GRI 302 Energy 2016	302-1	Energy consumption within the organization	3.2 Energy Management	66	
	302-3	Energy intensity	3.2 Energy Management	66	
	302-4	Reduction of energy consumption	3.2 Energy Management	66	
Material Topic - Climate Strategy					
GRI 3: Material Topics 2021	3-3	Management of material topics	Materiality Analysis 3.1 Climate Strategy	13 52	
GRI 201 Economic Performance 2016	201-2	Financial implications and other risks and opportunities due to climate change	3.1 Climate Strategy	52	
GRI 305 Emissions 2016	305-1	Direct (Scope 1) Greenhouse Gas (GHG) emissions	3.1 Climate Strategy	52	
	305-2	Energy indirect (Scope 2) Greenhouse Gas (GHG) emissions	3.1 Climate Strategy	52	
	305-3	Other indirect (Scope 3) Greenhouse Gas (GHG) emissions	3.1 Climate Strategy	52	
	305-4	Greenhouse Gas (GHG) emissions intensity	3.1 Climate Strategy	52	
	305-5	Reduction of Greenhouse Gas (GHG) emissions	3.1 Climate Strategy	52	
Material Topic -Waste Management					
GRI 3: Material Topics 2021	3-3	Management of material topics	Materiality Analysis 3.4 Waste Management	13 75	
GRI 306 Waste Material 2020	306-1	Waste generation and significant waste-related impacts	3.4 Waste Management	75	
	306-2	Management of significant waste-related impacts	3.4 Waste Management	75	
	306-3	Waste generated	3.4 Waste Management	75	
	306-4	Waste diverted from disposal	3.4 Waste Management	75	
	306-5	Waste directed to disposal	3.4 Waste Management	75	

GRI Standards Categories / Topics	Number	GRI Standards Disclosure Content	Corresponding Chapters	Page No.	Omitted/ Remarks
Material Topic - Air Pollution Control					
GRI 305 Emissions 2016	305-7	Nitrogen oxides (NOx), sulfur oxides (Sox), and other significant air emissions	3.5 Air Pollution Emissions Management	79	
Other Disclosures					
GRI 303 Water and Effluents 2018	303-3	Water withdrawal	3.3 Water Resources Management	70	
	303-4	Water discharge	3.3 Water Resources Management	70	
	303-5	Water consumption	3.3 Water Resources Management	70	
Social Aspect					
Material Topic - Occupational Health and Safety					
GRI 3: Material Topics 2021	3-3	Management of material topics	Economic Performance Materiality Analysis	13 100	
GRI 403 Occupational Health and Safety 2018	403-1	Occupational health and safety management system	4.4 Occupational Safety and Health Management	100	
	403-2	Hazard identification, risk management, and incident investigation	4.4 Occupational Safety and Health Management	100	
	403-3	Occupational health services	4.4 Occupational Safety and Health Management	100	
	403-4	Worker participation, consultation, and communication on occupational health and safety	4.4 Occupational Safety and Health Management	100	
	403-5	Worker training on occupational health and safety	4.4 Occupational Safety and Health Management	100	
	403-6	Promotion of worker health	4.4 Occupational Safety and Health Management	100	
	403-7	Prevention and mitigation of occupational health and safety impacts directly linked to business relationships	4.4 Occupational Safety and Health Management	100	

GRI Standards Categories / Topics	Number	GRI Standards Disclosure Content	Corresponding Chapters	Page No.	Omitted/ Remarks
GRI 403 Occupational Health and Safety 2018	403-8	Workers covered by the occupational health and safety management system	4.4 Occupational Safety and Health Management	100	
	403-9	Occupational injuries	4.4 Occupational Safety and Health Management	100	
	403-10	Work-related illnesses	4.4 Occupational Safety and Health Management	100	
Material Topic - Talent Attraction and Retention					
GRI 3: Material Topics 2021	3-3	Management of material topics	Materiality Analysis 4.1 Talent Attraction and Retention	13 85	
GRI 401 Employment 2016	401-1	New employee hires and employee turnover	4.1 Talent Attraction and Retention	85	
	401-2	Benefits provided to full-time employees that are not provided to temporary or part-time employees	4.1 Talent Attraction and Retention	85	
	401-3	Parental leave	4.1 Talent Attraction and Retention	85	
GRI 405 Diversity and Equal Opportunity 2016	405-1	Diversity of governance bodies and employees	4.1 Talent Attraction and Retention 6.1 Corporate Governance	85 119	
Material Topic - Talent Nurturing and Development					
GRI 3: Material Topics 2021	3-3	Management of material topics	Materiality Analysis 4.2 Talent Nurturing and Development	13 94	
GRI 402 Labor/ Management Relations 2016	402-1	Minimum notice periods regarding operational changes	4.2 Talent Nurturing and Development	94	
GRI 404 Training and Education 2016	404-1	Average hours of training per year per employee	4.2 Talent Nurturing and Development	94	
	404-3	Percentage of employees receiving regular performance and career development reviews	4.2 Talent Nurturing and Development	94	

GRI Standards Categories / Topics	Number	GRI Standards Disclosure Content	Corresponding Chapters	Page No.	Omitted/ Remarks
Other Disclosures					
GRI 406 Non-Discrimination 2016	406-1	Incidents of discrimination and corrective actions taken	4.3 Human Rights	99	

II : SASB Sustainability Accounting Standards Comparison Table(2023-12 Edition)

Disclosure Topic	Indicator Number	Disclosure Indicators	Characteristics	Description/Corresponding Chapter	Page No.
Greenhouse Gas Emissions	TC-SC-110a.1	- Gross global Scope 1 emissions - Total emissions from perfluorinated compounds	Quantitative	(1) Direct greenhouse gas emissions from Scope 1 were 69,935 tons of CO ₂ e. (2) Perfluorinated greenhouse gas emissions were 14,109 tons of CO ₂ e Please refer to the contents of Chapter 3.1 for details.	52
	TC-SC-110a.2	Discussion of long-term and short-term strategy and plans to manage Scope 1 emissions, emissions reduction targets, and an analysis of performance against those targets	Discussion and Analysis	In order to align with the international trend of carbon reduction, the Company has set reduction targets for Scope 1 and for fluorocarbons. Please refer to the contents of Chapter 3.1 for details. Note: The treatment and destruction of the gas from the intensification process covers the control of perfluorinated compounds.	52
Energy Management During the Manufacturing Process	TC-SC-130a.1	- Total energy consumption - Proportion of grid electricity - Proportion of renewable energy	Quantitative	(1) Total energy consumption was 5,449,380 GJ (2) Total purchased electricity use was 4,802,454 GJ, accounting for 88.04% of total energy use. (3) Total renewable energy use was 49,918.72 GJ, accounting for 0.92% of total energy use. Please refer to the contents of Chapter 3.2 for details.	66
Water Resources Management	TC-SC-140a.1	- Total water withdrawn; proportion of such in regions with high or extremely high baseline water stress - Total water consumed; proportion of such in regions with high or extremely high baseline water stress\	Quantitative	The total amount of water withdrawn was 7,397.62 thousand cubic meters, and the total amount of water consumed was 2,343.20 thousand cubic meters. None of the plants are classified as high risk areas for water resources. Please refer to the contents of Chapter 3.3 for details.	70
Waste Management	TC-SC-150a.1	- Hazardous waste generated during the manufacturing process weight - Proportion recycled	Quantitative	The total amount of hazardous waste generated was 7,868.64 tons, of which 92.10% was recycled and reused. Please refer to the contents of Chapter 3.4 for details.	75
Labor Health and Safety	TC-SC-320a.1	Description of efforts to assess, monitor, and reduce employee exposure to human health hazards	Discussion and Analysis	Please refer to the contents of Chapter 4.4 for details	100
	TC-SC-320a.2	Total monetary loss due to legal incidents related to violations of employee health and safety	Quantitative	In 2024, there were no legal proceedings related to violations of employee health and safety regulations.	-
Recruiting and Managing a Global and Skilled Workforce	TC-SC-330a.1	Percentage of employees requiring work visas	Quantitative	In 2024, the employees that are foreign nationals or located offshore accounted for 5.74% of the entire employees, and employees of the subsidiaries in Japan accounted for 0.42% of the entire employees.	-

Disclosure Topic	Indicator Number	Disclosure Indicators	Characteristics	Description/Corresponding Chapter	Page No.
Product Lifecycle Management	TC-SC-410a.1	• Percentage of revenue from products containing IEC 62474 substances	Quantitative	The percentage of revenue from products containing IEC 62474 substances was 0%.	-
	TC-SC-410a.2	• Processor energy efficiency at a system level for: (1) servers, (2) desktops, and (3) laptops	Quantitative	Not applicable	-
Materials Sourcing	TC-SC-440a.1	• Description of management for risks associated with the use of critical materials	Discussion and Analysis	Please refer to the contents of Chapter 2.1 for details.	44
Intellectual Property Protection & Competitive Behavior	TC-SC-520a.1	• Total monetary losses as a result of legal proceedings associated with anticompetitive behavior regulations	Quantitative	No such incidents occurred in 2024 and there were no monetary losses.	-
Activity Indicator	TC-SC-000.A	• Total production	Quantitative	107,205.94 (m ² /year)	-
	TC-SC-000.B	• Percentage of self-owned plant production	Quantitative	The percentage of self-owned plant production was 100%.	-

III : TCFD Climate Related Financial Disclosure Framework Comparison Table

Level	Disclosure Item	General Industry Metrics	Corresponding Chapter	Page No.
Governance	a	Describe the Board's oversight status of climate related risks and opportunities.	3.1 Climate Strategy	52
	b	Describe management's role in assessing and managing climate-related risks and opportunities.		
Strategy	a	Describe the short-term, medium-term, and long-term climate-related risks and opportunities identified by the organization.		
	b	Describe the impacts of climate related risks and opportunities on the organization's business, strategic, and financial planning.		
	c	Describe the organization's strategic resilience, taking into account physical climate risks for different climate-related scenarios, including 2°C or even more severe scenarios.		
Risk Management	a	Describe the organization's processes for identifying and assessing climate-related risks.		
	b	Describe the organization's process for managing climate-related risks.		
	c	Describe how the identification, assessment and management processes for climate related risks are integrated into the organization's overall risk management system.		
Metrics and Targets	a	Disclose the metrics used by the organization to conduct climate related risk and opportunity assessments in accordance with the strategy and the risk management process.		
	b	Disclosure of greenhouse gas emissions and associated risks in Scope 1, Scope 2 and Scope 3 (where applicable).		
	c	Describe the objectives being used by the organization to manage climate related risks and opportunities, and the performances in achieving those objectives.		

IV: International Sustainability Standards Board (ISSB) Sustainability Reporting Standards Content Index

International Financial Reporting Standard (IFRS) S1 General Requirements for Disclosure of Sustainability-related Financial Information

Core Content

Core Content	Disclosure Requirements	Corresponding Chapter	Page No.
 Governance	The governance processes, controls and procedures an entity uses to monitor, manage and oversee sustainability-related risks and opportunities.	Organizational Structure for Sustainable Development	10
		3.1 Climate Strategy	52
		6.5 Risk Management	129
 Strategy	An entity's strategy for managing sustainability-related risks and opportunities.	Sustainable Development Strategy	10
		3.1 Climate Strategy	13
		Materiality Analysis	52
 Risk Management	An entity's processes to identify, assess, prioritize and monitor sustainability-related risks and opportunities.	Materiality Analysis	13
		3.1 Climate Strategy	52
		6.5 Risk Management	129
 Metrics and Targets	An entity's performance in relation to its sustainability-related risks and opportunities, including progress toward any targets the entity has set, and any targets it is required to meet by law or regulation.	Sustainability Performance and Honors	8
		3.1 Climate Strategy	13
		Materiality Analysis	52

General Requirements

S1 Standard	Disclosure Requirements	Corresponding Chapter	Page No.
Sources of Guidance	Identifying entity-specific sustainability-related risks and opportunities, identifying applicable disclosure requirements, and disclosing information about sources of guidance.	About this Report	139
Location of disclosures	An entity is required to provide disclosures required by IFRS Sustainability Disclosure Standards as part of its general purpose financial reports.	About this Report	139
Timing of Reporting	The entity's sustainability-related financial disclosures shall cover the same reporting period as the related financial statements.	About this Report	139
Comparative Information	An entity shall disclose comparative information in respect of the preceding period for all amounts disclosed in the reporting period.	About this Report	139
Statement of Compliance	An entity whose sustainability-related financial disclosures comply with all the requirements of the IFRS Sustainability Disclosure Standards shall make an explicit and unreserved statement of compliance.	About this Report	139

Judgements, Uncertainties and Errors

S1 Standard	Disclosure Requirements	Corresponding Chapter	Page No.
Judgements	An entity shall disclose information to enable users of general purpose financial reports to understand the judgements that the entity has made in the process of preparing its sustainability-related financial disclosures and that have the most significant effect on the information included in those disclosures.	About this Report Materiality Analysis	13 139
Measurement Uncertainty	An entity shall disclose information to enable users of general purpose financial reports to understand the most significant uncertainties affecting the amounts reported in its sustainability-related financial disclosures.	About this Report	139
Errors	An entity shall correct material prior period errors by restating the comparative amounts for the prior period(s) disclosed unless it is impracticable to do so.	About this Report	139

Implementation Plan Schedule

	Phase 1	Phase 2	Phase 3	Phase 4
Tasks	1-1. Establish a cross-departmental project team for adopting the IFRS Sustainability Disclosure Standards. 1-2. Conduct a preliminary identification of significant differences and impacts between current sustainability information and the IFRS Sustainability Disclosure Standards. 1-3. Conduct a preliminary identification of reporting entities. 1-4. Develop an implementation plan.	2-1. Identify sustainability-related risks and opportunities, as well as financial impacts, and assess material sustainability-related financial information. 2-2. Identify and collect necessary data. 2-3. Revise and adjust corporate processes, including financial and non-financial reporting, information systems, supply chain management processes, internal controls, and daily operations across departments.	3-1. Trial preparation of the annual report's sustainability information section. 3-2. Continuously update the internal control operations manual in accordance with IFRS sustainability information and conduct training sessions.	4-1. Official disclosure and submission of the annual report's sustainability information section.
Expected Completion (or Actual Completion) Time	1-1. Completed in Q4 2024. 1-2. Completed in Q1 2025. 1-3. Completed in Q1 2025. 1-4. Completed in Q4 2024.	2-1. Expected completion in Q2 2025. 2-2. Expected completion in Q4 2025. 2-3. Expected completion in Q3 2026.	3-1. Expected completion in Q3 2026. 3-2. Expected completion in Q4 2026.	4-1. Expected completion in Q1 2027.

International Financial Reporting Standard (IFRS) S2 Climate-related Disclosures

The International Financial Reporting Standard (IFRS) S2 Climate-related Disclosures aims to require businesses to disclose information related to climate change risks and opportunities, covering transition risks, physical risks, and climate-related opportunities faced by companies. This standard is built upon the four core elements of the Task Force on Climate-related Financial Disclosures (TCFD) framework: Governance; Strategy; Risk Management; and Metrics & Targets. These ensure that stakeholders can effectively assess PSMC's climate-related risks and opportunities, thereby supporting resource allocation and decision-making. For detailed information on PSMC's climate-related disclosures, please refer to the [3.1 Climate Strategy](#) section of this Report.

To successfully implement the IFRS Sustainability Disclosure Standards, comply with regulatory requirements, and enhance information transparency, PSMC has aligned with the Roadmap for Taiwan Listed Companies to Align with IFRS Sustainability Disclosure Standards issued by the Financial Supervisory Commission (FSC) on August 17, 2023. Furthermore, PSMC has referred to the IFRS Sustainability Disclosure Standards Alignment Section and the exclusive roadmap for listed companies to facilitate this transition. By the end of 2024, PSMC established a cross-departmental project team and formulated an implementation plan and timeline. The Company has also engaged specialized institutions to assist in initiating the implementation plan. The rollout is progressing in accordance with the timeline, and the implementation status will be reported to the Board of Directors on a quarterly basis for monitoring and control.

V : Comparison of "The Operating Procedures for the Preparation and Filing of Sustainability Reports by Publicly Traded Companies" of the Taiwan Stock Exchange

Schedule 1.8 - Sustainability Disclosure Indicators - Semiconductor Industry

Number	Indicator Description	Description or Corresponding Chapter	Page No.
I	Total energy consumption, percentage of purchased electricity and renewable energy usage rate	Total energy consumption: 5,411,252GJ Percentage of purchased electricity: 88.04% Renewable energy usage rate: 0.93%	-
II	Total water withdrawal and total water consumption	Total water withdrawal: 7,397.62 thousand cubic meters Total water consumption: 2,343.20 thousand cubic meters	-
III	Weight of hazardous waste generated and percentage recycled	Total amount of hazardous waste: 7,868.64 tons Percentage of waste recycled and reused: 92.10%	-
IV	Explain the type, number of people, and percentage of occupational disasters	Please refer to the contents of Chapter 4.4 for details.	100
V	Disclosure of product lifecycle management: weight and percentage of recycling, including scrapped products and electronic waste	Total waste generation of approximately 18,895.99 metric tons in 2024, with a recycling rate of 95.79%	-
VI	Description of risk management related to the use of critical materials	Please refer to the contents of Chapter 2.1 for details.	44
VII	Total monetary losses due to lawsuits related to anti-competitive behavior regulations	No such incidents occurred in 2024, and there were no monetary losses	-
VIII	Production of major products by product category	107,205.94 (m ² /year)	-

Schedule II – Climate Related Information for Publicly Traded Companies -1- The Risks and Opportunities that climate change poses to the company and the countermeasures taken by the company

Item	Execution Status
Describe oversight status and governance of the Board of Directors and the management role for climate-related risks and opportunities.	Please refer to the contents of Chapter 3.1 for details.
Describe how the impacts of identified climate risks and opportunities on the organization's business, strategic, and financial planning. (Short-term, medium-term and long-term)	
Describe the impacts of extreme weather events and transformation actions on the financial planning.	
Describe how the identification, assessment and management processes for climate-related risks are integrated into the organization's overall risk management system.	
Describe the use of scenarios, parameters, assumptions, analysis factors and the major impacts on financial planning, if resilience of climate change risk is based on the assessment of scenario analysis.	Please refer to materiality analysis and the contents of Chapter 3.1 for details.
Describe the details of transformation planning, indicators and targets used by the organization to assess and manage physical risk and transformation risk, if there is transformation planning in response to the management of climate related risk.	
Describe the pricing basis, if planning tool is on the basis of internal carbon pricing.	Evaluate the future introduction of an internal carbon pricing mechanism
If climate-related goals are set, the activities covered, the scope of greenhouse gas emissions, the planning schedule, annual achievement progress and other information should be described, if carbon offsets or renewable energy certificates (RECs) are used to achieve relevant goals, the source and quantity of offset carbon reduction credits or quantity of renewable energy certificates (RECs) should be described.	Please refer to the contents of Chapter 3.1 for details.
Greenhouse gas inventory and assurance, reduction targets, strategies and specific action plans	Also fill in 1-1 and 1-2 of Schedule II

Schedule II – Climate Related Information for Publicly Traded Companies 1-1 - Greenhouse Gas Inventory and Assurance Results in the last two years

Item	Execution Status / Corresponding Chapter				
1-1-1 GHG Inventory Information: Describe the GHG emission volume (metric tons CO ₂ e), intensity (metric tons CO ₂ e / NTD\$ million) in the last two years, and the scope of data coverage.	Data Year	2023 年		2024 年	
	Scope of data coverage	Parent company + Subsidiaries		Parent company + Subsidiaries	
	Item	Total emissions (metric tons CO ₂ e)	The intensity (metric tons CO ₂ e/NT\$1,000)	Total emissions (metric tons CO ₂ e)	The intensity (metric tons CO ₂ e/NT\$1,000)
	Scope I	49,145	0.0011	65,935	0.0015
	Scope II	534,354	0.0121	640,752	0.0143
	Total	583,499	0.0133	706,687	0.0158
	Scope III	171,863	0.0039	214,549	0.0048
1-1-2 GHG Assurance Information: Describe the assurance overview in the last two years, including assurance scope, assurance institution, assurance criteria and assurance opinion.	Data Year	2023		2024	
	Assurance scope	Parent company		Parent company	
	Assurance institution	The British Standards Institution (BSI)		The British Standards Institution (BSI)	
	Assurance criteria	ISO 14064-3		ISO 14064-3	
	Assurance opinion	Please refer to our official website for ISO 14064-1 certificate.		Please refer to our official website for ISO 14064-1 certificate.	

Schedule II – Climate Related Information for Publicly Traded Companies 1-2-Greenhouse Gas Reduction Goals, Strategies and Specific Action Plans

Item	Execution Status
Describe the greenhouse gas reduction base year and its data, reduction targets, strategies, specific action plans and achievement of reduction targets.	Please refer to materiality analysis and the contents of Chapter 3.1 for details.

VI : United Nations Global Compact (UNGC) Comparison Table

Item No.	Article of Principle	Corresponding Chapter	Page No.
Human Rights Area			
1	Businesses should support and respect the protection of internationally proclaimed human rights	4.3 Human Rights	99
2	Businesses should make sure that they are not complicit in human rights abuses	4.3 Human Rights	99
Labor Area			
3	Businesses should uphold the freedom of association and the effective recognition of the right to collective bargaining	4.4 Occupational health and safety	100
4	Elimination of all forms of forced and compulsory labor	4.3 Human Rights	99
5	Effective abolition of child labor	4.3 Human Rights	99
6	Elimination of discrimination in respect of employment and occupation	4.1 Talent Attraction and Retention 4.3 Human Rights	85 99
Environmental Area			
7	Support a precautionary approach to environmental challenges	3.1 Climate Strategy 3.2 Energy Management 3.3 Water Resource Management 3.4 Waste Management 3.5 Air Pollution Emission Management 3.6 Biodiversity Management	52 66 70 75 79 81

Item No.	Article of Principle	Corresponding Chapter	Page No.
8	Undertake initiatives to promote greater environmental responsibility	3.1 Climate Strategy 3.2 Energy Management 3.3 Water Resource Management 3.4 Waste Management 3.5 Air Pollution Emission Management 3.6 Biodiversity Management	52 66 70 75 79 81
9	Encourage the development and diffusion of environmentally friendly technologies	3.1 Climate Strategy 3.2 Energy Management 3.3 Water Resource Management 3.4 Waste Management 3.5 Air Pollution Emission Management	52 66 70 75 79
Anti-corruption Area			
10	Businesses should work against corruption in all its forms, including extortion and bribery	6.2 Integrity Management	123



VII : ISO 26000 Guidance on Social Responsibility Standards Comparison Table

Core Subjects	Article of Principle	Corresponding Chapter	Page No.	
Organizational Governance	A system for making and implementing decisions for the execu-tion of the organization's objectives	Company Profile	4	
Human Rights	Verification of compliance with regulations and avoidance of risks caused by human rights issues	6.3 Regulatory Compliance	127	
	Human rights risk situations	6.5 Risk Management	129	
	Avoidance of complicity — direct, beneficial, and silent con-spiratorial relationships (avoidance of complicity)	6.2 Integrity Management	123	
	Resolving grievances (resolving gripes and complaints)	4.3 Human Rights	99	
	Discrimination and vulnerable groups	4.3 Human Rights	99	
	Civil and political rights	4.1 Talent Attraction and Re-tention	85	
	Economic, social, and cultural rights	4.1 Talent Attraction and Re-tention	85	
	Fundamental principles and rights at work	4.1 Talent Attraction and Re-tention 4.2 Talent Nurturing and De-velopment	85 94	
	Employment and employment relationships	4.1 Talent Attraction and Re-tention 4.2 Talent Nurturing and De-velopment	85 94	
	Conditions of work and social protections	4.1 Talent Attraction and Re-tention 4.2 Talent Nurturing and De-velopment	85 94	
Labor Practices	Social dialogue	4.1 Talent Attraction and Re-tention 4.2 Talent Nurturing and De-velopment	85 94	
	Health and safety at work	4.4 Occupational health and safety	100	
	Human resources development and training in the workplace	4.2 Talent Nurturing and De-velopment	94	
	The Environment	Prevention of pollution	3.2 Energy Management	66
			3.3 Water Resource Man-agement	70
3.4 Waste Management			75	
3.5 Air Pollution Emission Management			79	
Sustainable resource use		3.2 Energy Management	66	
		3.3 Water Resource Man-agement	70	
	3.4 Waste Management	75		

Core Subjects	Article of Principle	Corresponding Chapter	Page No.
The Environment	Climate change mitigation and adaptation	3.1 Climate Strategy	52
	Protection of the environment, biodiversity, and restoration of natural habitats	3.6 Biodiversity Management	81
Fair Operating Practices	Anti-corruption	6.2 Integrity Management	123
	Responsible political involvement	6.2 Integrity Management	123
	Fair competition	6.2 Credibility Management	123
	Promoting social responsibility in the value chain	6.1 Integrity Management	119
		6.2 Integrity Management	123
		6.3 Regulatory Compliance	127
		6.4 Responsible Taxation	128
	Respect for property rights	1.1 Innovation and Intellec-tual Property Management	25
Consumer Issues	Fair marketing, factual and unbiased information, and fair contractual practices	2.2 Sustainable Supply Chain Management	46
	Protecting consumers' health and safety	1.3 Customer Relationship Management	40
	Sustainable consumption	2.2 Sustainable Supply Chain Management	46
	Consumer service, support, and complaint and dispute resolu-tion	1.3 Customer Relationship Management	40
	Consumer data protection and privacy	6.6 Information Security	135
	Access to essential services	1.3 Customer Relationship Management	40
	Education and Awareness	4.2 Talent Nurturing and De-velopment	94
	Community Involvement	5.1 Social Influence	110
	Education and culture	5.1 Social Influence	110
	Employment creation and skills development	4.2 Talent Nurturing and De-velopment	94
Community Involvement and Development	Technology development	1.1 Innovation and Intellec-tual Property Management	25
	Wealth and income creation	Economic Performance	7
		1.1 Innovation and Intellec-tual Property Management	25
	Health	4.4 Occupational health and safety	100
	Social investment	5.1 Social Influence	110

Independent Third Party Verification Statement



INDEPENDENT ASSURANCE OPINION STATEMENT

Powerchip Semiconductor Manufacturing Corporation 2024 Sustainability Report

The British Standards Institution is independent to Powerchip Semiconductor Manufacturing Corporation (hereafter referred to as PSMC in this statement) and has no financial interest in the operation of PSMC other than for the assessment and verification of the sustainability statements contained in this report.

This independent assurance opinion statement has been prepared for the stakeholders of PSMC only for the purposes of assuring its statements relating to its sustainability report, more particularly described in the Scope below. It was not prepared for any other purpose. The British Standards Institution will not, in providing this independent assurance opinion statement, accept or assume responsibility (legal or otherwise) or accept liability for or in connection with any other purpose for which it may be used, or to any person by whom the independent assurance opinion statement may be read.

This independent assurance opinion statement is prepared on the basis of review by the British Standards Institution of information presented to it by PSMC. The review does not extend beyond such information and is solely based on it. In performing such review, the British Standards Institution has assumed that all such information is complete and accurate.

Any queries that may arise by virtue of this independent assurance opinion statement or matters relating to it should be addressed to PSMC only.

Scope

The scope of engagement agreed upon with PSMC includes the followings:

1. The assurance scope is consistent with the description of Powerchip Semiconductor Manufacturing Corporation.
2. The evaluation of the nature and extent of the PSMC's adherence to AA1000 AccountAbility Principles (2018) and the reliability of specified sustainability performance information in this report as conducted in accordance with type 2 of AA1000AS v3 sustainability assurance engagement.
3. The assessment of disclosure to be in conformance with the applicable SASB industry standard(s) in this report as conducted in accordance with type 1 of AA1000AS v3 sustainability assurance engagement.

This statement was prepared in English and translated into Chinese for reference only.

Opinion Statement

We conclude that the Powerchip Semiconductor Manufacturing Corporation provides a fair view of the PSMC sustainability programmes and performances during 2024. The sustainability report subject to assurance is materially correct without voluntary omissions based upon testing within the limitations of the scope of the assurance, the information and data provided by the PSMC and the sample taken. We believe that the performance information of Environment, Social and Governance (ESG) are correctly represented. The sustainability performance information disclosed in the report demonstrate PSMC's efforts recognized by its stakeholders.

Our work was carried out by a team of sustainability report assurers in accordance with the AA1000AS v3. We planned and performed this part of our work to obtain the necessary information and explanations we considered to provide sufficient evidence that PSMC's description of their approach to AA1000AS v3 and their self-declaration in accordance with GRI Standards and SASB Standard(s) were fairly stated.

Methodology

Our work was designed to gather evidence on which to base our conclusion. We undertook the following activities:

- a top level review of issues raised by external parties that could be relevant to PSMC's policies to provide a check on the appropriateness of statements made in the report.
- discussion with managers on PSMC's approach to stakeholder engagement. Moreover, we had sampled 2 external stakeholders to conduct interview.
- interview with 49 staffs involved in sustainability management, report preparation and provision of report information were carried out.
- review of materiality assessment process.
- review of key organizational developments.
- review of the extent and maturity of the relevant accounting systems for financial and non-financial reports.
- review of the findings of internal audits.
- the verification of performance data and claims made in the report through meeting with managers responsible for gathering data.
- review of the processes for gathering and ensuring the accuracy of data, followed data trails to initial aggregated source and checked sample data to greater depth during site visits.
- the consolidated financial data are based on audited financial data, we checked that this data was consistently reproduced.
- review of supporting evidence for claims made in the reports.
- an assessment of the organization's reporting and management processes concerning this reporting against the principles of Inclusivity, Materiality, Responsiveness, and Impact as described in the AA1000AP (2018).
- an assessment of the organization's use of metrics or targets of SASB Standard to assess and manage topic-related risks and opportunities.

Conclusions

A detailed review against the Inclusivity, Materiality, Responsiveness, and Impact of AA1000AP (2018) and sustainability performance information as well as GRI Standards and SASB Standard(s) is set out below:

Inclusivity

In this report, it reflects that PSMC has continually sought the engagement of its stakeholders and established material sustainability topics, as the participation of stakeholders has been conducted in developing and achieving an accountable and strategic response to sustainability. There are fair reporting and disclosures for the information of Environment, Social and Governance (ESG) in this report, so that appropriate planning and target-setting can be supported. In our professional opinion the report covers the PSMC's inclusivity issues and has demonstrated sustainable conduct supported by top management and implemented in all levels among organization.

Materiality

The PSMC publishes material topics that will substantively influence and impact the assessments, decisions, actions and performance of PSMC and its stakeholders. The sustainability information disclosed enables its stakeholders to make informed judgements about the PSMC's management and performance. In our professional opinion the report covers the PSMC's materiality assessment process and material issues.

Responsiveness

PSMC has implemented the practice to respond to the expectations and perceptions of its stakeholders. An Ethical Policy for the PSMC is developed and continually provides the opportunity to further enhance PSMC's responsiveness to stakeholder concerns. Topics that stakeholder concern about have been responded timely. In our professional opinion the report covers the PSMC's responsiveness issues.

Impact

PSMC has identified and fairly represented impacts that were measured and disclosed in probably balanced and effective way. PSMC has established processes to monitor, measure, evaluate, and manage impacts that lead to more effective decision-making and results-based management within an organization. In our professional opinion the report covers the PSMC's impact issues.

Performance information

Based on our work described in this statement and with no conflict of interest with the PSMC in relation to providing the assurance of ESG performance information, which has been assured, specified sustainability performance information such as GRI Standards disclosures disclosed in this report, PSMC and BSI have agreed upon to include in the scope. In our view, the data and information contained within Powerchip Semiconductor Manufacturing Corporation 2024 Sustainability Report are reliable.

GRI Sustainability Reporting Standards (GRI Standards)

PSMC provided us with their self-declaration of in accordance with GRI Standards 2021 (For each material topic covered in the applicable GRI Sector Standard and relevant GRI Topic Standard, including the disclosures of applicable economic, environmental, and social information, comply with all reporting requirements for disclosures). Based on our review, we confirm that sustainable development disclosures with reference to GRI Standards' disclosures are reported, partially reported, or omitted. In our professional opinion the self-declaration covers the PSMC's sustainability topics.

SASB Standards

PSMC provided us with their self-declaration of in accordance with SASB Standard(s) (Semiconductors Sustainability Accounting Standard, version 2023-12). Based on our review, we confirm that the sustainability disclosure topics & accounting metrics of SASB Standard(s) (Semiconductors Sustainability Accounting Standard, version 2023-12) are reported, partially reported, or omitted. In our professional opinion, the self-declaration covers disclosure topics, associated accounting metrics and activity metrics for applicable SASB industry standard(s).

Assurance level

The high level assurance provided is in accordance with AA1000AS v3 in our review, as defined by the scope and methodology described in this statement.

The moderate level assurance provided is in accordance with AA1000AS v3 in our review of SASB Standard(s).

Responsibility

This sustainability report is the responsibility of the PSMC's chairman as declared in his responsibility letter. Our responsibility is to provide an independent assurance opinion statement to stakeholders giving our professional opinion based on the scope and methodology described.

Competency and Independence

The assurance team was composed of auditors experienced in relevant sectors, and trained in a range of sustainability, environmental and social standards including AA1000AS, ISO 14001, ISO 45001, ISO 14064, and ISO 9001. BSI is a leading global standards and assessment body founded in 1901. The assurance is carried out in line with the BSI Fair Trading Code of Practice.



...making excellence a habit.™

Statement No: SRA-TW-804452
2025-04-24

For and on behalf of BSI:


Peter Pu, Managing Director BSI Taiwan

Taiwan Headquarters: 2nd Floor, No. 37, Ji-Hua Rd., Ni-Hu Dist., Taipei 114, Taiwan, R.O.C.
A Member of the BSI Group of Companies.



PSMC

Powerchip Semiconductor Manufacturing Corp.

No. 18, Lixing 1st Rd, Hsinchu Science Park, Hsinchu City, Taiwan

TEL : 886-3-5795000

FAX : 886-3-5788565

